PCOS ULTRASOUND CLASSIFICATION — MODELS_DL: EVERY RUN

Every result printed in the notebooks of Models_DL.zip (17 notebooks): 58 runs, including repeated runs of the same model + classifier. All of these were single exploratory runs (no fixed seed controlling the network training, or one seed without a repeat run) and are reported for completeness. Test set: 1,468 images (1,052 Appears abnormal, 416 Appears normal).

1. Overall Comparison — Best to Worst by Test Accuracy

Rank	Model	Classifier / run	Test Acc. (%)	Prec. (%)	Recall (%)	F1 (%)	Source notebook (.ipynb)
1	MobileNetV2	AdaBoost (dedicated AdaBoost notebook)	93.26	93.19	93.26	93.18	mobilenet with adaboost.ipynb (cell 3)
2	MobileNetV2	AdaBoost	93.12	93.12	93.12	93.12	MobileNet with ML.ipynb (cell 10)
3	EfficientNet-B3	AdaBoost	92.98	92.91	92.98	92.91	efficinetnet b3 baseline and ml.ipynb (cell 11)
4	EfficientNet-B3	SVM	92.37	92.53	92.37	92.43	efficinetnet b3 baseline and ml.ipynb (cell 15)
5	EfficientNet-B3	Random Forest	92.37	92.40	92.37	92.17	efficinetnet b3 baseline and ml.ipynb (cell 5)
6	EfficientNet-B3	XGBoost	92.17	92.19	92.17	91.96	efficinetnet b3 baseline and ml.ipynb (cell 13)
7	MobileNetV2	Direct (fine-tuned CNN) (Random Forest section run)	92.10	92.07	92.10	92.08	MobileNet with ML.ipynb (cell 4)
8	MobileNetV2	Direct (fine-tuned CNN) (Gaussian Naive Bayes section run)	92.03	91.97	92.03	91.88	MobileNet with ML.ipynb (cell 8)
9	InceptionV3	Direct (fine-tuned CNN) (baseline)	91.76	91.80	91.76	91.52	Inception v3 baseline and ml.ipynb (cell 3)
10	MobileNetV2	Direct (fine-tuned CNN) (normal split)	91.49	91.46	91.49	91.27	MobileNet with Preprocessing.ipynb (cell 6)
11	MobileNetV2	Gaussian Naive Bayes	91.42	91.85	91.42	91.54	MobileNet with ML.ipynb (cell 8)
12	MobileNetV2	Direct (fine-tuned CNN) (Dice Loss baseline)	91.21	91.37	91.21	90.88	MobileNet with DiceLoss (BaseLine).ipynb (cell 3)
13	EfficientNet-B3	Direct (fine-tuned CNN) (baseline)	90.67	91.15	90.67	90.18	efficinetnet b3 baseline and ml.ipynb (cell 3)
14	ResNet18	Random Forest	90.60	90.54	90.60	90.56	ResNet with ML.ipynb (cell 4)
15	MobileNetV2	Random Forest	90.40	90.26	90.40	90.29	MobileNet with ML.ipynb (cell 4)
16	MobileNetV2	Direct (fine-tuned CNN) (AdaBoost notebook run)	90.40	90.48	90.40	90.04	mobilenet with adaboost.ipynb (cell 3)
17	VGG16	Direct (fine-tuned CNN) (Dice Loss baseline)	90.33	90.56	90.33	89.90	Vgg with Dice Loss (Baseline).ipynb (cell 5)
18	VGG16	SVM	90.05	90.55	90.05	90.20	Xgboost and SVM and all 4 models.ipynb (cell 27)
19	EfficientNet-B3	KNN	90.05	89.97	90.05	89.78	efficinetnet b3 baseline and ml.ipynb (cell 7)
20	MobileNetV2	KNN	89.51	89.40	89.51	89.21	MobileNet with ML.ipynb (cell 6)
21	InceptionV3	SVM	89.24	89.40	89.24	89.30	Inception v3 baseline and ml.ipynb (cell 16)
22	MobileNetV2	SVM	88.96	89.40	88.96	89.11	Xgboost and SVM and all 4 models.ipynb (cell 15)
23	ResNet18	KNN	88.90	88.70	88.90	88.71	ResNet with ML.ipynb (cell 6)
24	ResNet18	XGBoost	88.69	88.60	88.69	88.64	Xgboost and SVM and all 4 models.ipynb (cell 7)

Rank	Model	Classifier / run	Test Acc. (%)	Prec. (%)	Recall (%)	F1 (%)	Source notebook (.ipynb)
25	ResNet18	Direct (fine-tuned CNN) (Random Forest section run)	88.42	88.74	88.42	87.76	ResNet with ML.ipynb (cell 4)
26	ResNet18	Direct (fine-tuned CNN) (Gaussian Naive Bayes section run)	88.28	88.09	88.28	87.93	ResNet with ML.ipynb (cell 8)
27	MobileNetV2	Direct (fine-tuned CNN) (KNN section run)	88.15	88.53	88.15	87.42	MobileNet with ML.ipynb (cell 6)
28	MobileNetV2	Direct (fine-tuned CNN) (AdaBoost section run)	88.15	88.69	88.15	87.36	MobileNet with ML.ipynb (cell 10)
29	InceptionV3	Random Forest	87.74	88.02	87.74	87.85	Inception v3 baseline and ml.ipynb (cell 6)
30	VGG16	KNN	87.47	87.36	87.47	86.92	Vgg with ML.ipynb (cell 6)
31	ResNet18	AdaBoost	87.19	87.43	87.19	87.29	ResNet with ML.ipynb (cell 10)
32	MobileNetV2	Direct — 5-fold CV	87.19	87.76	87.19	86.26	MobileNet with Kfold.ipynb (cell 3)
33	InceptionV3	AdaBoost	87.06	87.37	87.06	87.18	Inception v3 baseline and ml.ipynb (cell 12)
34	VGG16	XGBoost	86.92	86.64	86.92	86.67	Xgboost and SVM and all 4 models.ipynb (cell 25)
35	InceptionV3	XGBoost	86.78	86.82	86.78	86.80	Inception v3 baseline and ml.ipynb (cell 14)
36	ResNet18	SVM	86.72	88.67	86.72	87.14	Xgboost and SVM and all 4 models.ipynb (cell 9)
37	MobileNetV2	XGBoost	86.72	86.65	86.72	86.68	Xgboost and SVM and all 4 models.ipynb (cell 13)
38	AlexNet	Direct — 5-fold CV	86.58	86.44	86.58	85.95	AlexNet with kfold.ipynb (cell 3)
39	EfficientNet-B3	Gaussian Naive Bayes	86.31	87.69	86.31	84.95	efficinetnet b3 baseline and ml.ipynb (cell 9)
40	ResNet18	Direct (fine-tuned CNN) (AdaBoost section run)	86.10	85.76	86.10	85.68	ResNet with ML.ipynb (cell 10)
41	AlexNet	XGBoost	85.56	86.08	85.56	85.75	Xgboost and SVM and all 4 models.ipynb (cell 19)
42	AlexNet	Direct (fine-tuned CNN) (Dice Loss baseline)	85.42	85.49	85.42	85.45	AlexNet with Dice Loss (Baseline).ipynb (cell 3)
43	VGG16	Direct — 5-fold CV with preprocessing	85.35	86.79	85.35	83.78	MobileNet with Preprocessing.ipynb (cell 4)
44	InceptionV3	KNN	84.95	85.37	84.95	85.11	Inception v3 baseline and ml.ipynb (cell 8)
45	AlexNet	SVM	84.74	87.11	84.74	85.26	Xgboost and SVM and all 4 models.ipynb (cell 21)
46	VGG16	Direct — 5-fold CV	84.20	85.94	84.20	82.25	Vgg kfold.ipynb (cell 3)
47	ResNet18	Direct (fine-tuned CNN) (Dice Loss baseline)	83.45	83.18	83.45	83.29	ResNet with Dice Loss (BaseLine).ipynb (cell 3)
48	InceptionV3	Gaussian Naive Bayes	83.38	86.40	83.38	84.01	Inception v3 baseline and ml.ipynb (cell 10)
49	VGG16	Gaussian Naive Bayes	83.04	82.70	83.04	82.82	Vgg with ML.ipynb (cell 8)
50	ResNet18	Direct — 5-fold CV	82.63	82.35	82.63	81.29	ResNet wthi Kfold.ipynb (cell 3)
51	AlexNet	KNN (second run; notebook section titled "Adaboost")	82.36	82.24	82.36	80.79	AlexNet with ML.ipynb (cell 10)
52	ResNet18	Direct (fine-tuned CNN) (KNN section run)	82.02	84.31	82.02	79.22	ResNet with ML.ipynb (cell 6)
53	AlexNet	Random Forest	81.61	81.05	81.61	80.25	AlexNet with ML.ipynb (cell 4)

Rank	Model	Classifier / run	Test Acc. (%)	Prec. (%)	Recall (%)	F1 (%)	Source notebook (.ipynb)
54	AlexNet	KNN	80.25	79.79	80.25	78.23	AlexNet with ML.ipynb (cell 6)
55	VGG16	AdaBoost	76.63	76.34	76.63	72.32	Vgg with ML.ipynb (cell 10)
56	AlexNet	Gaussian Naive Bayes	75.07	83.51	75.07	76.30	AlexNet with ML.ipynb (cell 8)
57	VGG16	Random Forest	74.73	80.21	74.73	66.63	Vgg with ML.ipynb (cell 4)
58	ResNet18	Gaussian Naive Bayes	67.57	74.25	67.57	69.12	ResNet with ML.ipynb (cell 8)

- 58 runs in total; none is merged or dropped, so the same model + classifier can appear several times (different notebooks, or repeated runs inside one notebook).
- Precision, Recall and F1-score are class-weighted averages.
- Section 3 lists all further notes.

2. Results, Confusion Matrices and Graphs — Table Order

1. MobileNetV2 + AdaBoost (dedicated AdaBoost notebook)

Source notebook: mobilenet with adaboost.ipynb (inside Models_DL.zip)

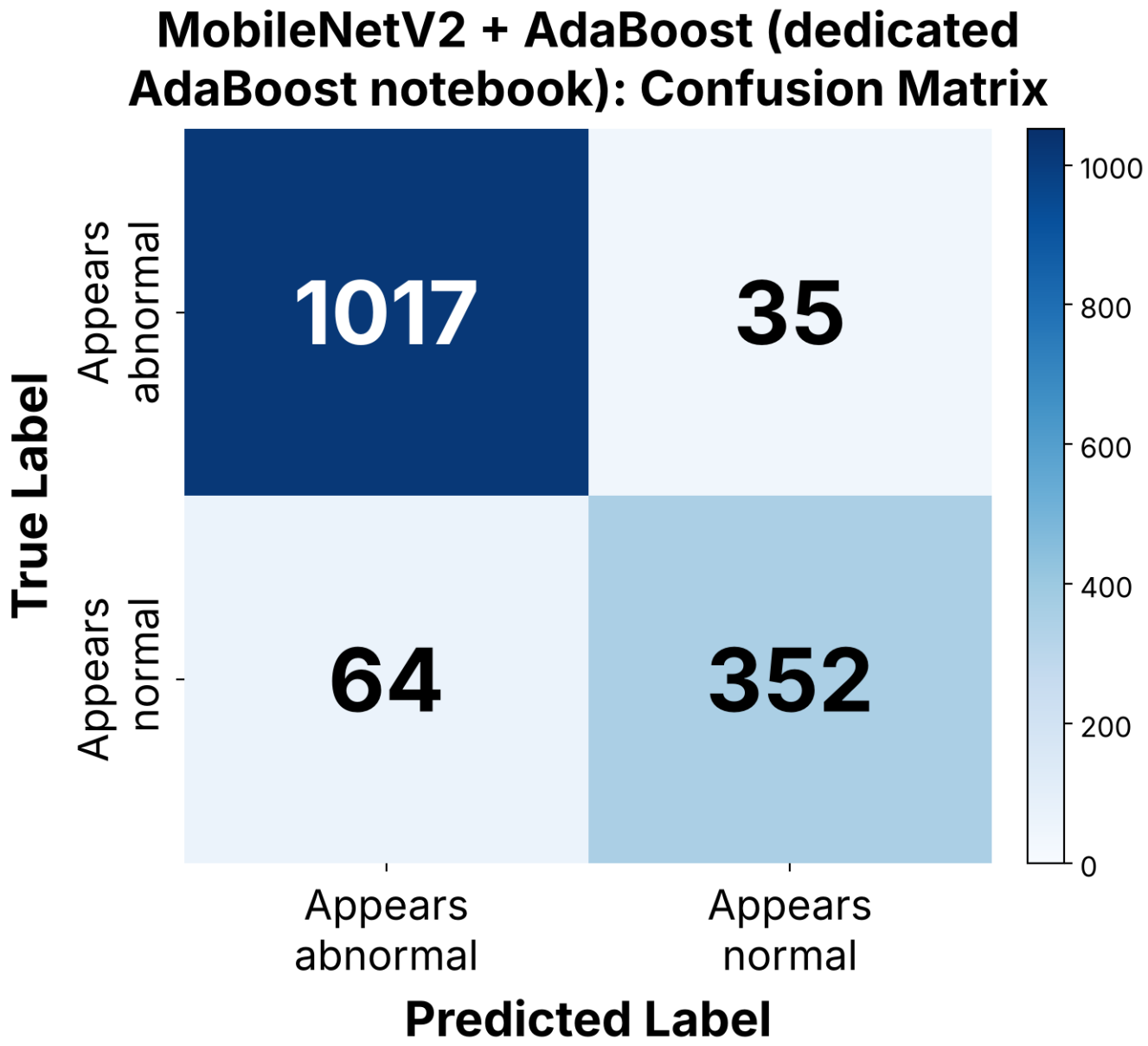
Accuracy: 93.26%

Precision: 93.19%

Recall: 93.26%

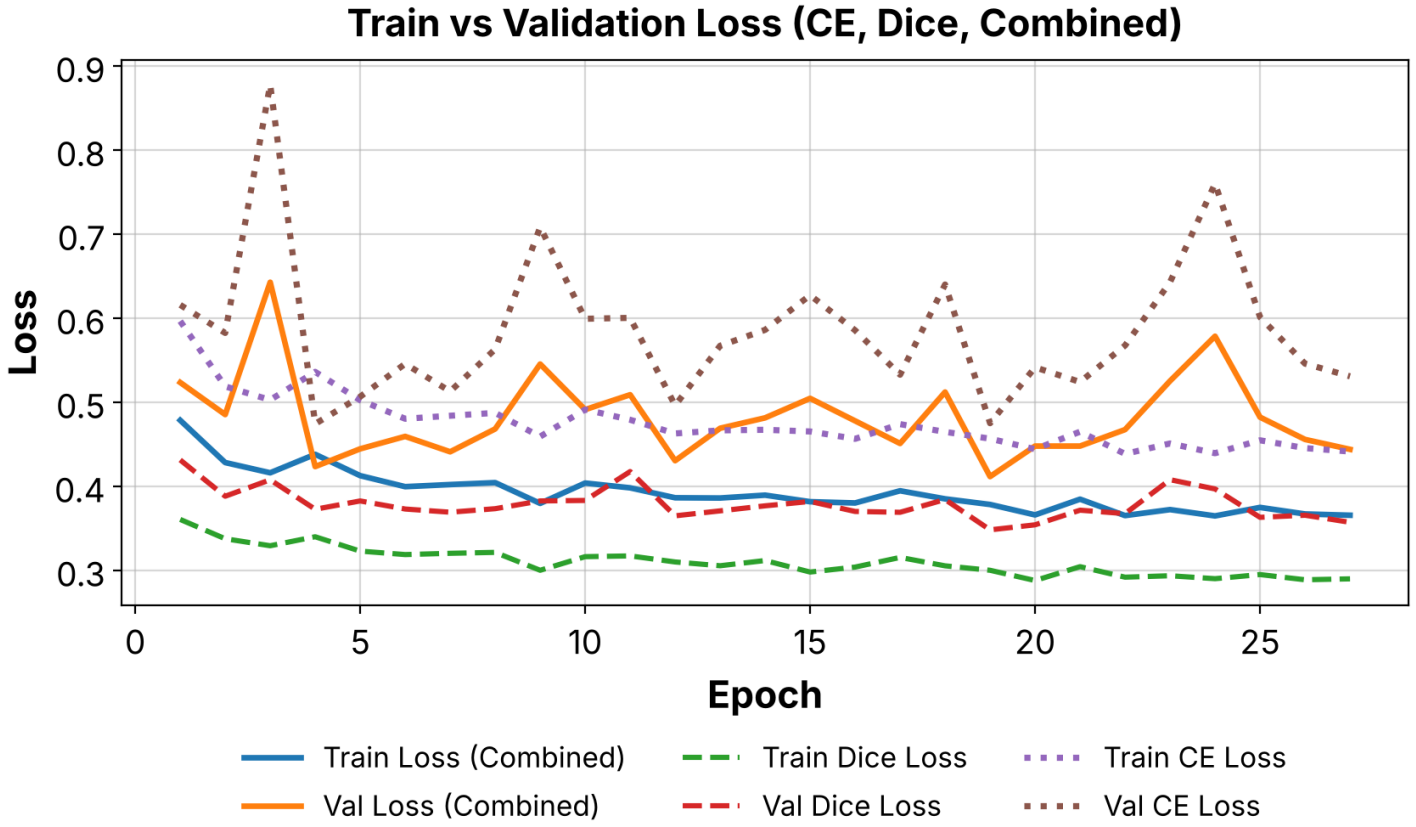
F1-score: 93.18%

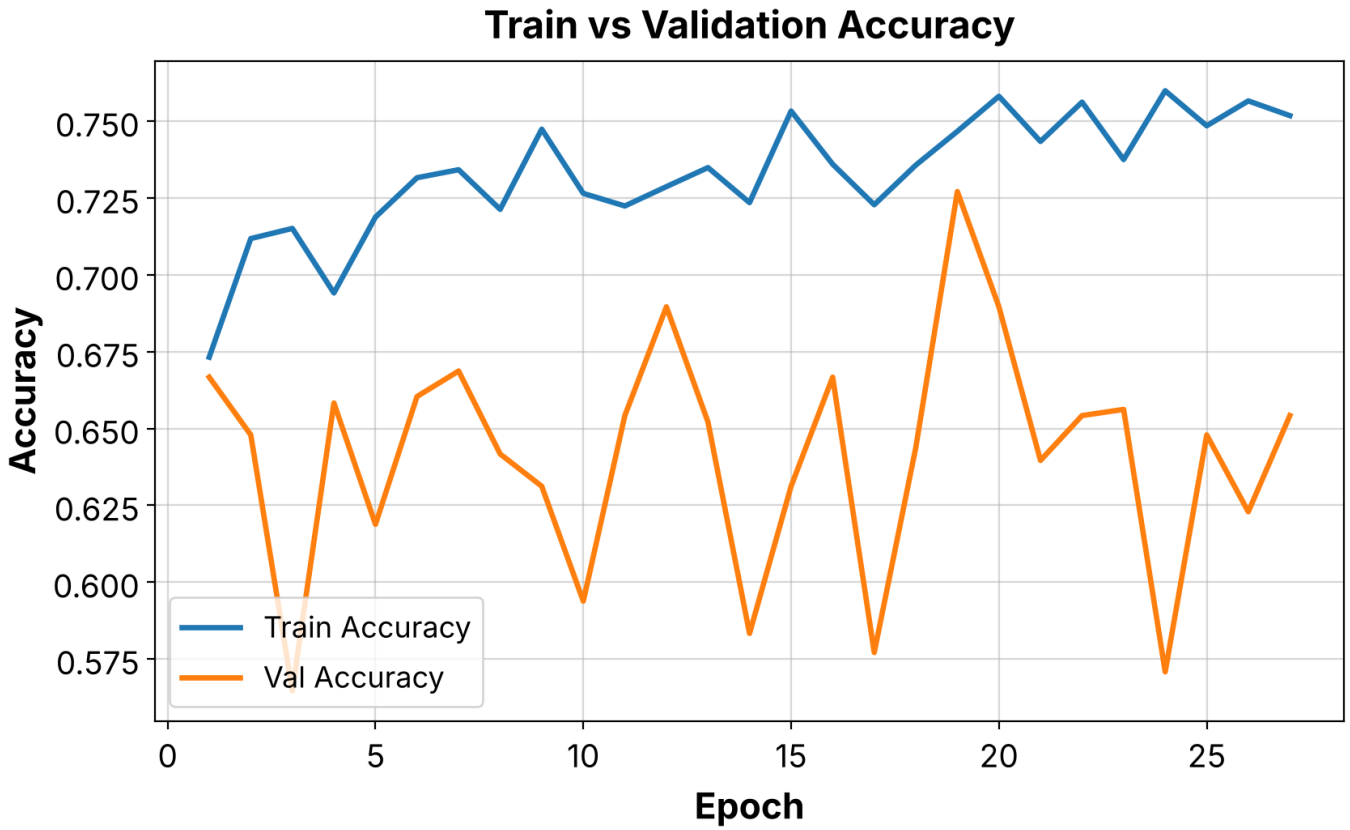
Confusion Matrix



1. MobileNetV2 + AdaBoost (dedicated AdaBoost notebook) — graphs

Training / Validation Graphs





2. MobileNetV2 + AdaBoost

Source notebook: MobileNet with ML.ipynb (inside Models_DL.zip)

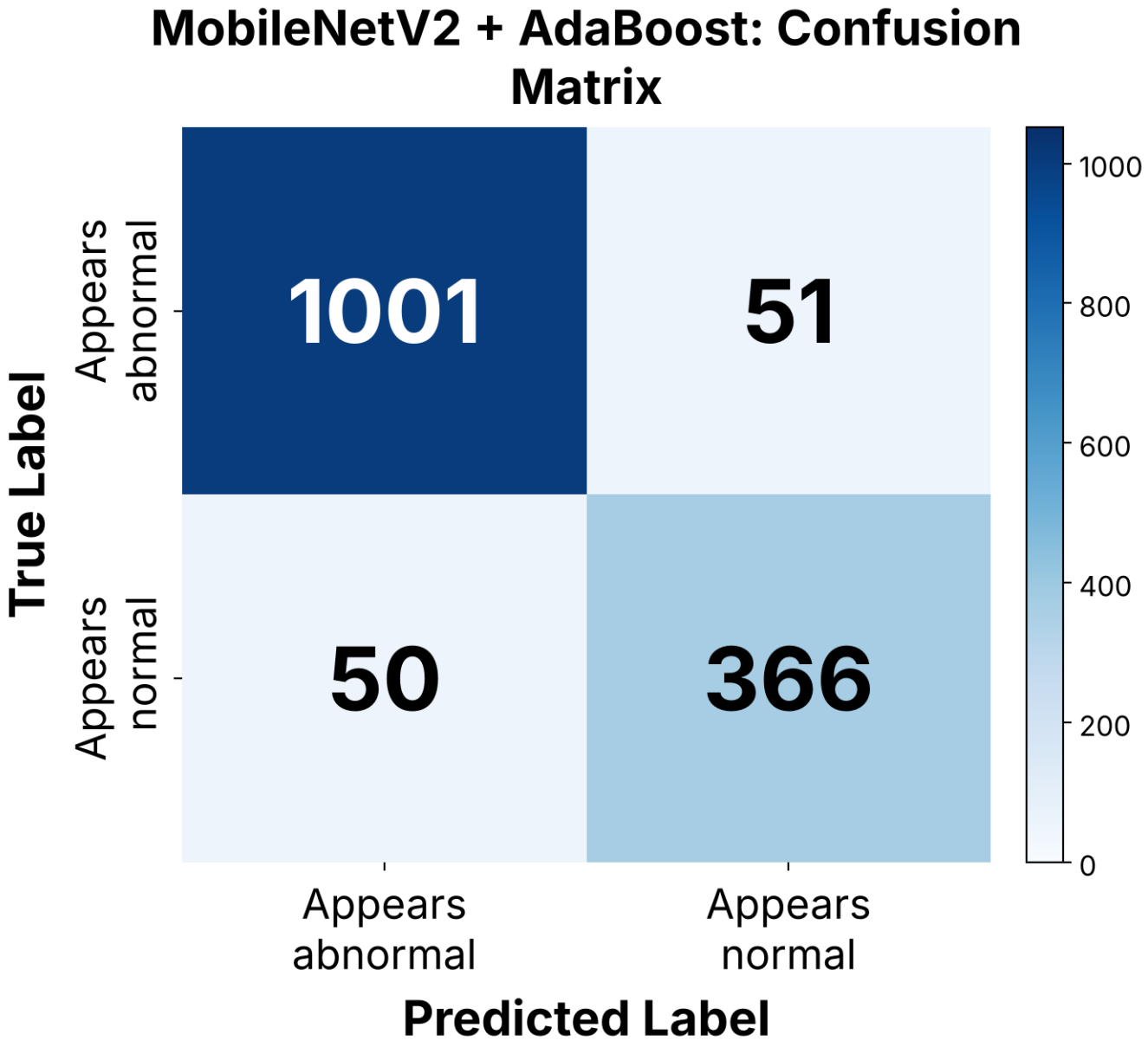
Accuracy: 93.12%

Precision: 93.12%

Recall: 93.12%

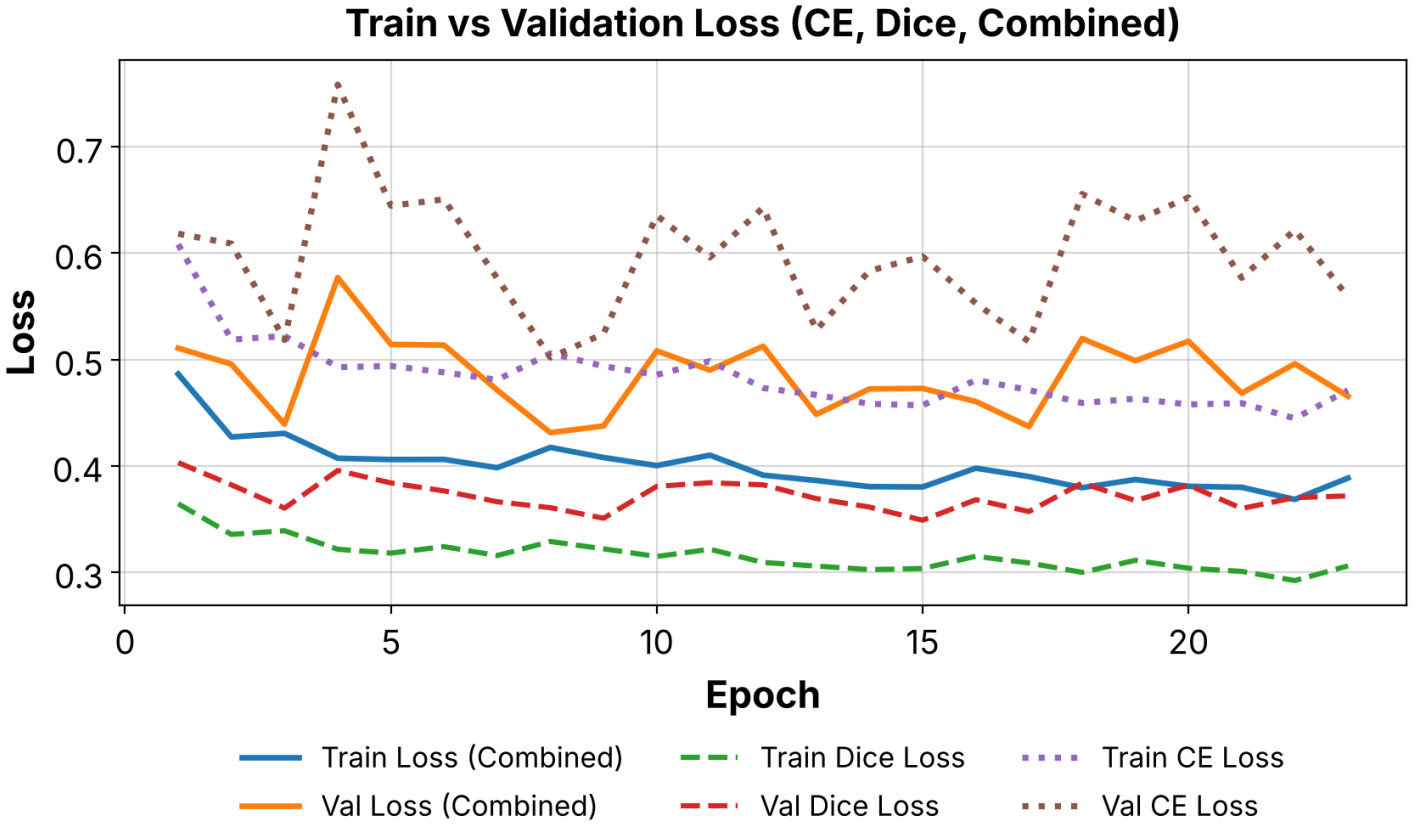
F1-score: 93.12%

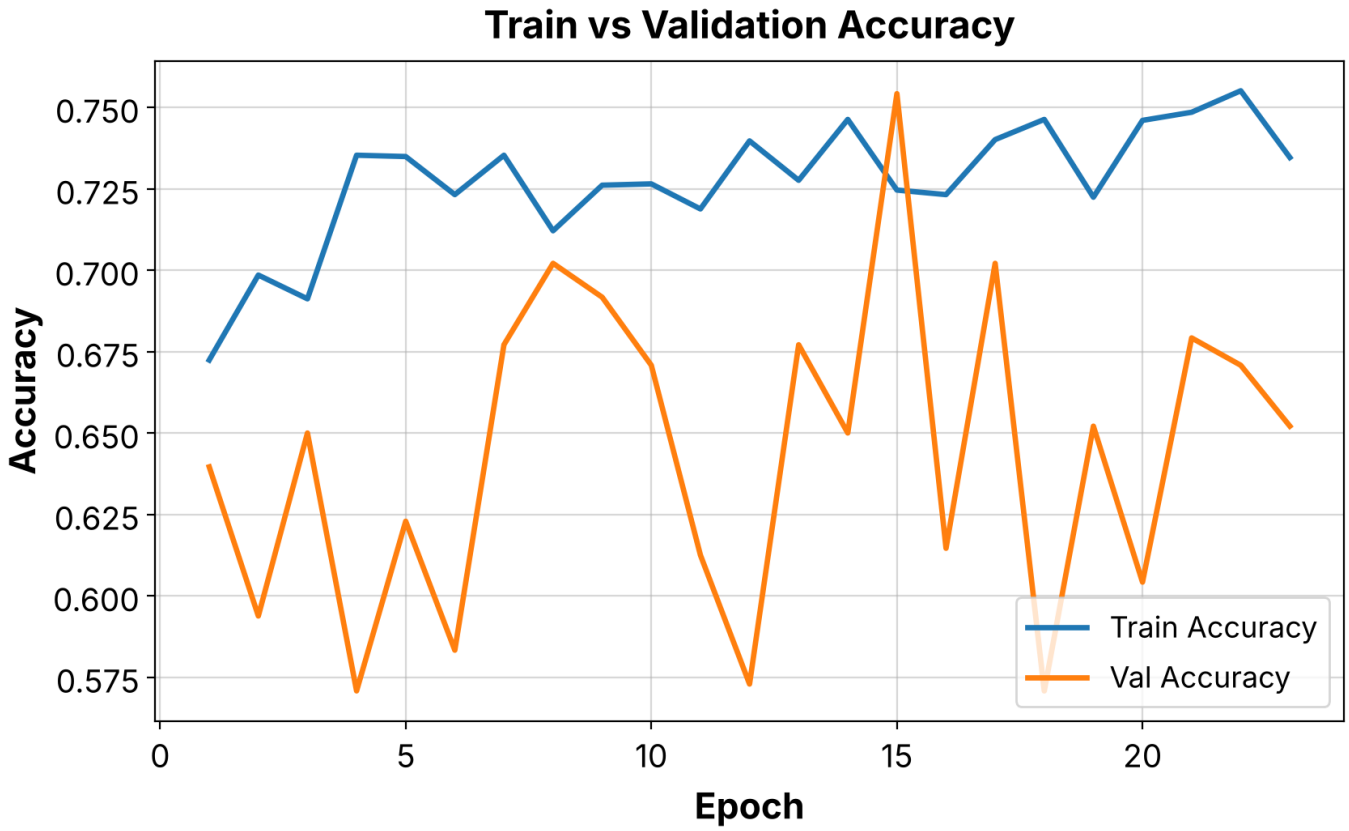
Confusion Matrix



2. MobileNetV2 + AdaBoost — graphs

Training / Validation Graphs





3. EfficientNet-B3 + AdaBoost

Source notebook: efficinetnet b3 baseline and ml.ipynb (inside Models_DL.zip)

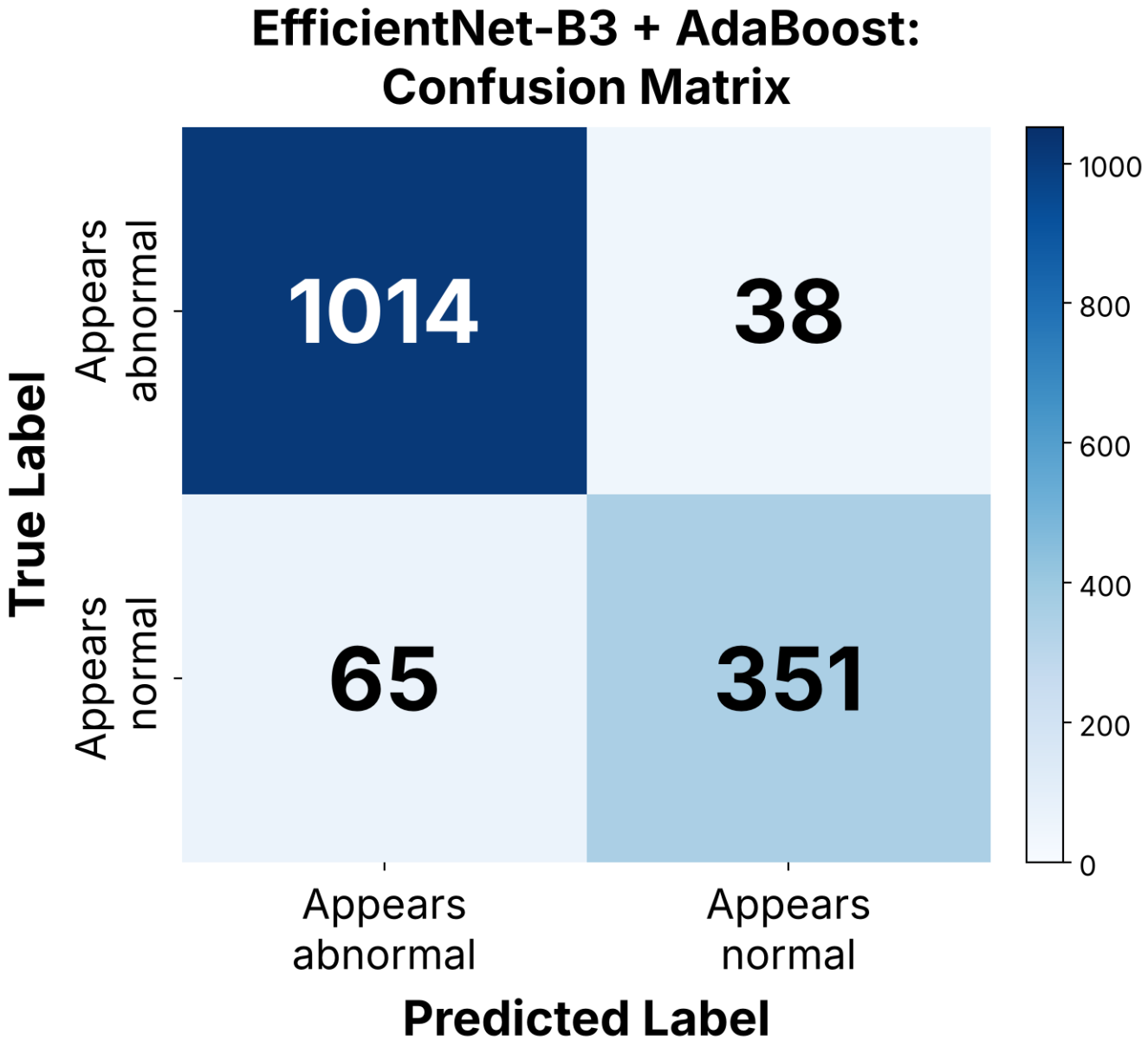
Accuracy: 92.98%

Precision: 92.91%

Recall: 92.98%

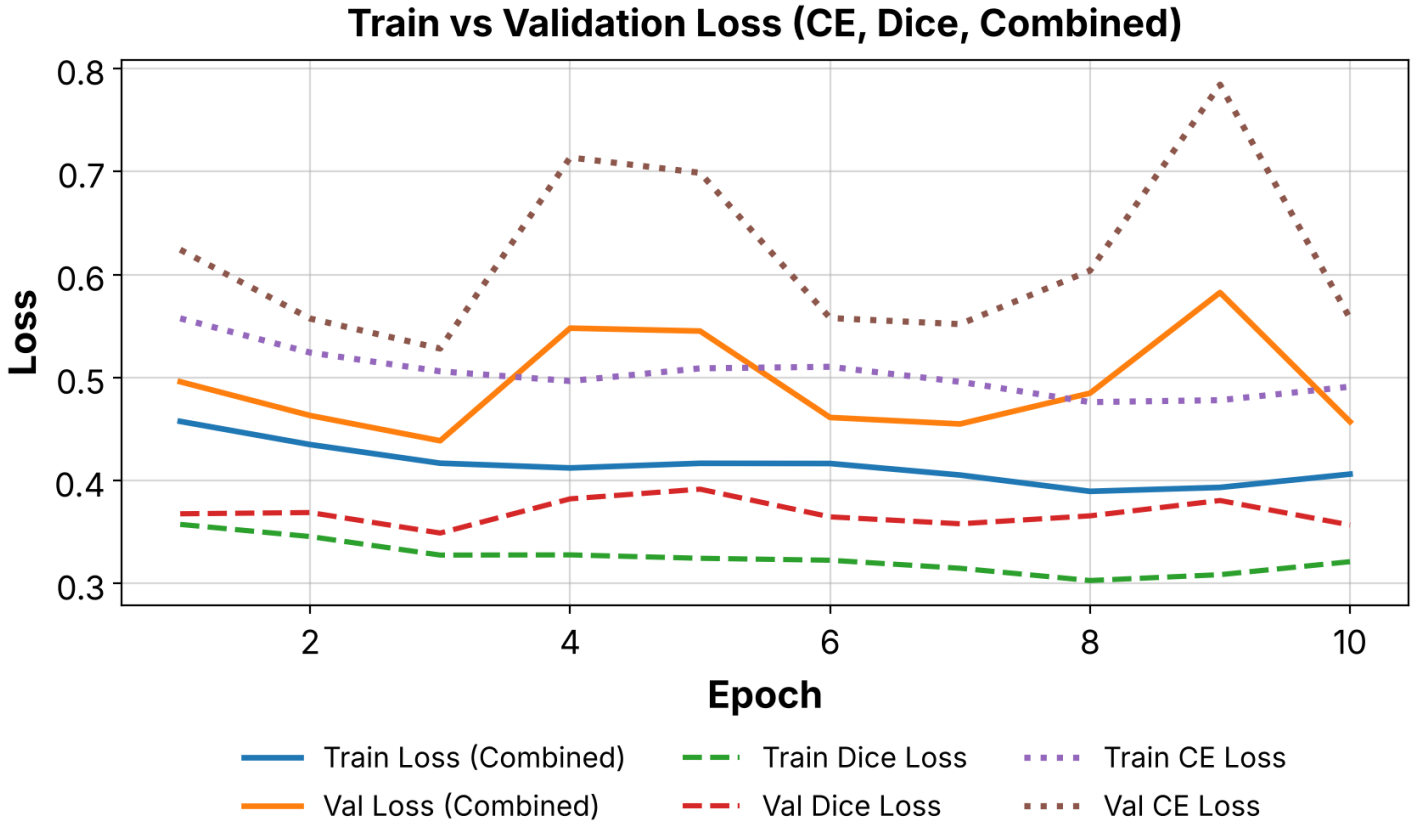
F1-score: 92.91%

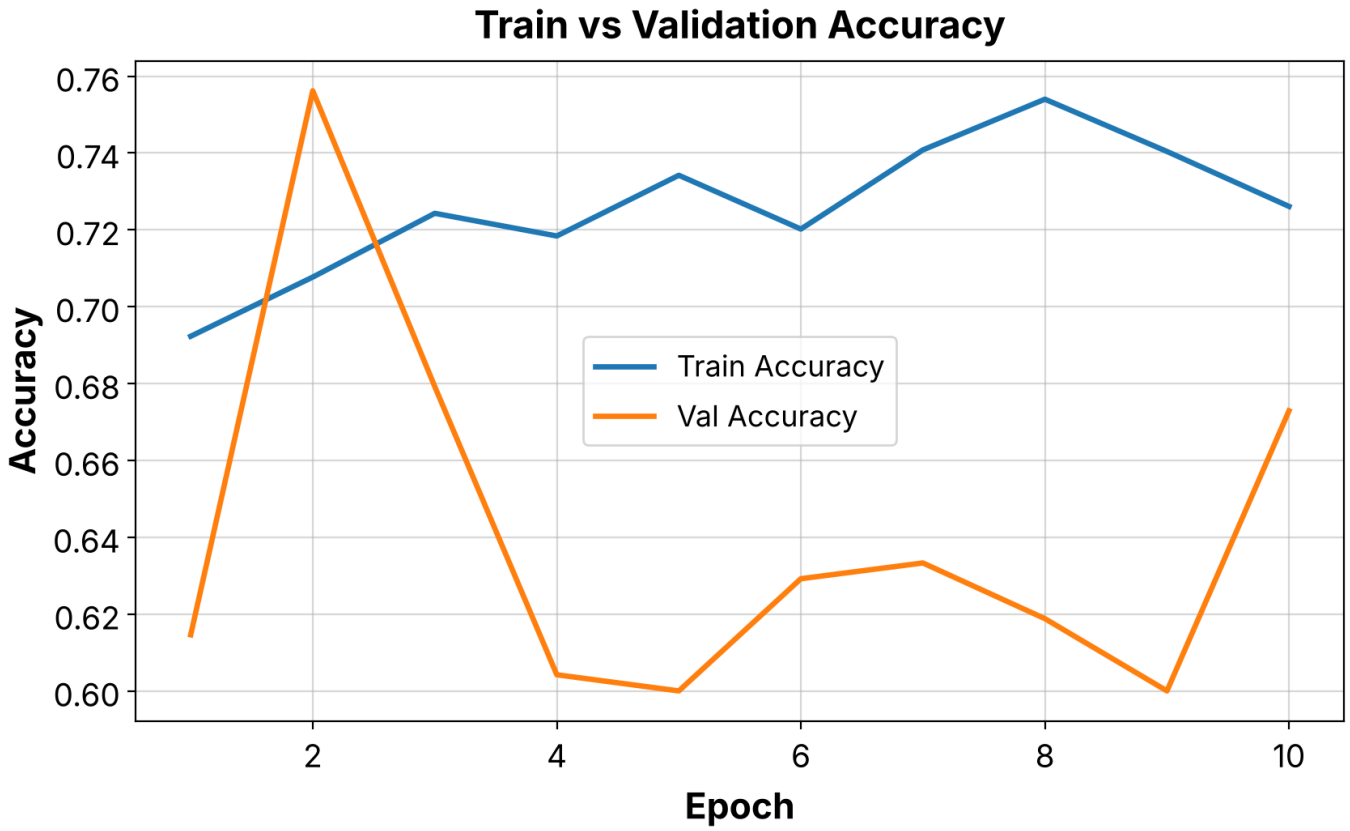
Confusion Matrix



3. EfficientNet-B3 + AdaBoost — graphs

Training / Validation Graphs





4. EfficientNet-B3 + SVM

Source notebook: efficinetnet b3 baseline and ml.ipynb (inside Models_DL.zip)

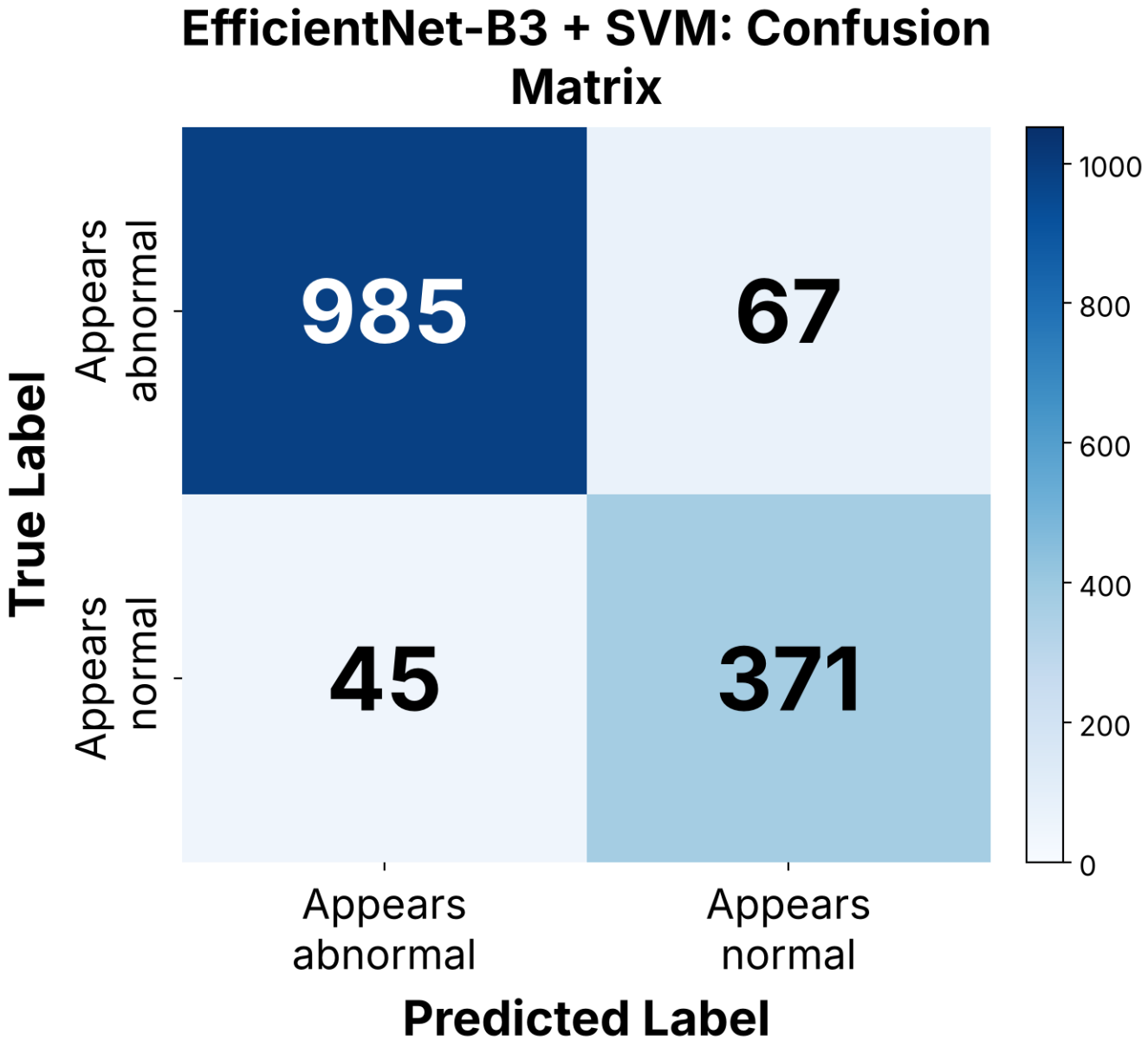
Accuracy: 92.37%

Precision: 92.53%

Recall: 92.37%

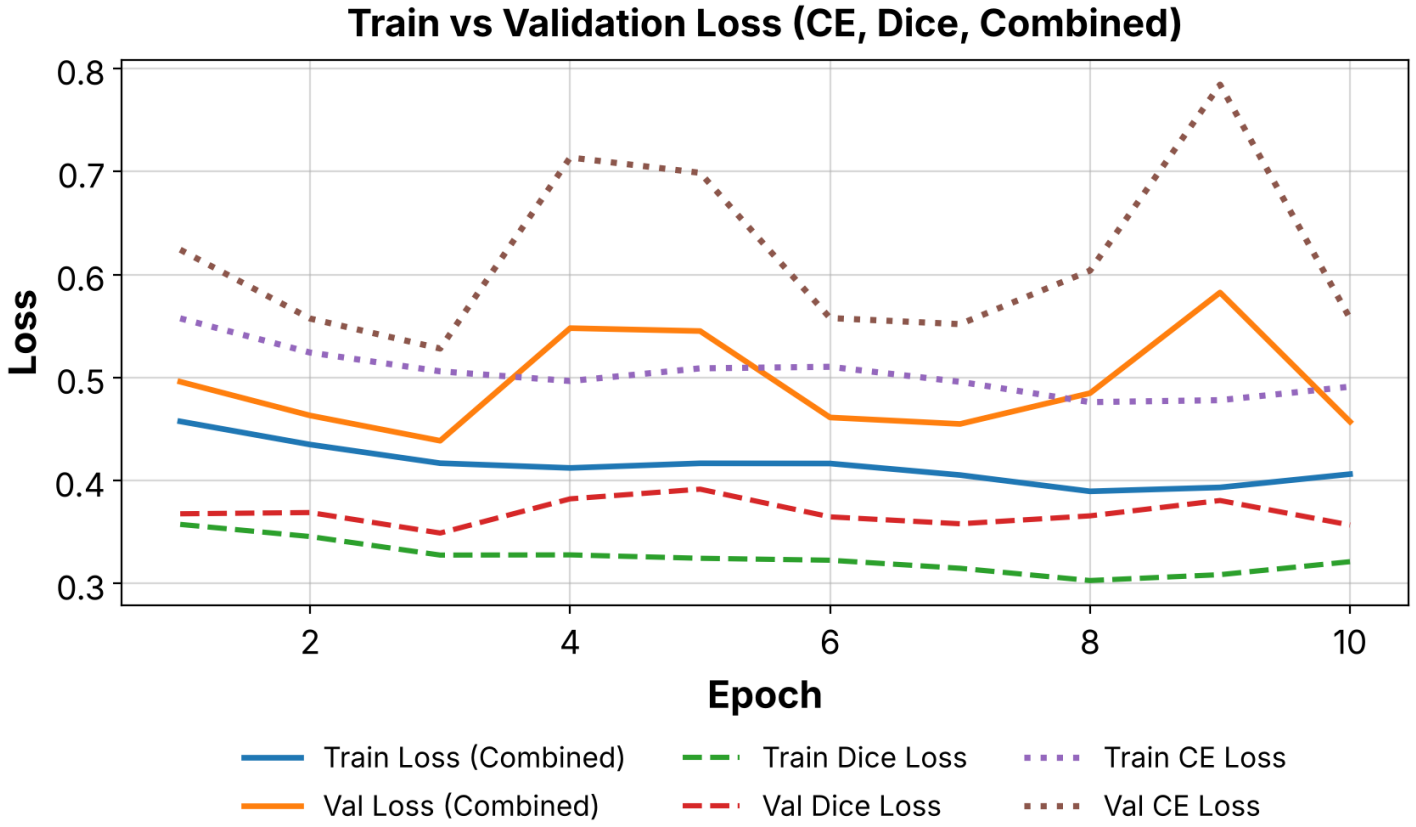
F1-score: 92.43%

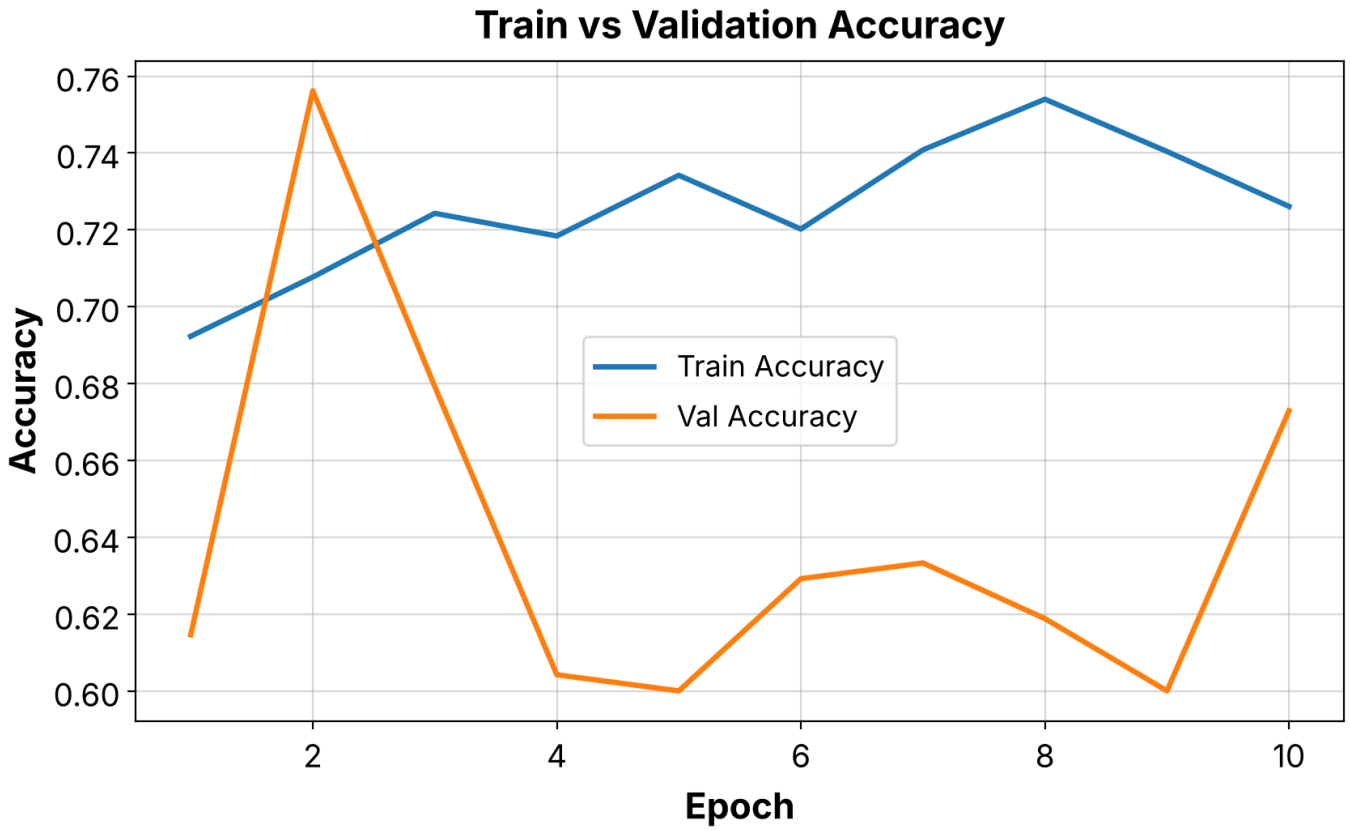
Confusion Matrix



4. EfficientNet-B3 + SVM — graphs

Training / Validation Graphs





5. EfficientNet-B3 + Random Forest

Source notebook: efficinetnet b3 baseline and ml.ipynb (inside Models_DL.zip)

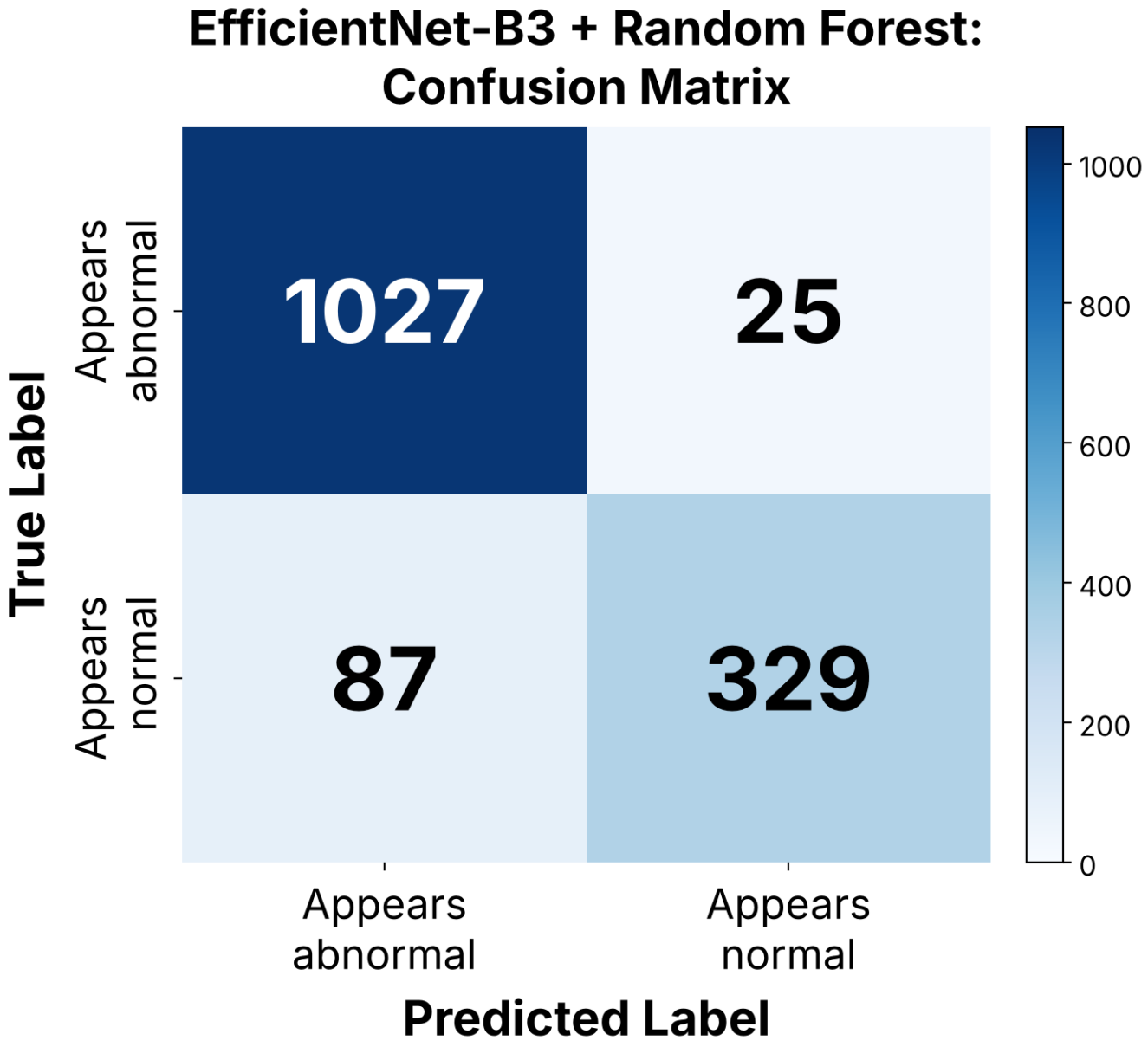
Accuracy: 92.37%

Precision: 92.40%

Recall: 92.37%

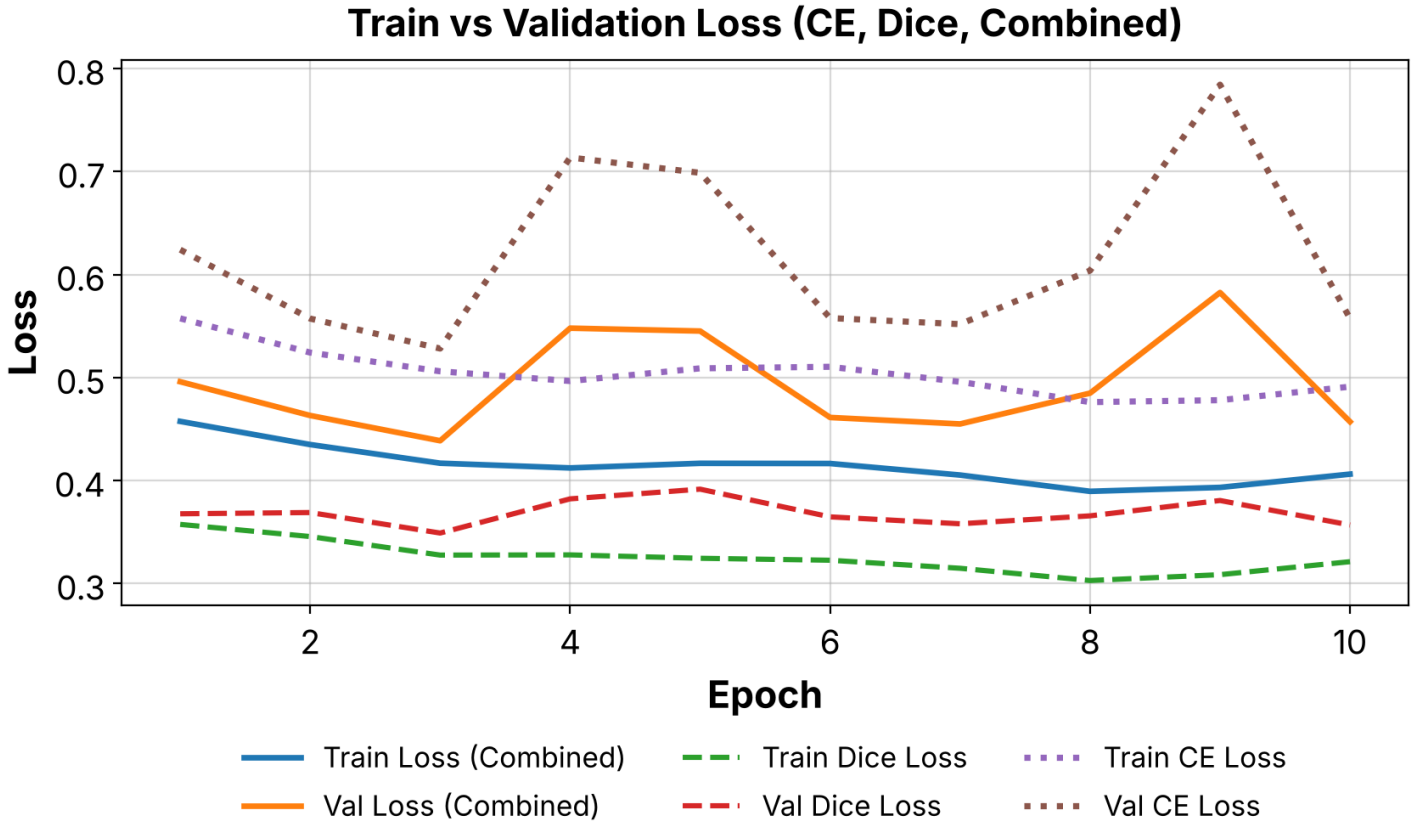
F1-score: 92.17%

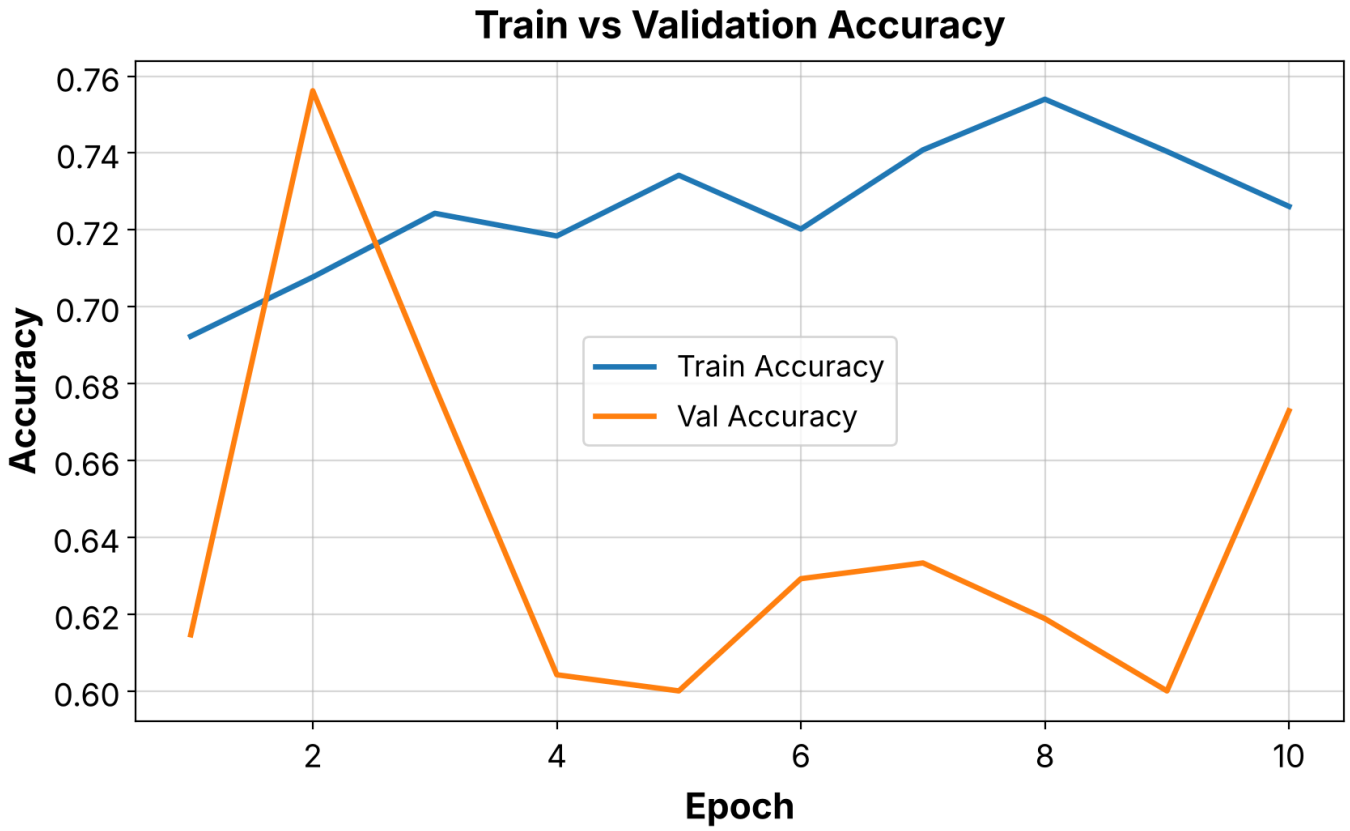
Confusion Matrix



5. EfficientNet-B3 + Random Forest — graphs

Training / Validation Graphs





6. EfficientNet-B3 + XGBoost

Source notebook: efficinetnet b3 baseline and ml.ipynb (inside Models_DL.zip)

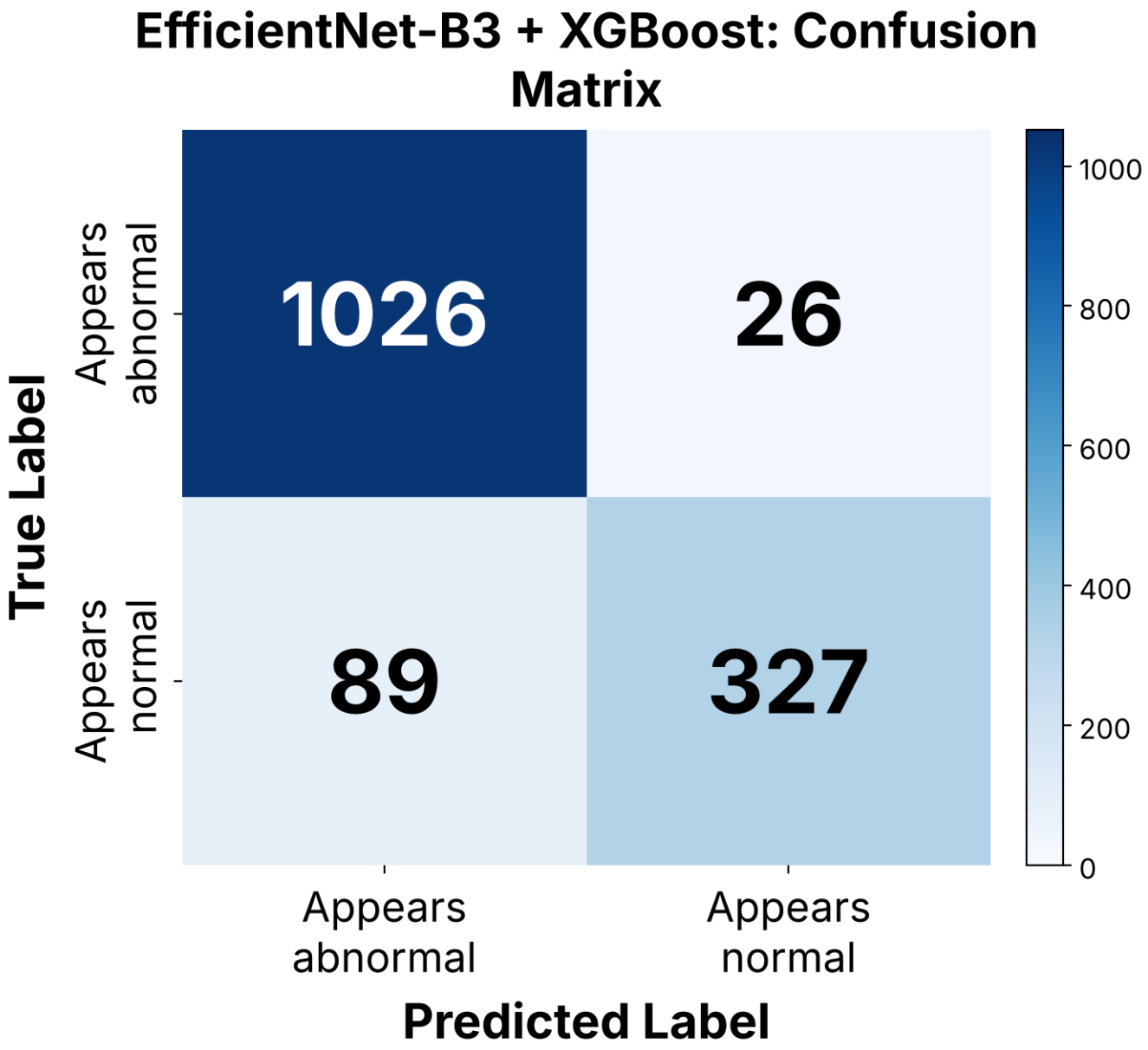
Accuracy: 92.17%

Precision: 92.19%

Recall: 92.17%

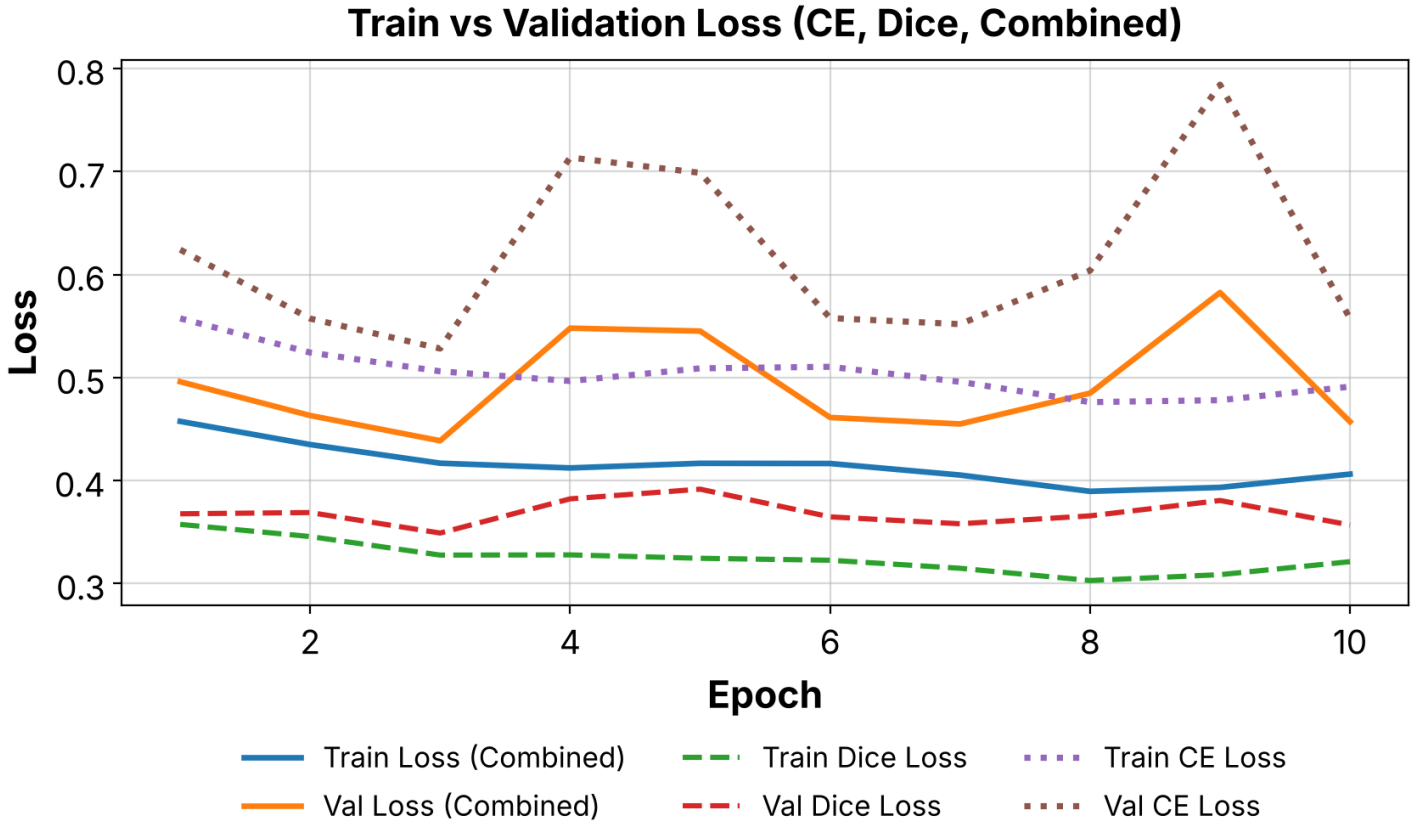
F1-score: 91.96%

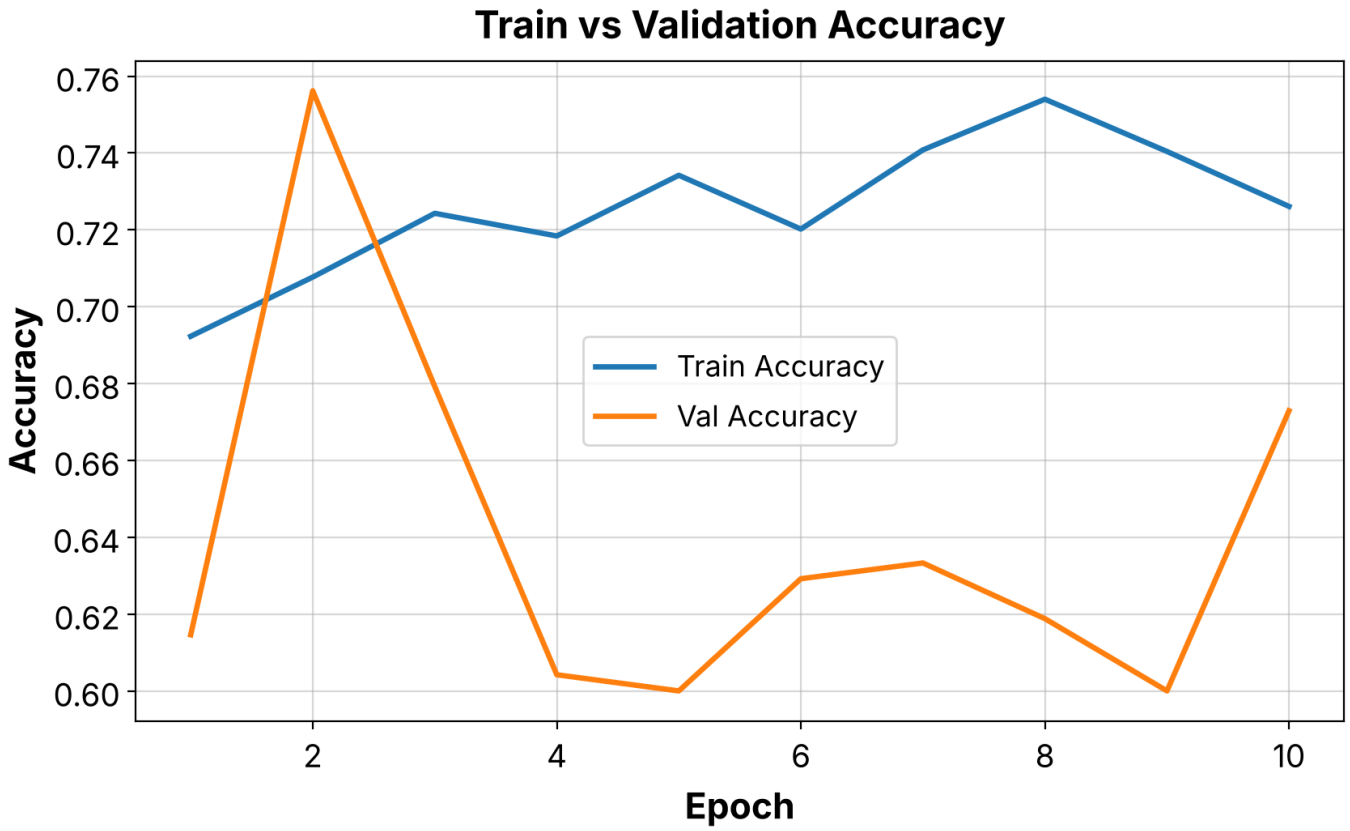
Confusion Matrix



6. EfficientNet-B3 + XGBoost — graphs

Training / Validation Graphs





7. MobileNetV2 — Direct Classification (Random Forest section run)

Source notebook: MobileNet with ML.ipynb (inside Models_DL.zip)

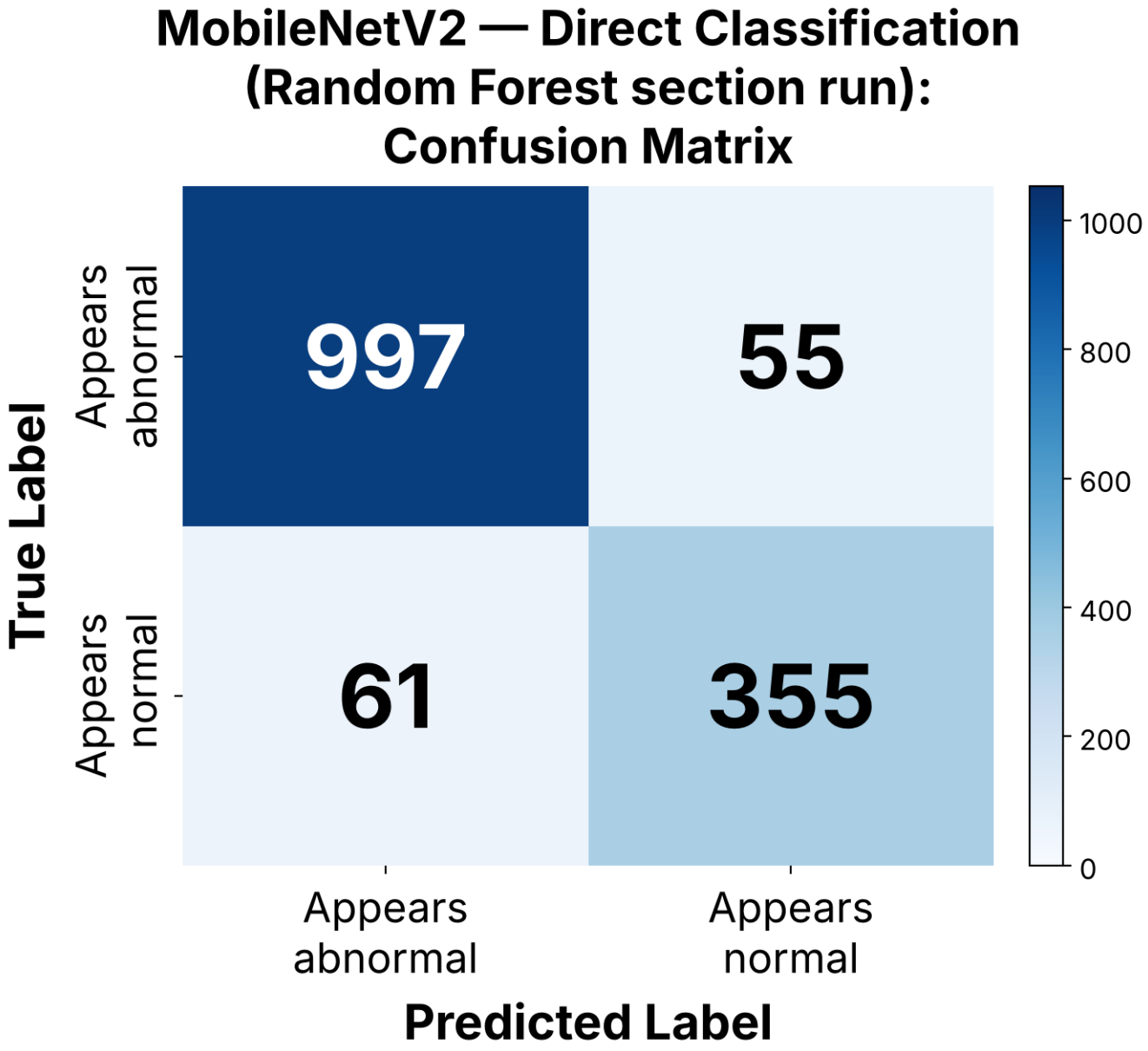
Accuracy: 92.10%

Precision: 92.07%

Recall: 92.10%

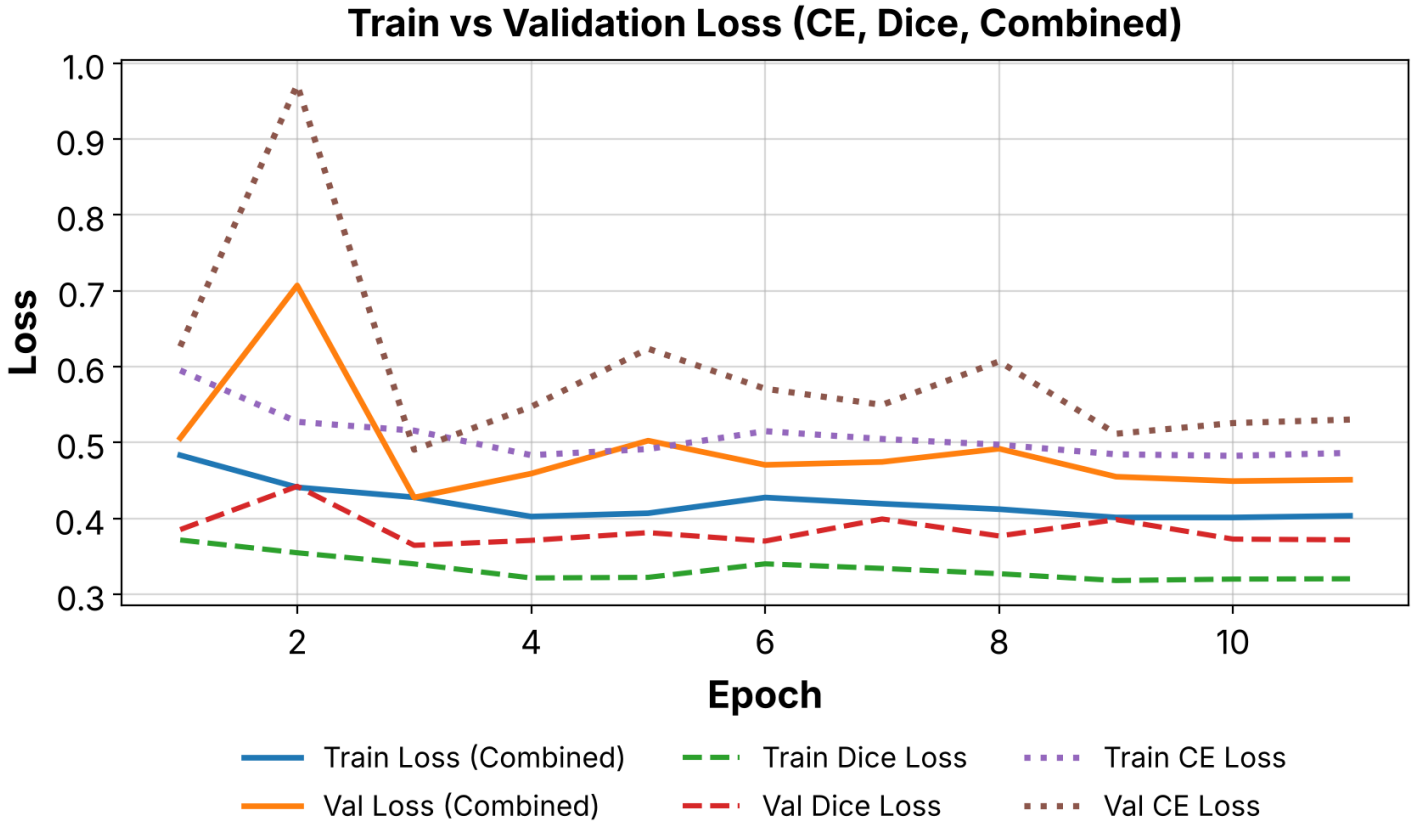
F1-score: 92.08%

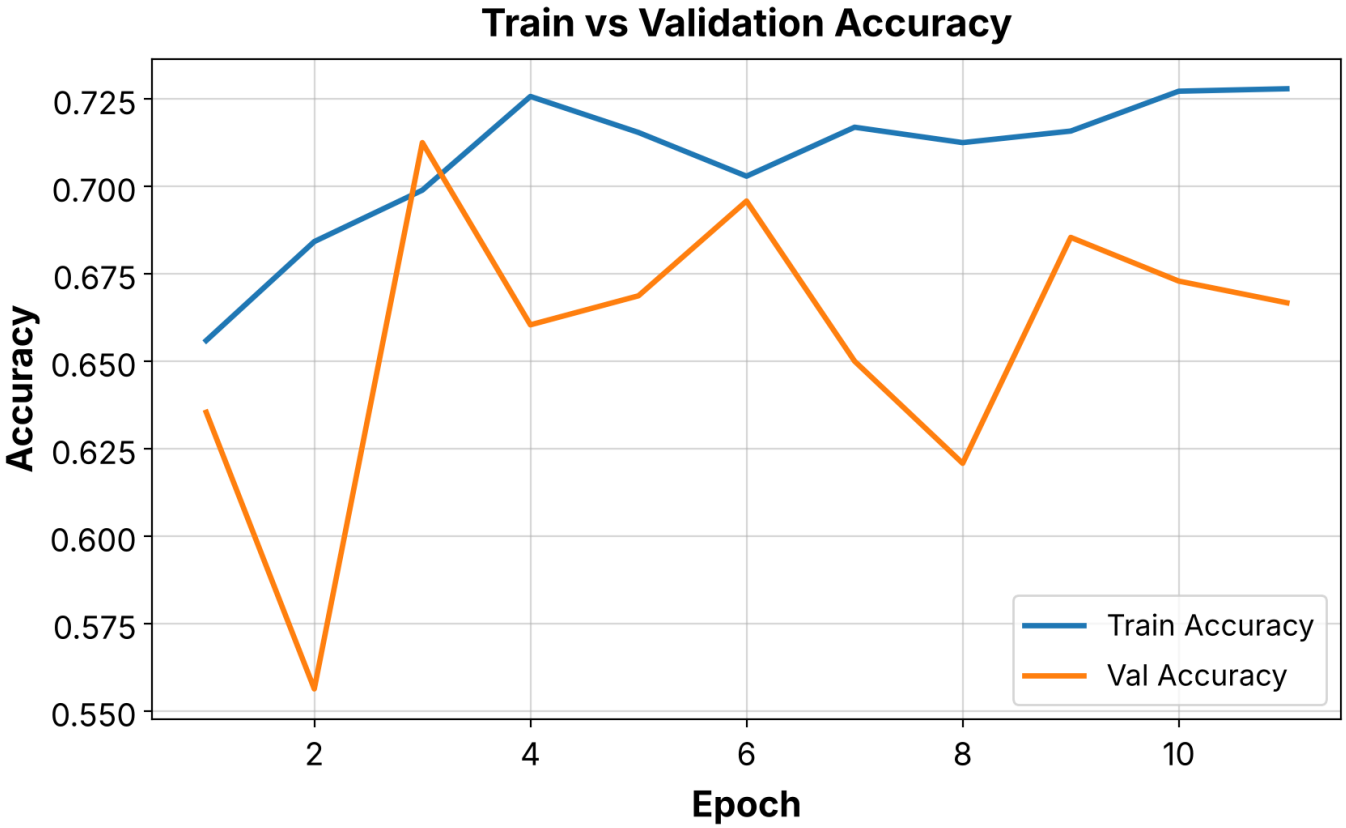
Confusion Matrix



7. MobileNetV2 — Direct Classification (Random Forest section run) — graphs

Training / Validation Graphs





8. MobileNetV2 — Direct Classification (Gaussian Naive Bayes section run)

Source notebook: MobileNet with ML.ipynb (inside Models_DL.zip)

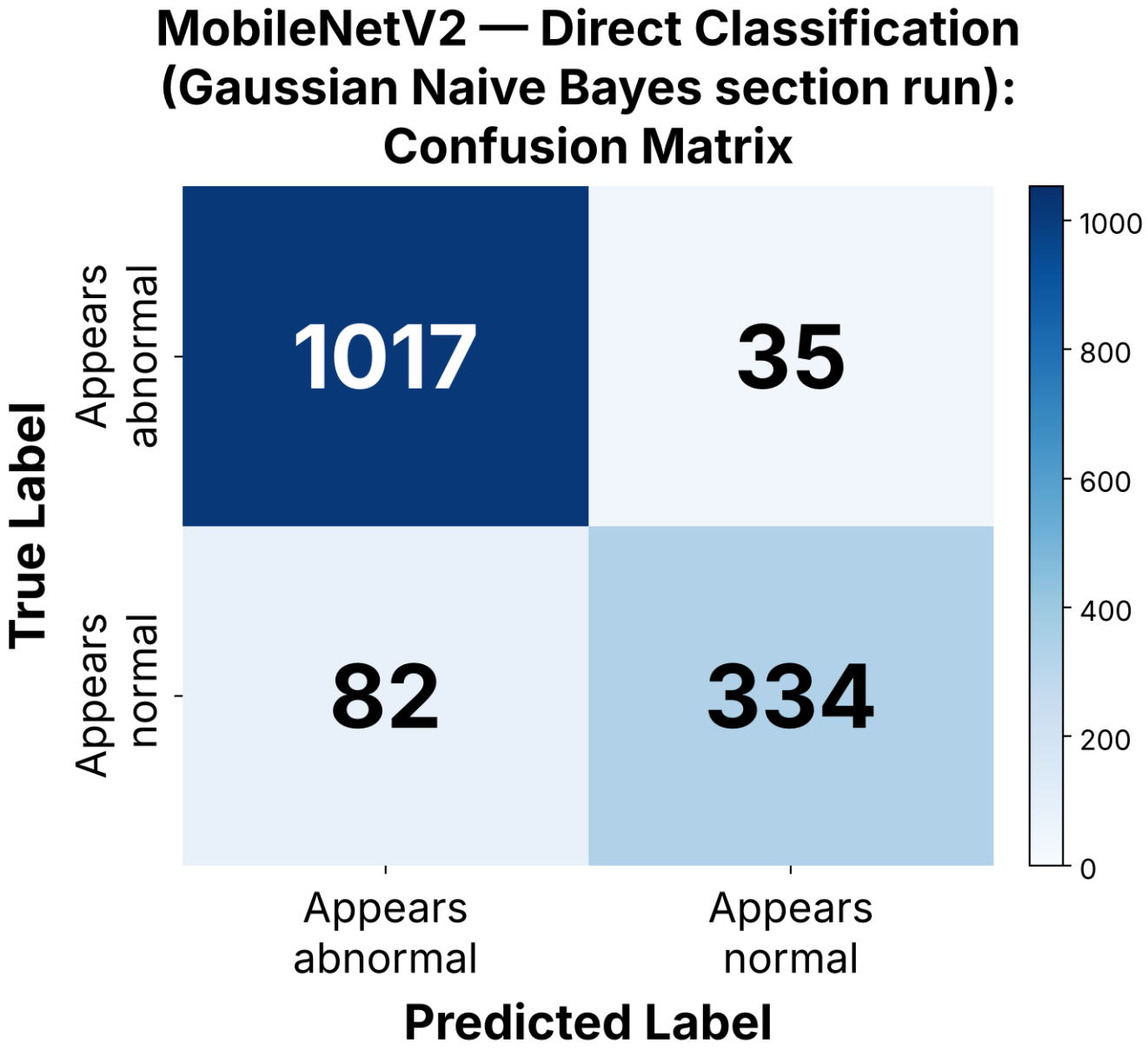
Accuracy: 92.03%

Precision: 91.97%

Recall: 92.03%

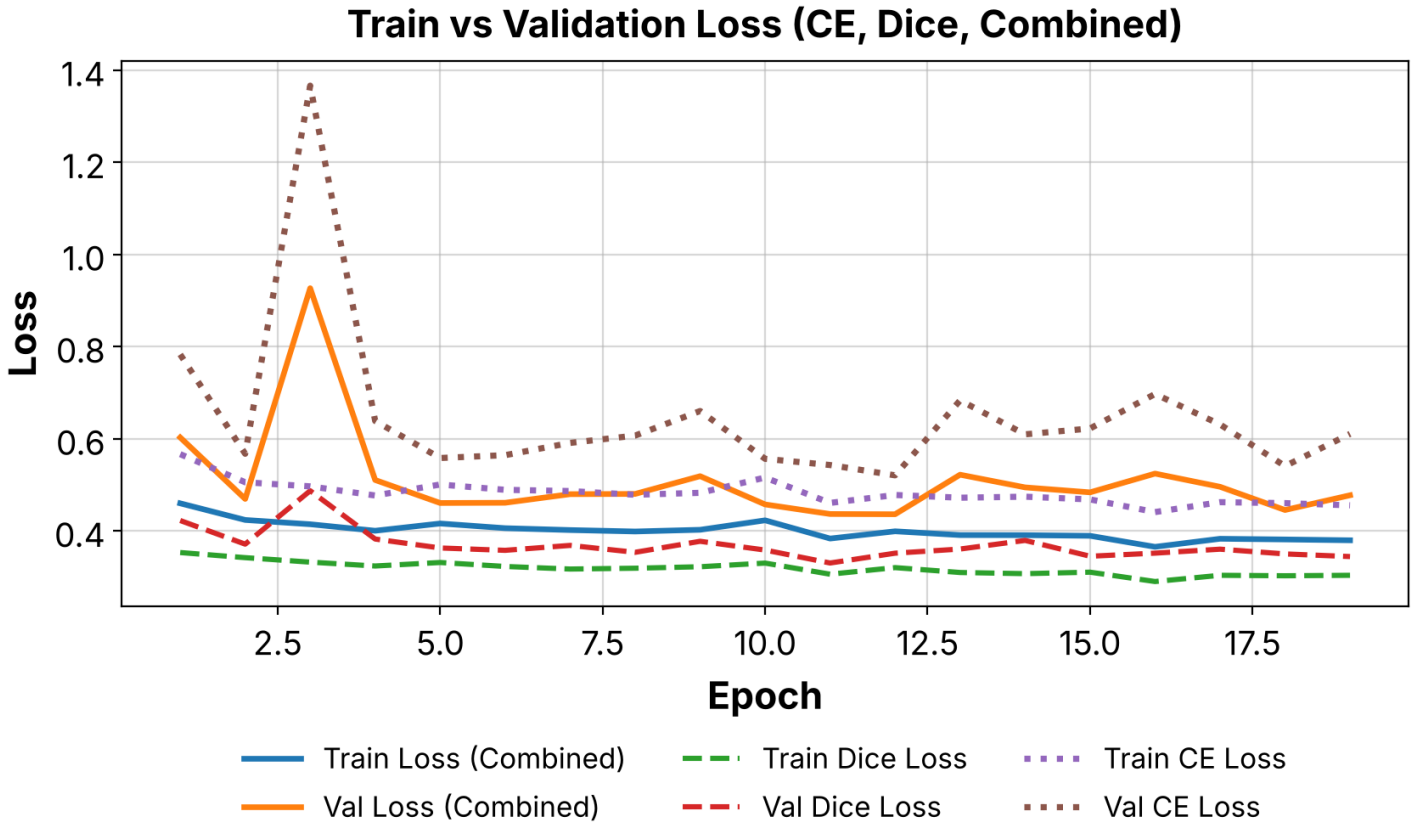
F1-score: 91.88%

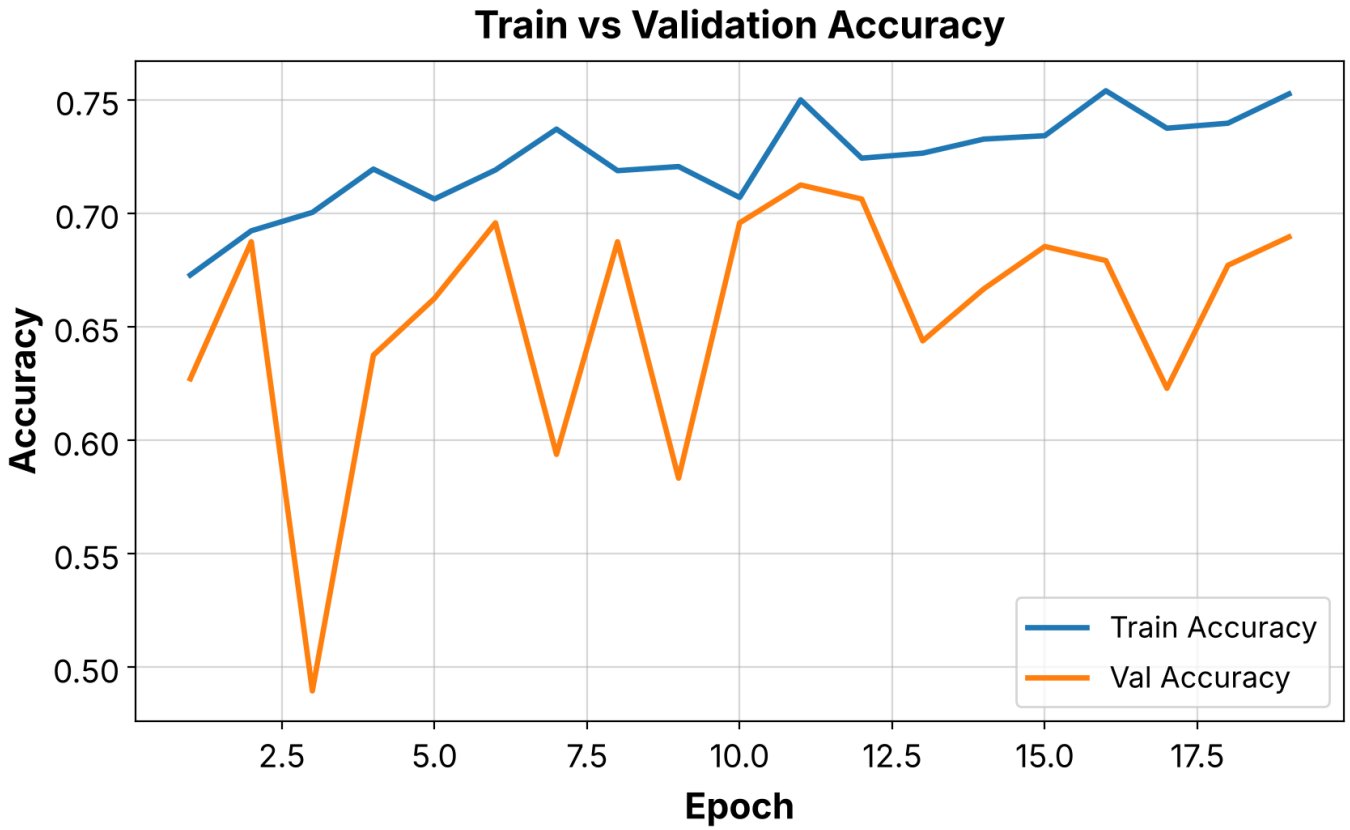
Confusion Matrix



8. MobileNetV2 — Direct Classification (Gaussian Naive Bayes section run) — graphs

Training / Validation Graphs





9. InceptionV3 — Direct Classification (baseline)

Source notebook: Inception v3 baseline and ml.ipynb (inside Models_DL.zip)

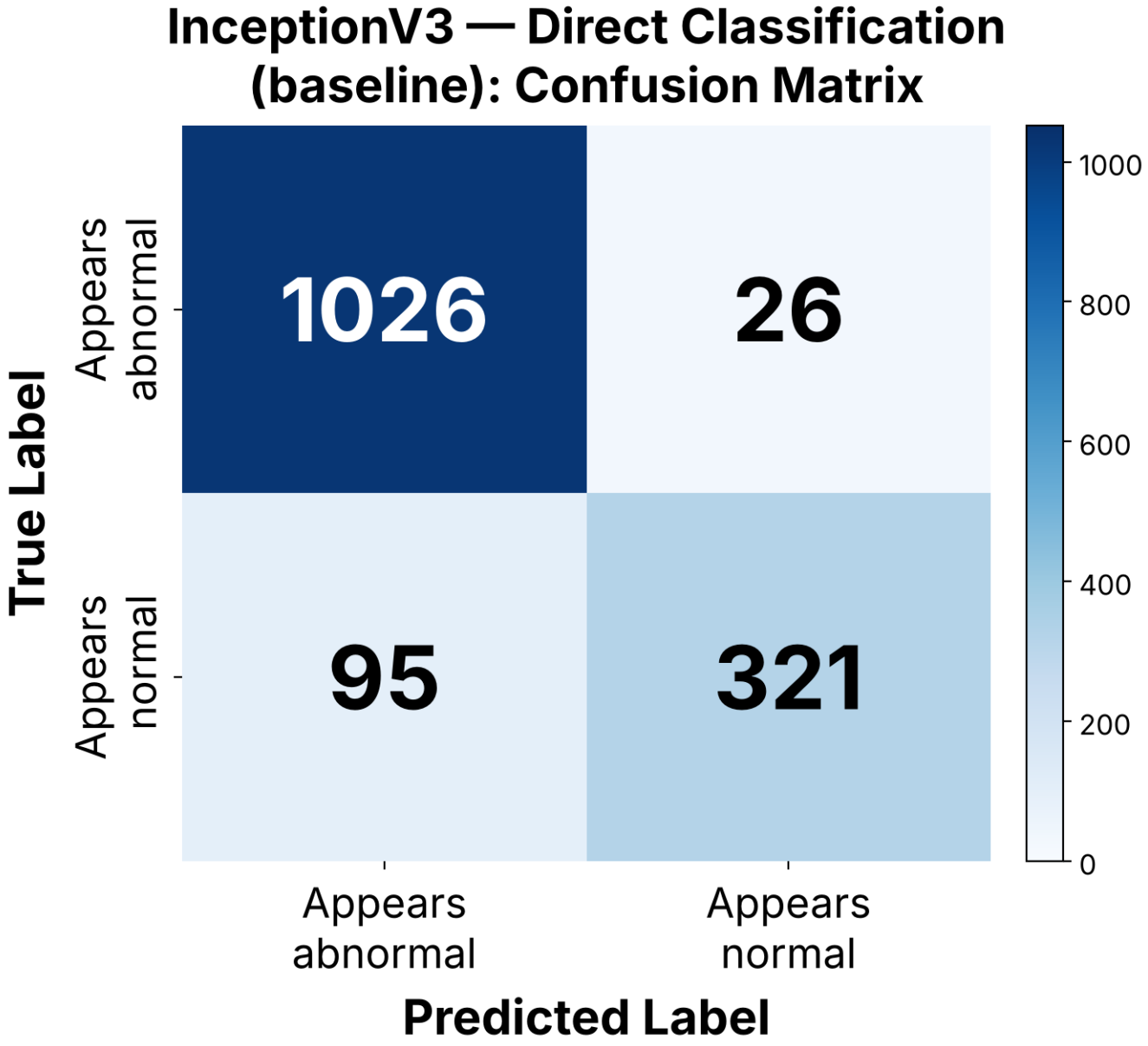
Accuracy: 91.76%

Precision: 91.80%

Recall: 91.76%

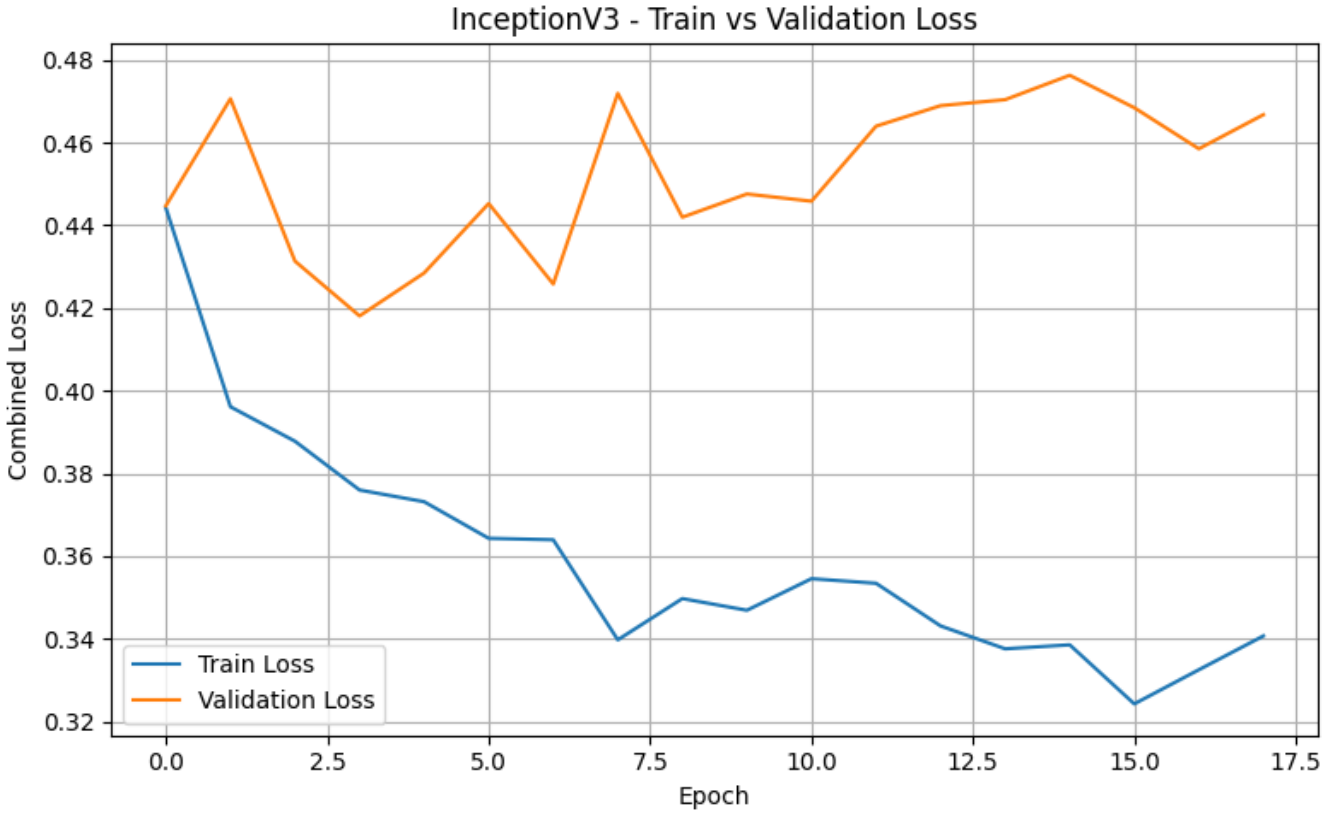
F1-score: 91.52%

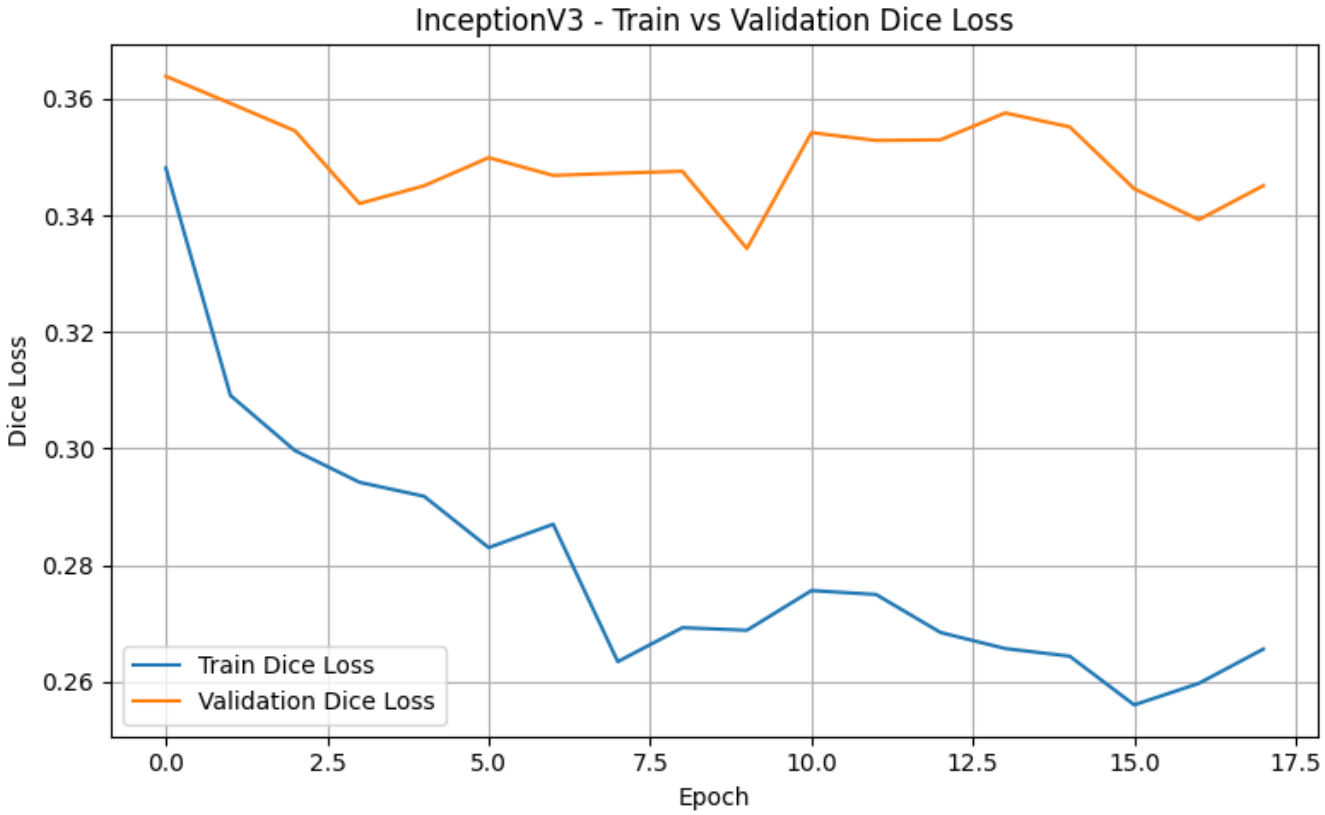
Confusion Matrix



9. InceptionV3 — Direct Classification (baseline) — graphs (1/2)

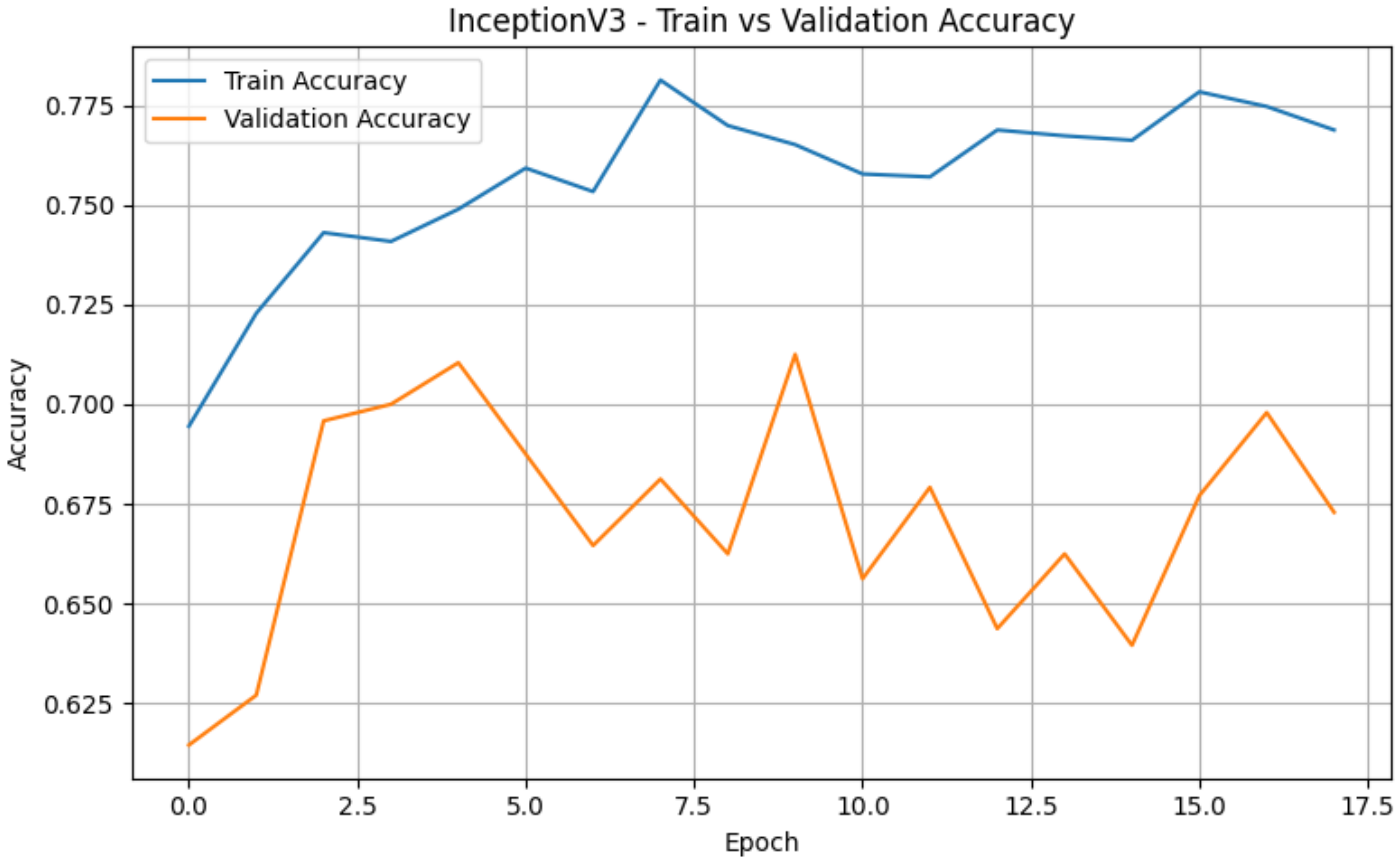
Training / Validation Graphs





9. InceptionV3 — Direct Classification (baseline) — graphs (2/2)

Training / Validation Graphs



10. MobileNetV2 — Direct Classification with Preprocessing (normal split)

Source notebook: MobileNet with Preprocessing.ipynb (inside Models_DL.zip)

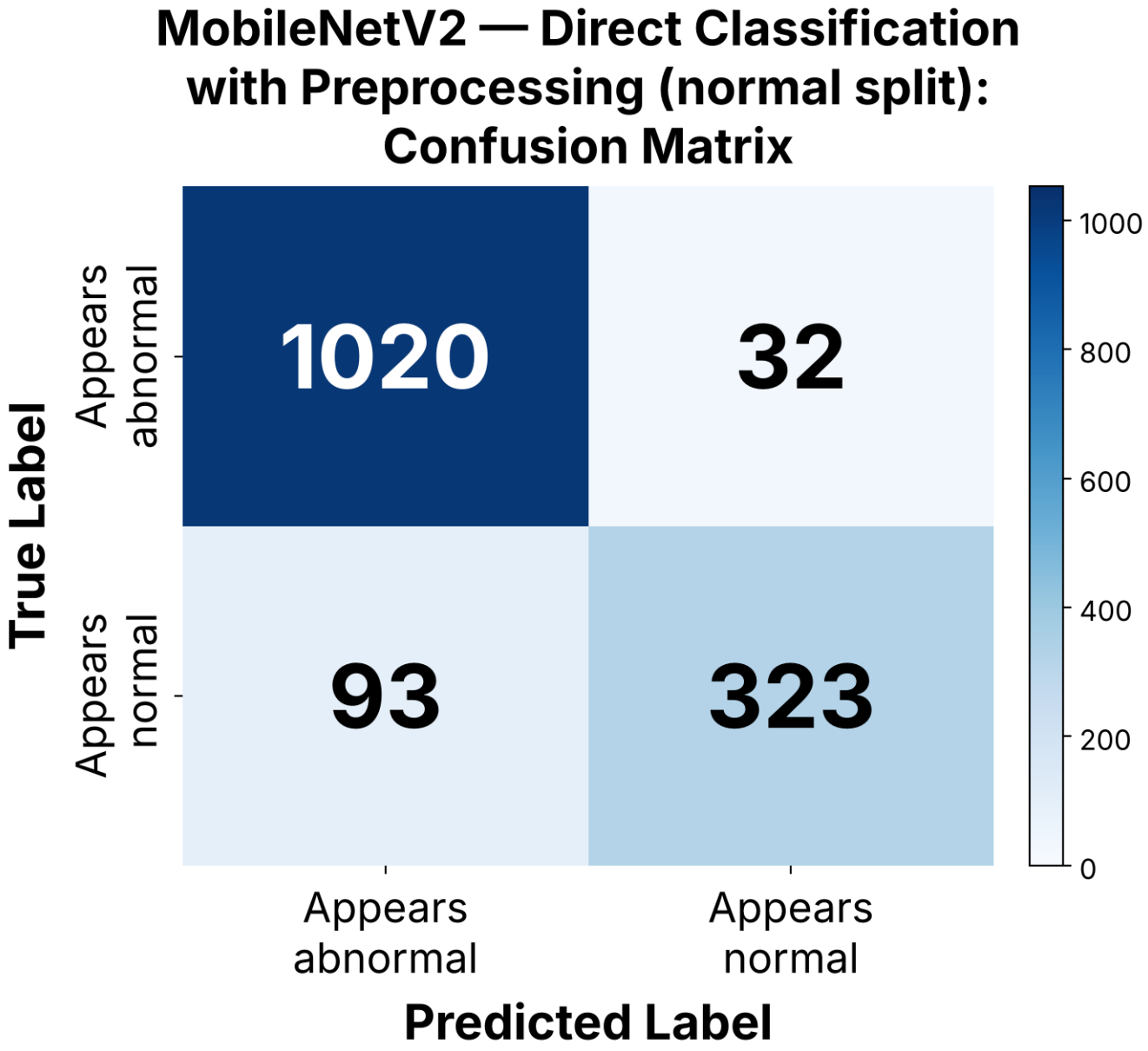
Accuracy: 91.49%

Precision: 91.46%

Recall: 91.49%

F1-score: 91.27%

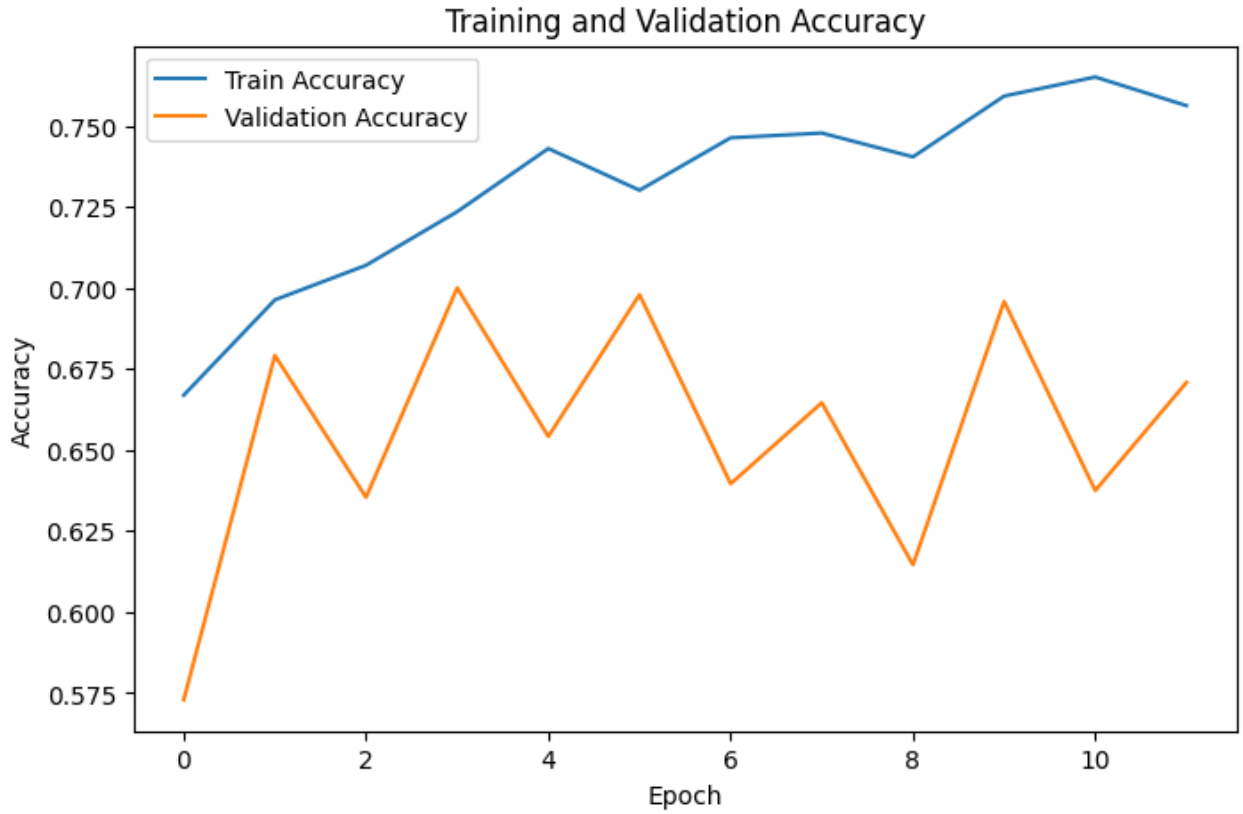
Confusion Matrix



10. MobileNetV2 — Direct Classification with Preprocessing (normal split) — graphs

Training / Validation Graphs





11. MobileNetV2 + Gaussian Naive Bayes

Source notebook: MobileNet with ML.ipynb (inside Models_DL.zip)

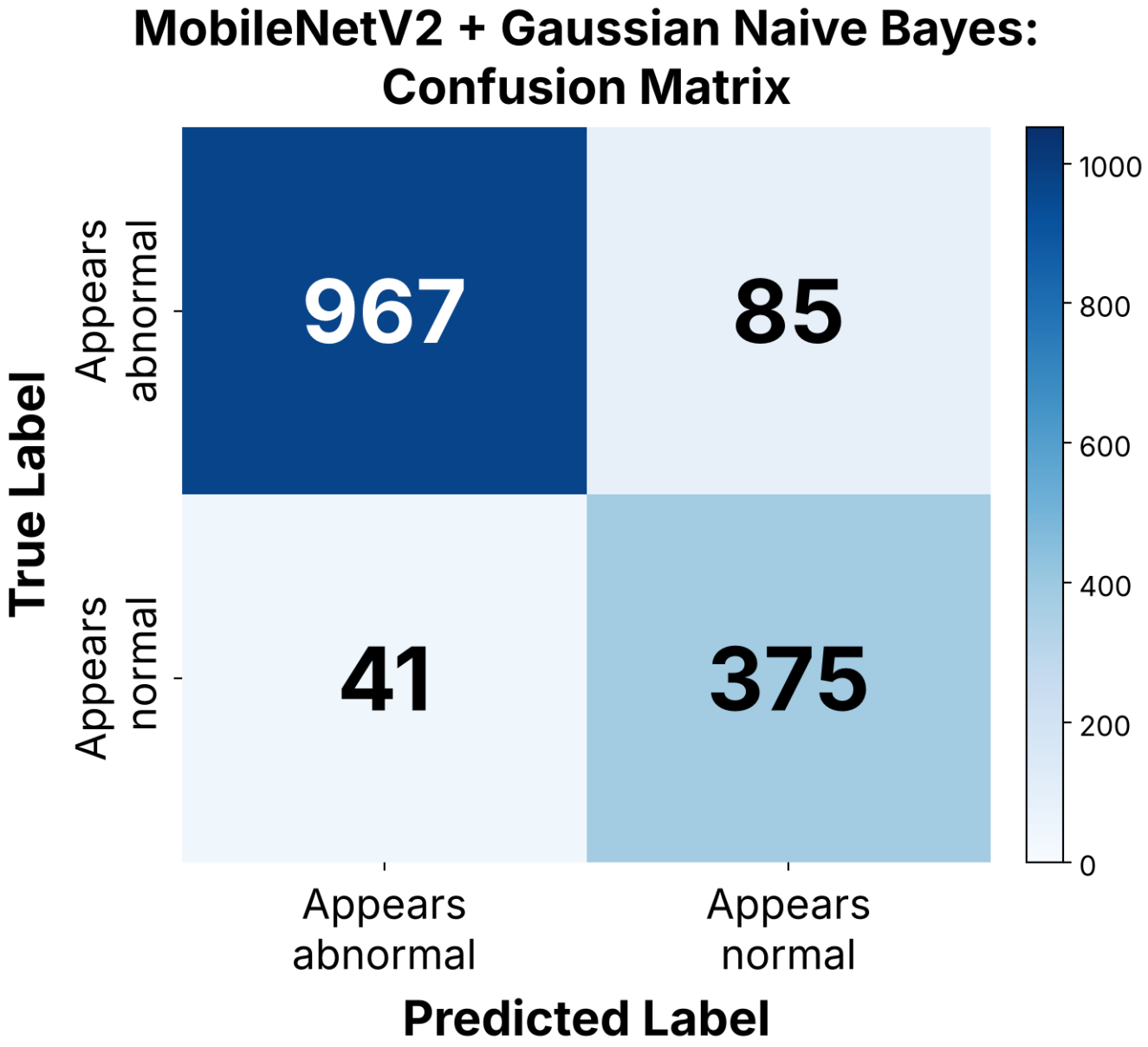
Accuracy: 91.42%

Precision: 91.85%

Recall: 91.42%

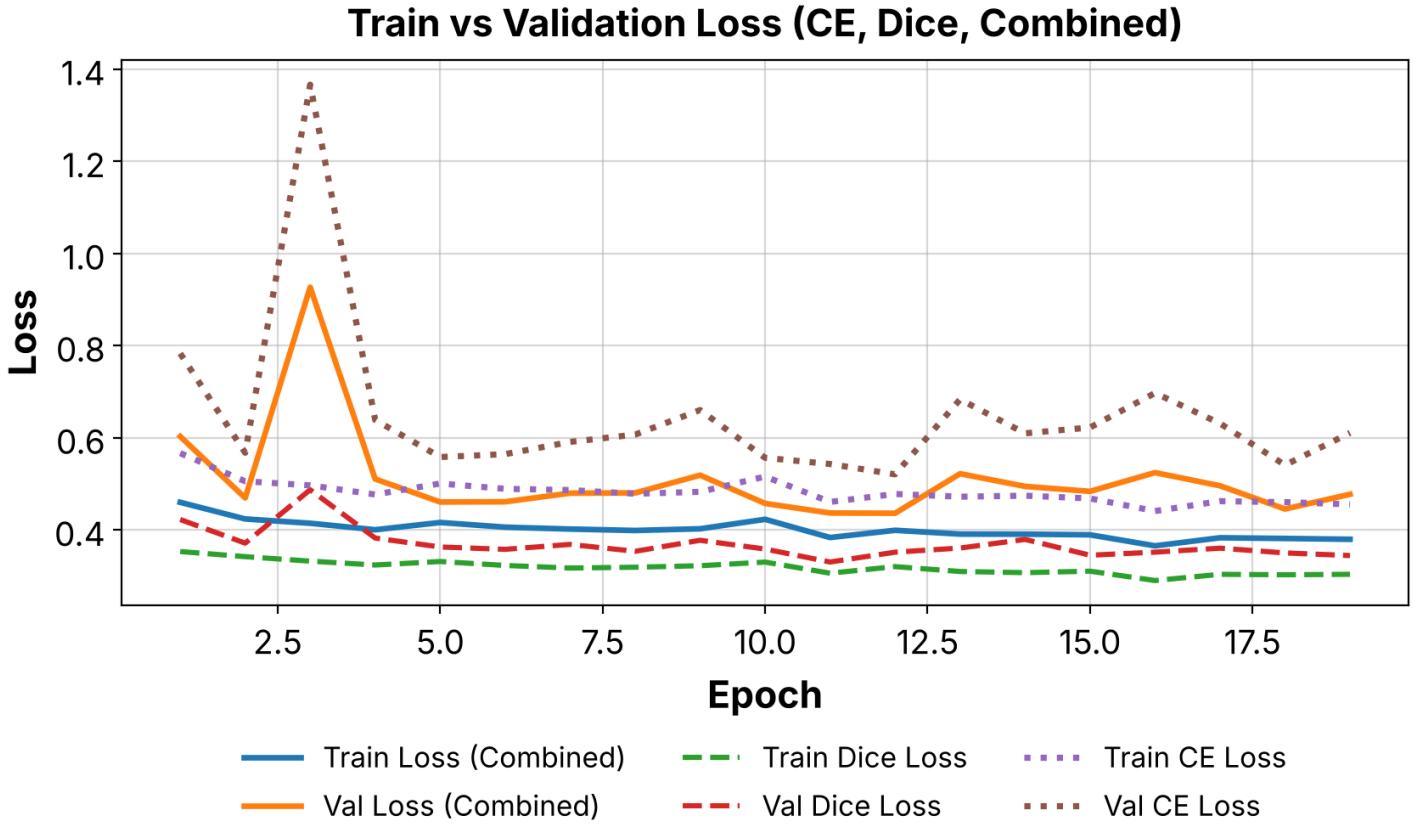
F1-score: 91.54%

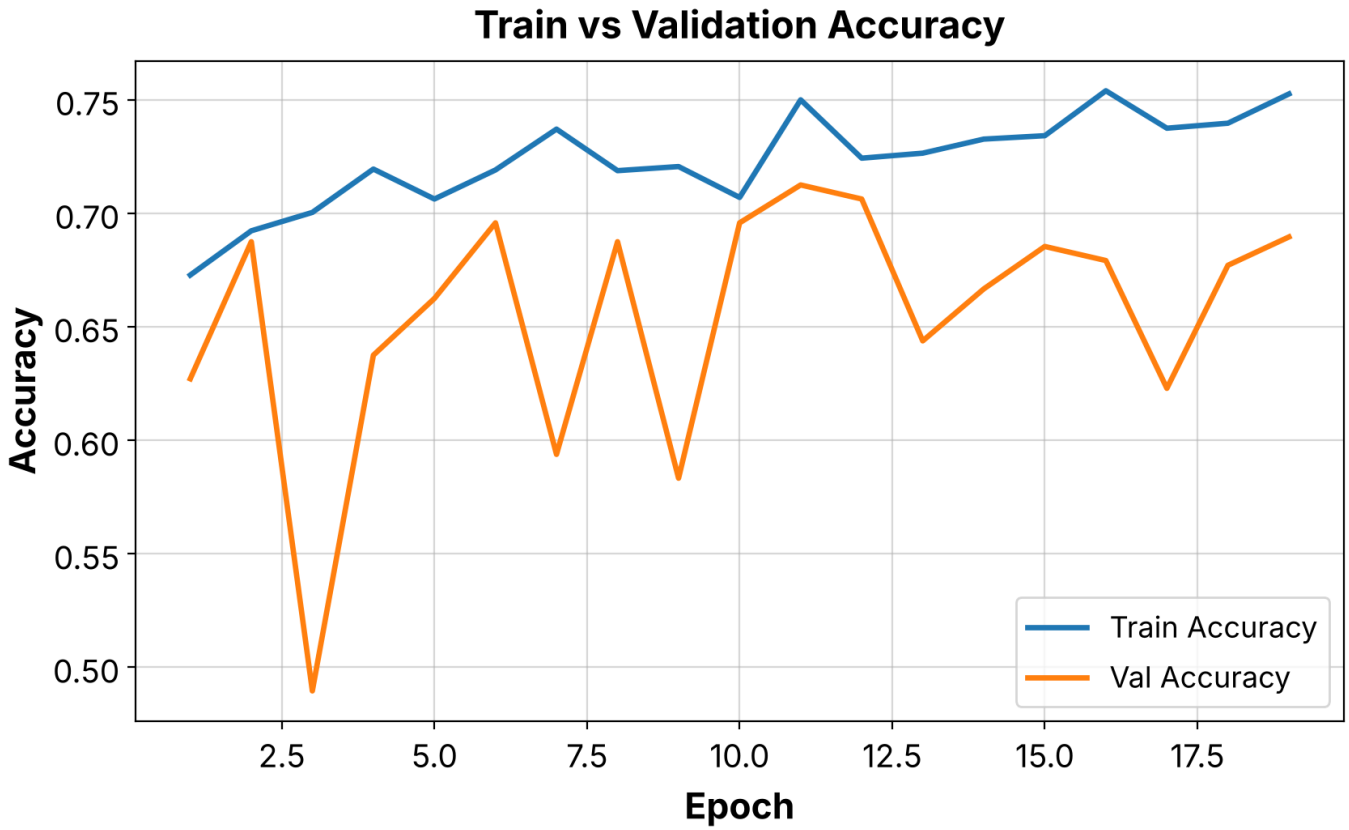
Confusion Matrix



11. MobileNetV2 + Gaussian Naive Bayes — graphs

Training / Validation Graphs





12. MobileNetV2 — Direct Classification (Dice Loss baseline)

Source notebook: MobileNet with DiceLoss (BaseLine).ipynb (inside Models_DL.zip)

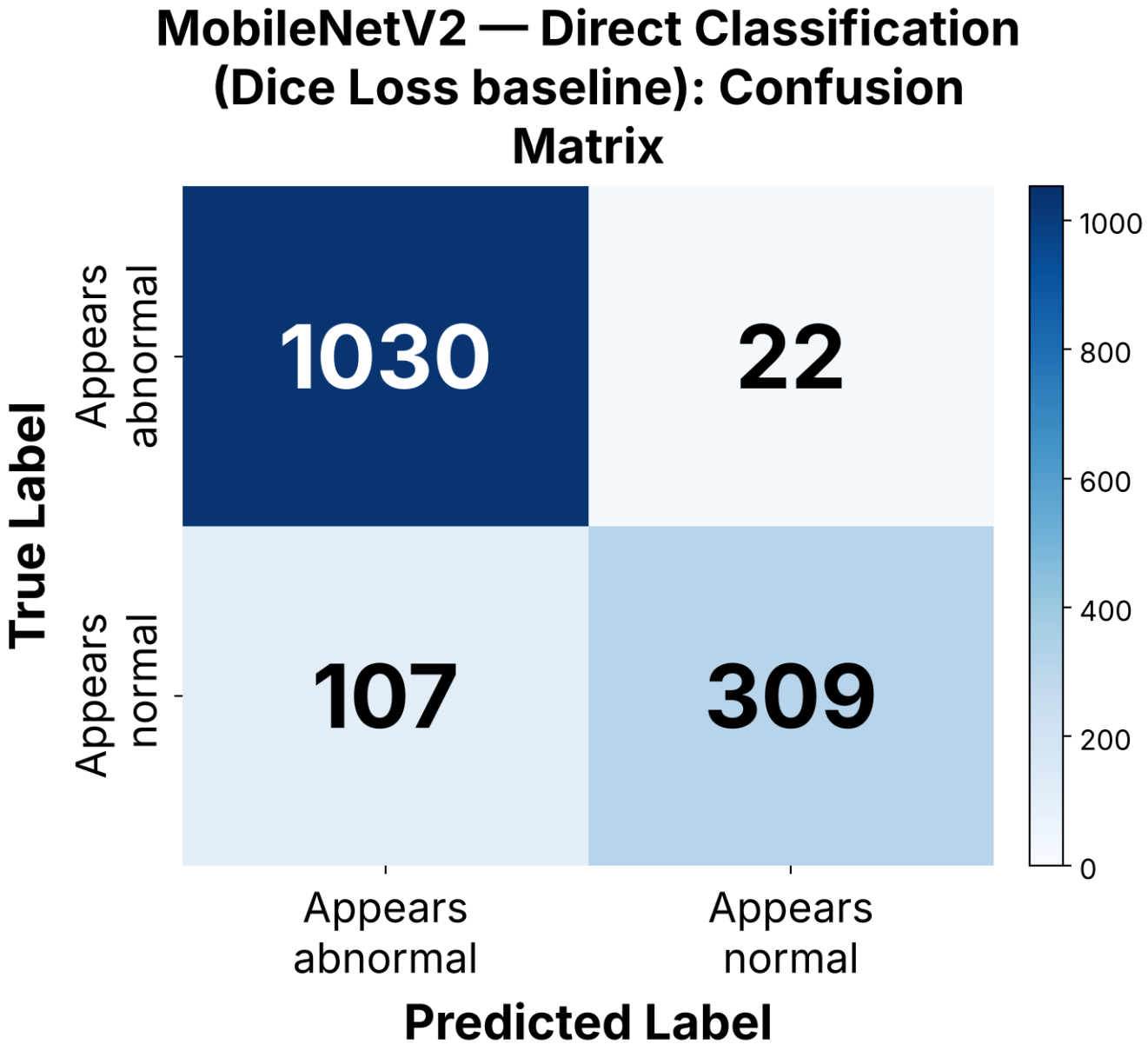
Accuracy: 91.21%

Precision: 91.37%

Recall: 91.21%

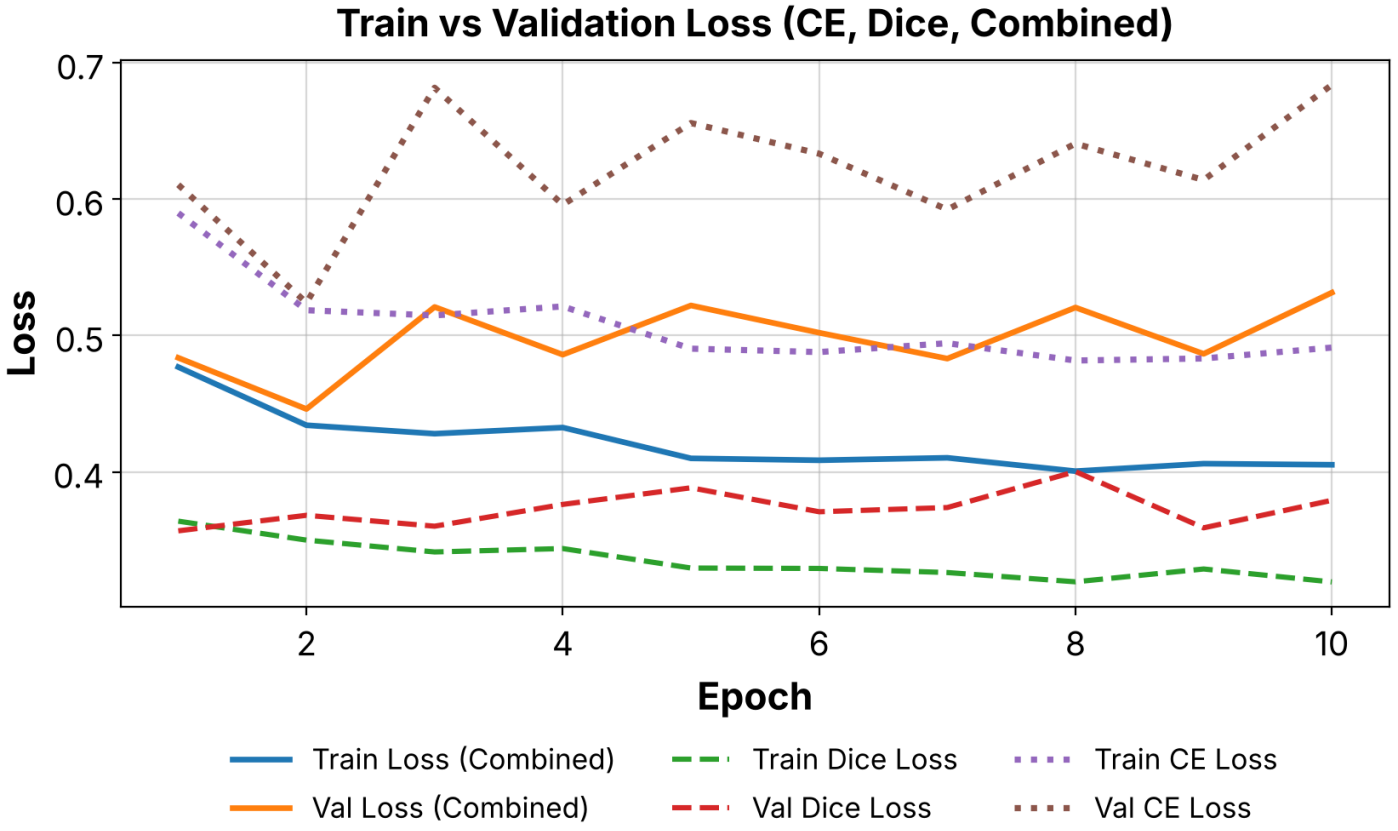
F1-score: 90.88%

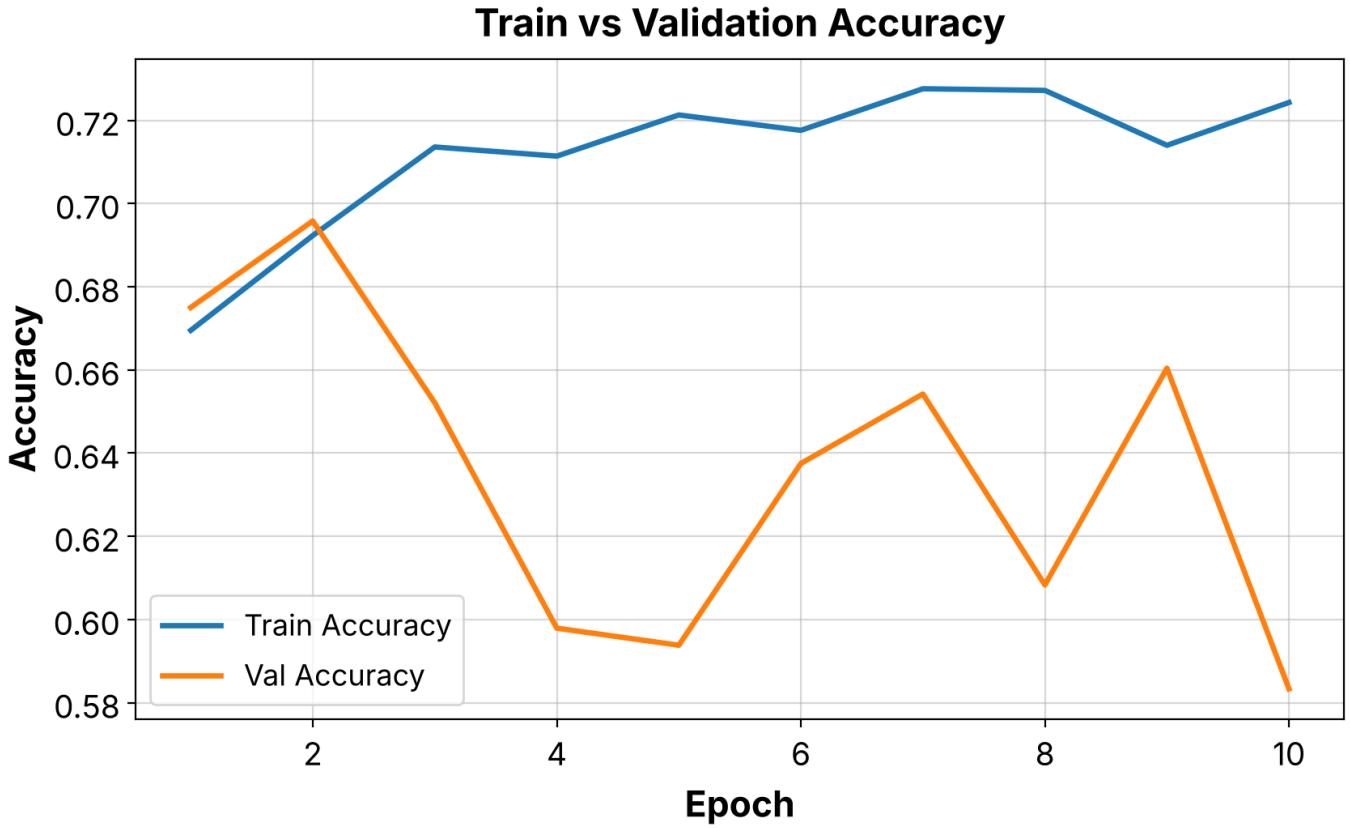
Confusion Matrix



12. MobileNetV2 — Direct Classification (Dice Loss baseline) — graphs

Training / Validation Graphs





13. EfficientNet-B3 — Direct Classification (baseline)

Source notebook: efficinetnet b3 baseline and ml.ipynb (inside Models_DL.zip)

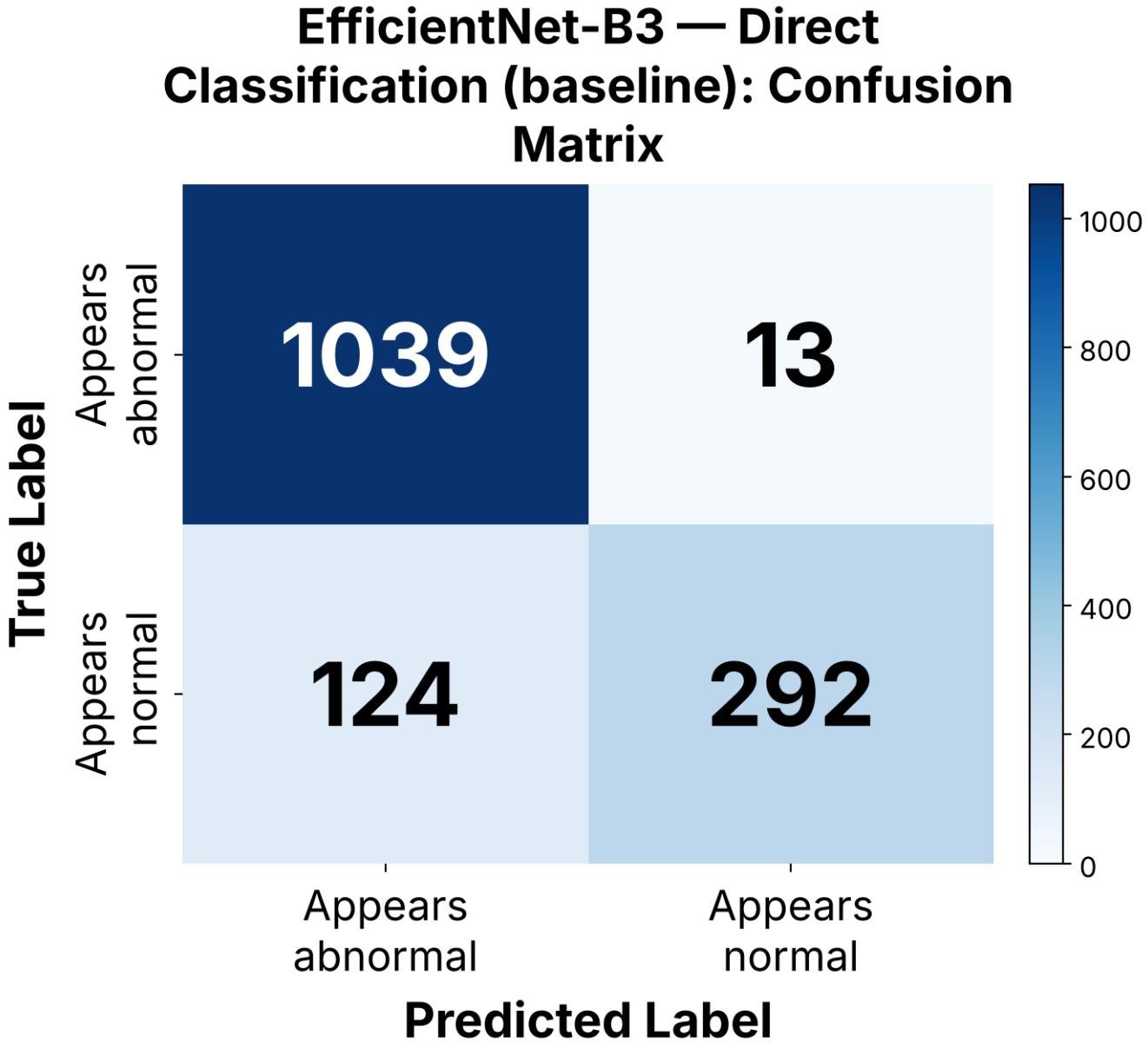
Accuracy: 90.67%

Precision: 91.15%

Recall: 90.67%

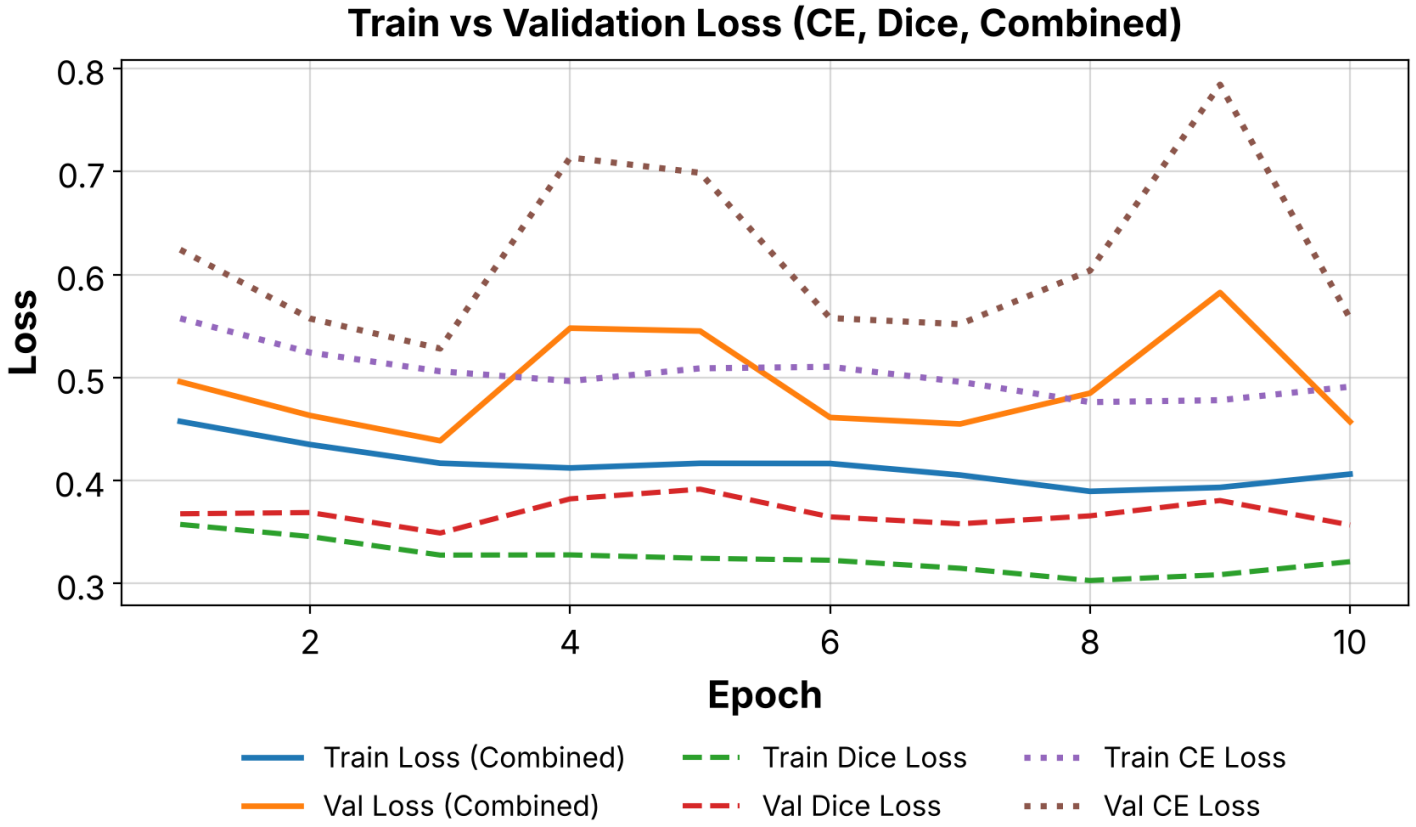
F1-score: 90.18%

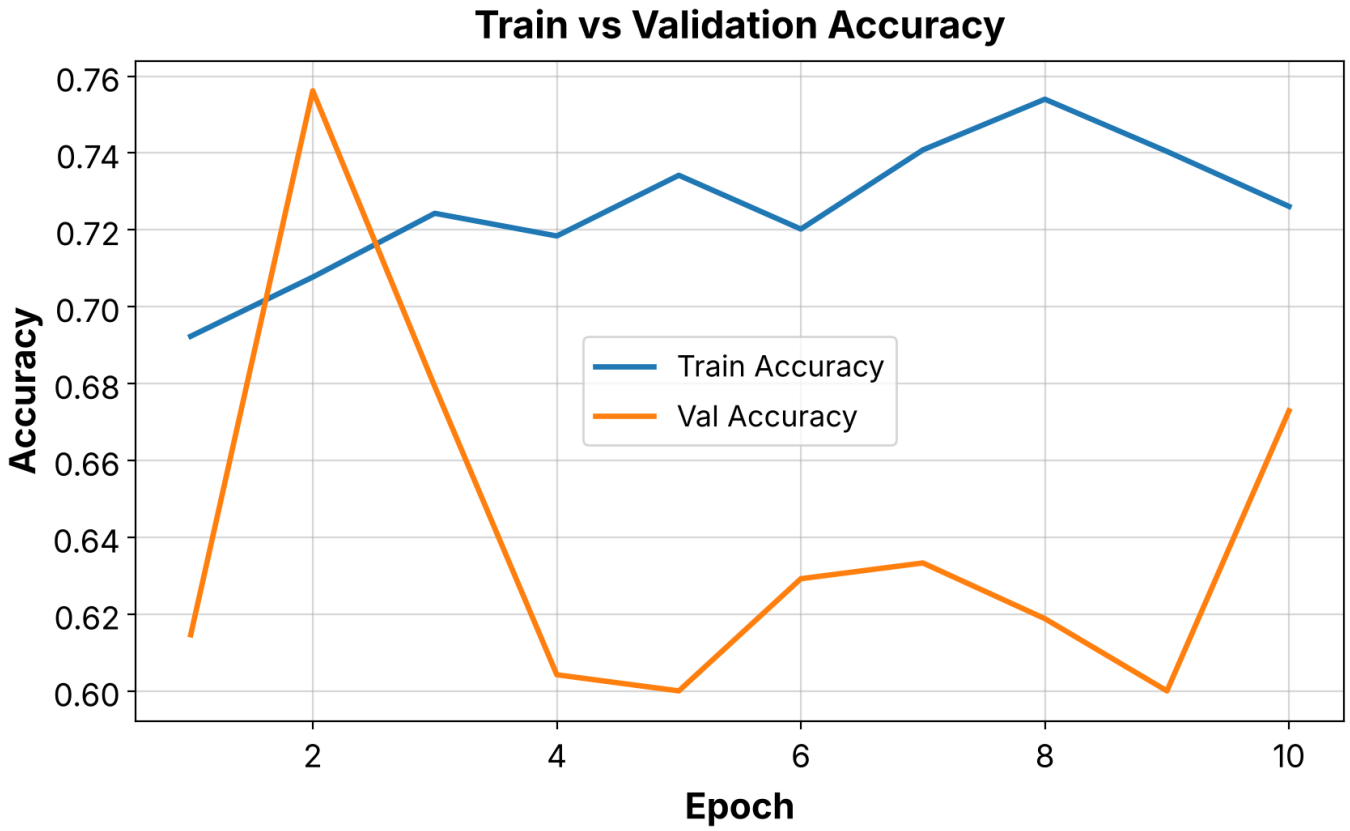
Confusion Matrix



13. EfficientNet-B3 — Direct Classification (baseline) — graphs

Training / Validation Graphs





14. ResNet18 + Random Forest

Source notebook: ResNet with ML.ipynb (inside Models_DL.zip)

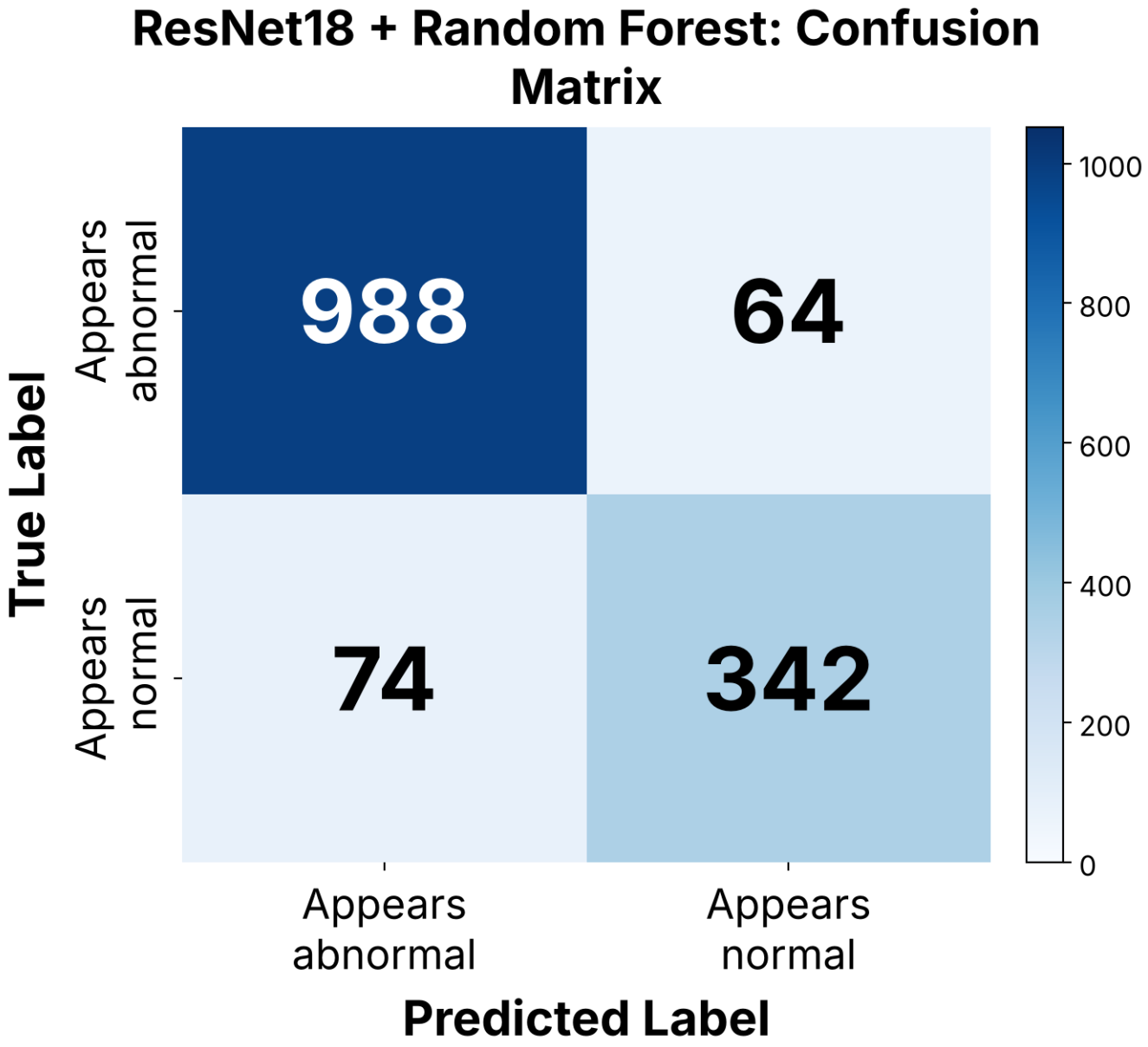
Accuracy: 90.60%

Precision: 90.54%

Recall: 90.60%

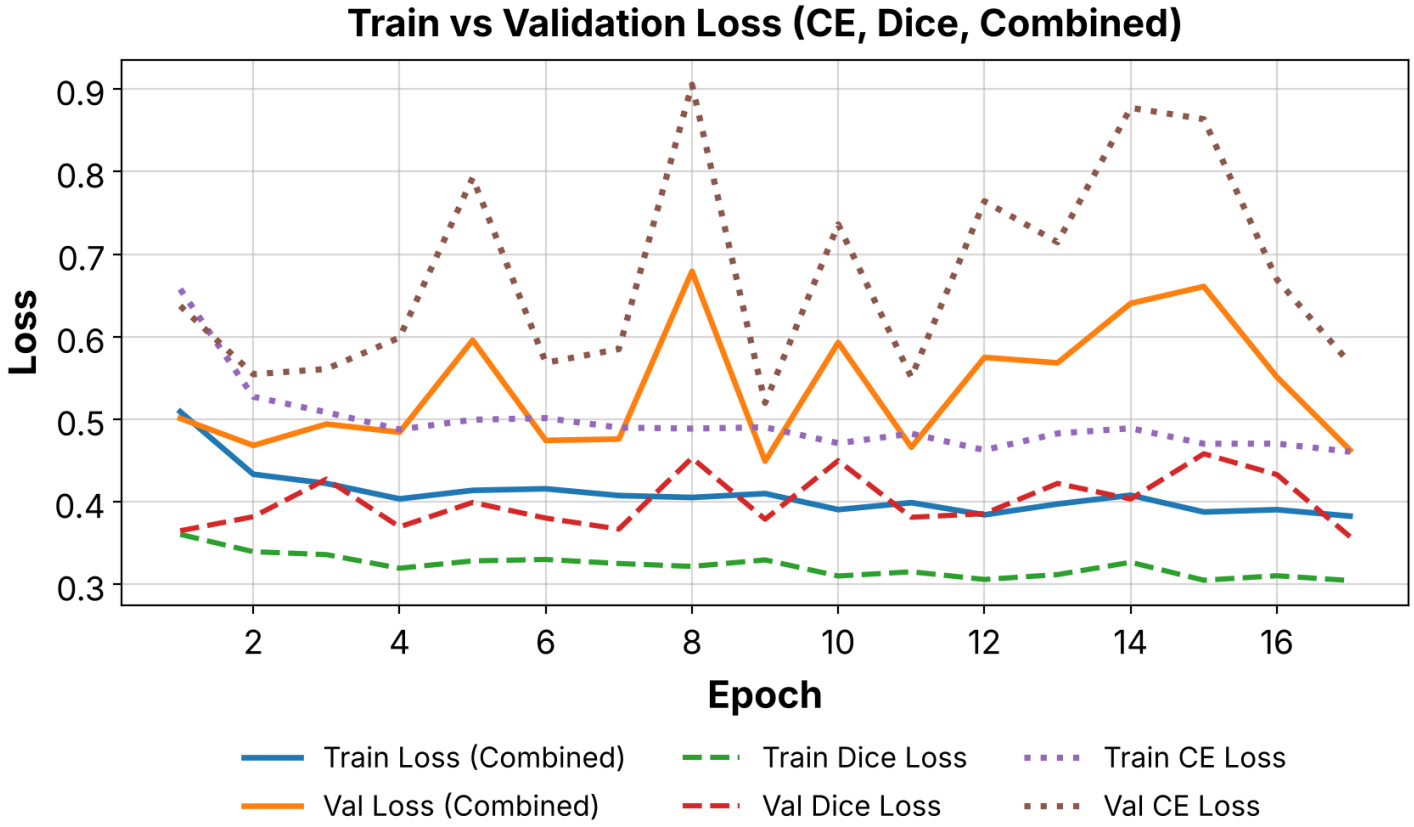
F1-score: 90.56%

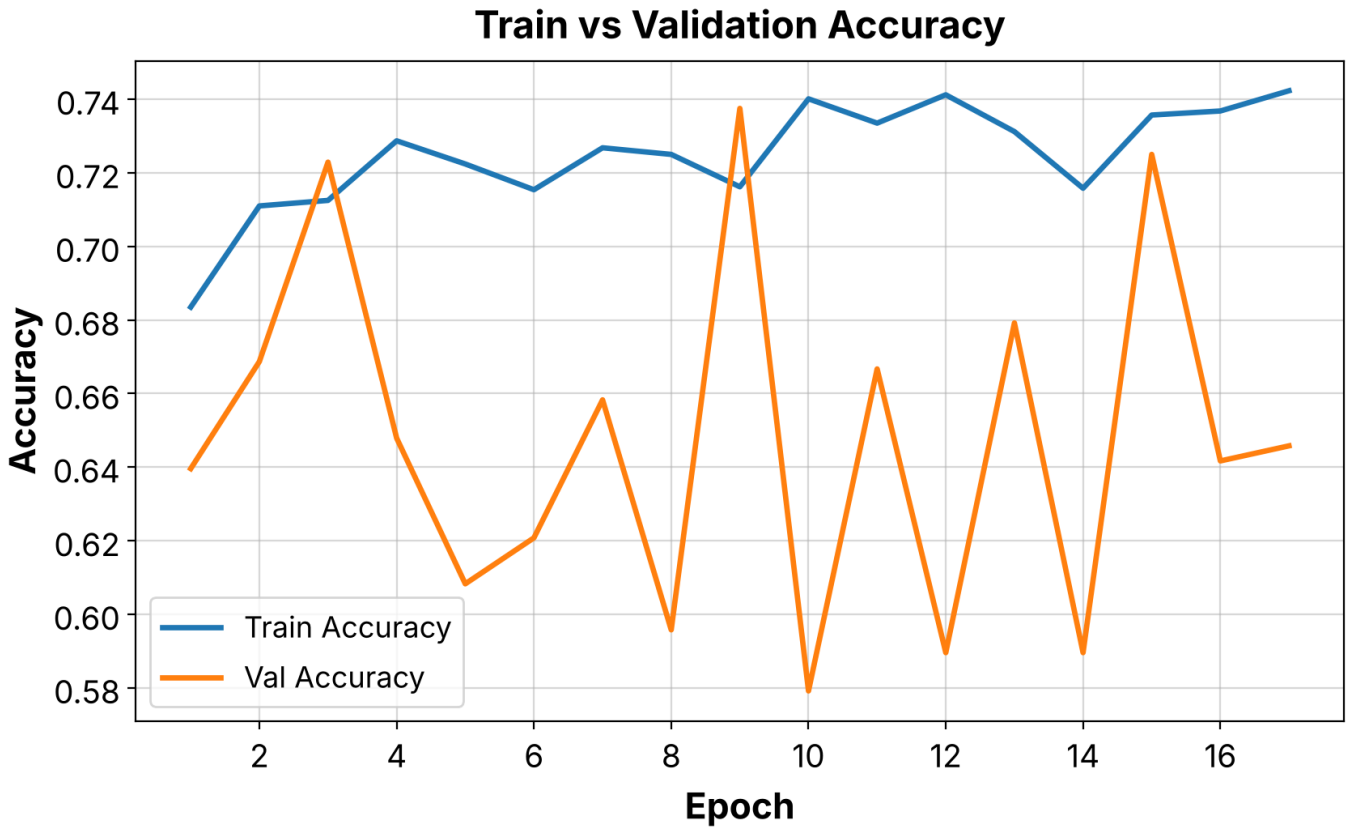
Confusion Matrix



14. ResNet18 + Random Forest — graphs

Training / Validation Graphs





15. MobileNetV2 + Random Forest

Source notebook: MobileNet with ML.ipynb (inside Models_DL.zip)

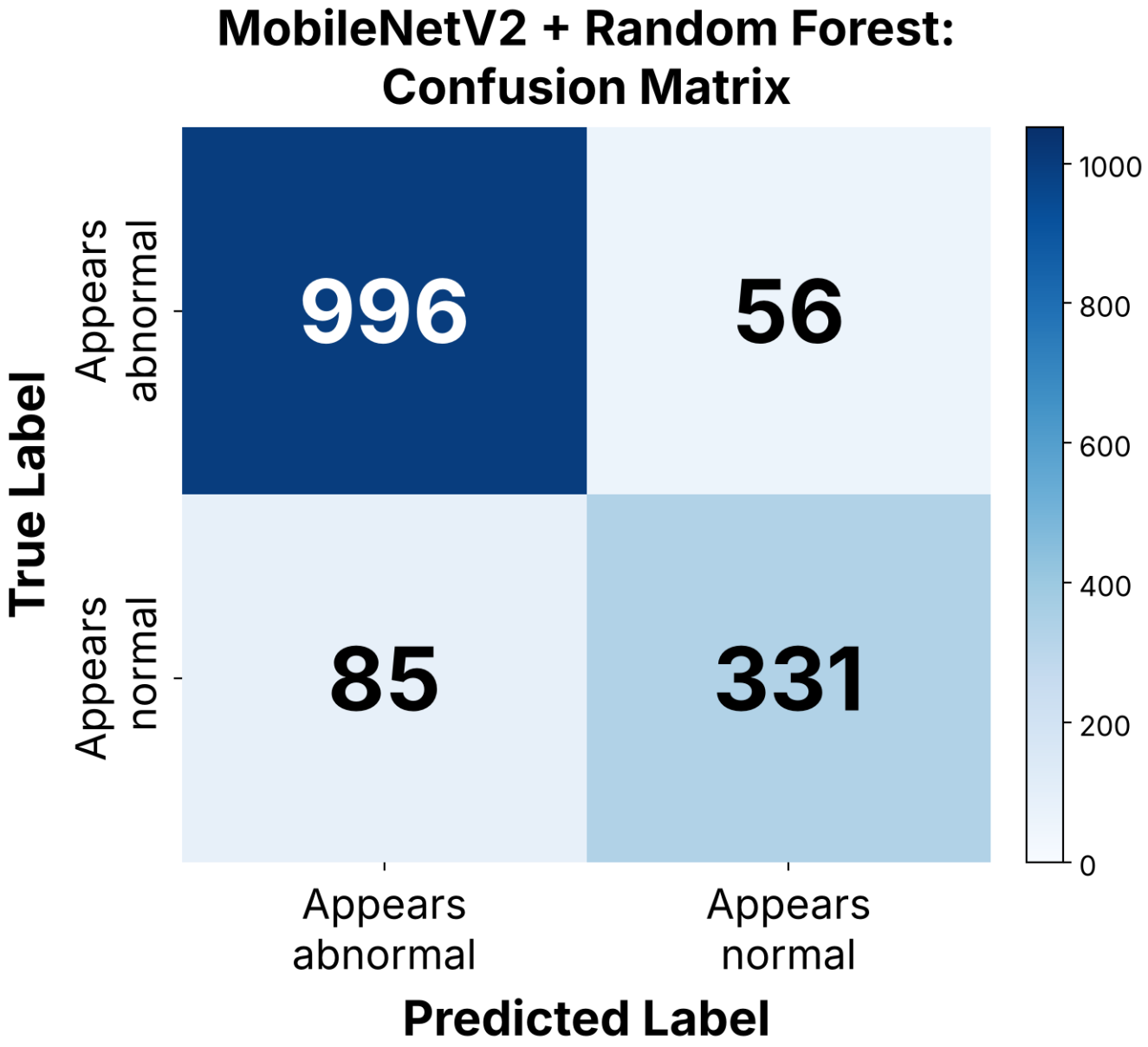
Accuracy: 90.40%

Precision: 90.26%

Recall: 90.40%

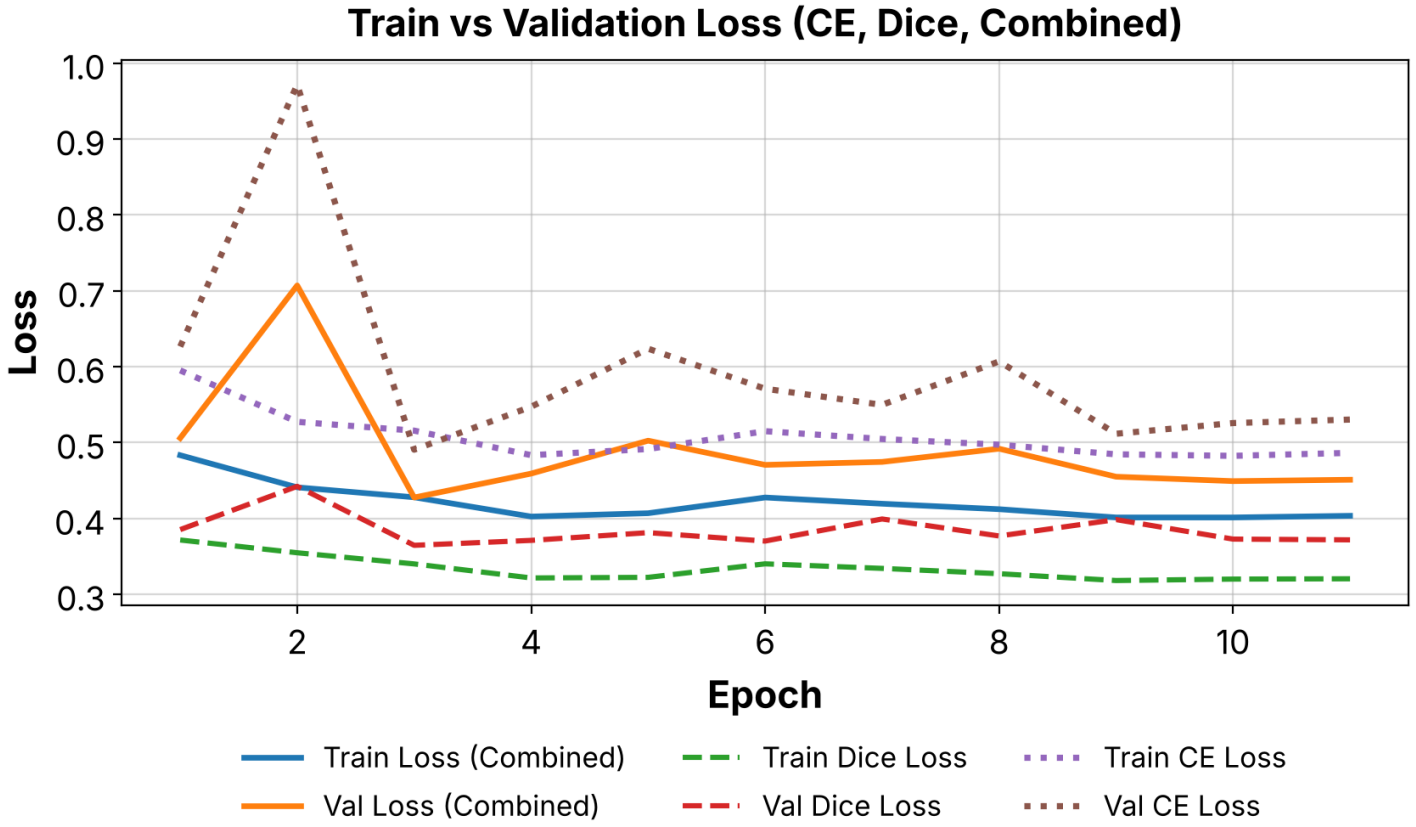
F1-score: 90.29%

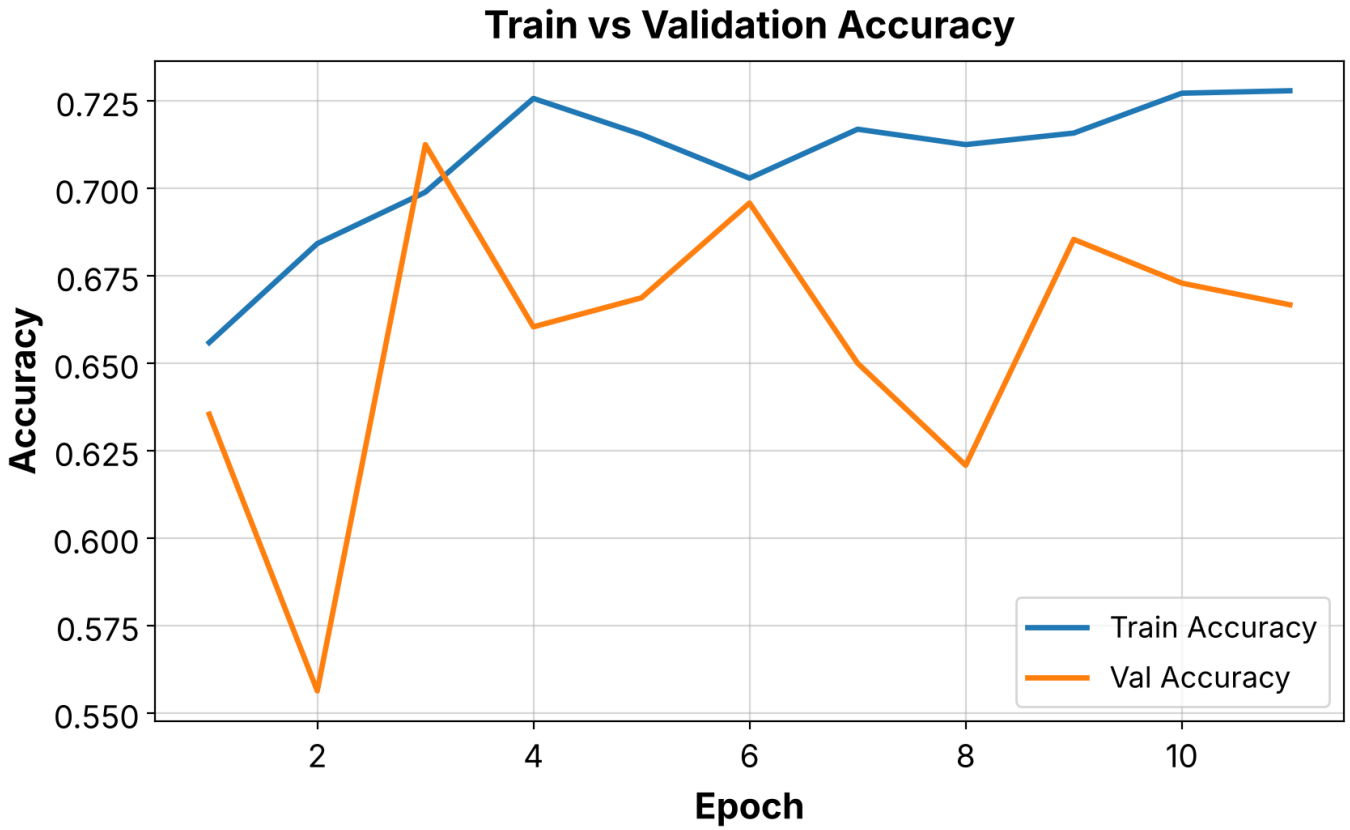
Confusion Matrix



15. MobileNetV2 + Random Forest — graphs

Training / Validation Graphs





16. MobileNetV2 — Direct Classification (AdaBoost notebook run)

Source notebook: mobilenet with adaboost.ipynb (inside Models_DL.zip)

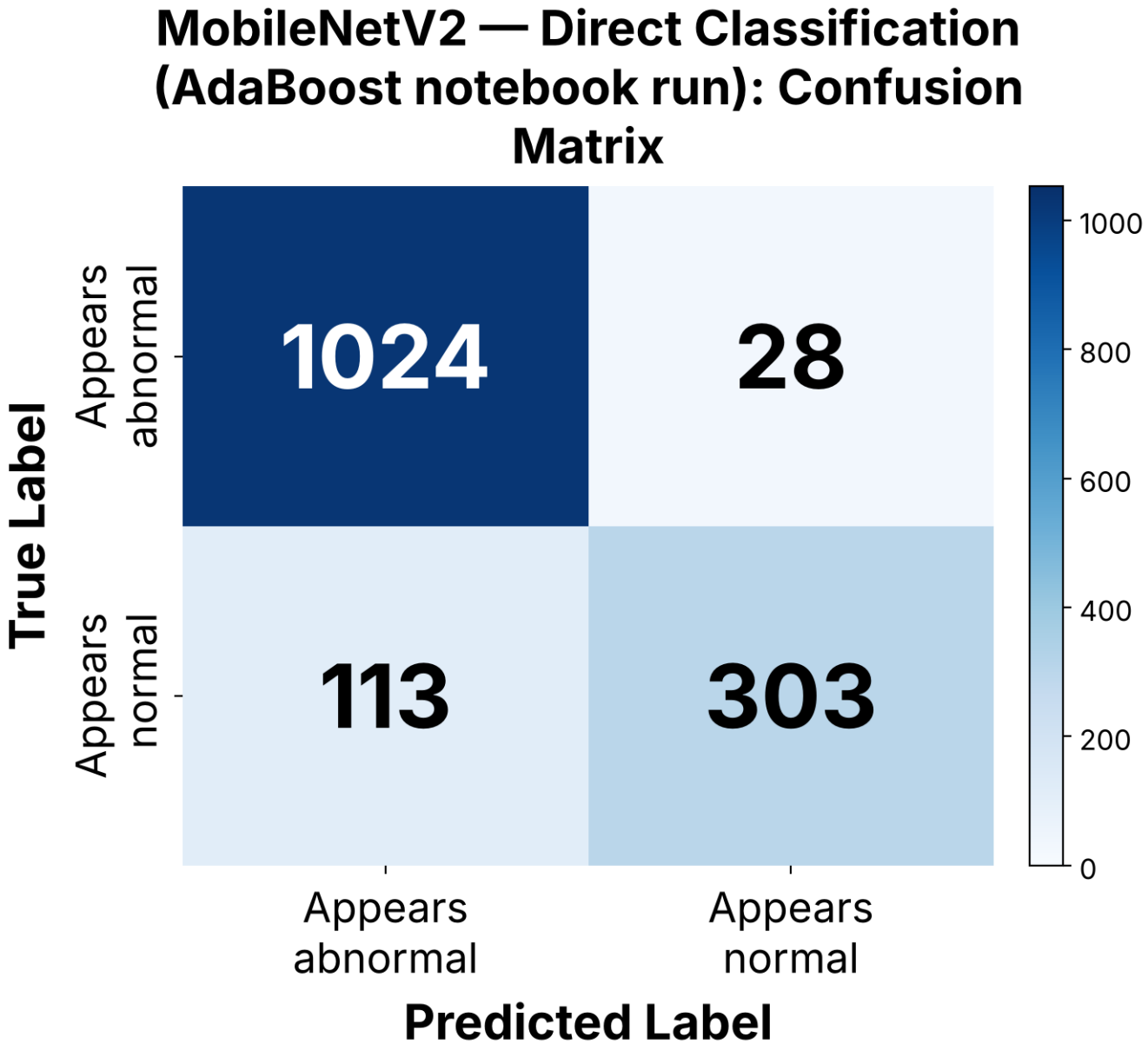
Accuracy: 90.40%

Precision: 90.48%

Recall: 90.40%

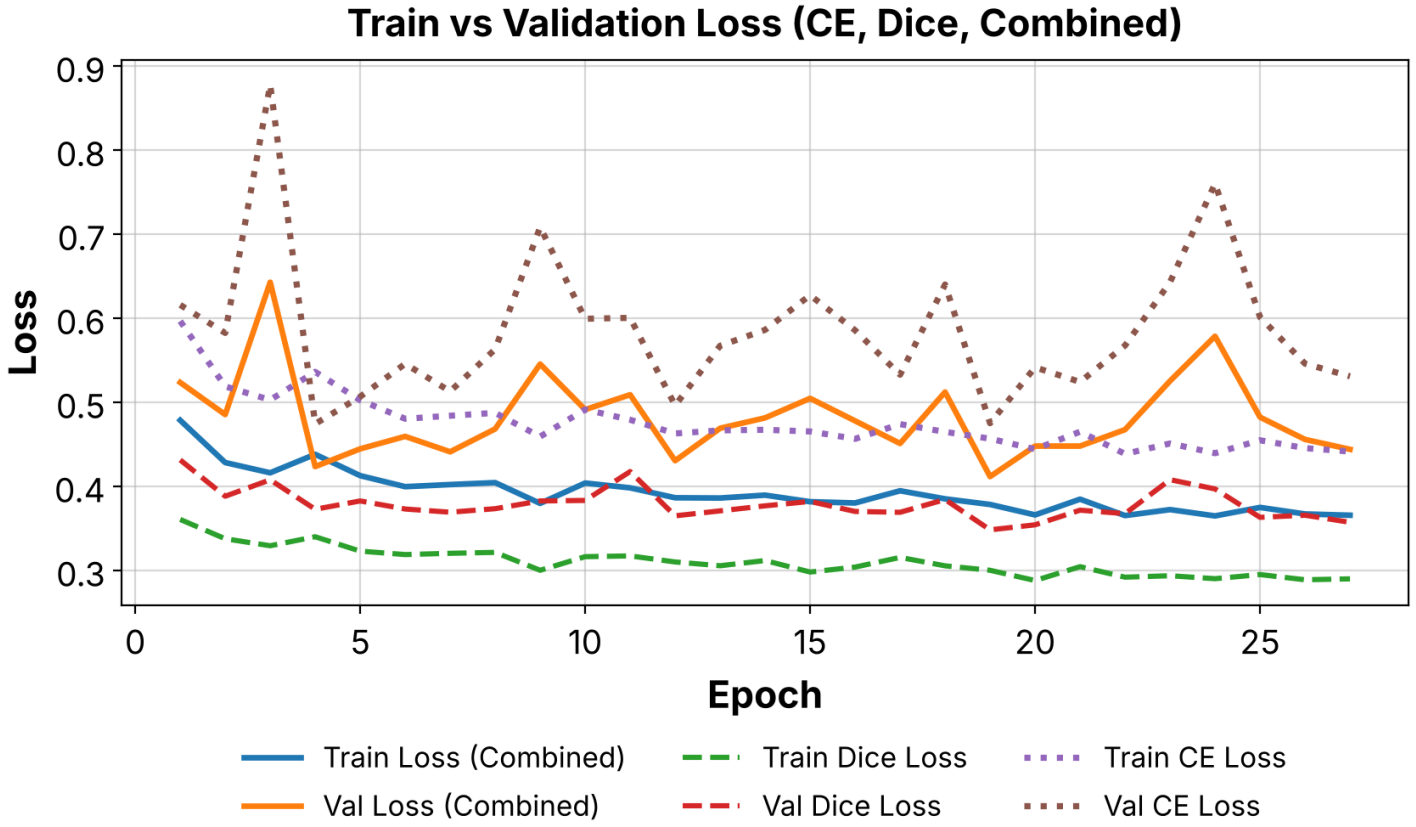
F1-score: 90.04%

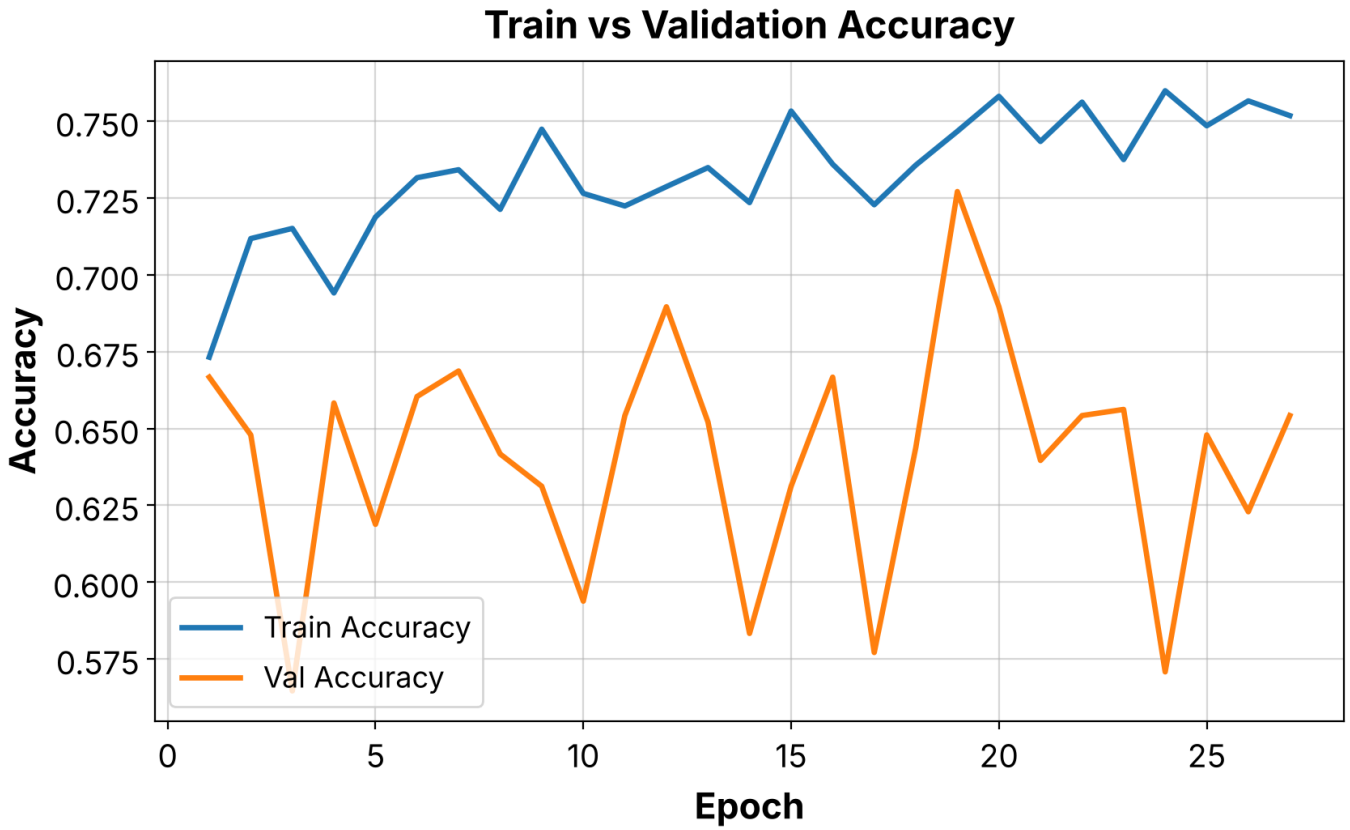
Confusion Matrix



16. MobileNetV2 — Direct Classification (AdaBoost notebook run) — graphs

Training / Validation Graphs





17. VGG16 — Direct Classification (Dice Loss baseline)

Source notebook: Vgg with Dice Loss (Baseline).ipynb (inside Models_DL.zip)

Accuracy: 90.33%

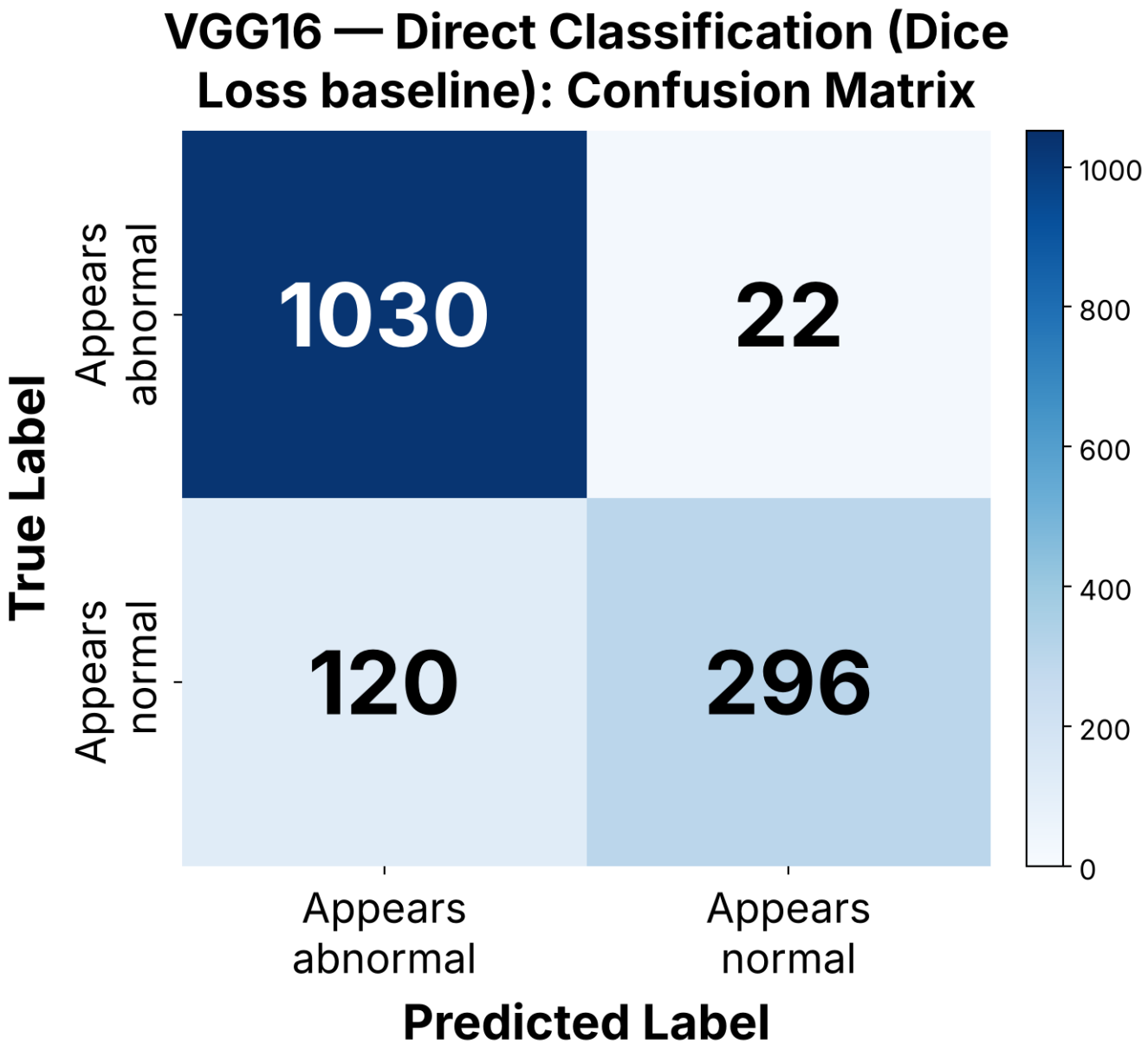
Precision: 90.56%

Recall: 90.33%

F1-score: 89.90%

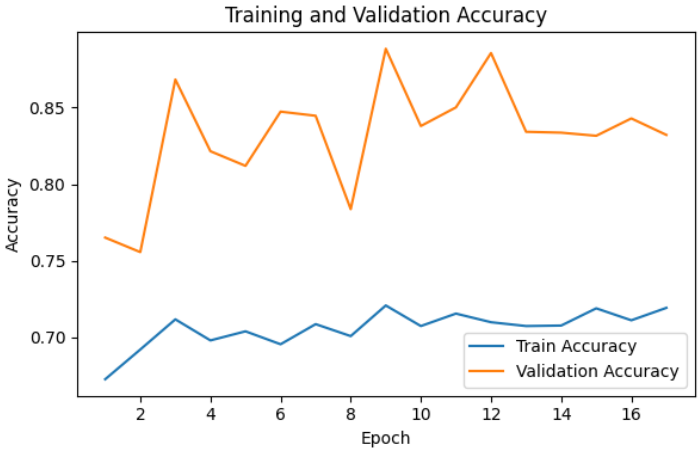
Note: the notebook prints 88.84% as its test / “Final Test Accuracy”, which does not agree with the notebook’s own printed confusion matrix and weighted recall (this printed number is not the test accuracy). The test accuracy shown here is the one implied by the confusion matrix and equals the weighted recall.

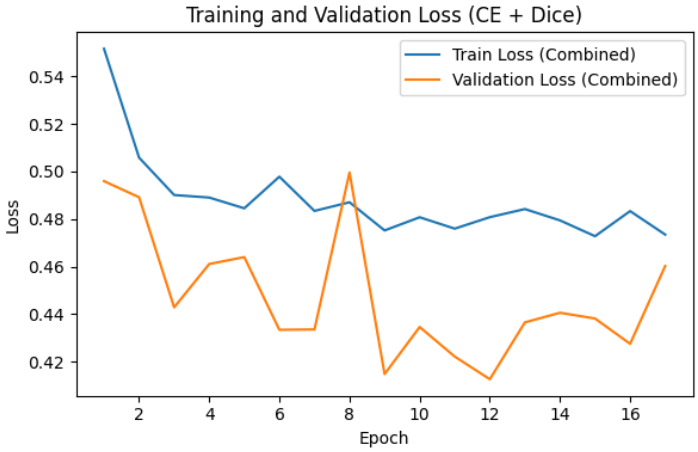
Confusion Matrix

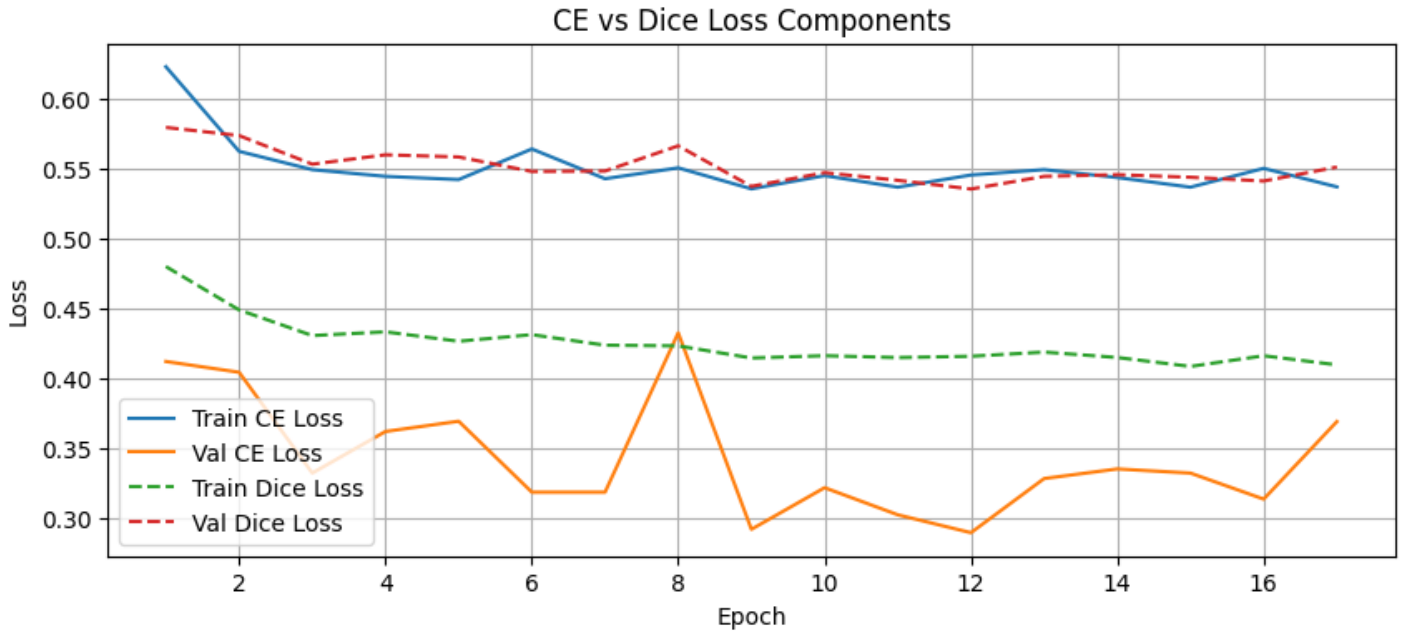


17. VGG16 — Direct Classification (Dice Loss baseline) — graphs

Training / Validation Graphs







18. VGG16 + SVM

Source notebook: Xgboost and SVM and all 4 models.ipynb (inside Models_DL.zip)

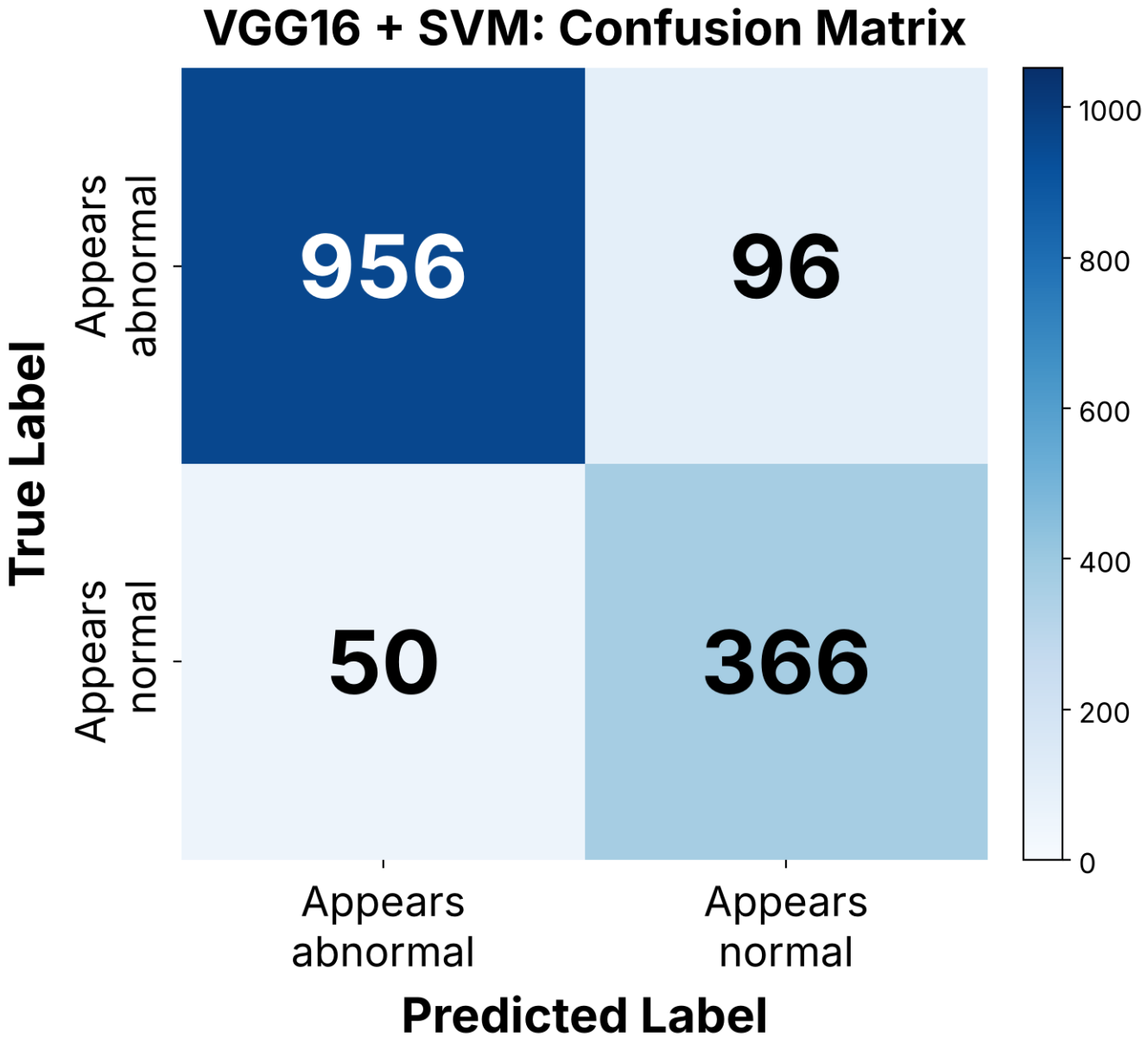
Accuracy: 90.05%

Precision: 90.55%

Recall: 90.05%

F1-score: 90.20%

Confusion Matrix



No training graphs: this notebook reloads an already-trained network and prints no epoch log, so there is nothing to plot for this row.

19. EfficientNet-B3 + KNN

Source notebook: efficinetnet b3 baseline and ml.ipynb (inside Models_DL.zip)

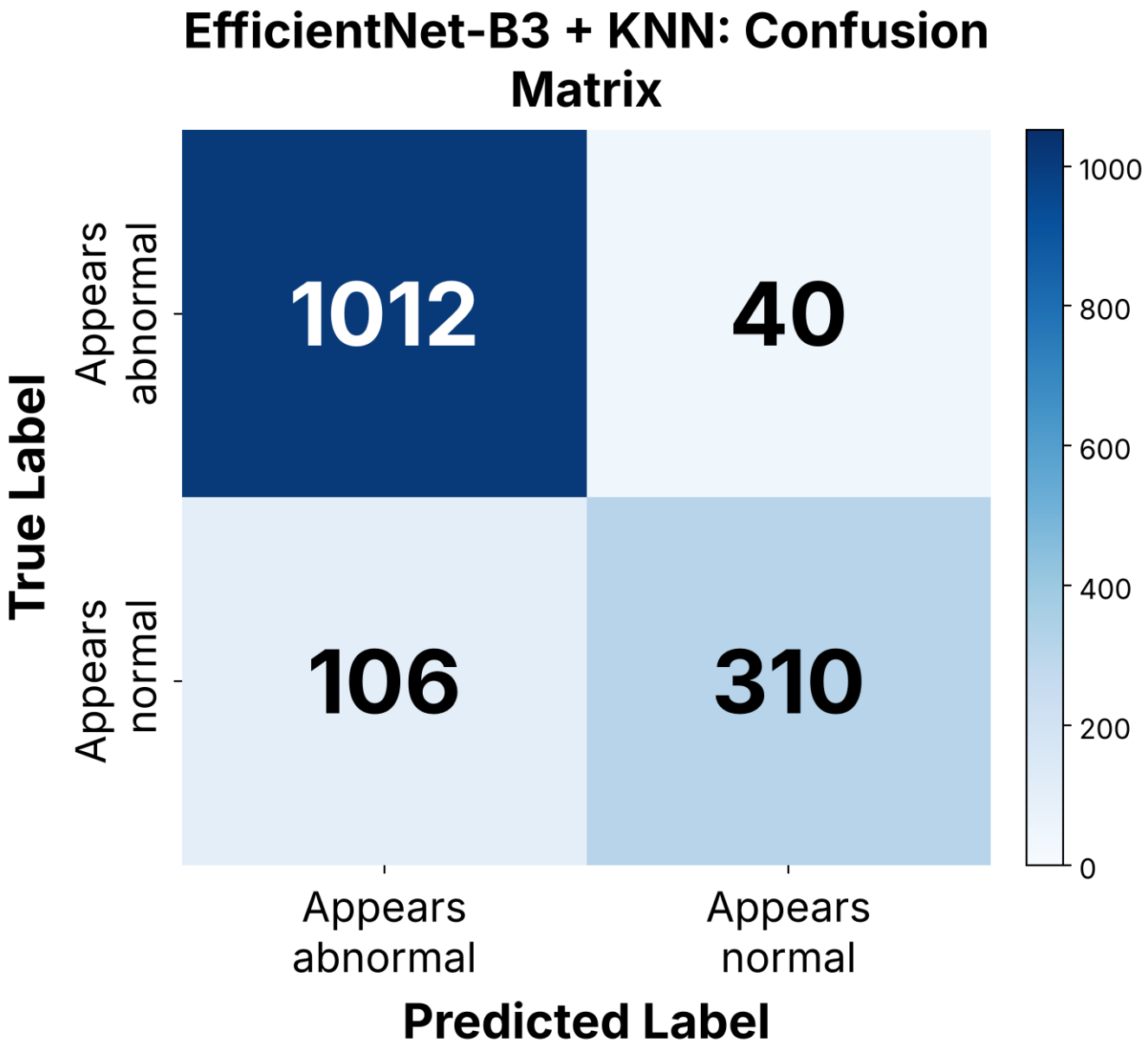
Accuracy: 90.05%

Precision: 89.97%

Recall: 90.05%

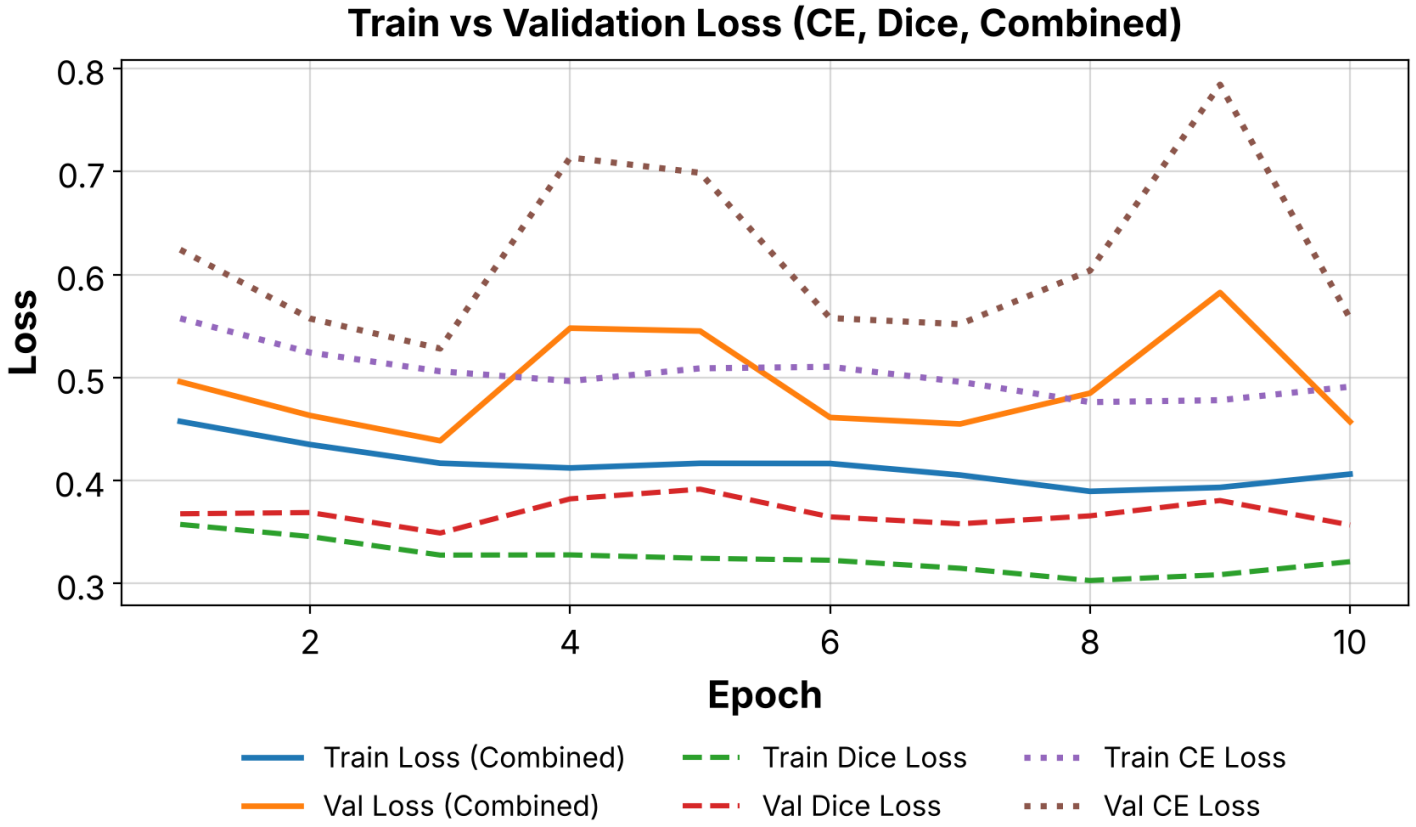
F1-score: 89.78%

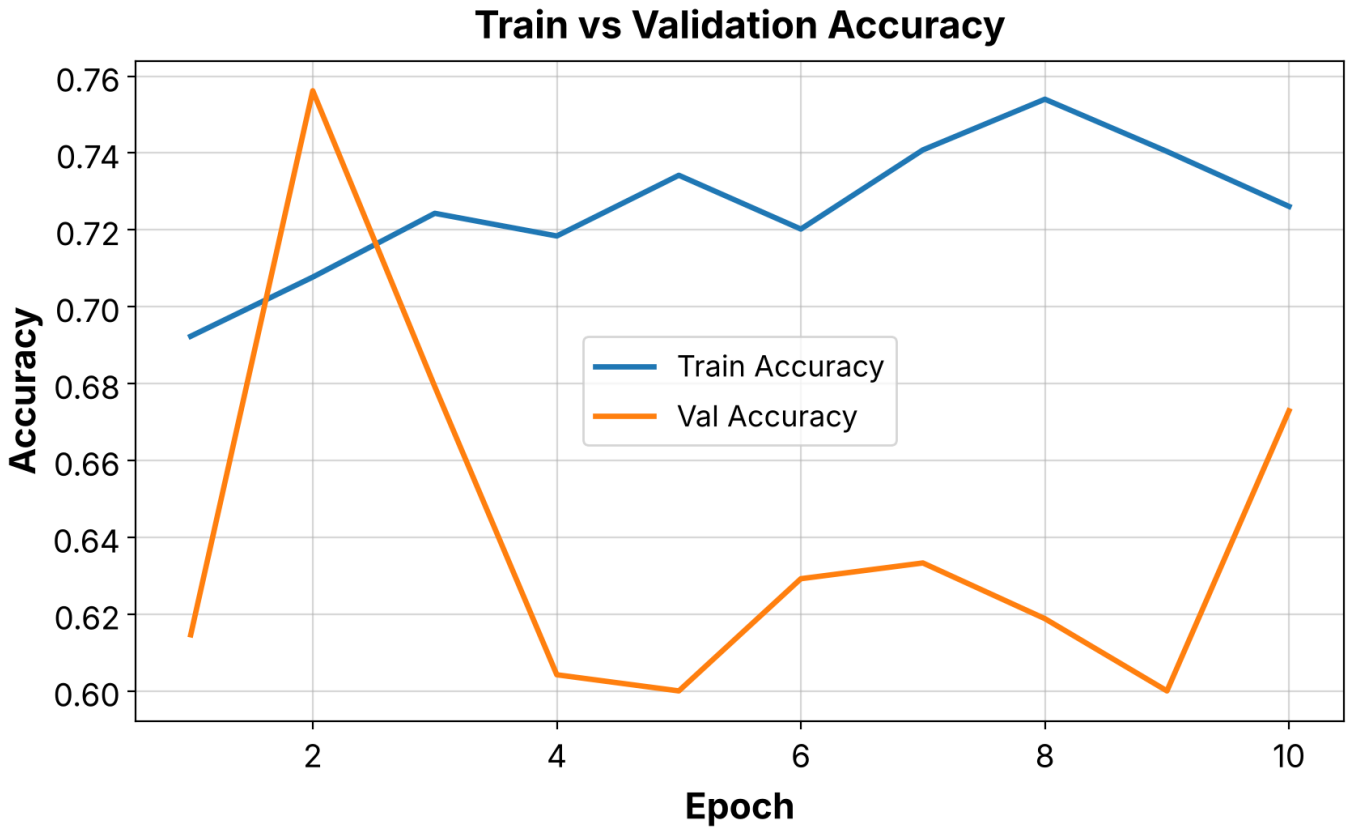
Confusion Matrix



19. EfficientNet-B3 + KNN — graphs

Training / Validation Graphs





20. MobileNetV2 + KNN

Source notebook: MobileNet with ML.ipynb (inside Models_DL.zip)

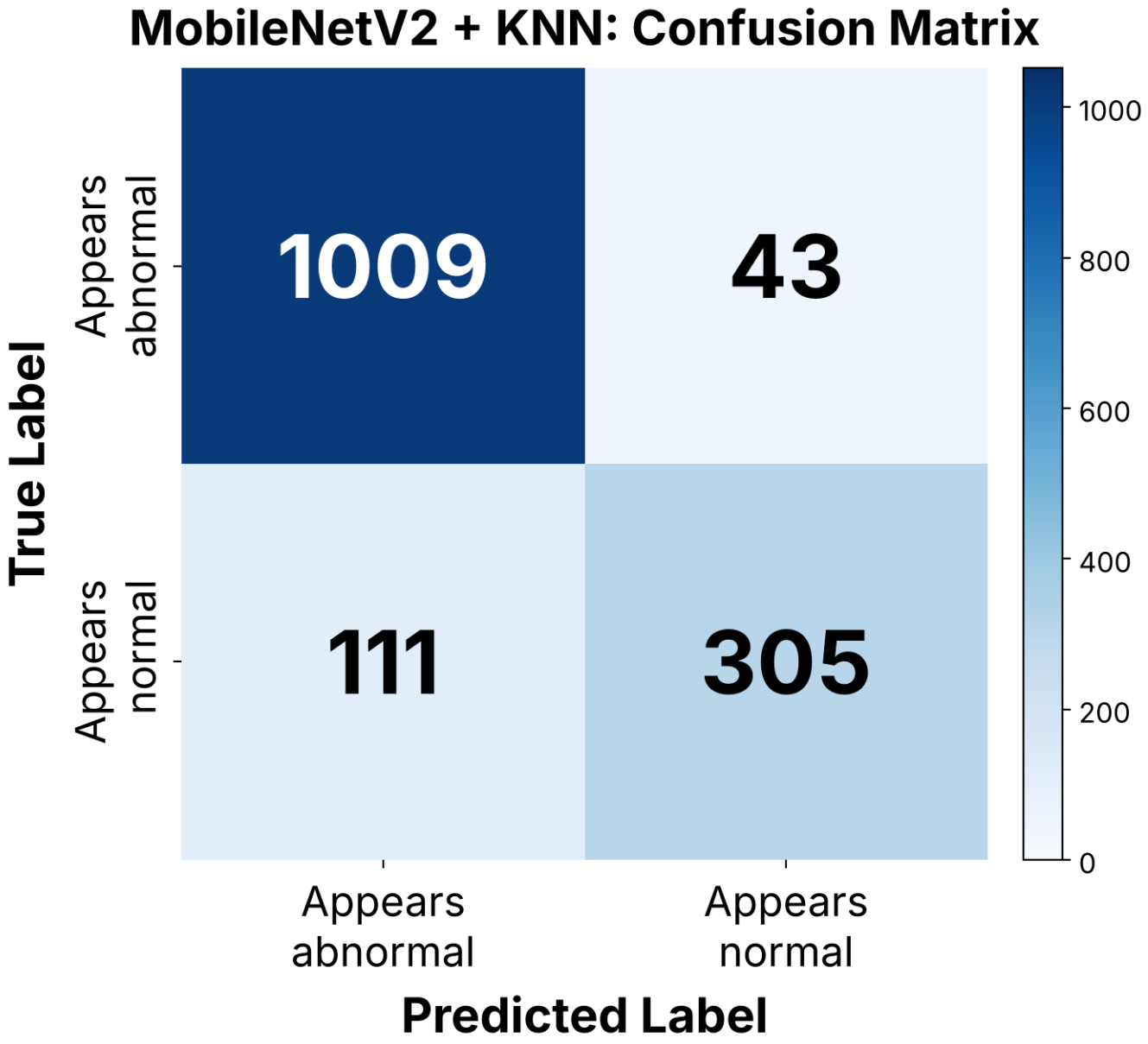
Accuracy: 89.51%

Precision: 89.40%

Recall: 89.51%

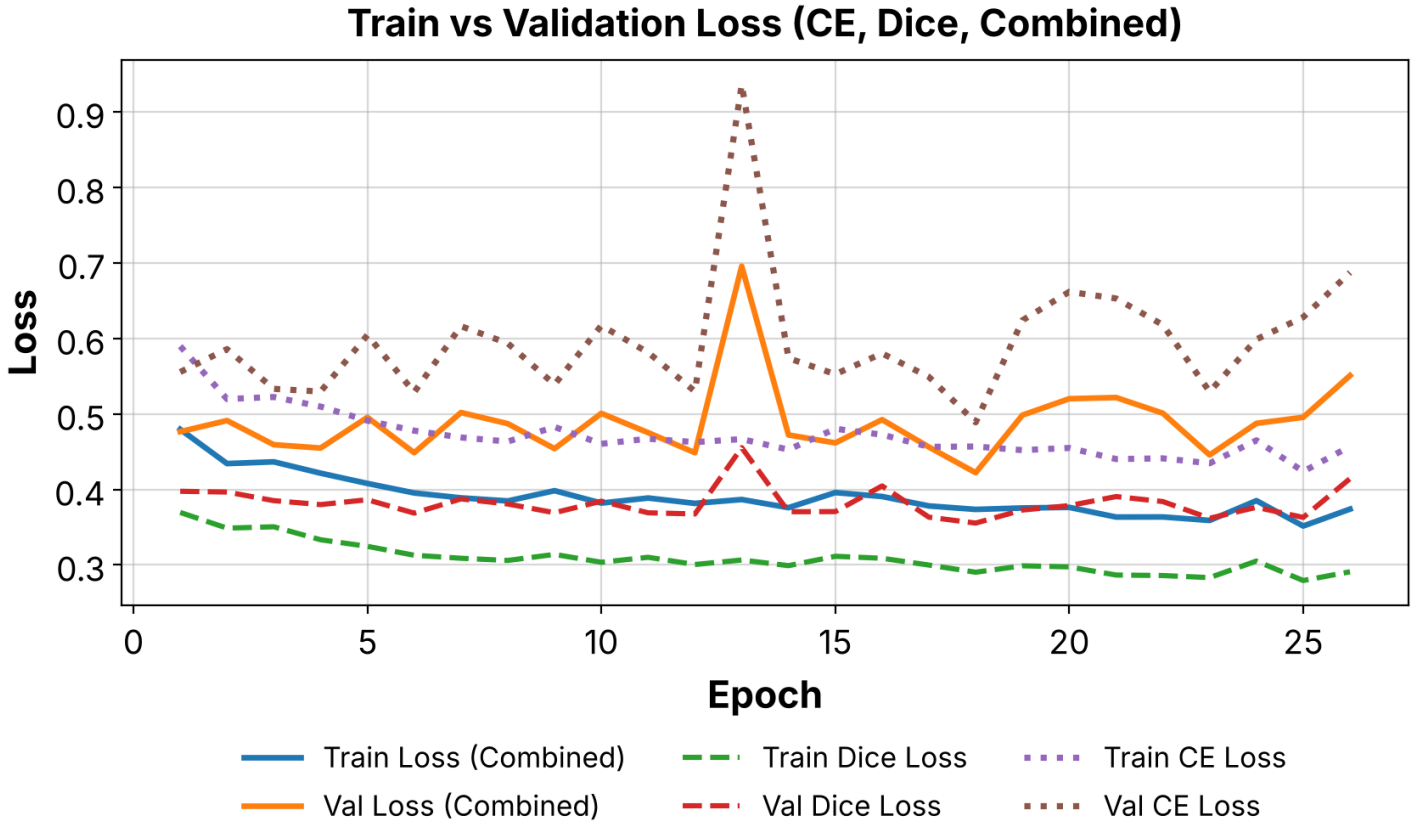
F1-score: 89.21%

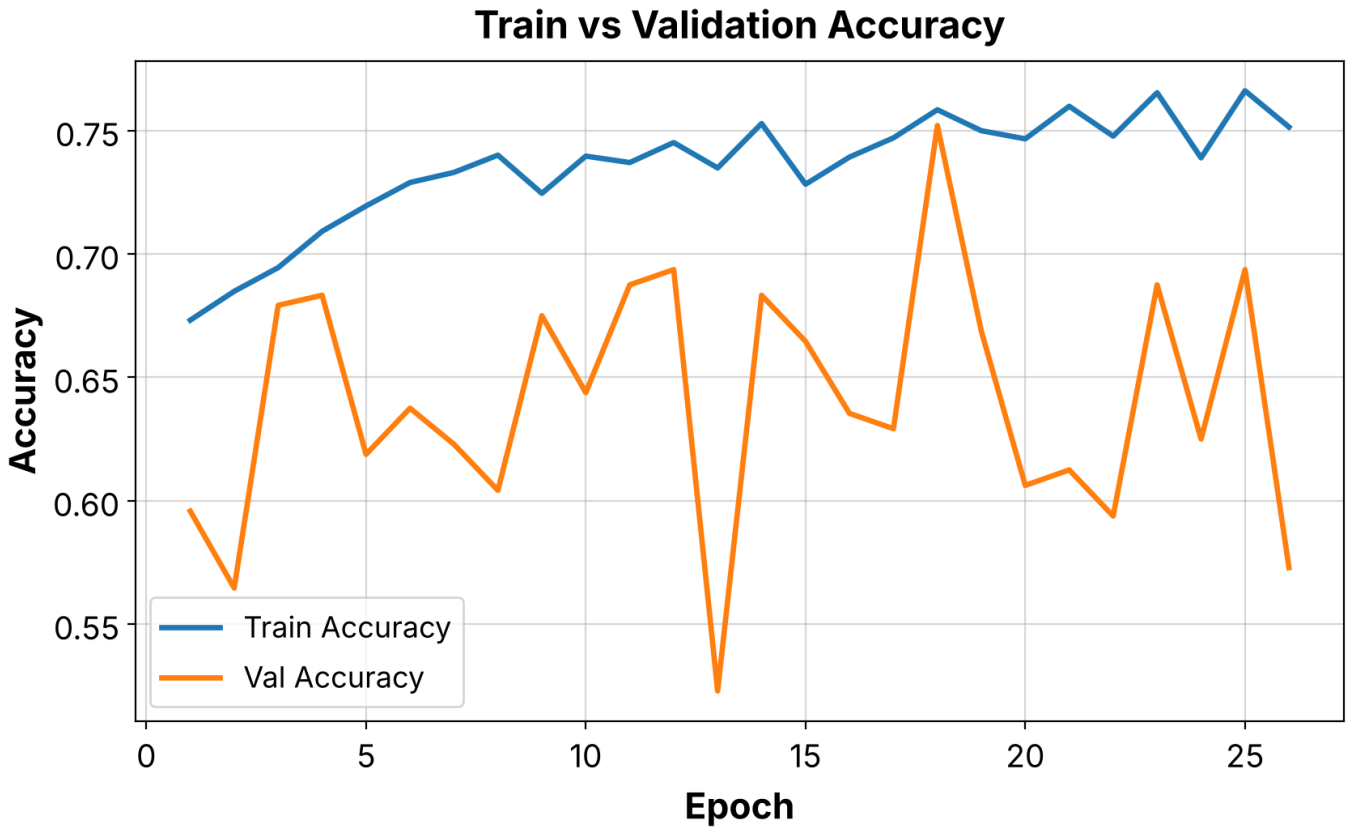
Confusion Matrix



20. MobileNetV2 + KNN — graphs

Training / Validation Graphs





21. InceptionV3 + SVM

Source notebook: Inception v3 baseline and ml.ipynb (inside Models_DL.zip)

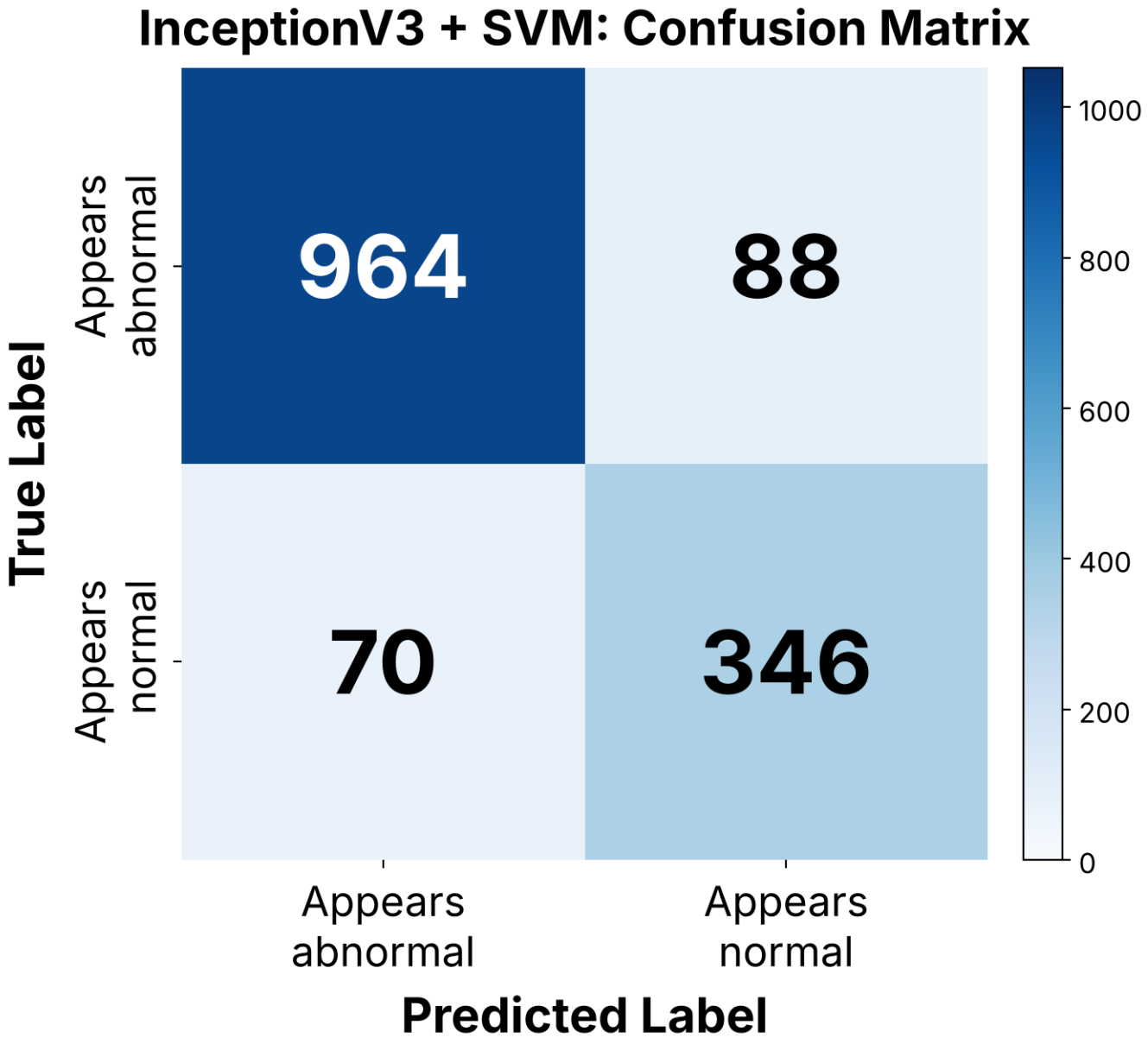
Accuracy: 89.24%

Precision: 89.40%

Recall: 89.24%

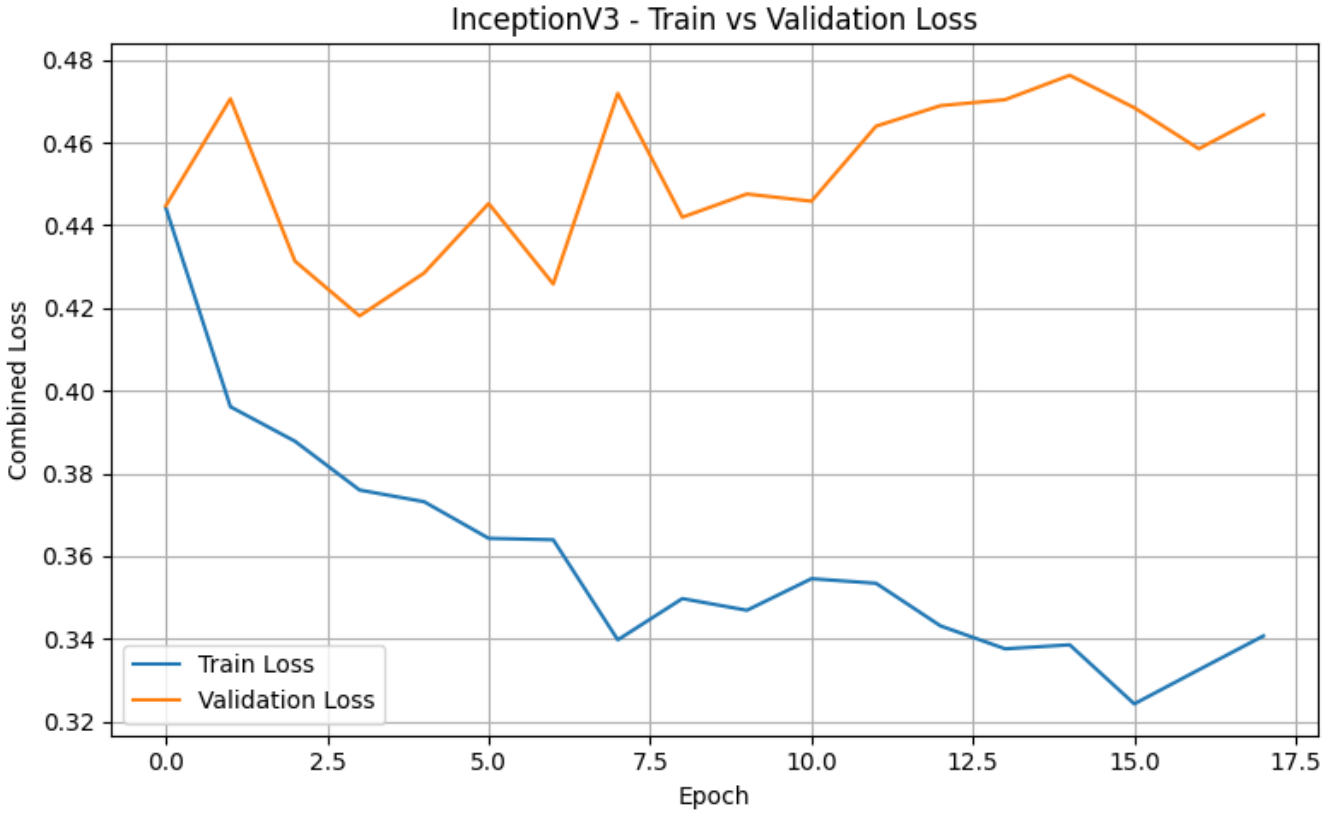
F1-score: 89.30%

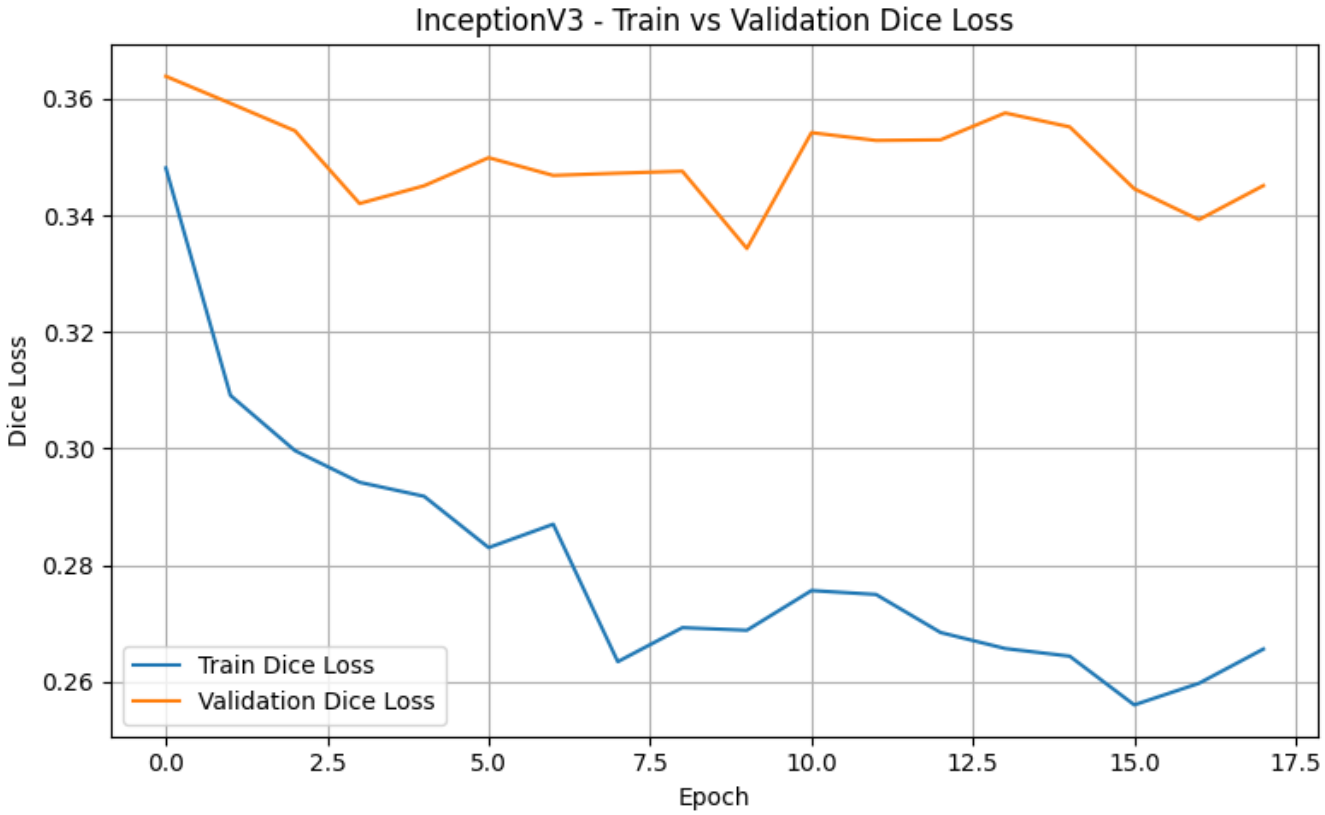
Confusion Matrix



21. InceptionV3 + SVM — graphs (1/2)

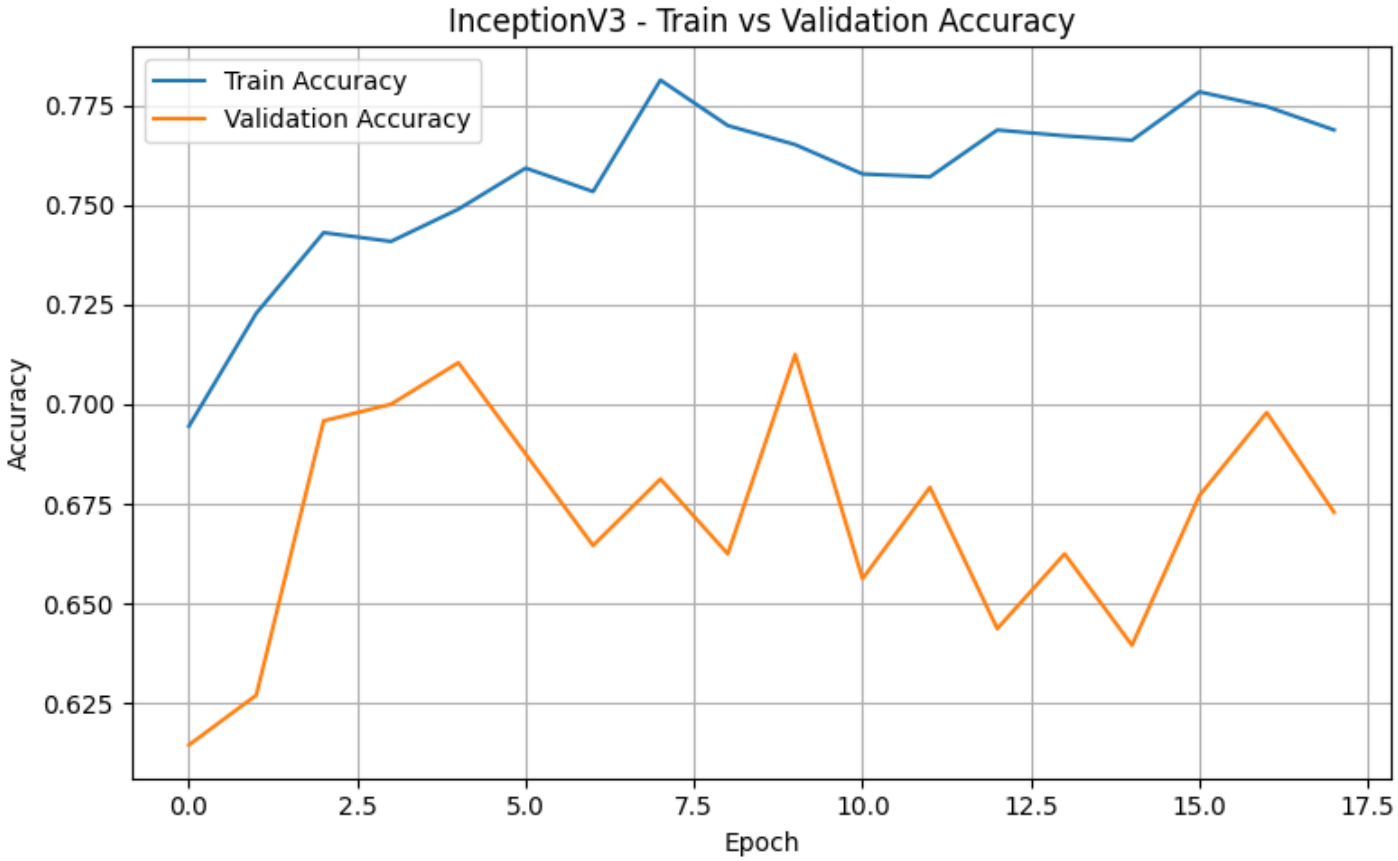
Training / Validation Graphs





21. InceptionV3 + SVM — graphs (2/2)

Training / Validation Graphs



22. MobileNetV2 + SVM

Source notebook: Xgboost and SVM and all 4 models.ipynb (inside Models_DL.zip)

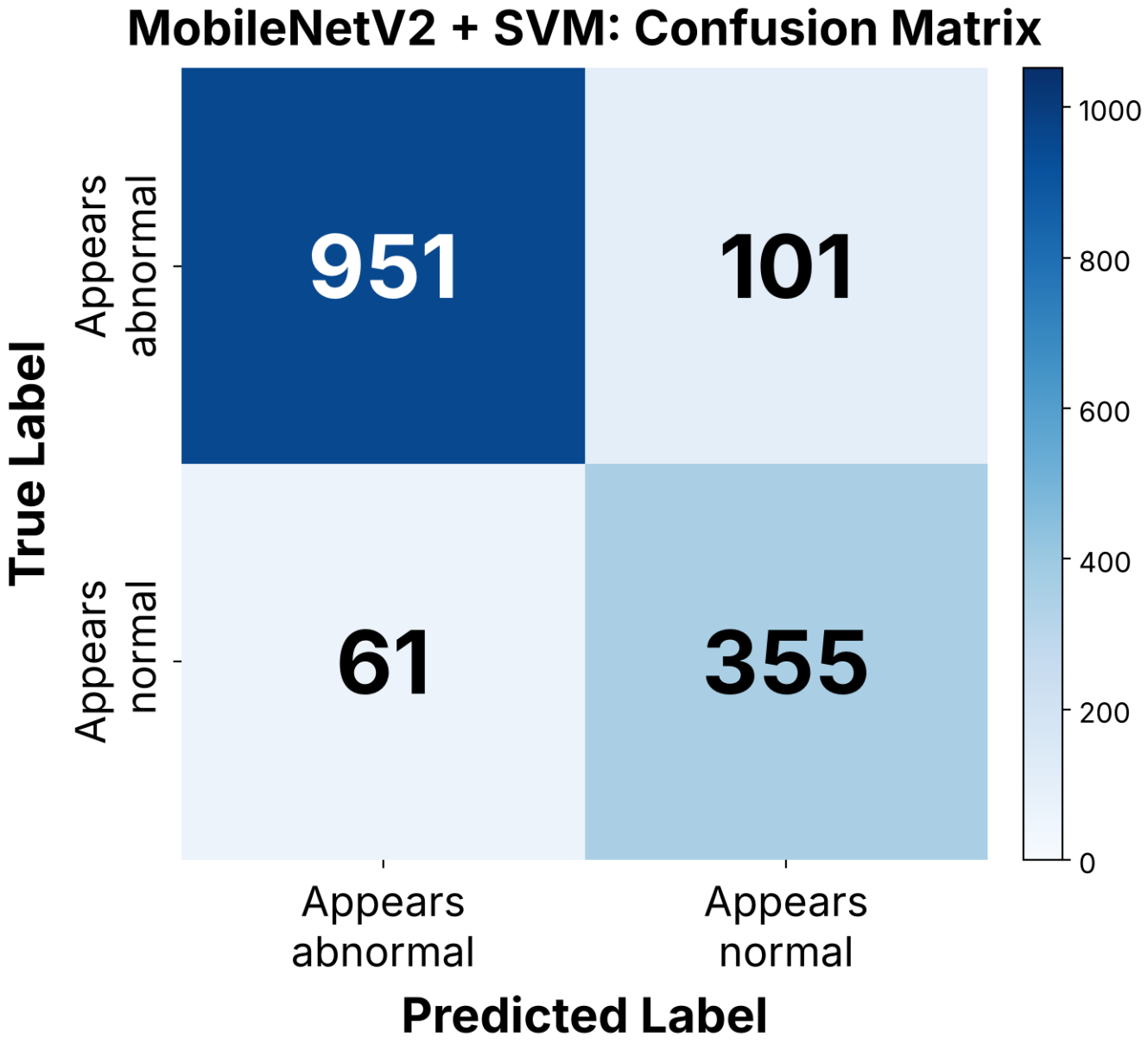
Accuracy: 88.96%

Precision: 89.40%

Recall: 88.96%

F1-score: 89.11%

Confusion Matrix



No training graphs: this notebook reloads an already-trained network and prints no epoch log, so there is nothing to plot for this row.

23. ResNet18 + KNN

Source notebook: ResNet with ML.ipynb (inside Models_DL.zip)

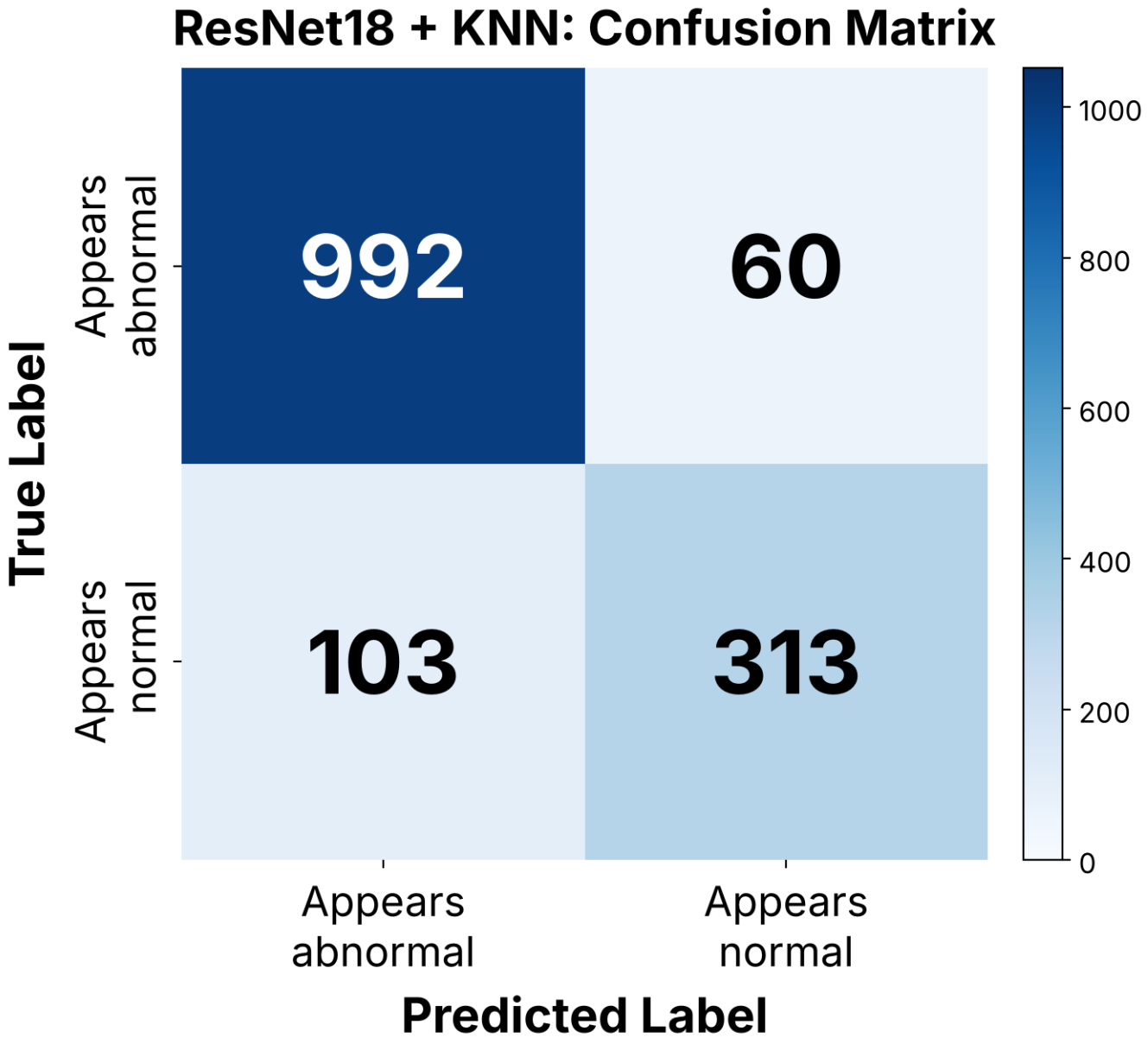
Accuracy: 88.90%

Precision: 88.70%

Recall: 88.90%

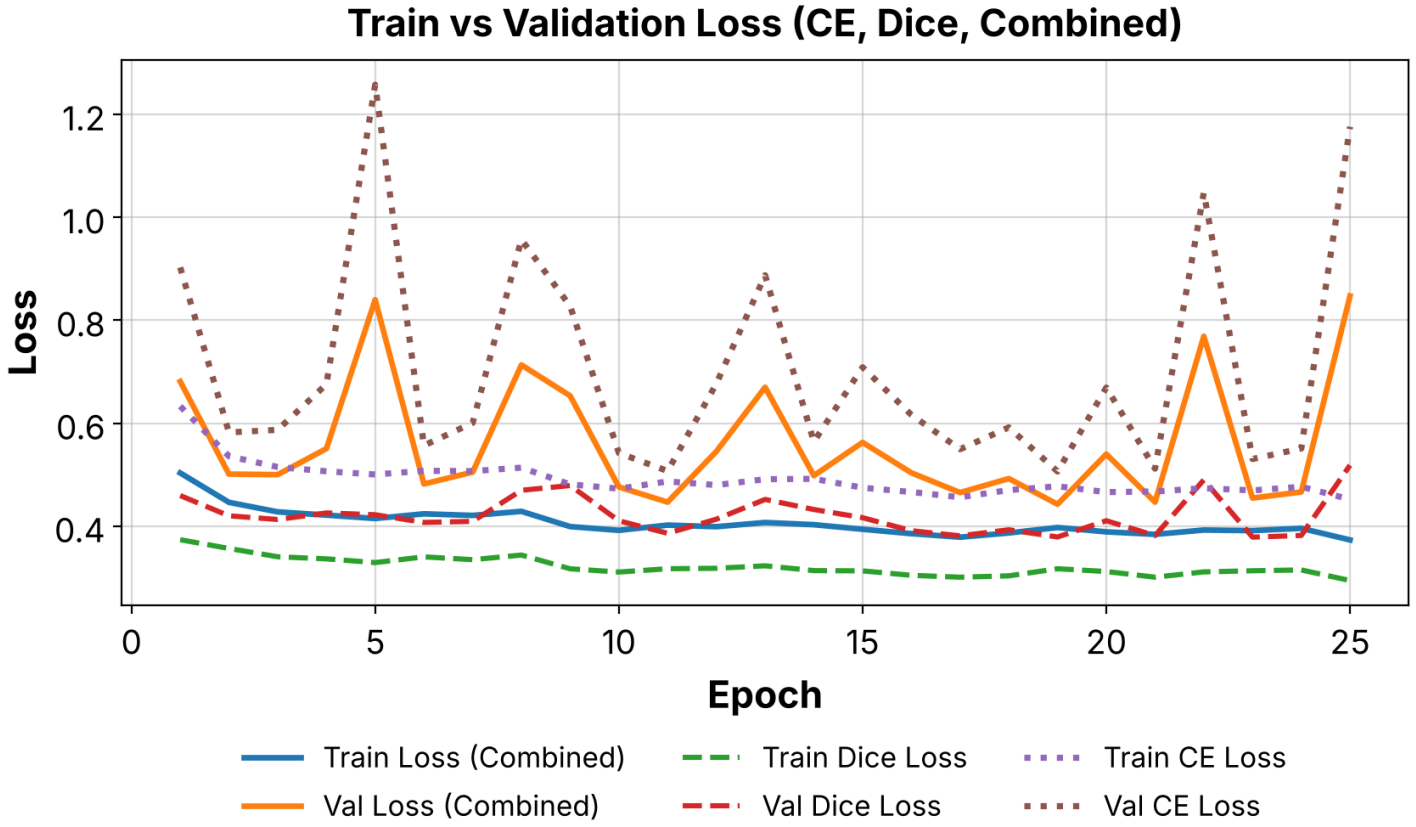
F1-score: 88.71%

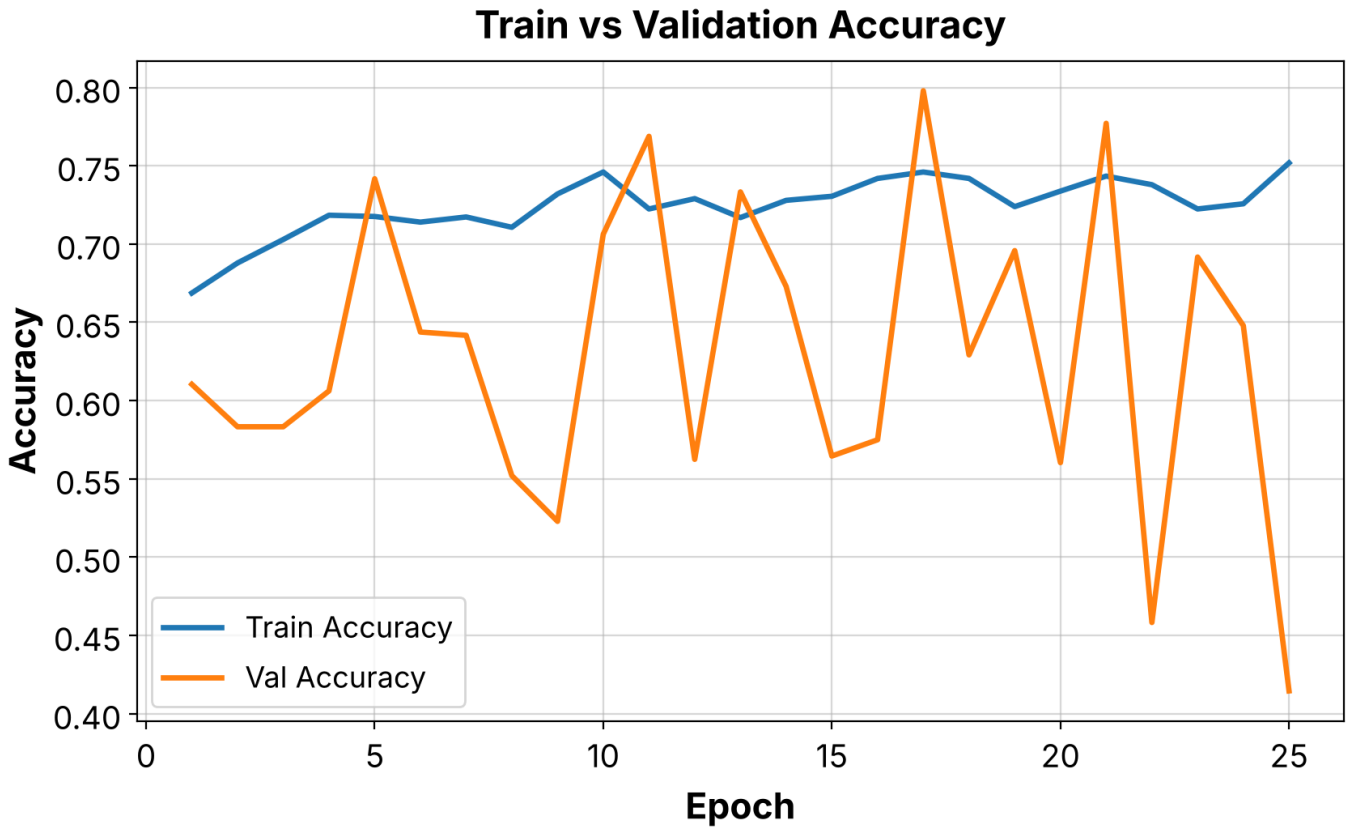
Confusion Matrix



23. ResNet18 + KNN — graphs

Training / Validation Graphs





24. ResNet18 + XGBoost

Source notebook: Xgboost and SVM and all 4 models.ipynb (inside Models_DL.zip)

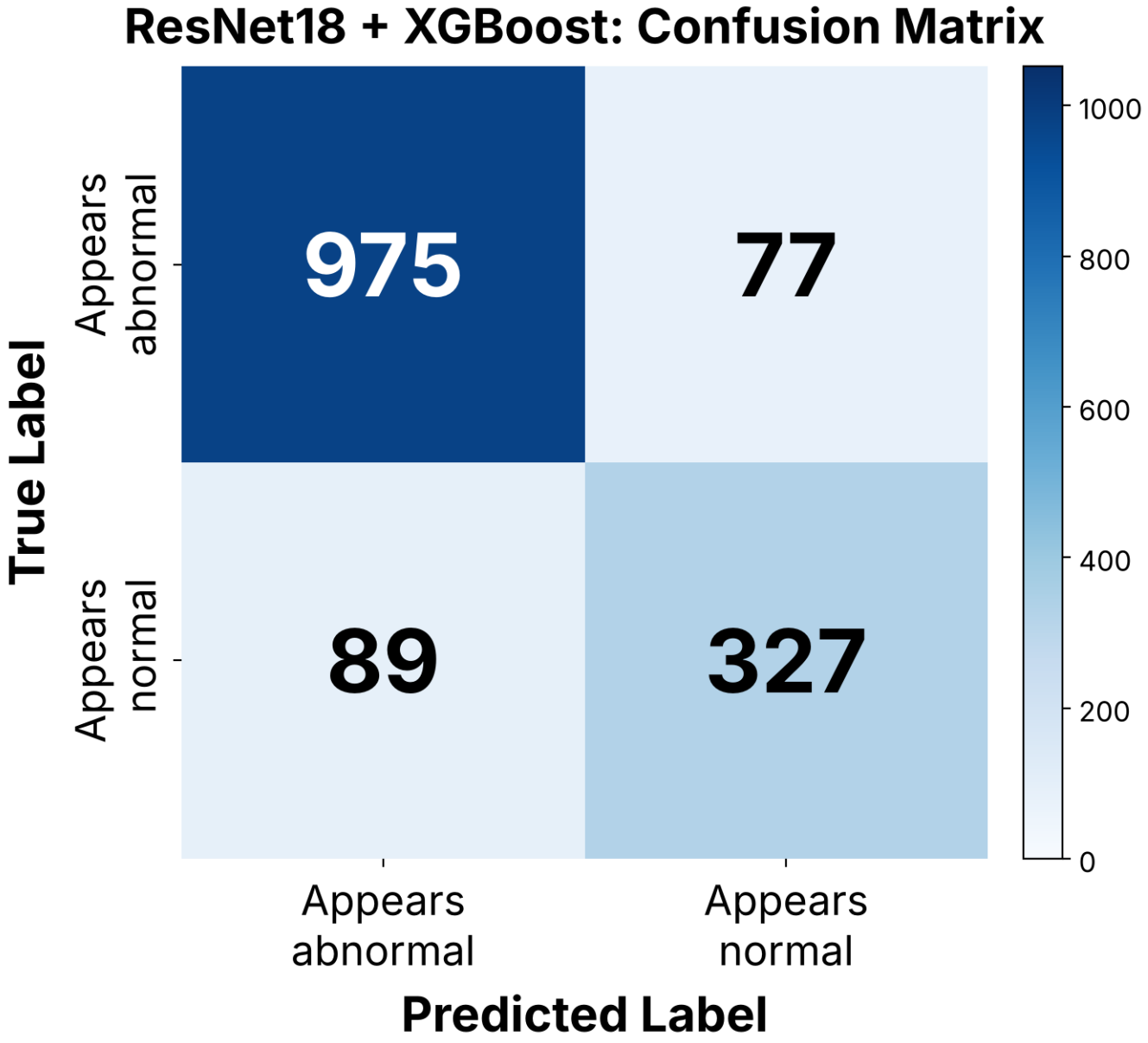
Accuracy: 88.69%

Precision: 88.60%

Recall: 88.69%

F1-score: 88.64%

Confusion Matrix



No training graphs: this notebook reloads an already-trained network and prints no epoch log, so there is nothing to plot for this row.

25. ResNet18 — Direct Classification (Random Forest section run)

Source notebook: ResNet with ML.ipynb (inside Models_DL.zip)

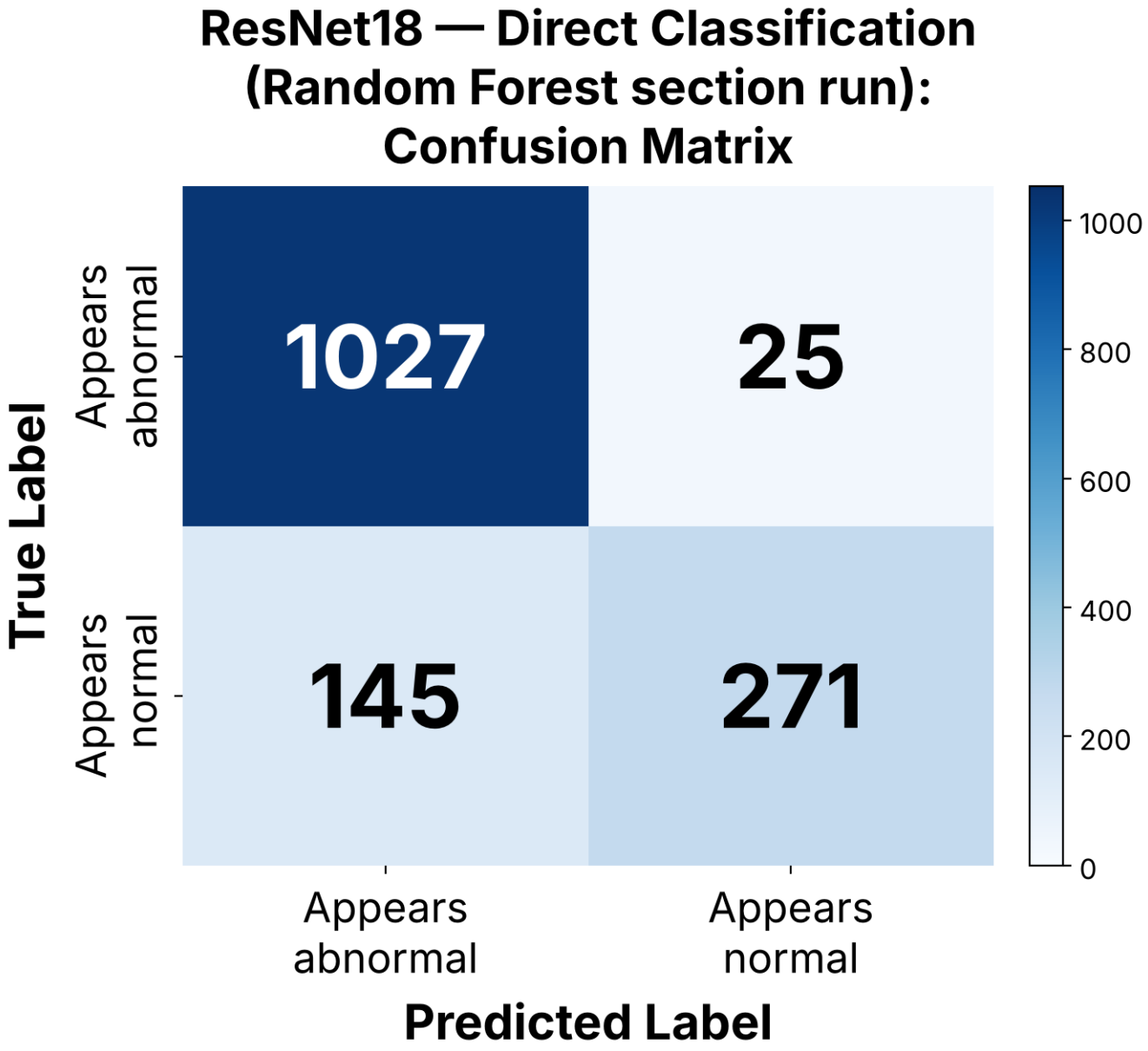
Accuracy: 88.42%

Precision: 88.74%

Recall: 88.42%

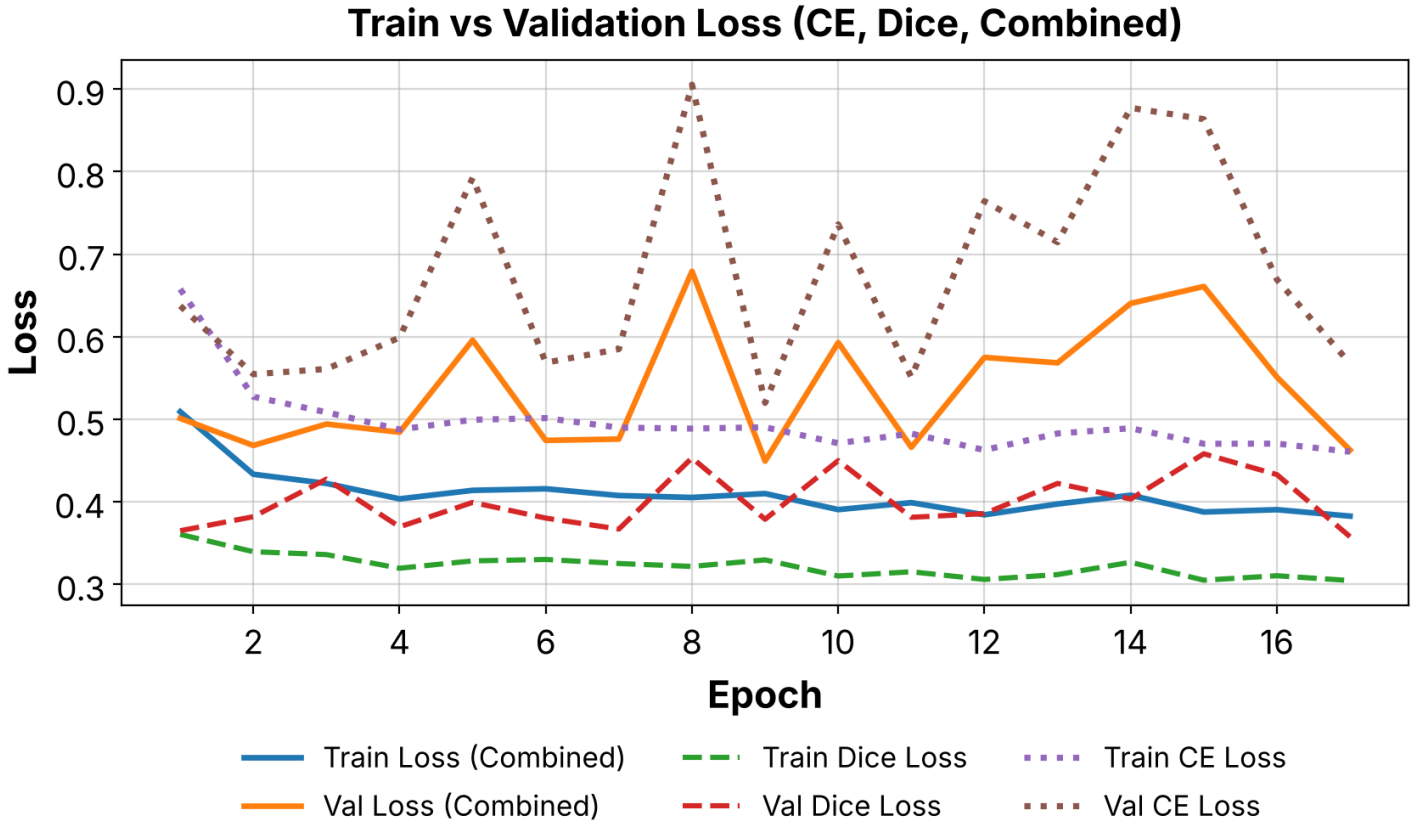
F1-score: 87.76%

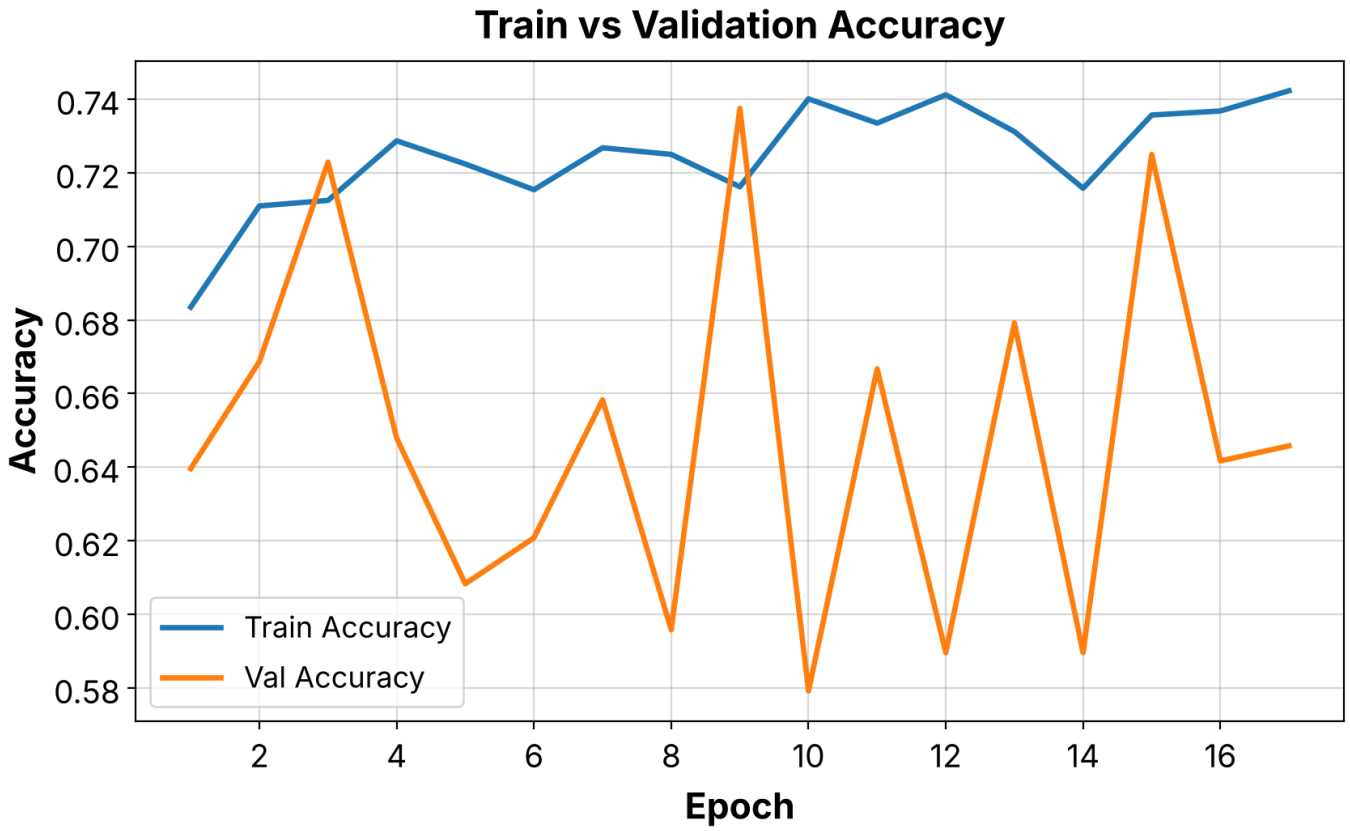
Confusion Matrix



25. ResNet18 — Direct Classification (Random Forest section run) — graphs

Training / Validation Graphs





26. ResNet18 — Direct Classification (Gaussian Naive Bayes section run)

Source notebook: ResNet with ML.ipynb (inside Models_DL.zip)

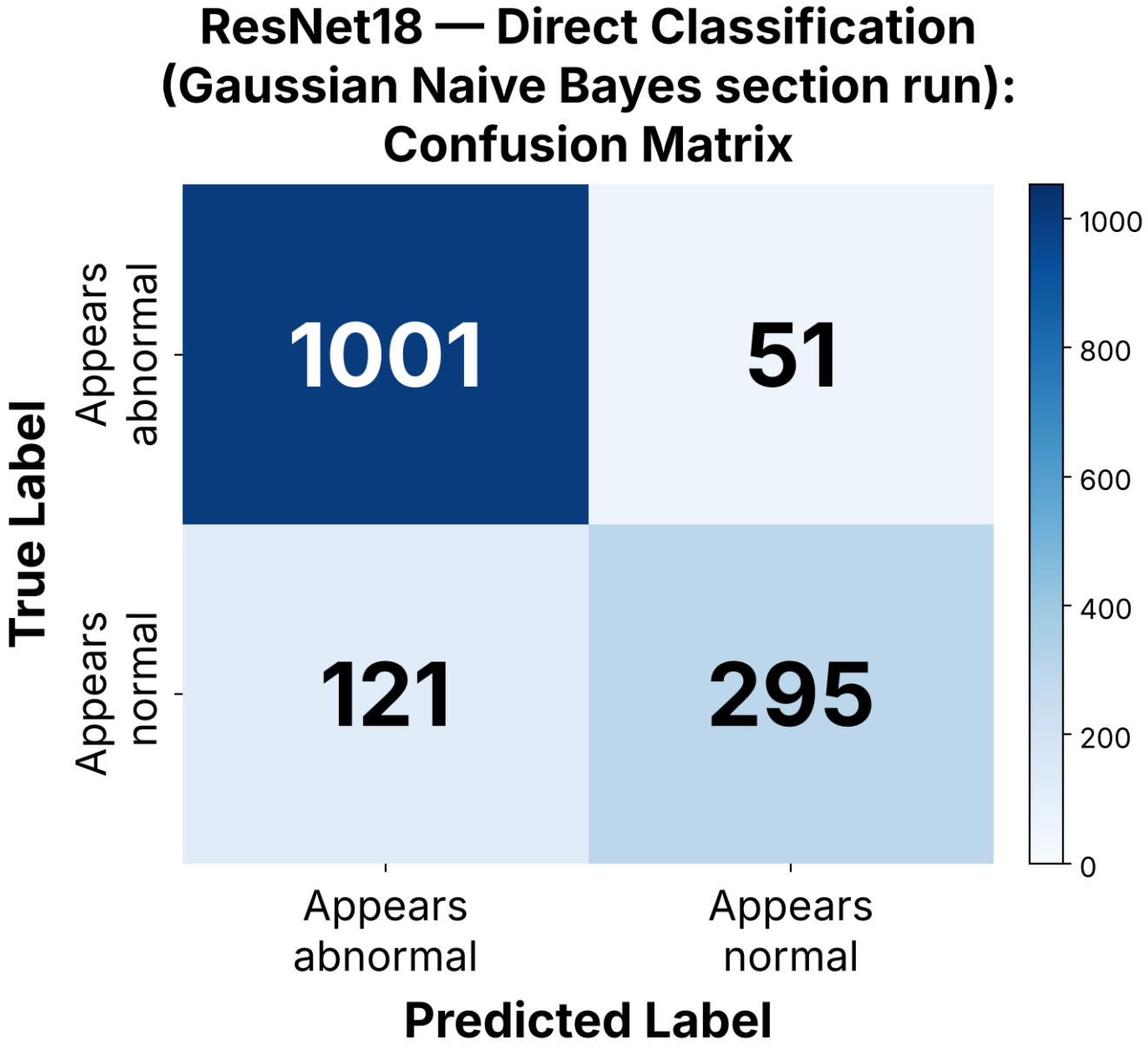
Accuracy: 88.28%

Precision: 88.09%

Recall: 88.28%

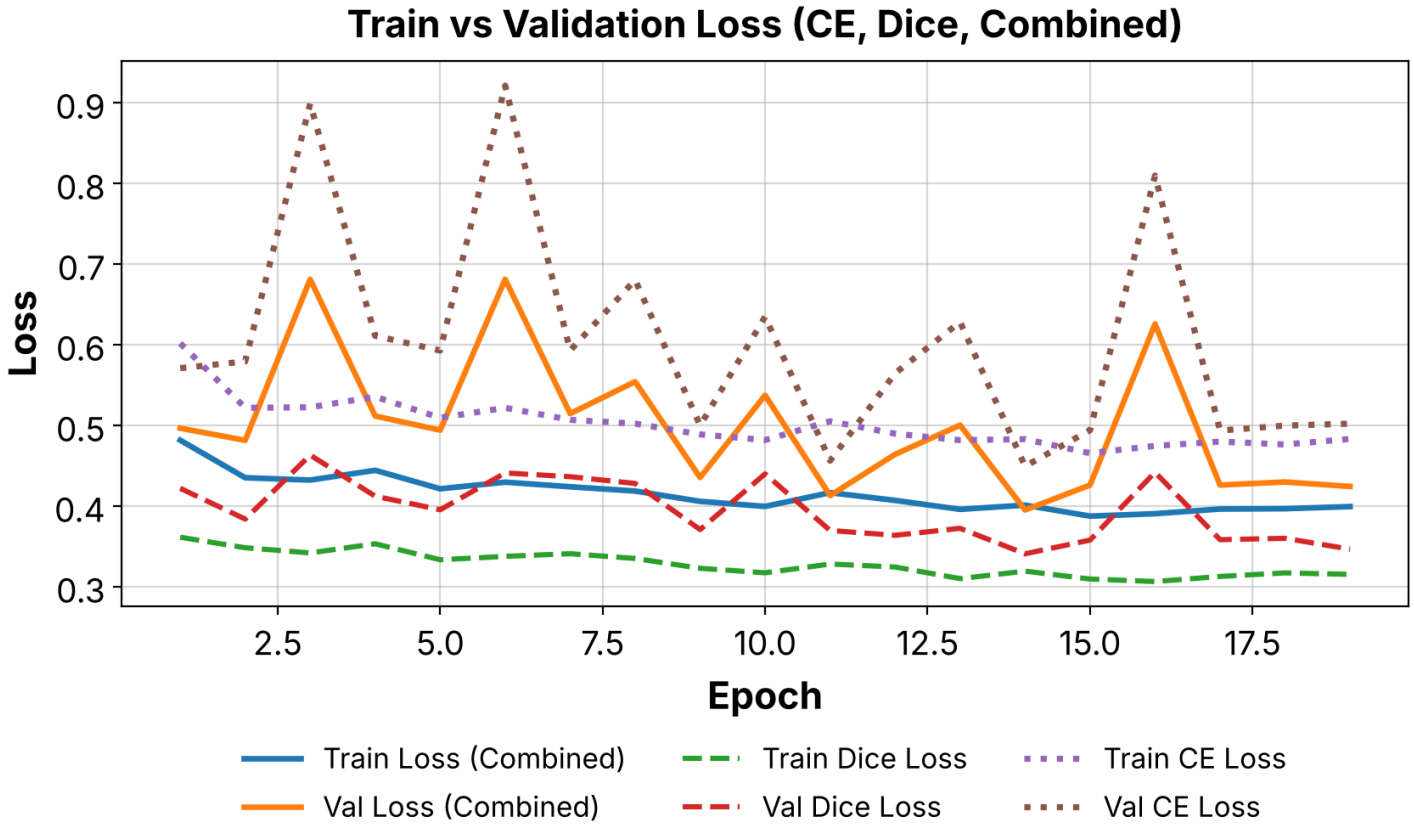
F1-score: 87.93%

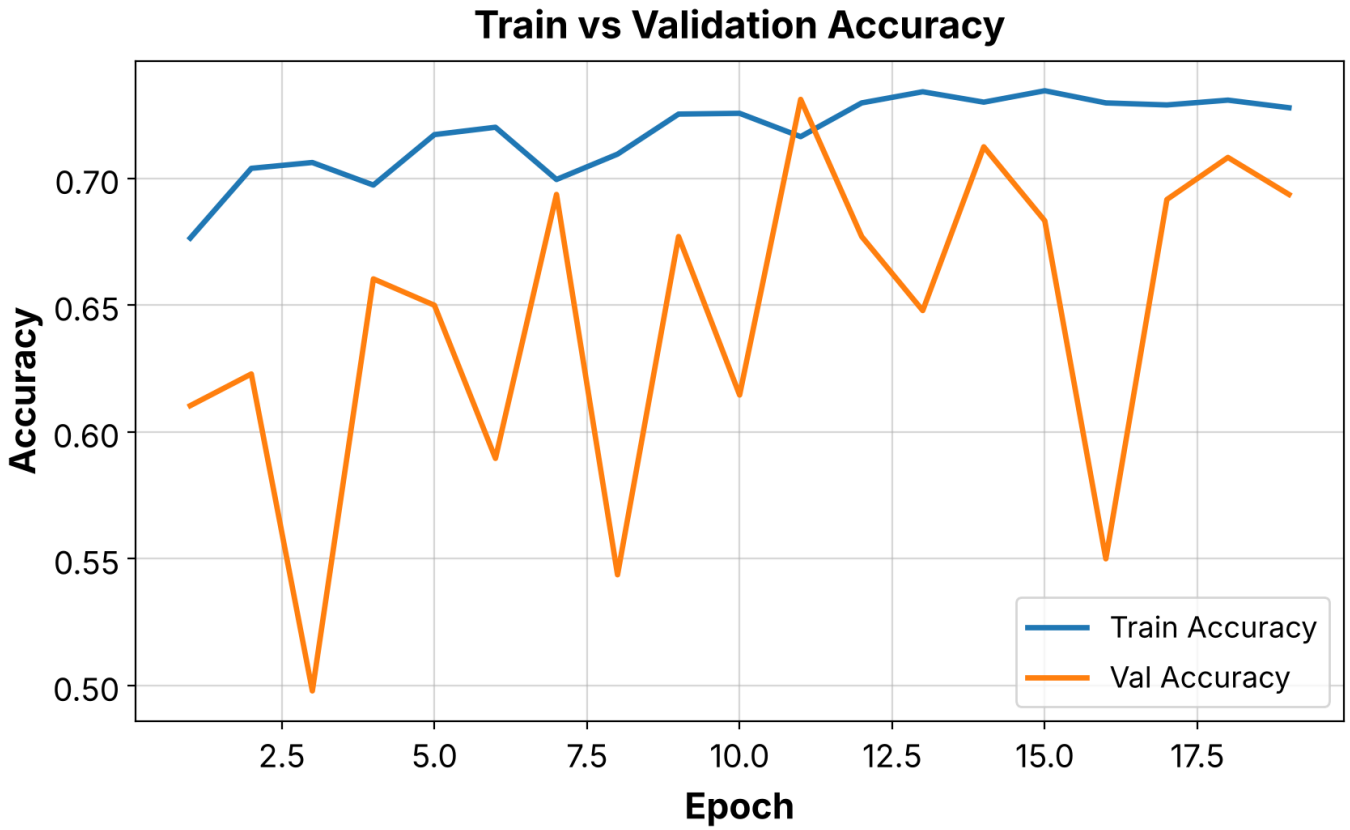
Confusion Matrix



26. ResNet18 — Direct Classification (Gaussian Naive Bayes section run) — graphs

Training / Validation Graphs





27. MobileNetV2 — Direct Classification (KNN section run)

Source notebook: MobileNet with ML.ipynb (inside Models_DL.zip)

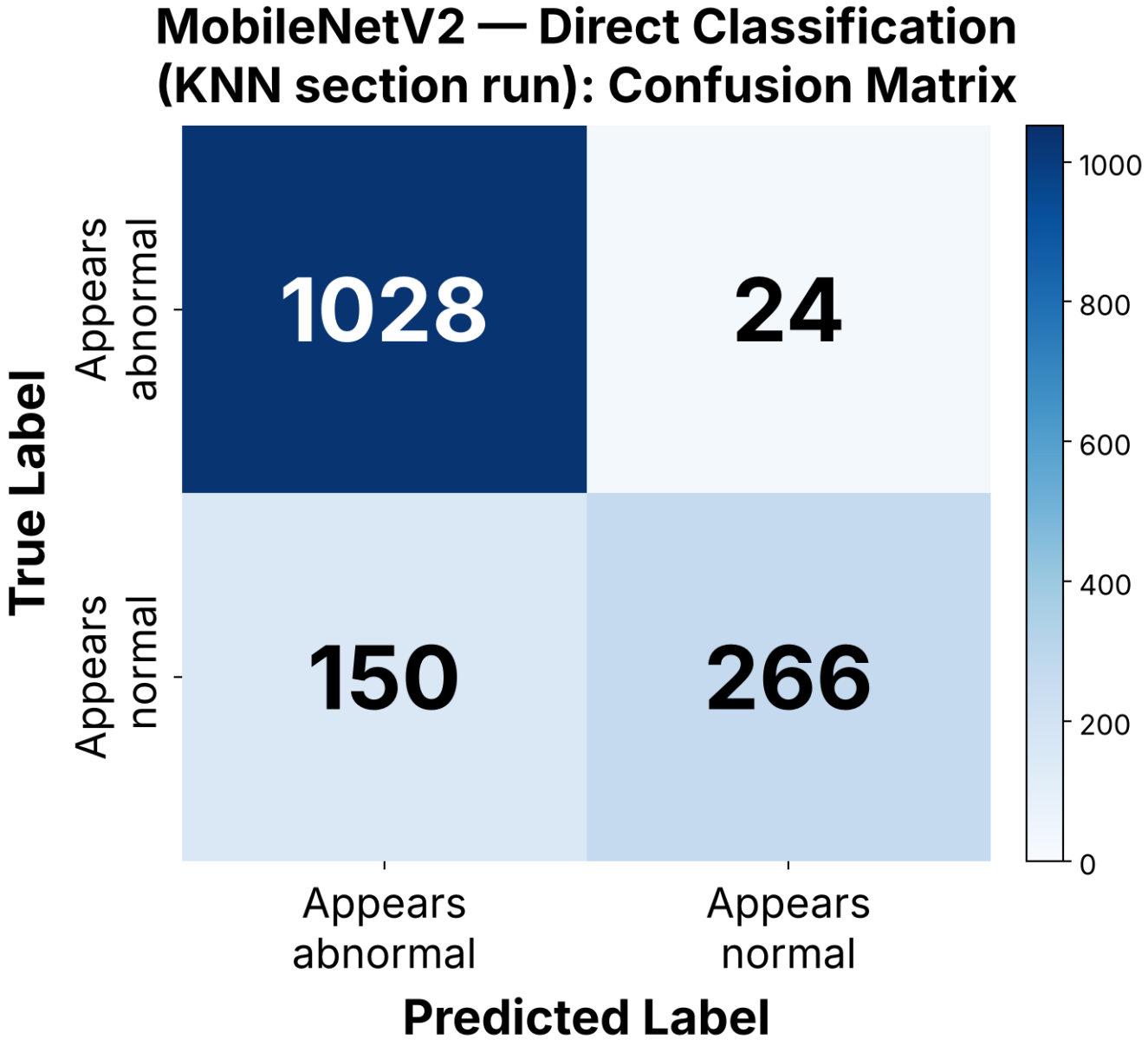
Accuracy: 88.15%

Precision: 88.53%

Recall: 88.15%

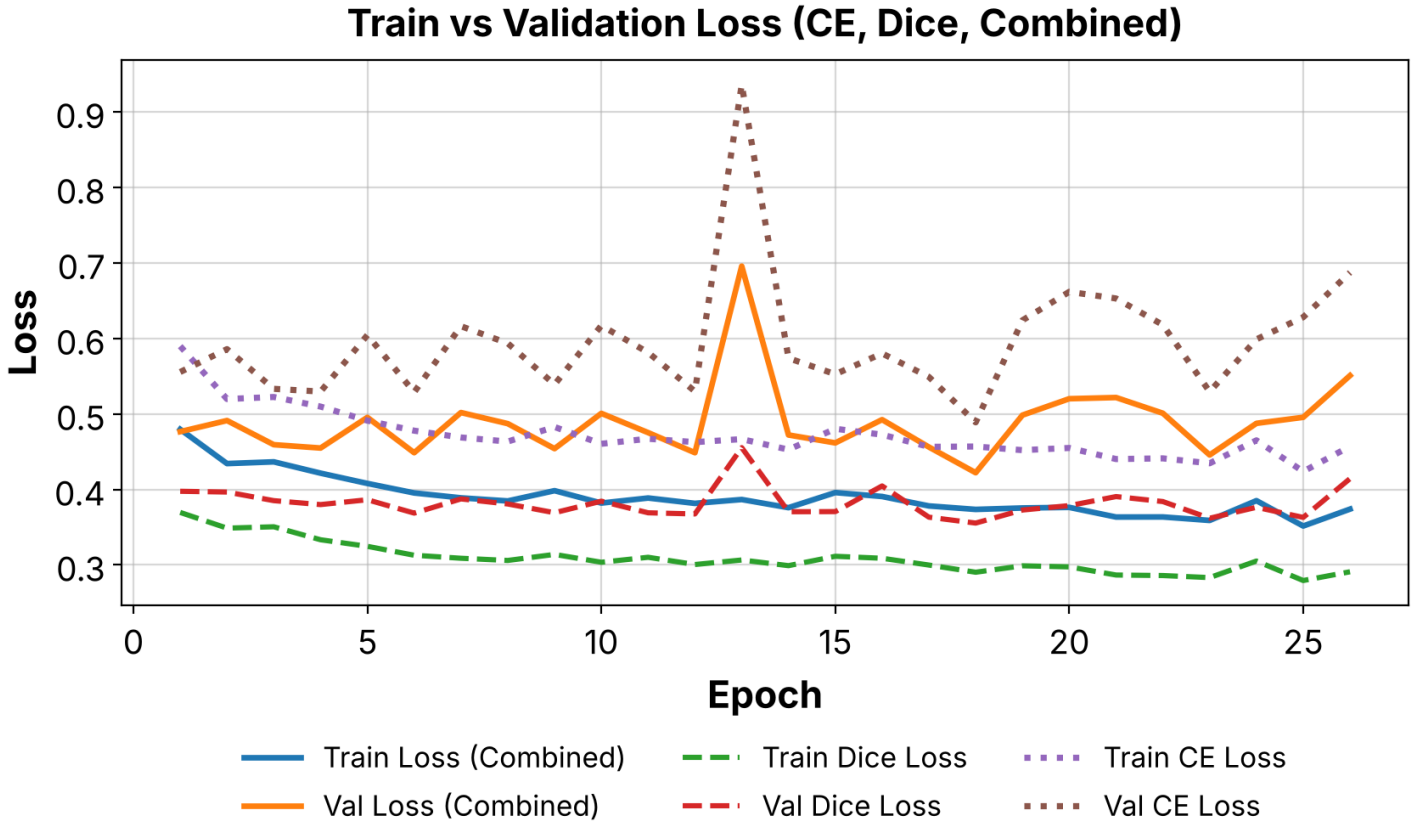
F1-score: 87.42%

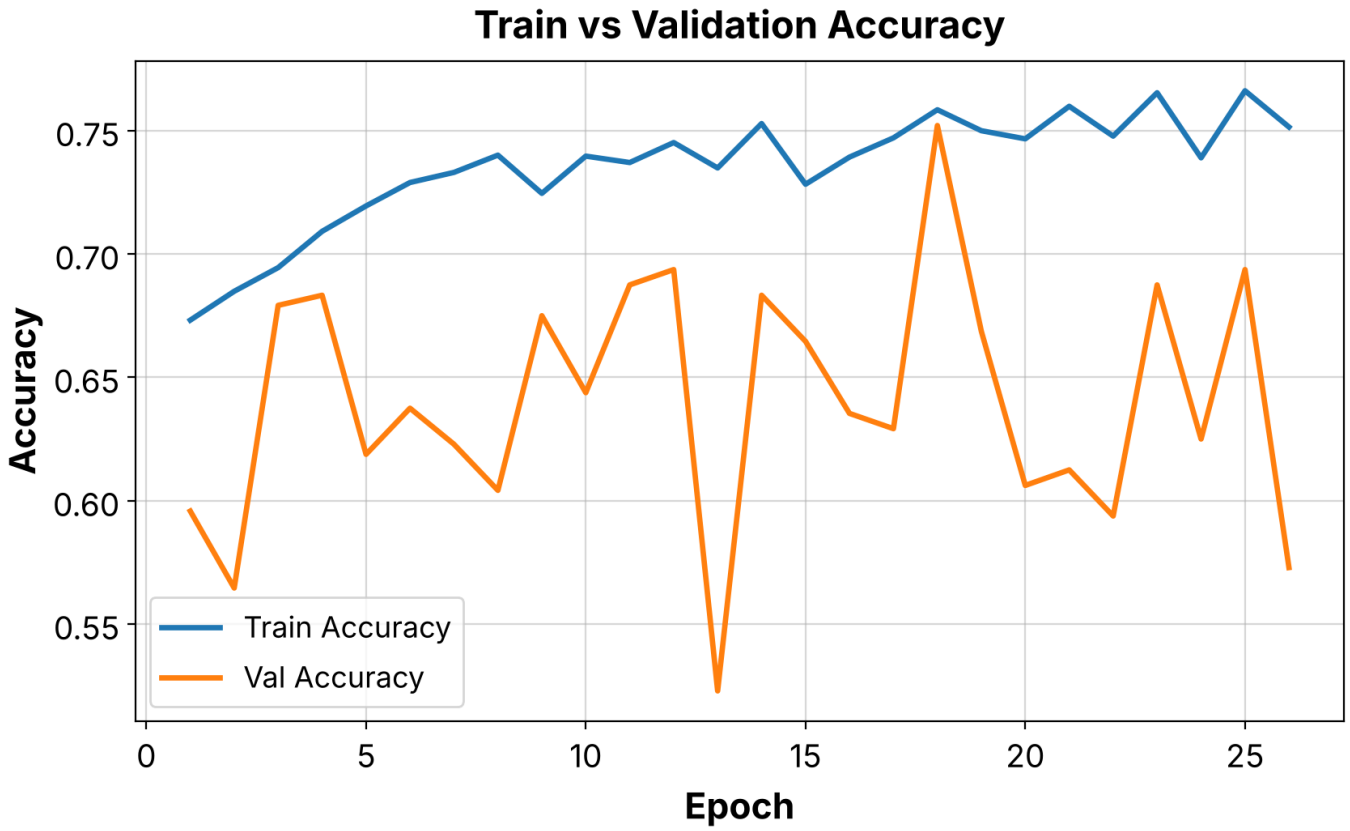
Confusion Matrix



27. MobileNetV2 — Direct Classification (KNN section run) — graphs

Training / Validation Graphs





28. MobileNetV2 — Direct Classification (AdaBoost section run)

Source notebook: MobileNet with ML.ipynb (inside Models_DL.zip)

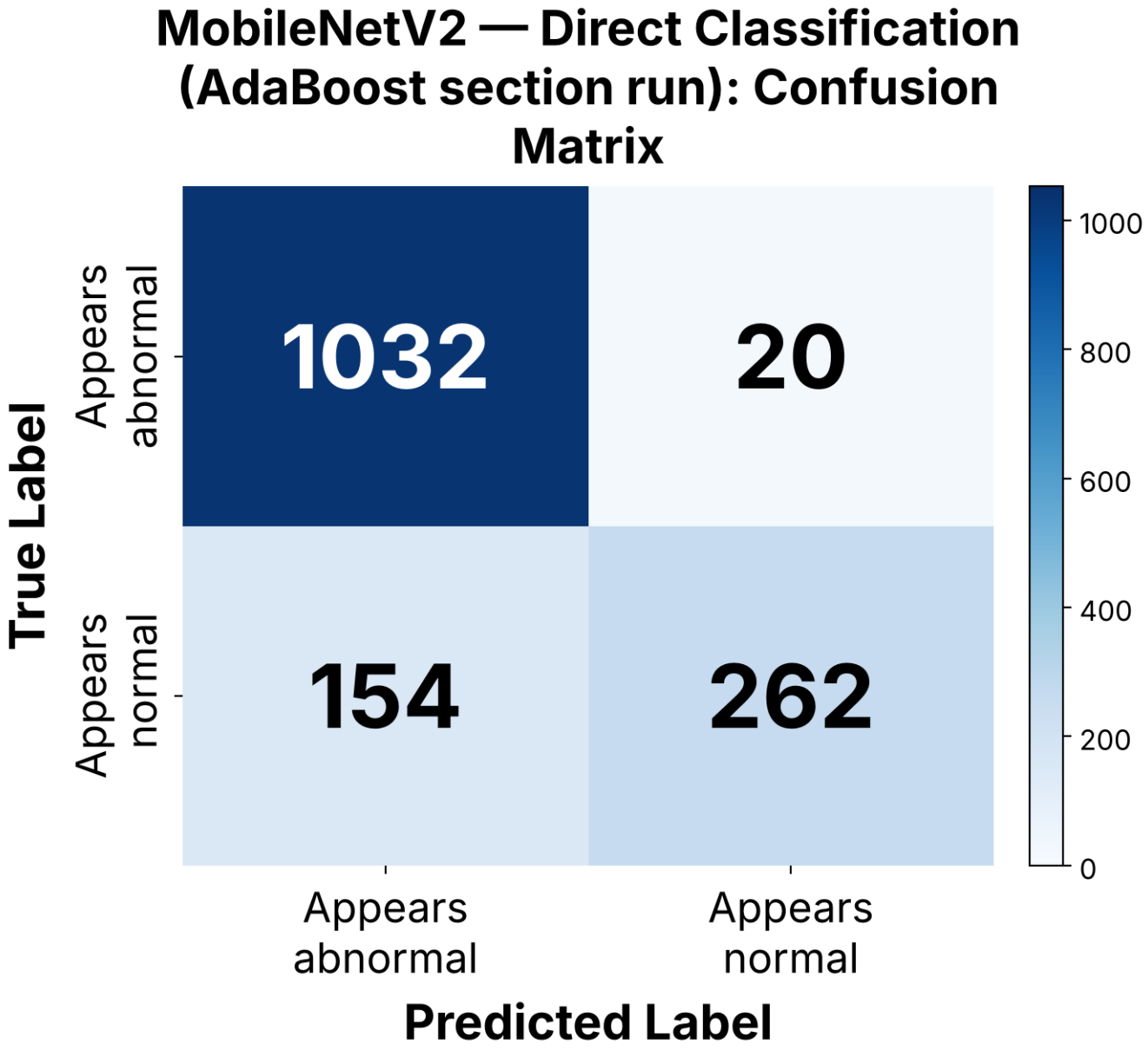
Accuracy: 88.15%

Precision: 88.69%

Recall: 88.15%

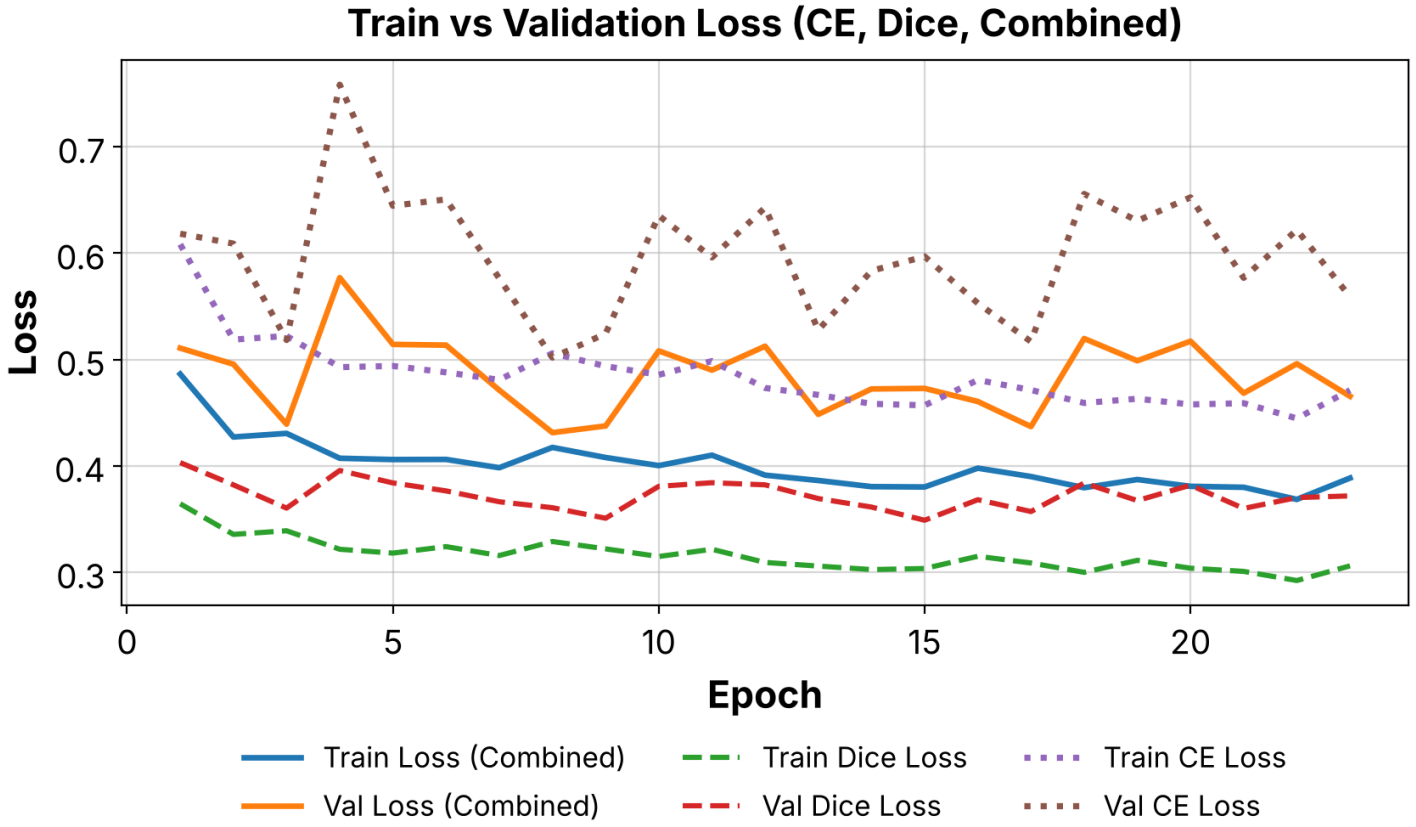
F1-score: 87.36%

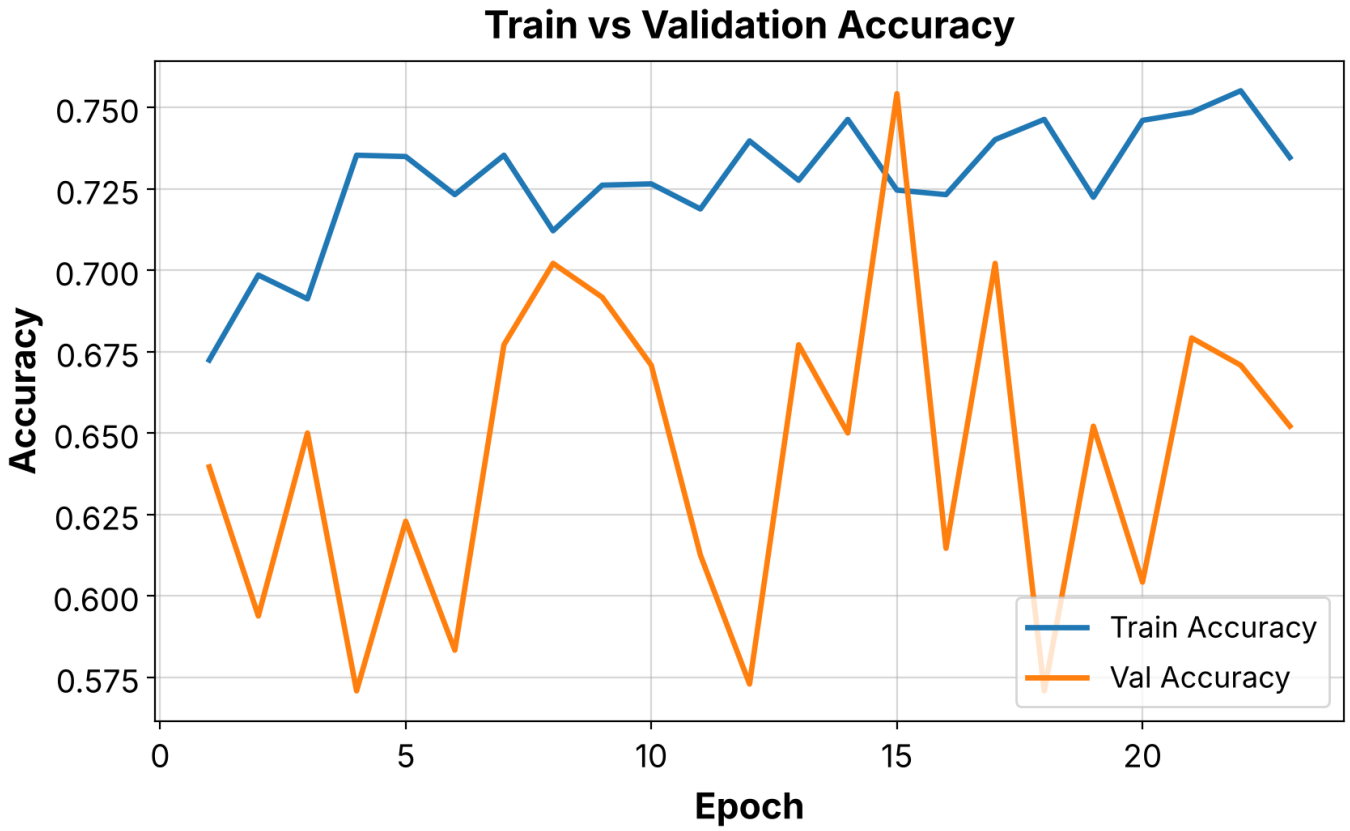
Confusion Matrix



28. MobileNetV2 — Direct Classification (AdaBoost section run) — graphs

Training / Validation Graphs





29. InceptionV3 + Random Forest

Source notebook: Inception v3 baseline and ml.ipynb (inside Models_DL.zip)

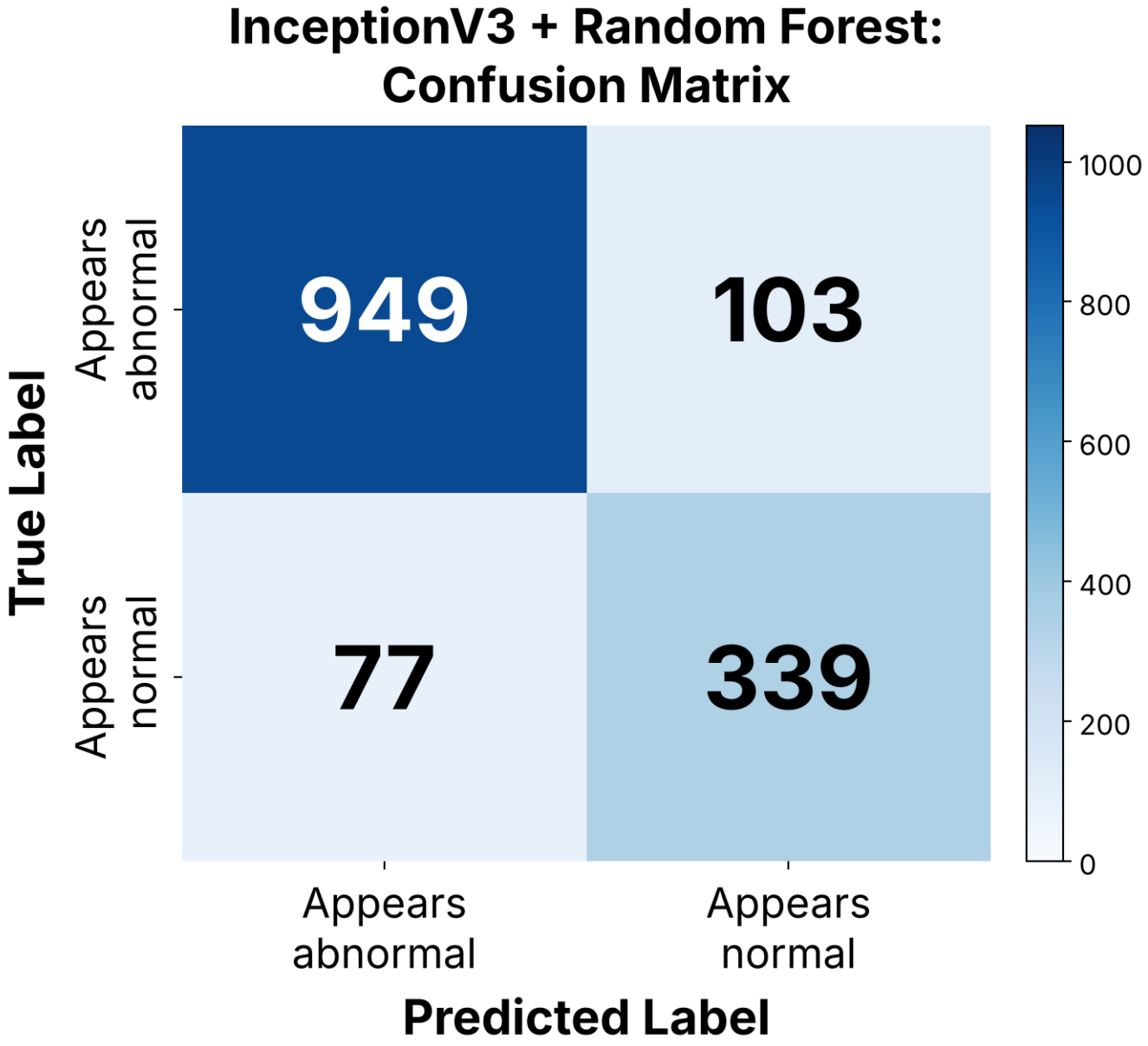
Accuracy: 87.74%

Precision: 88.02%

Recall: 87.74%

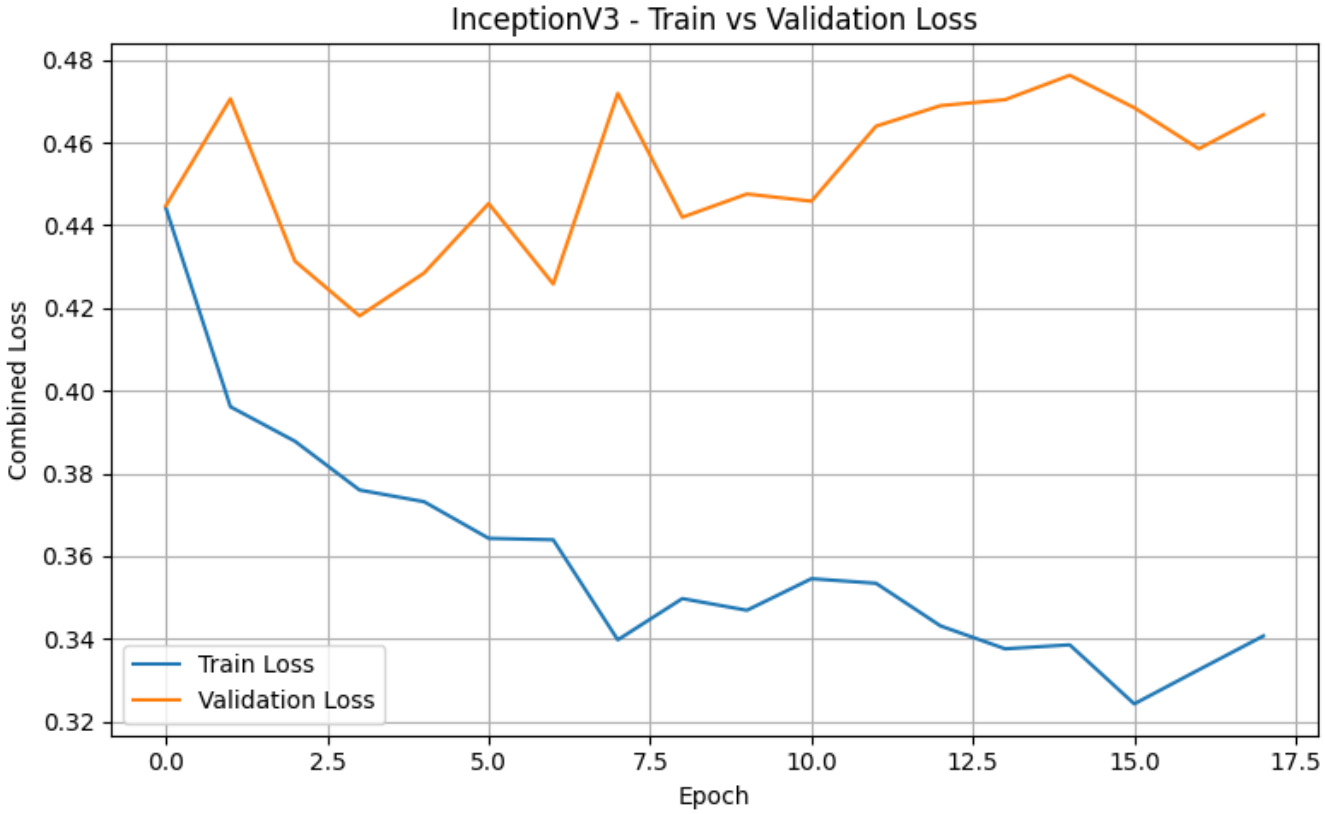
F1-score: 87.85%

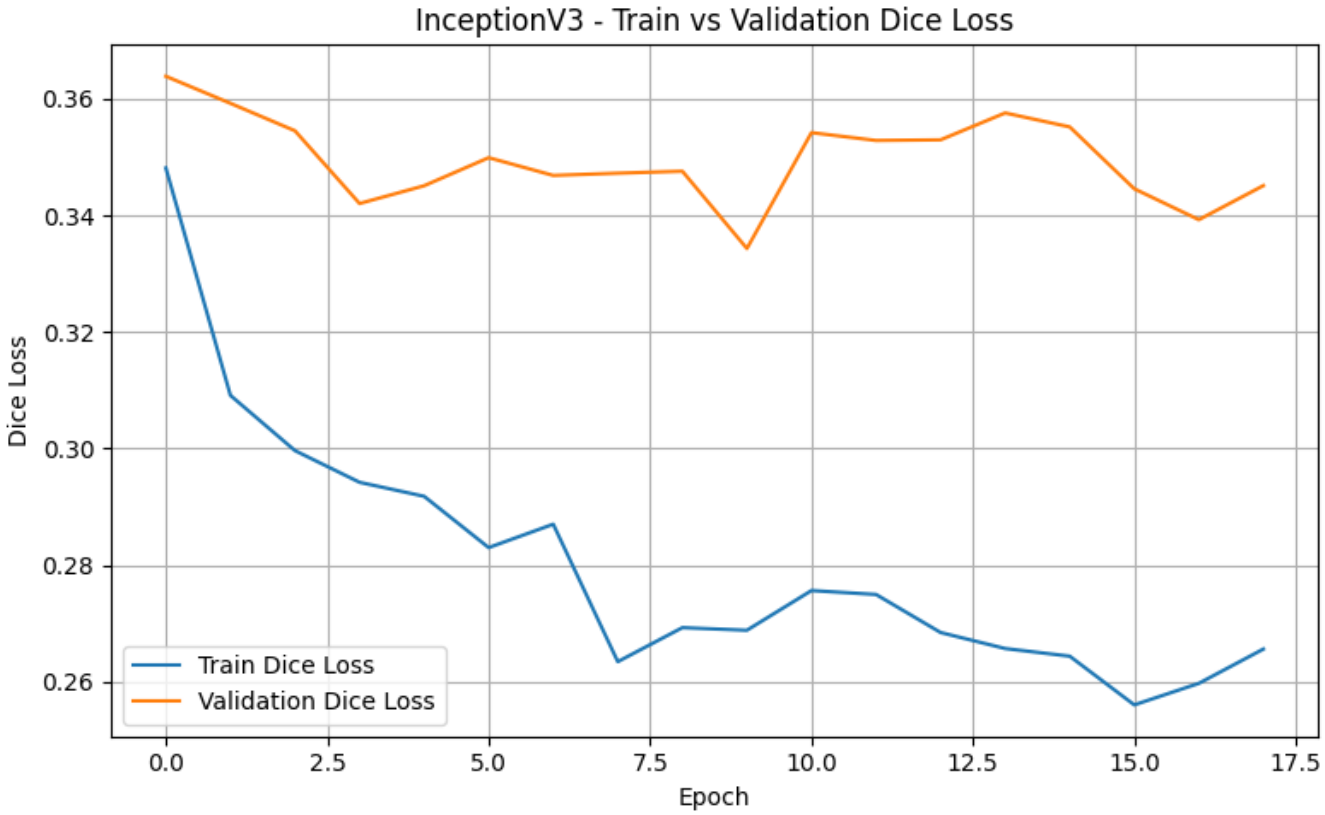
Confusion Matrix



29. InceptionV3 + Random Forest — graphs (1/2)

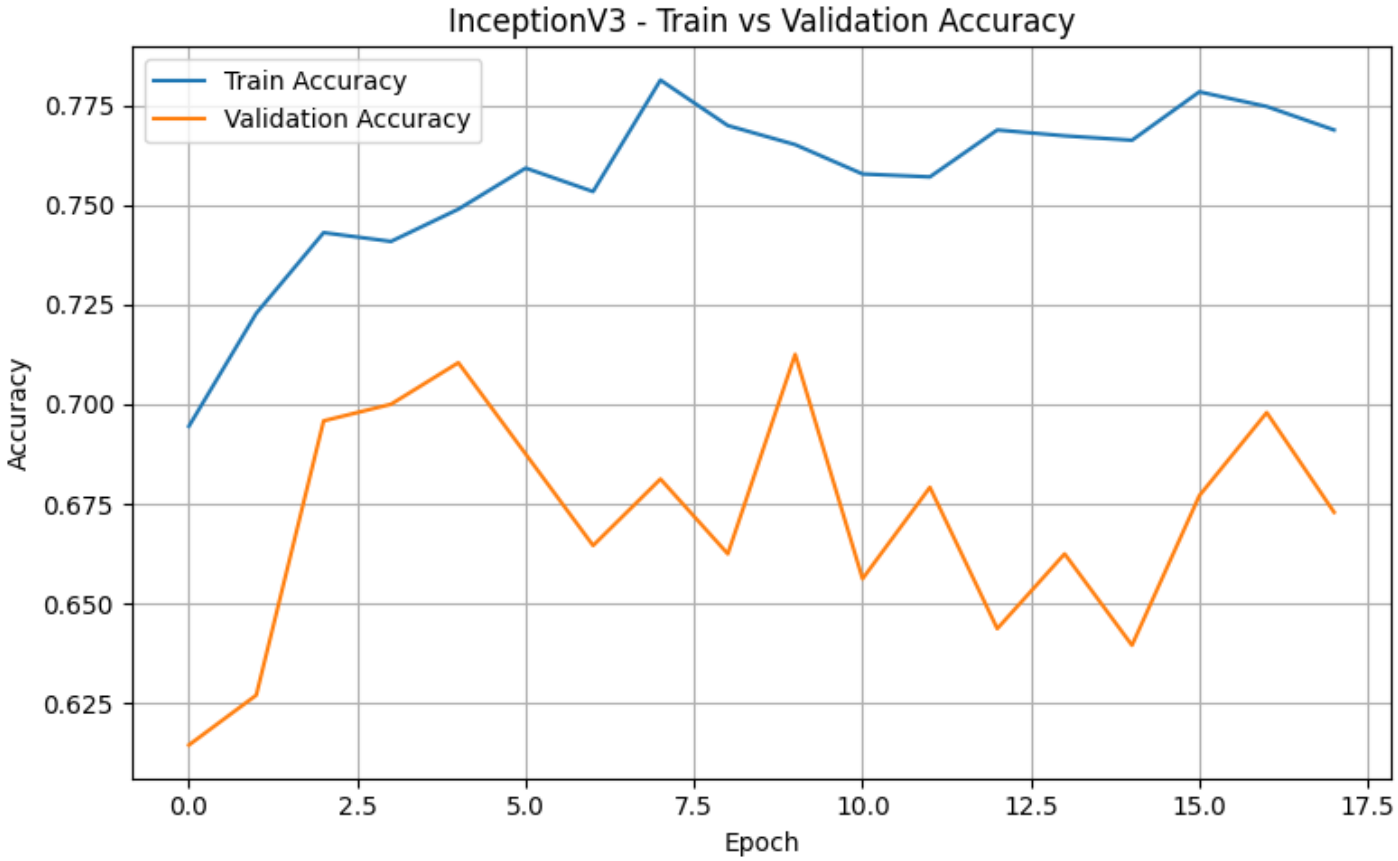
Training / Validation Graphs





29. InceptionV3 + Random Forest — graphs (2/2)

Training / Validation Graphs



30. VGG16 + KNN

Source notebook: Vgg with ML.ipynb (inside Models_DL.zip)

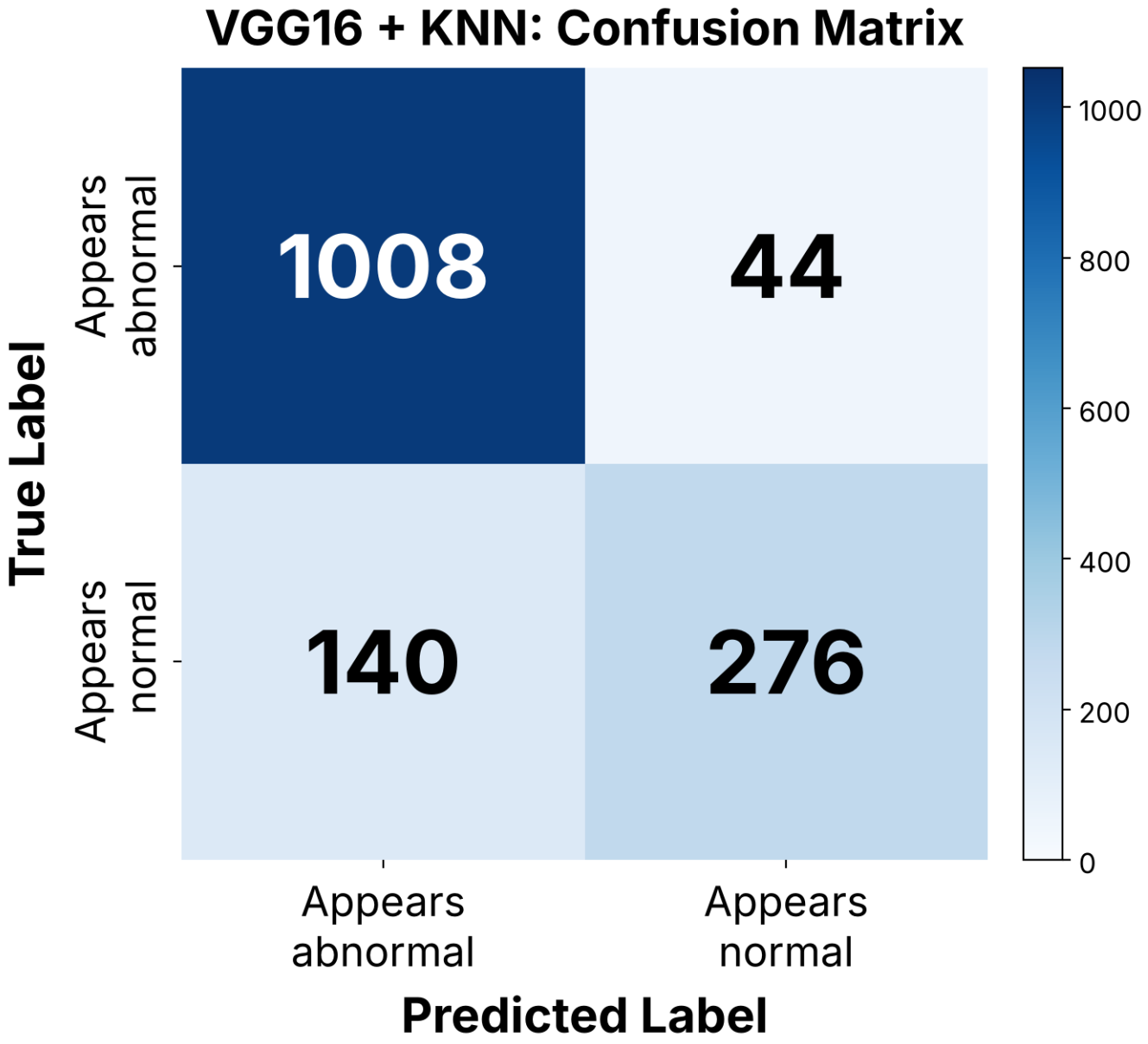
Accuracy: 87.47%

Precision: 87.36%

Recall: 87.47%

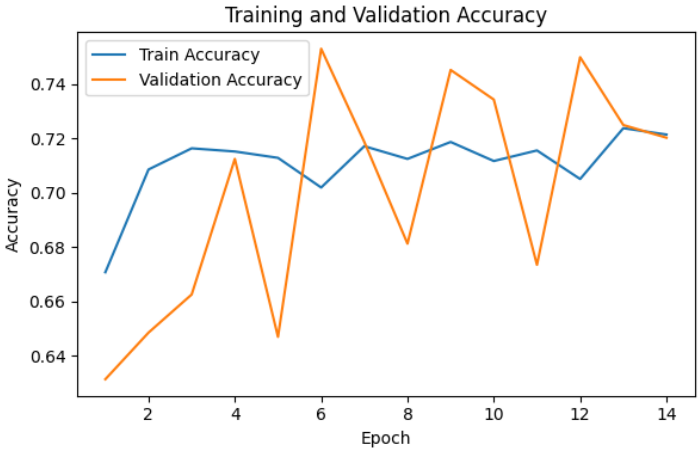
F1-score: 86.92%

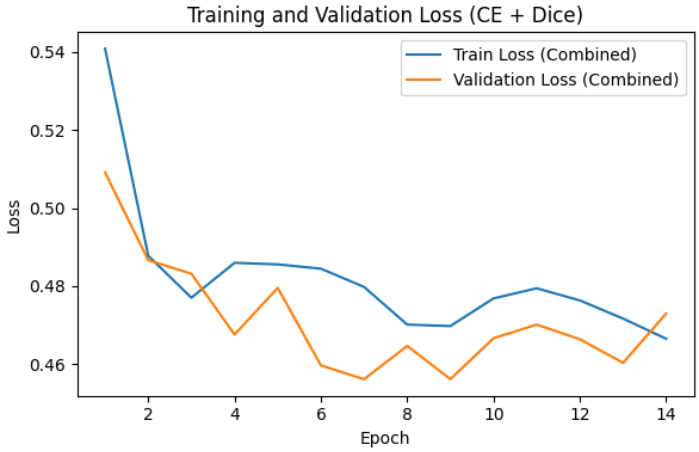
Confusion Matrix

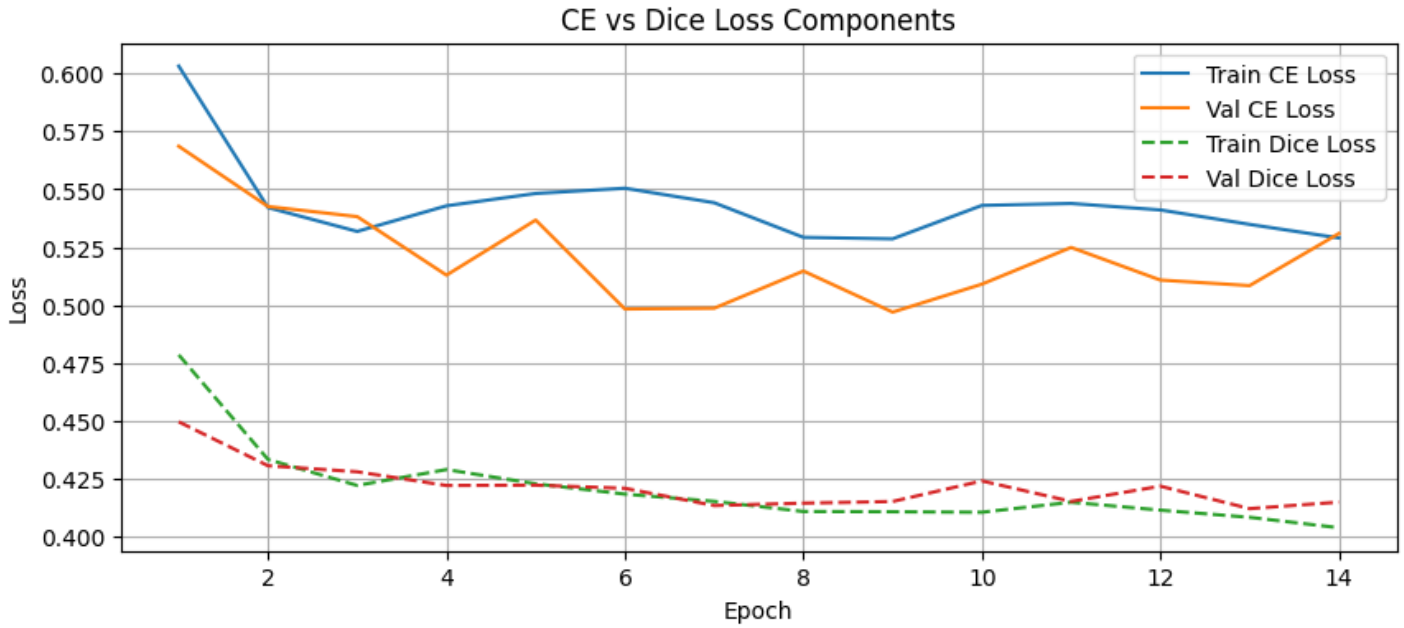


30. VGG16 + KNN — graphs

Training / Validation Graphs







31. ResNet18 + AdaBoost

Source notebook: ResNet with ML.ipynb (inside Models_DL.zip)

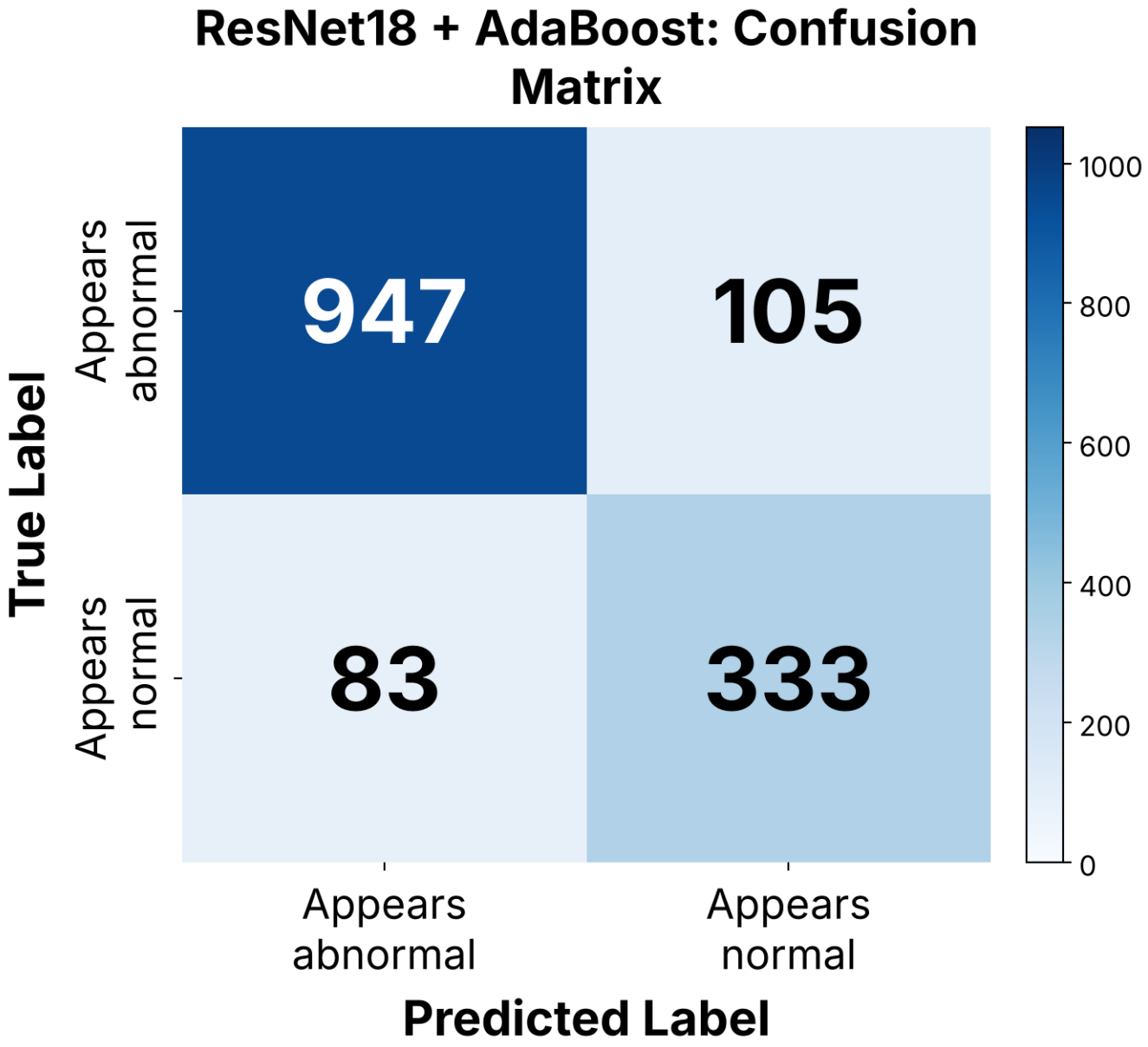
Accuracy: 87.19%

Precision: 87.43%

Recall: 87.19%

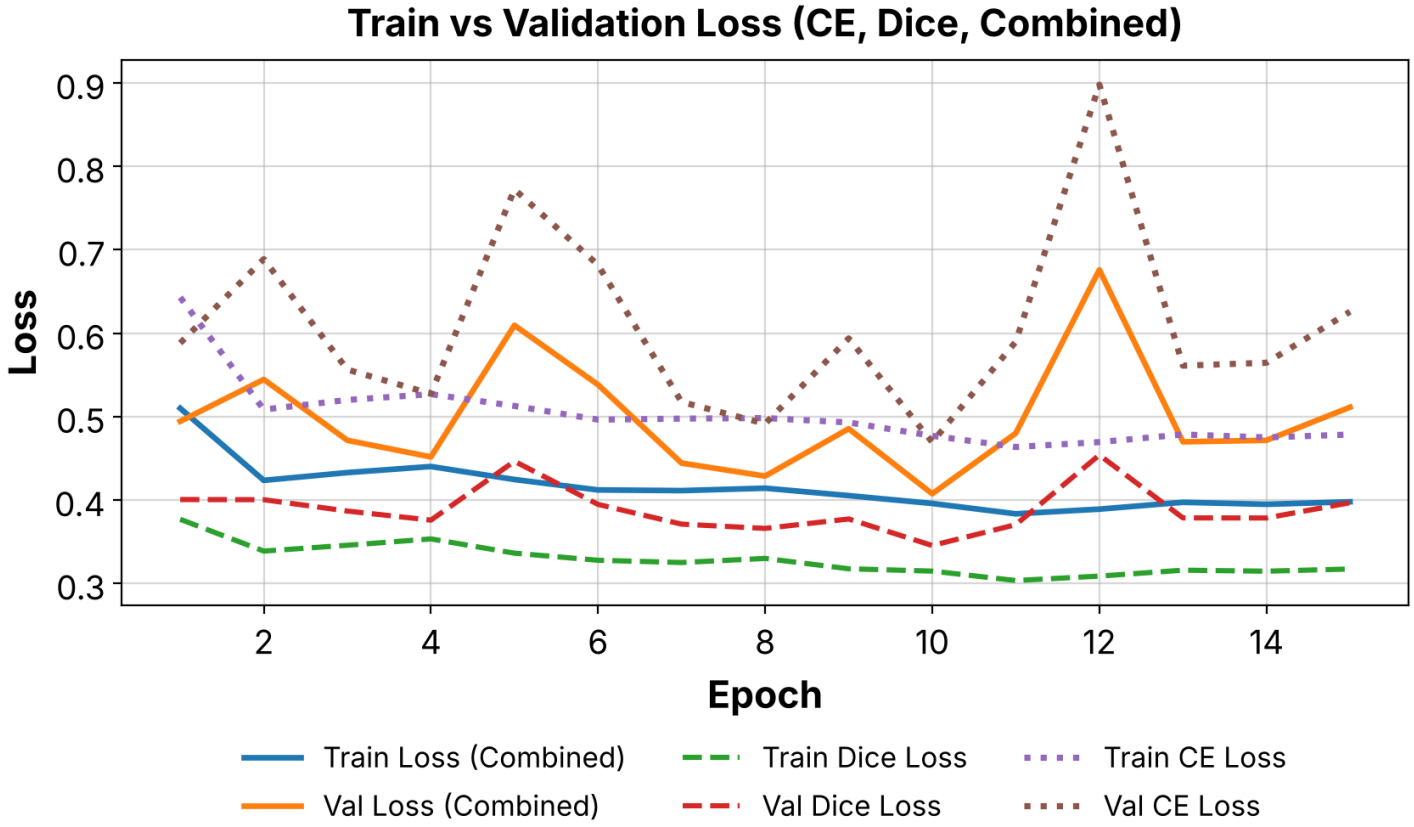
F1-score: 87.29%

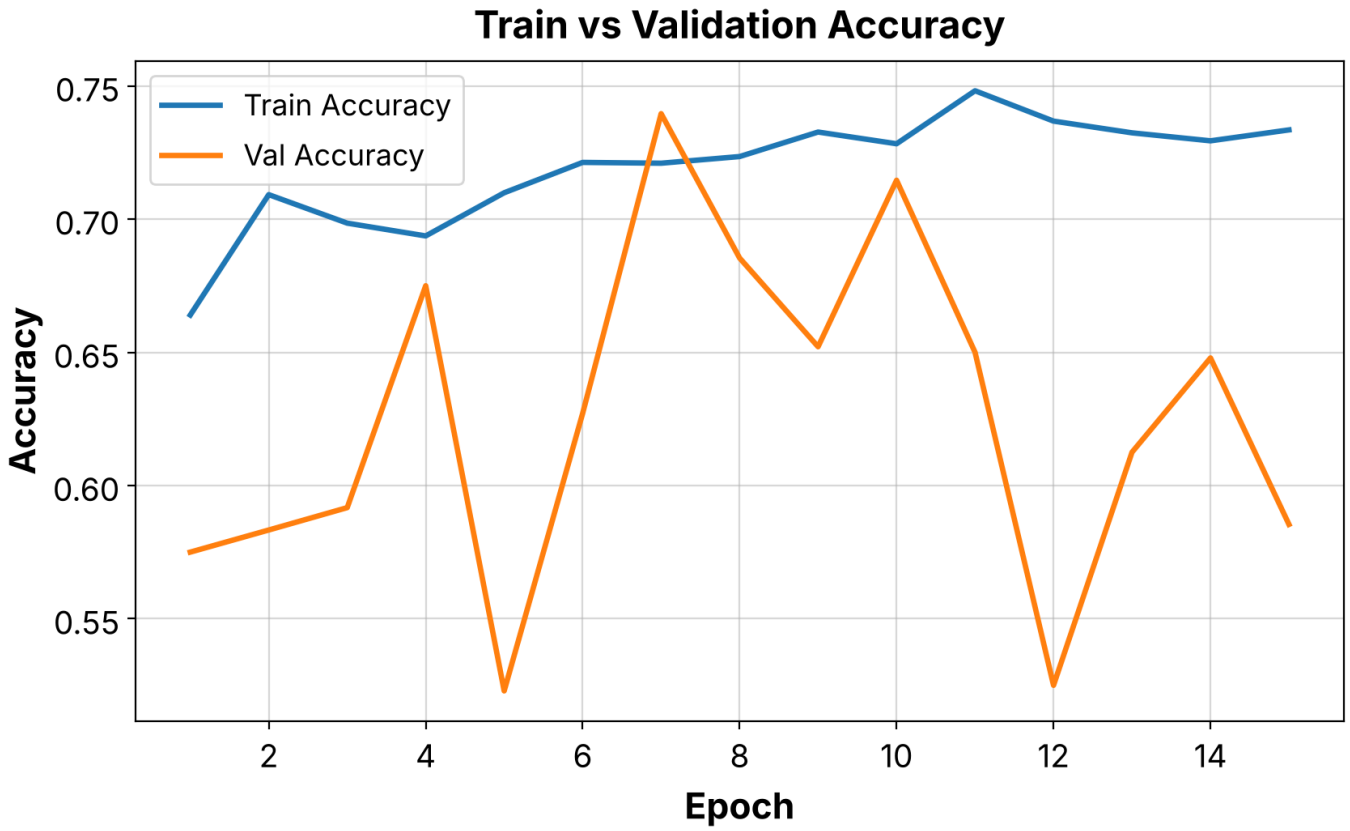
Confusion Matrix



31. ResNet18 + AdaBoost — graphs

Training / Validation Graphs





32. MobileNetV2 — 5-Fold CV, Direct Classification

Source notebook: MobileNet with Kfold.ipynb (inside Models_DL.zip)

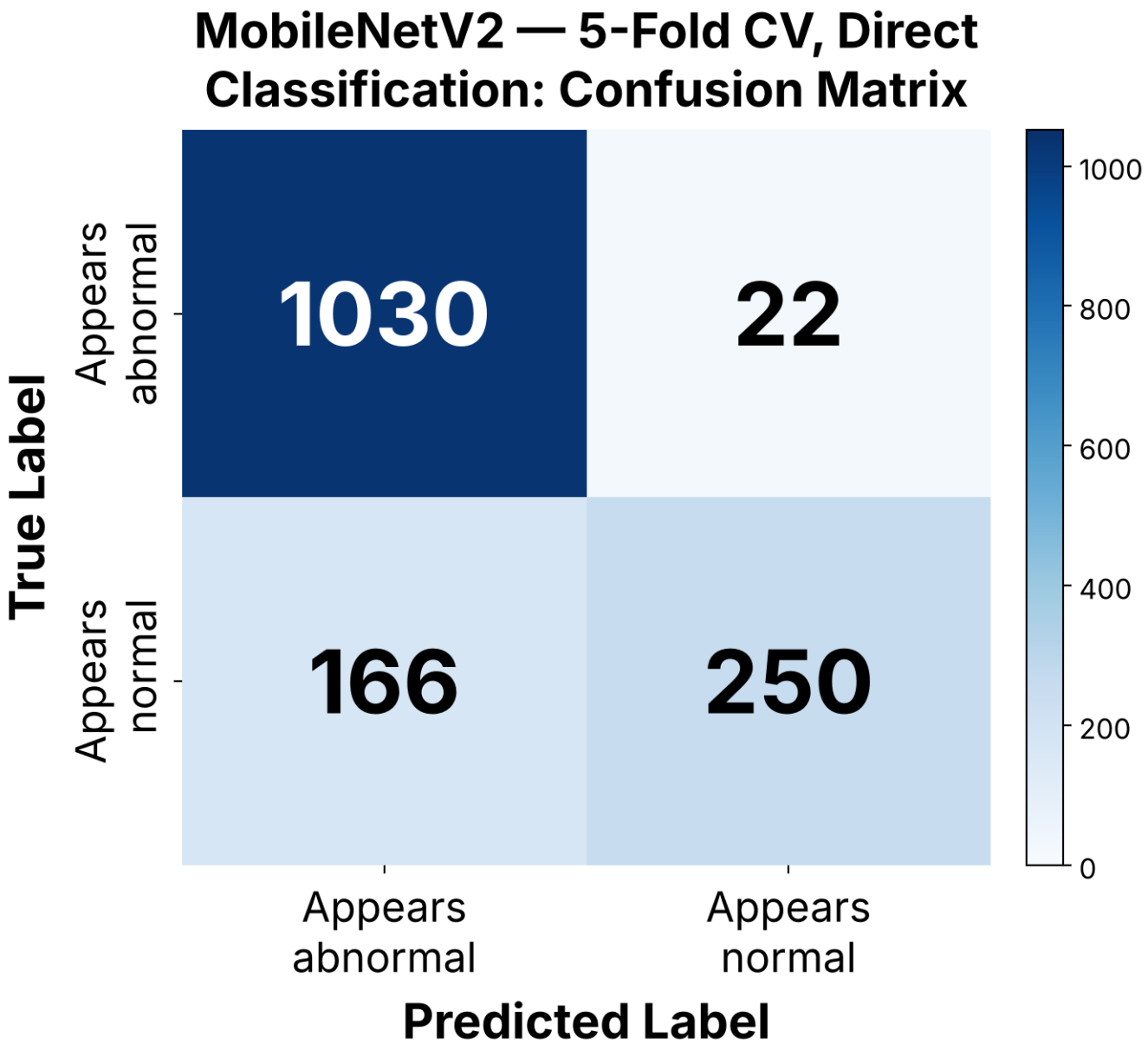
Accuracy: 87.19%

Precision: 87.76%

Recall: 87.19%

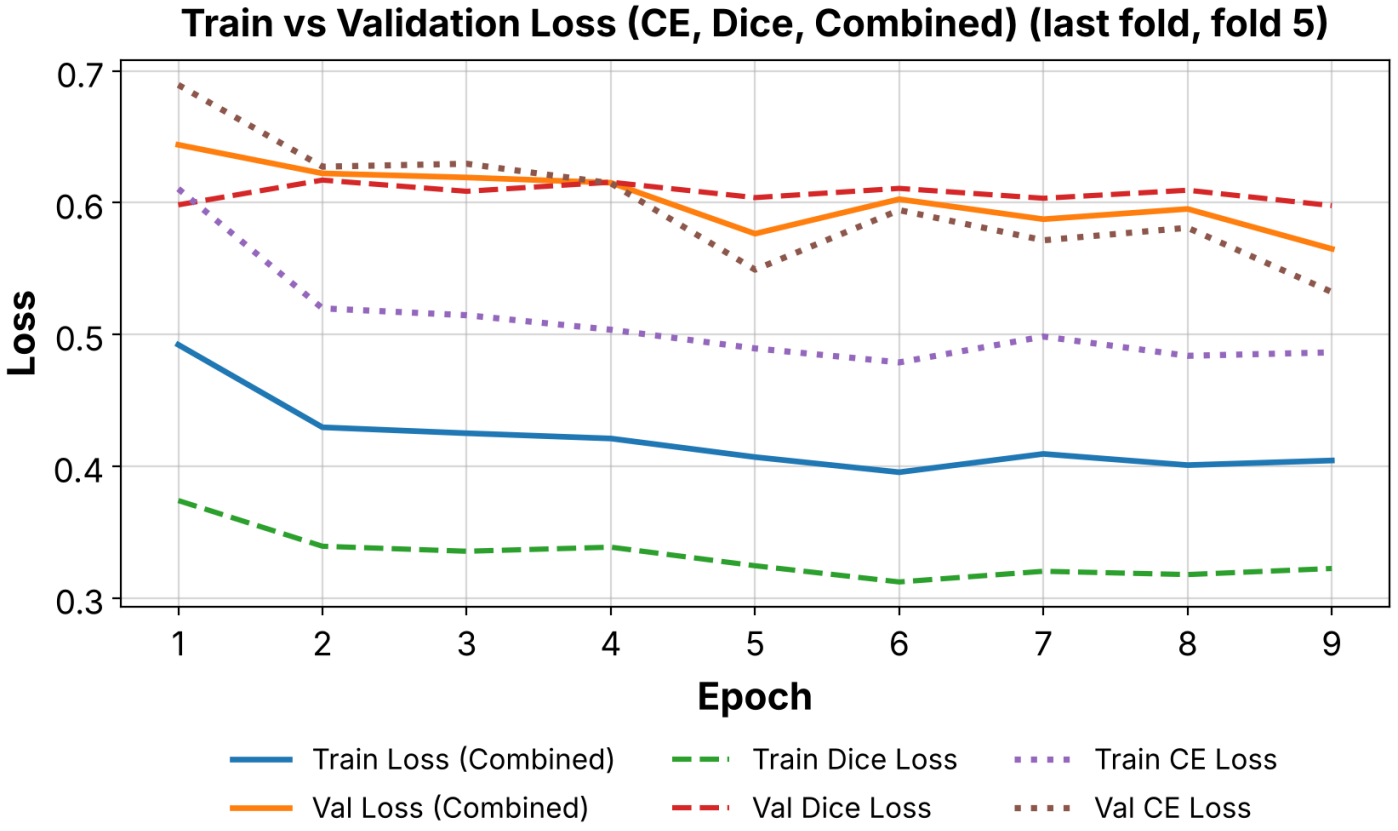
F1-score: 86.26%

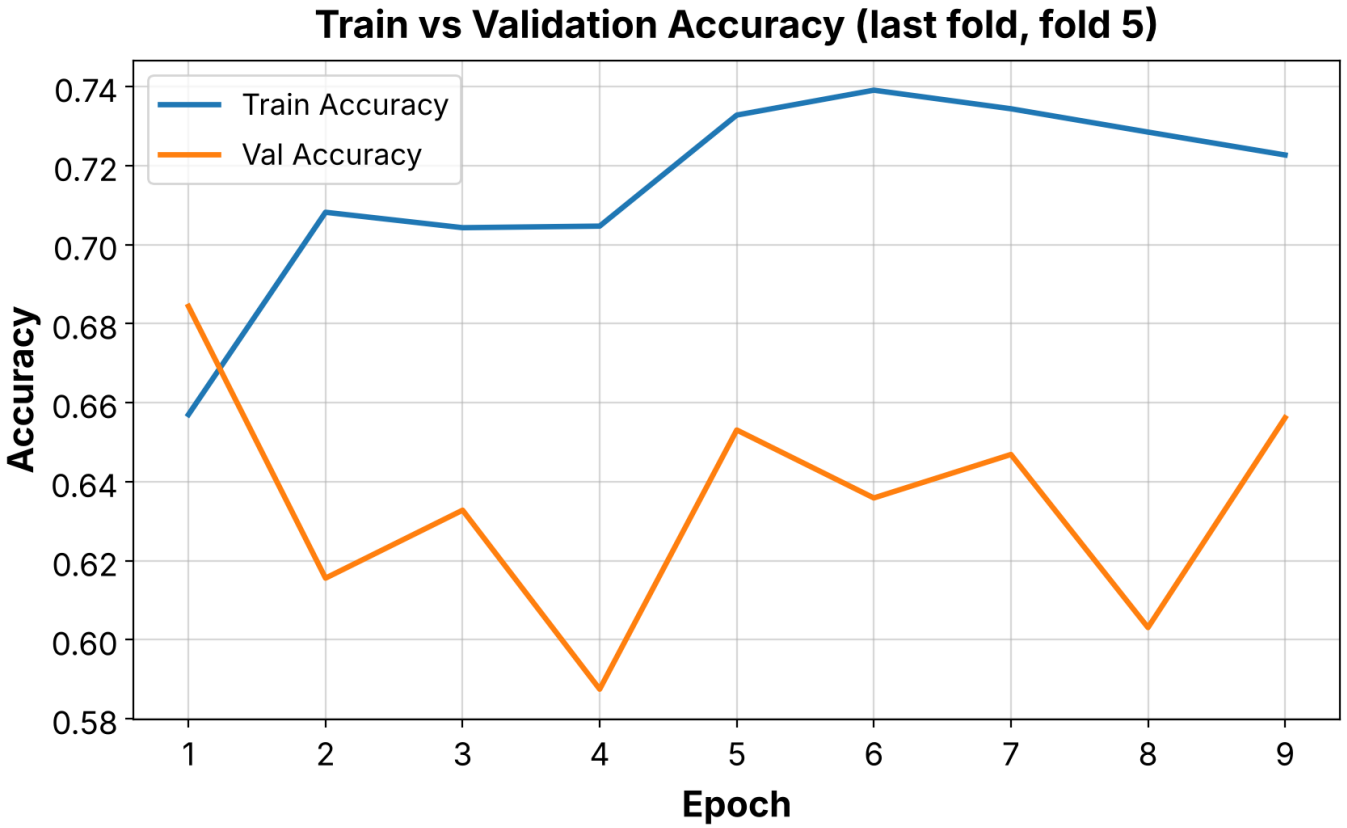
Confusion Matrix



32. MobileNetV2 — 5-Fold CV, Direct Classification — graphs (1/2)

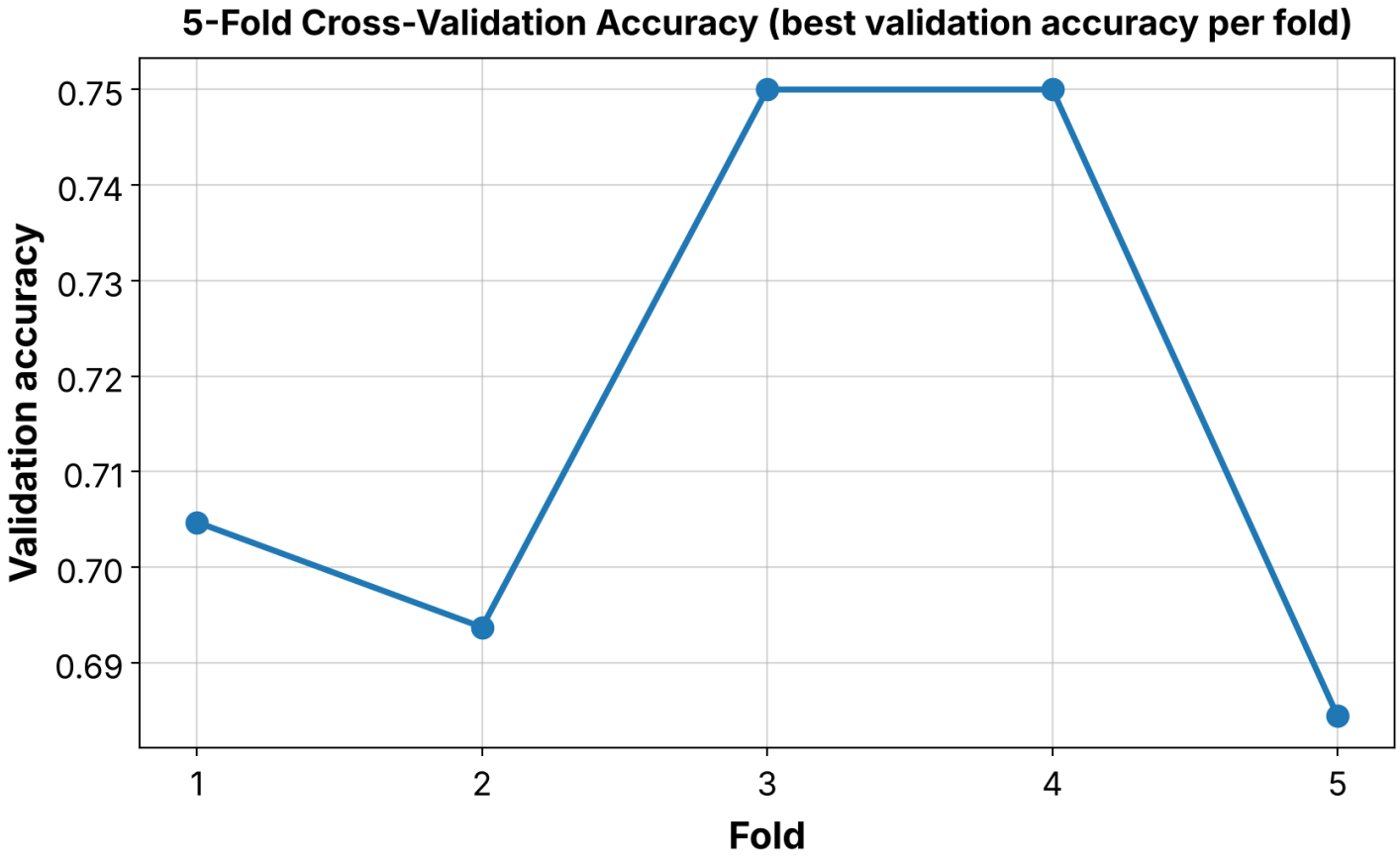
Training / Validation Graphs





32. MobileNetV2 — 5-Fold CV, Direct Classification — graphs (2/2)

Training / Validation Graphs



33. InceptionV3 + AdaBoost

Source notebook: Inception v3 baseline and ml.ipynb (inside Models_DL.zip)

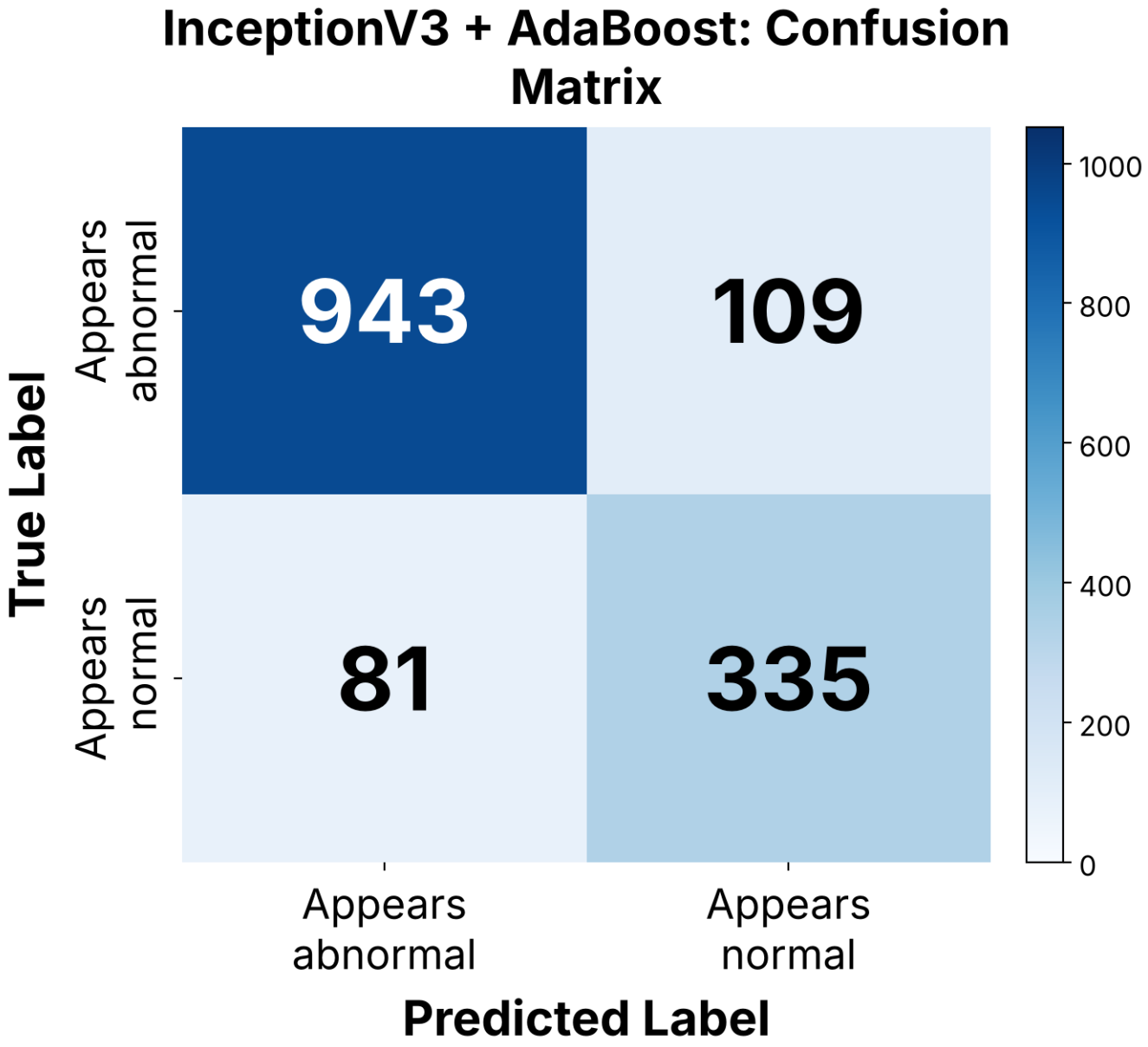
Accuracy: 87.06%

Precision: 87.37%

Recall: 87.06%

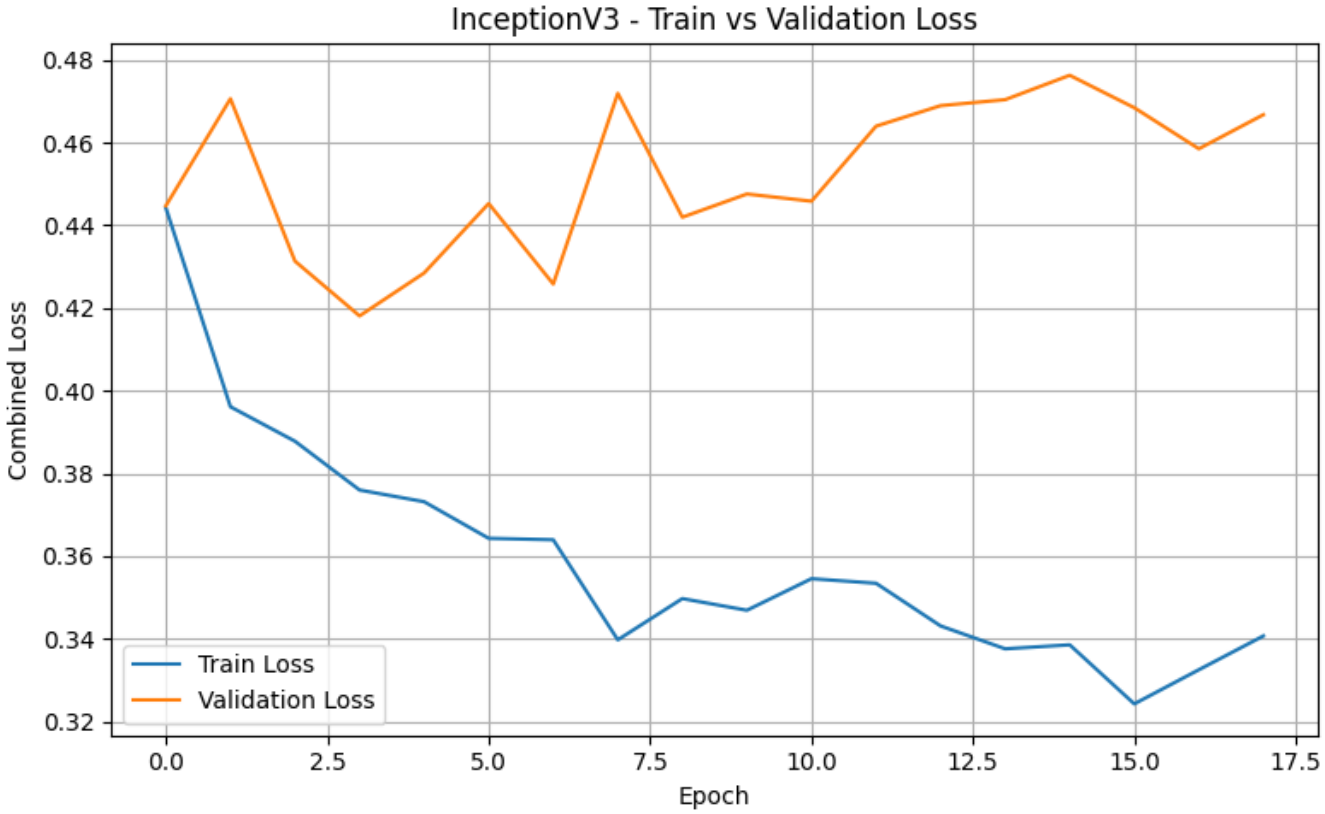
F1-score: 87.18%

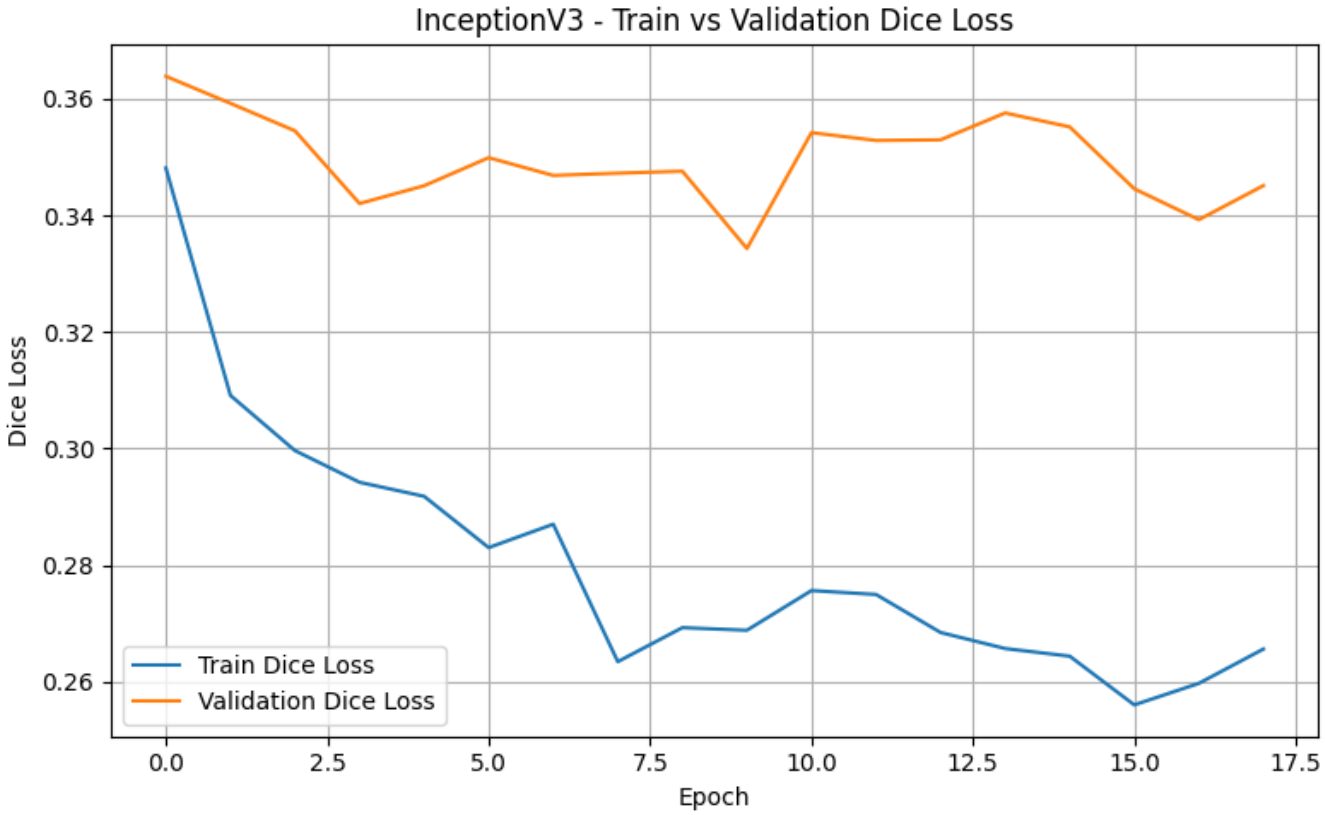
Confusion Matrix



33. InceptionV3 + AdaBoost — graphs (1/2)

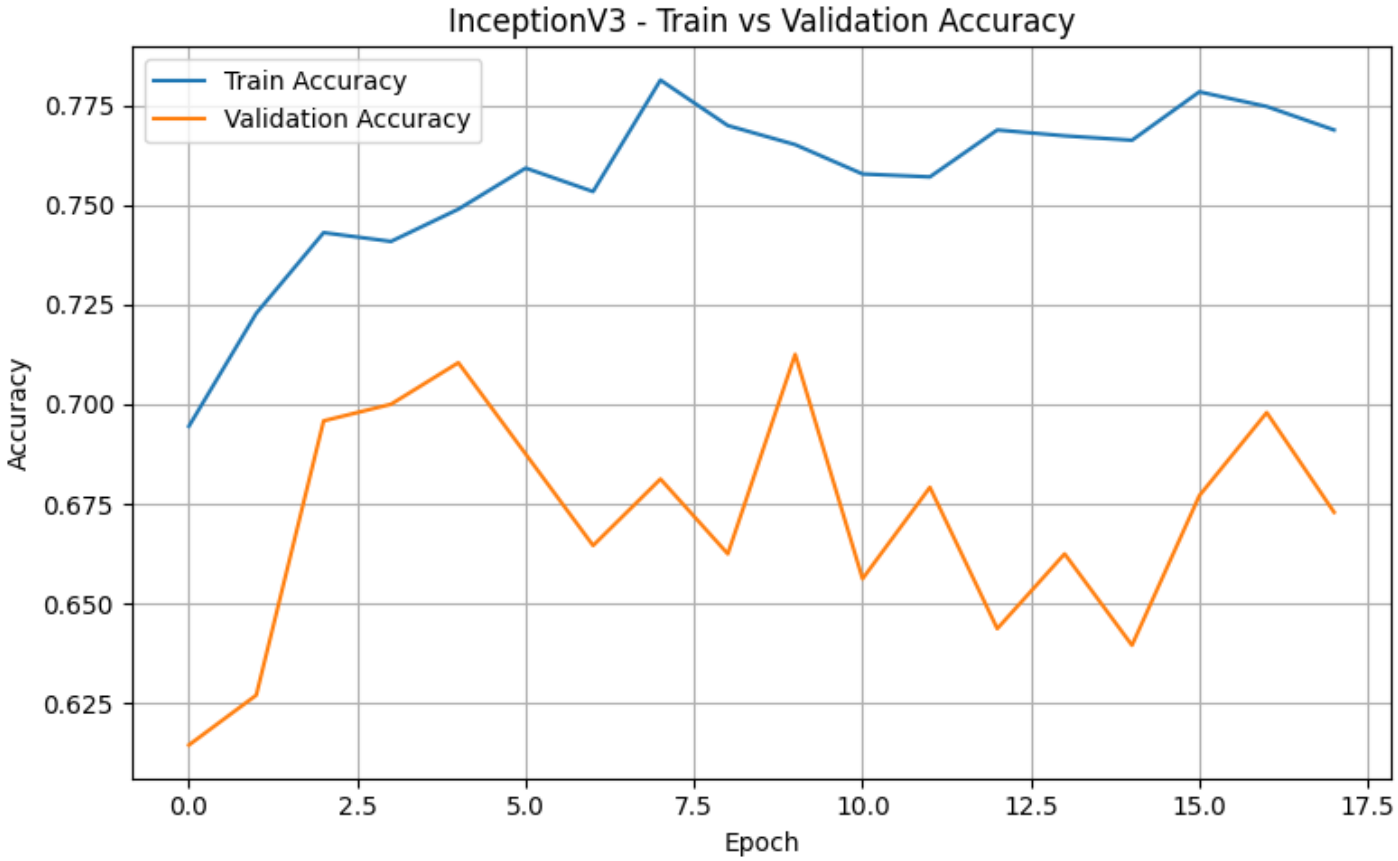
Training / Validation Graphs





33. InceptionV3 + AdaBoost — graphs (2/2)

Training / Validation Graphs



34. VGG16 + XGBoost

Source notebook: Xgboost and SVM and all 4 models.ipynb (inside Models_DL.zip)

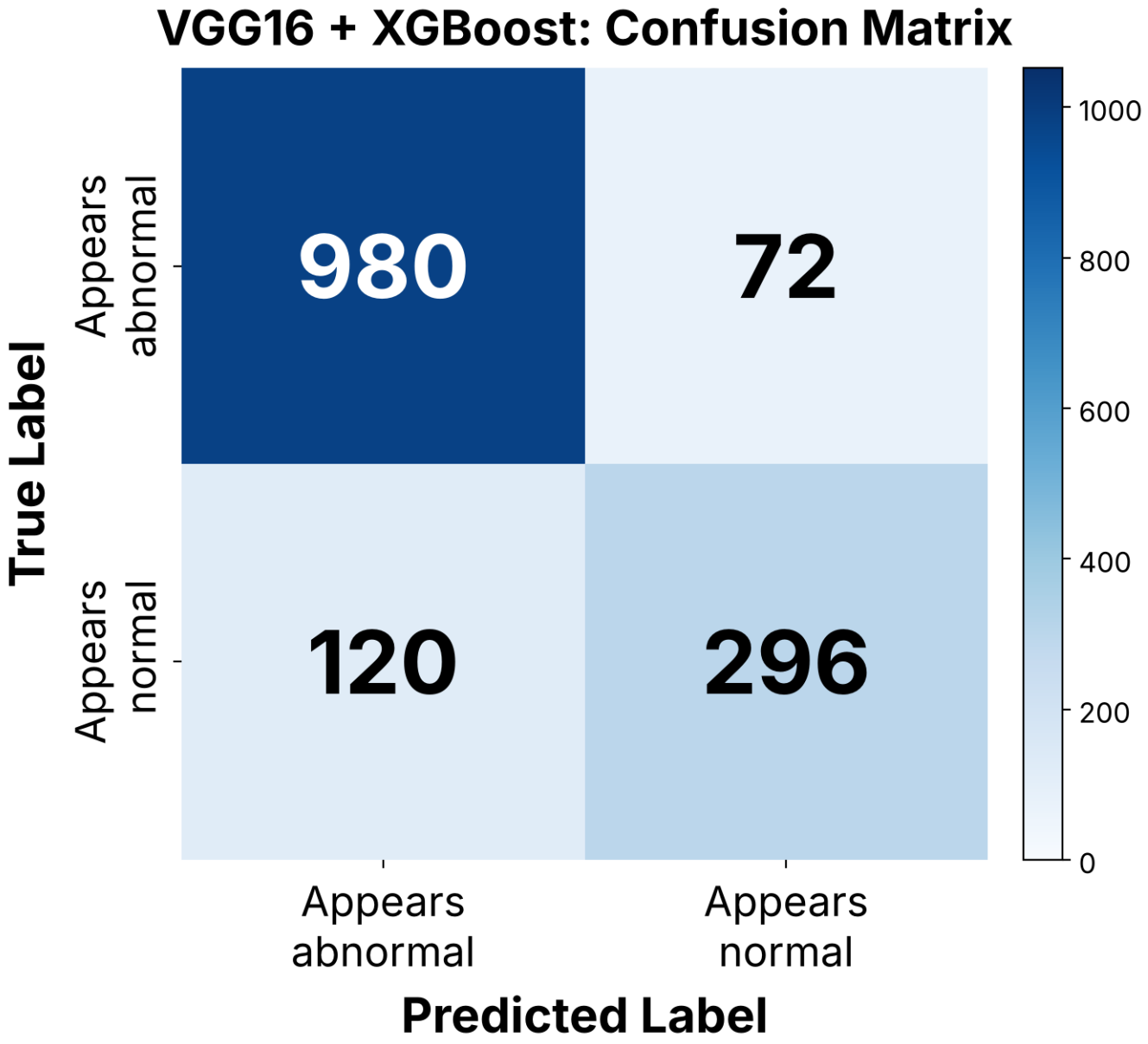
Accuracy: 86.92%

Precision: 86.64%

Recall: 86.92%

F1-score: 86.67%

Confusion Matrix



No training graphs: this notebook reloads an already-trained network and prints no epoch log, so there is nothing to plot for this row.

35. InceptionV3 + XGBoost

Source notebook: Inception v3 baseline and ml.ipynb (inside Models_DL.zip)

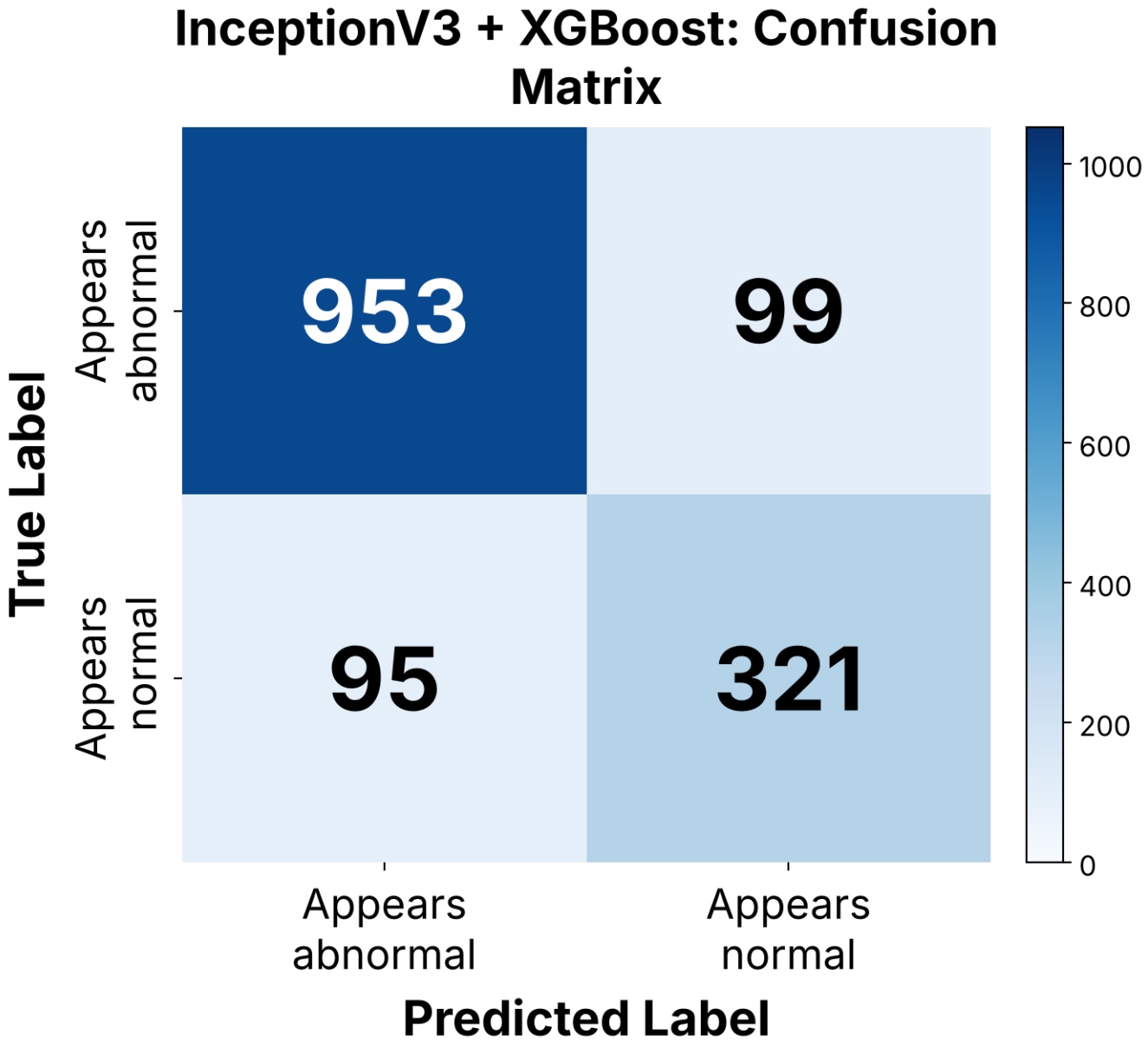
Accuracy: 86.78%

Precision: 86.82%

Recall: 86.78%

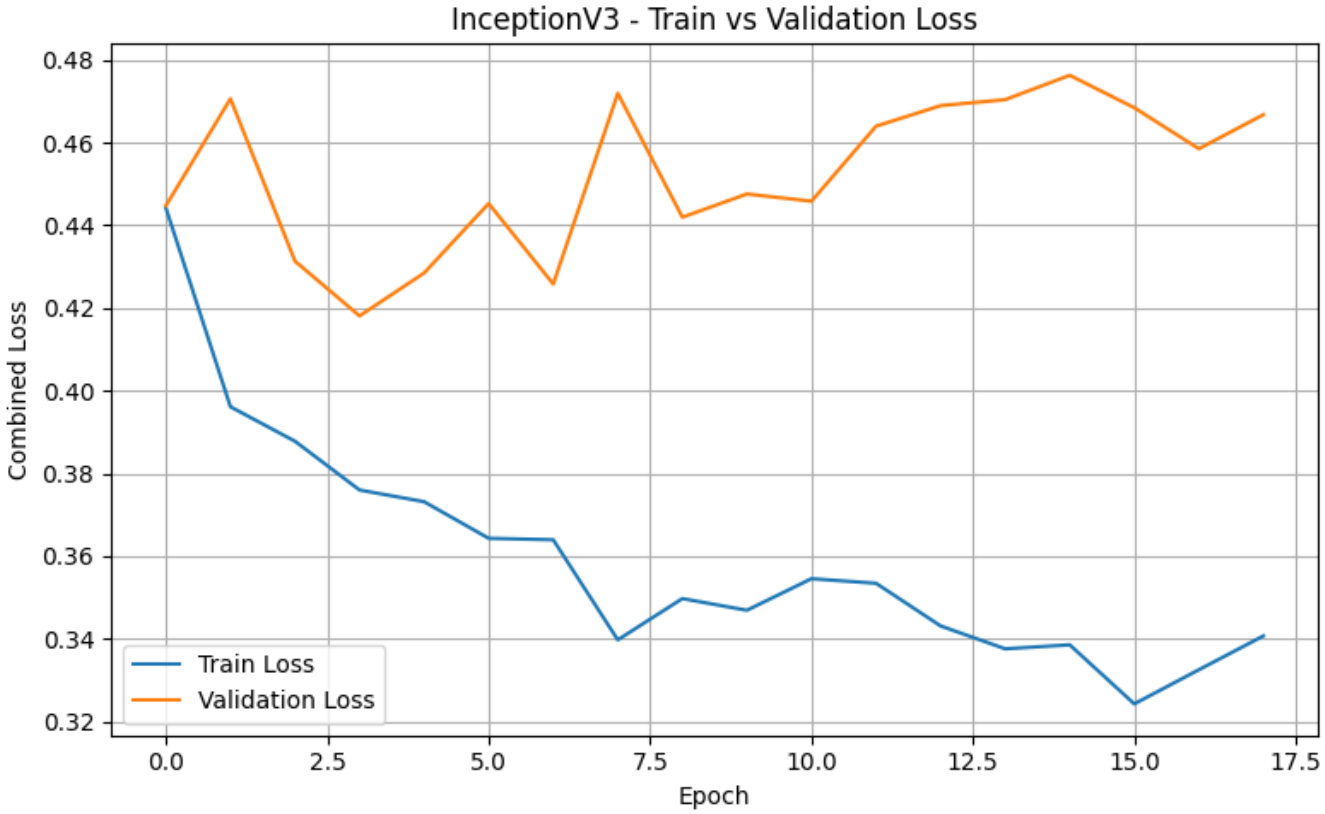
F1-score: 86.80%

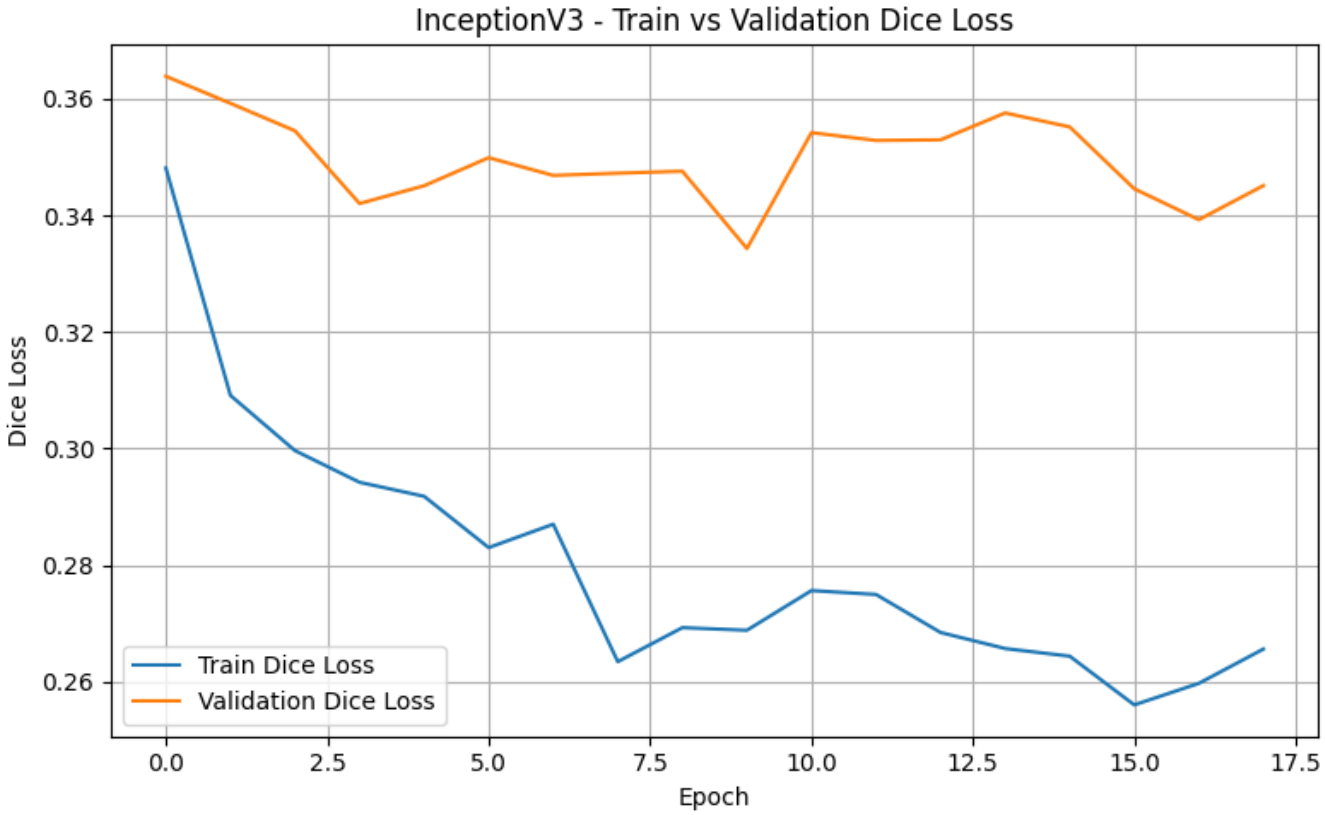
Confusion Matrix



35. InceptionV3 + XGBoost — graphs (1/2)

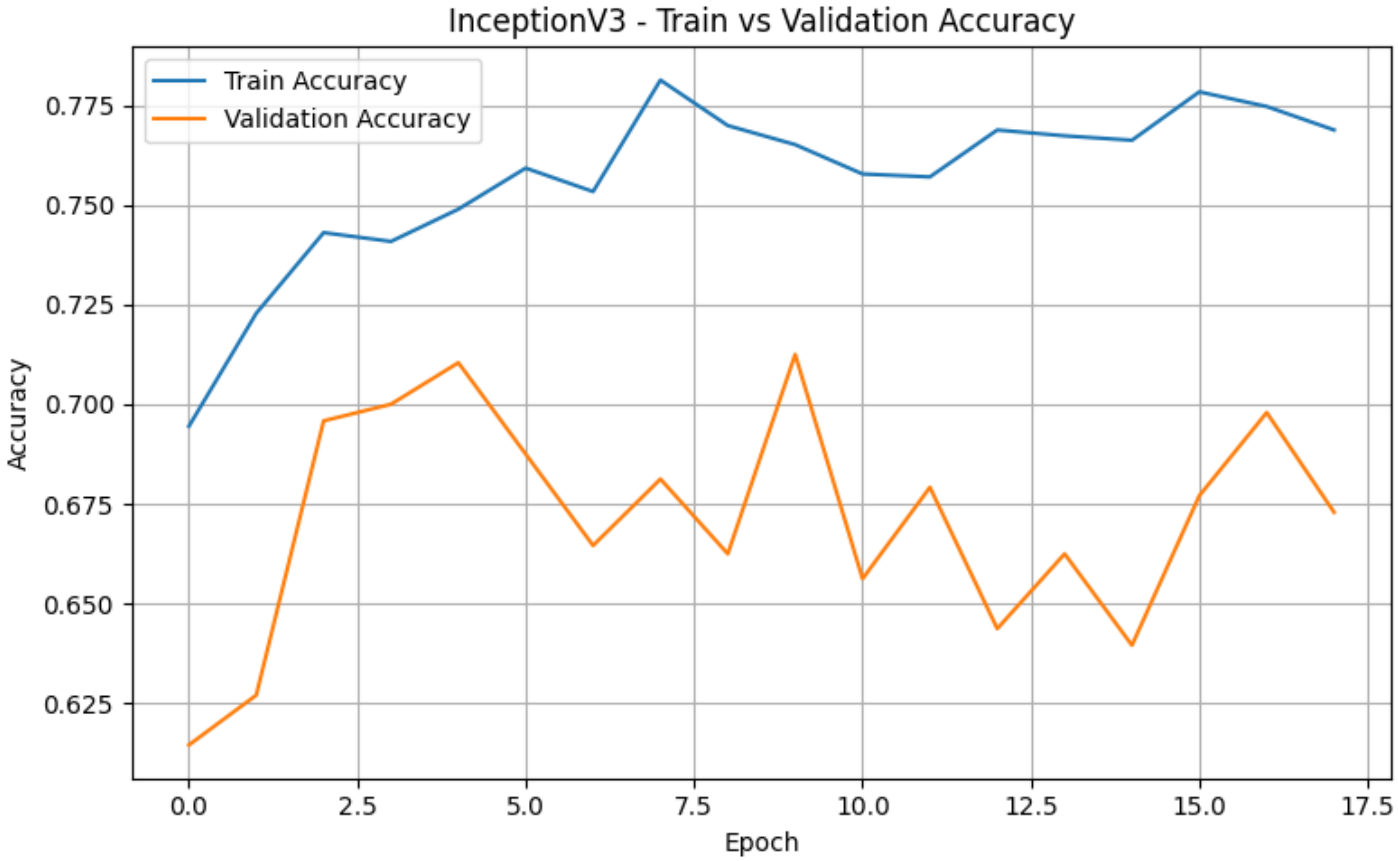
Training / Validation Graphs





35. InceptionV3 + XGBoost — graphs (2/2)

Training / Validation Graphs



36. ResNet18 + SVM

Source notebook: Xgboost and SVM and all 4 models.ipynb (inside Models_DL.zip)

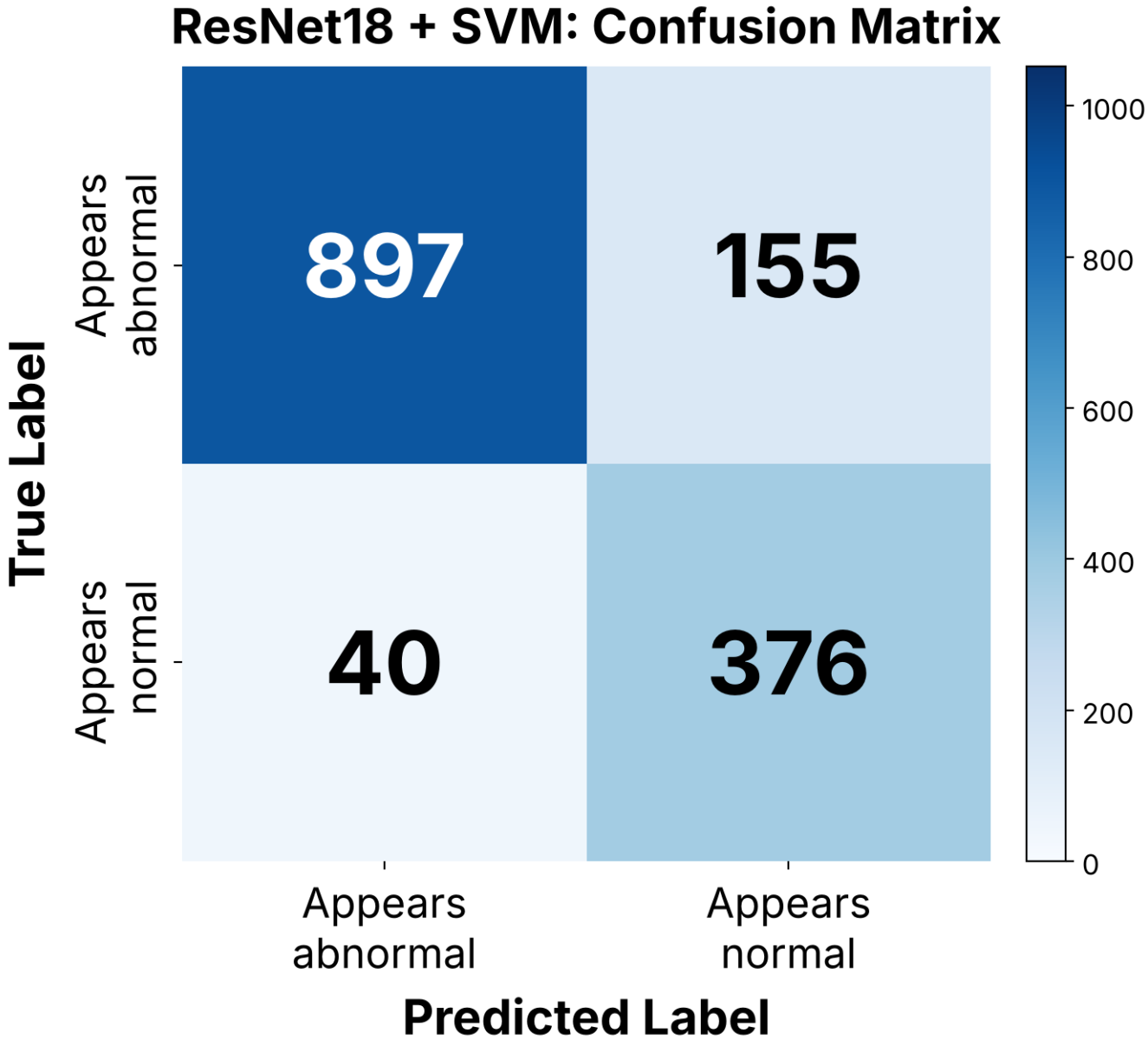
Accuracy: 86.72%

Precision: 88.67%

Recall: 86.72%

F1-score: 87.14%

Confusion Matrix



No training graphs: this notebook reloads an already-trained network and prints no epoch log, so there is nothing to plot for this row.

37. MobileNetV2 + XGBoost

Source notebook: Xgboost and SVM and all 4 models.ipynb (inside Models_DL.zip)

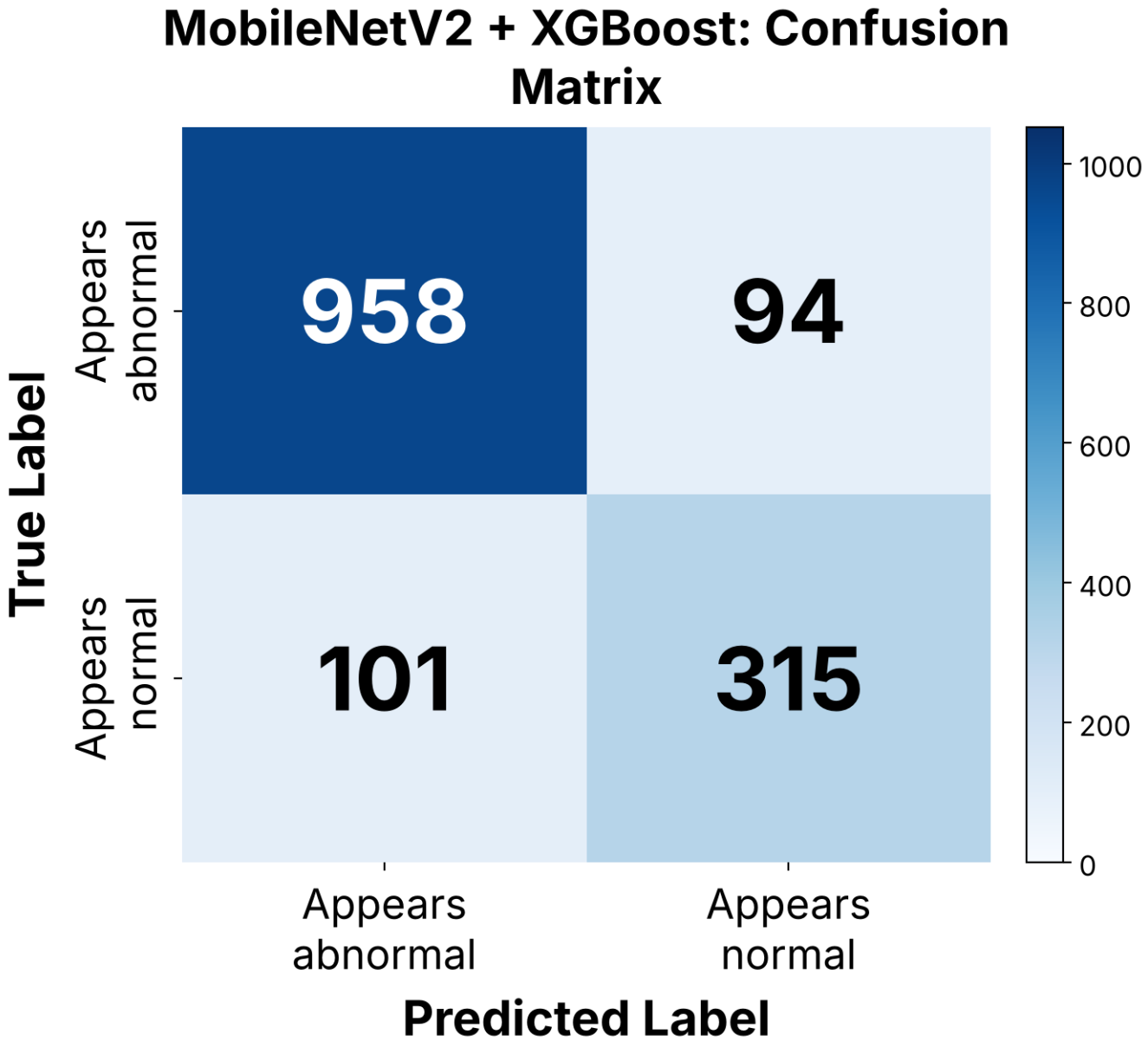
Accuracy: 86.72%

Precision: 86.65%

Recall: 86.72%

F1-score: 86.68%

Confusion Matrix



No training graphs: this notebook reloads an already-trained network and prints no epoch log, so there is nothing to plot for this row.

38. AlexNet — 5-Fold CV, Direct Classification

Source notebook: AlexNet with kfold.ipynb (inside Models_DL.zip)

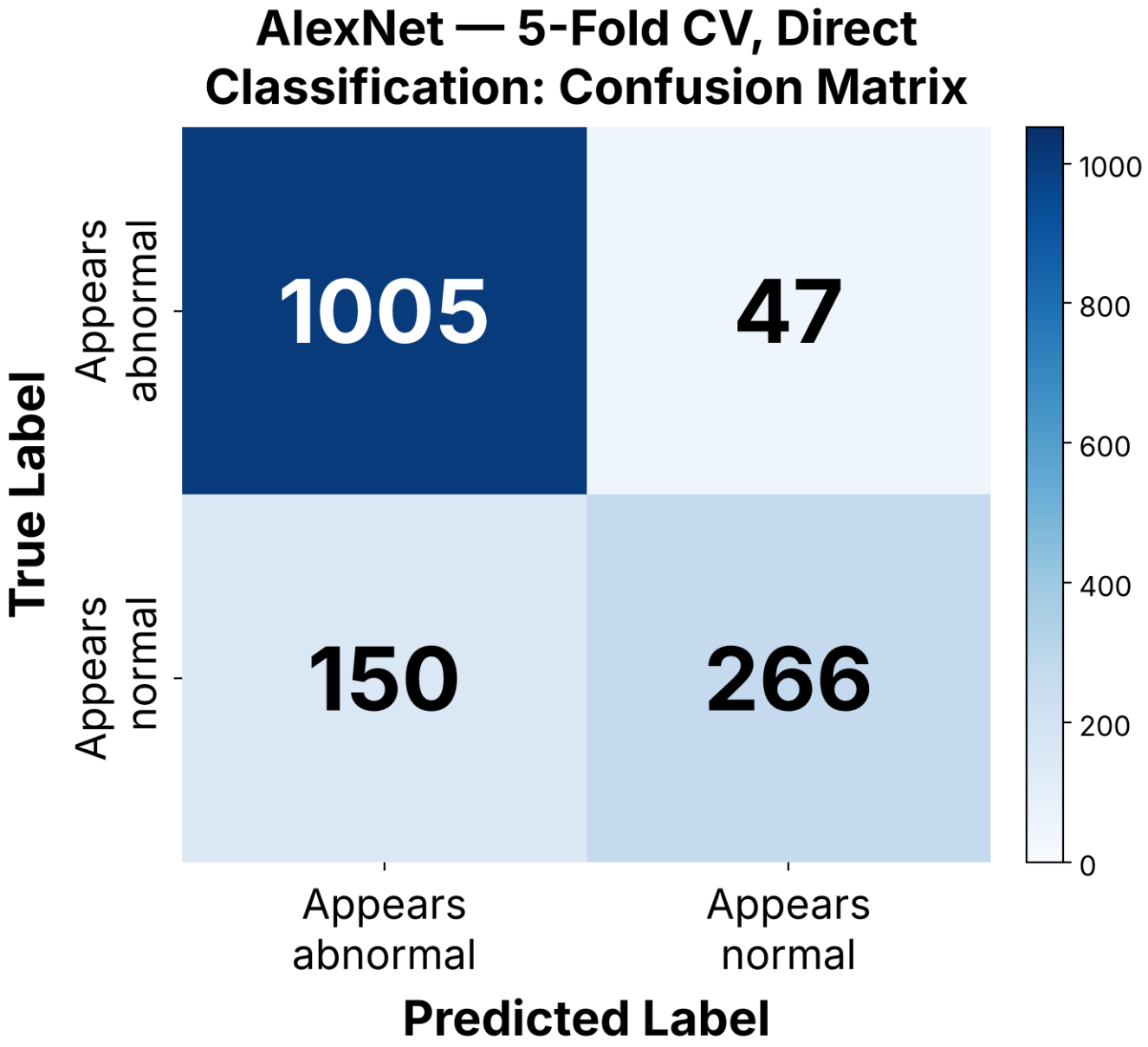
Accuracy: 86.58%

Precision: 86.44%

Recall: 86.58%

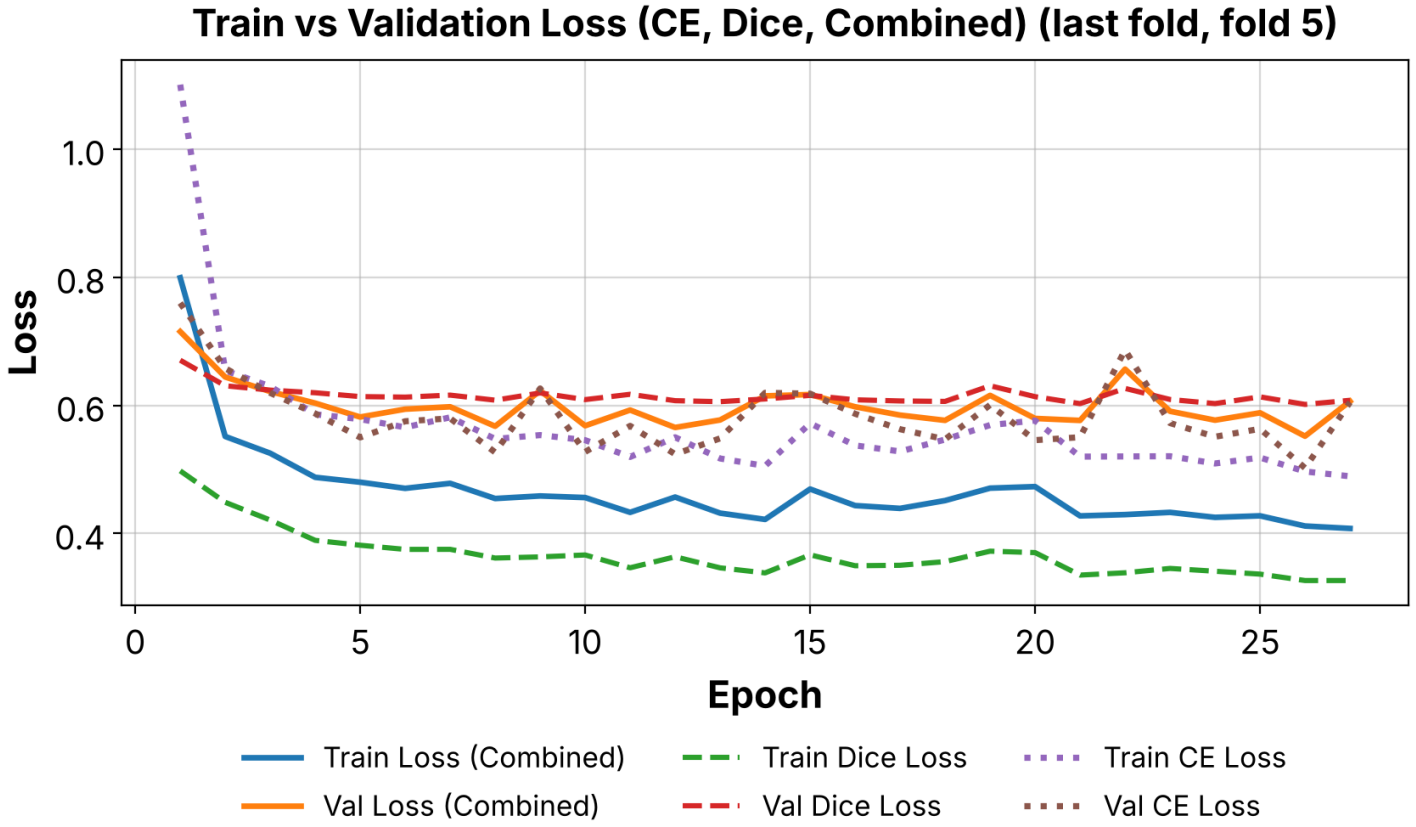
F1-score: 85.95%

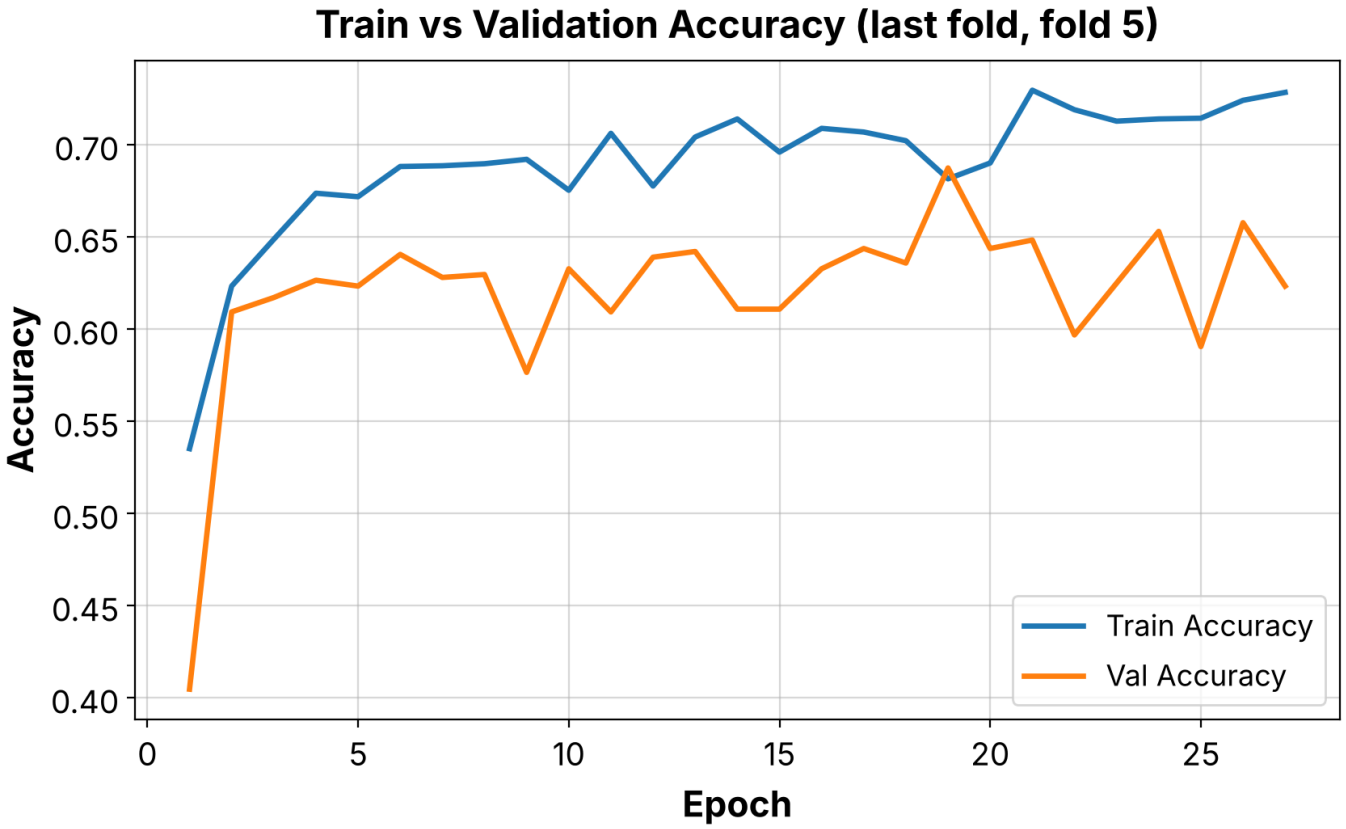
Confusion Matrix



38. AlexNet — 5-Fold CV, Direct Classification — graphs (1/2)

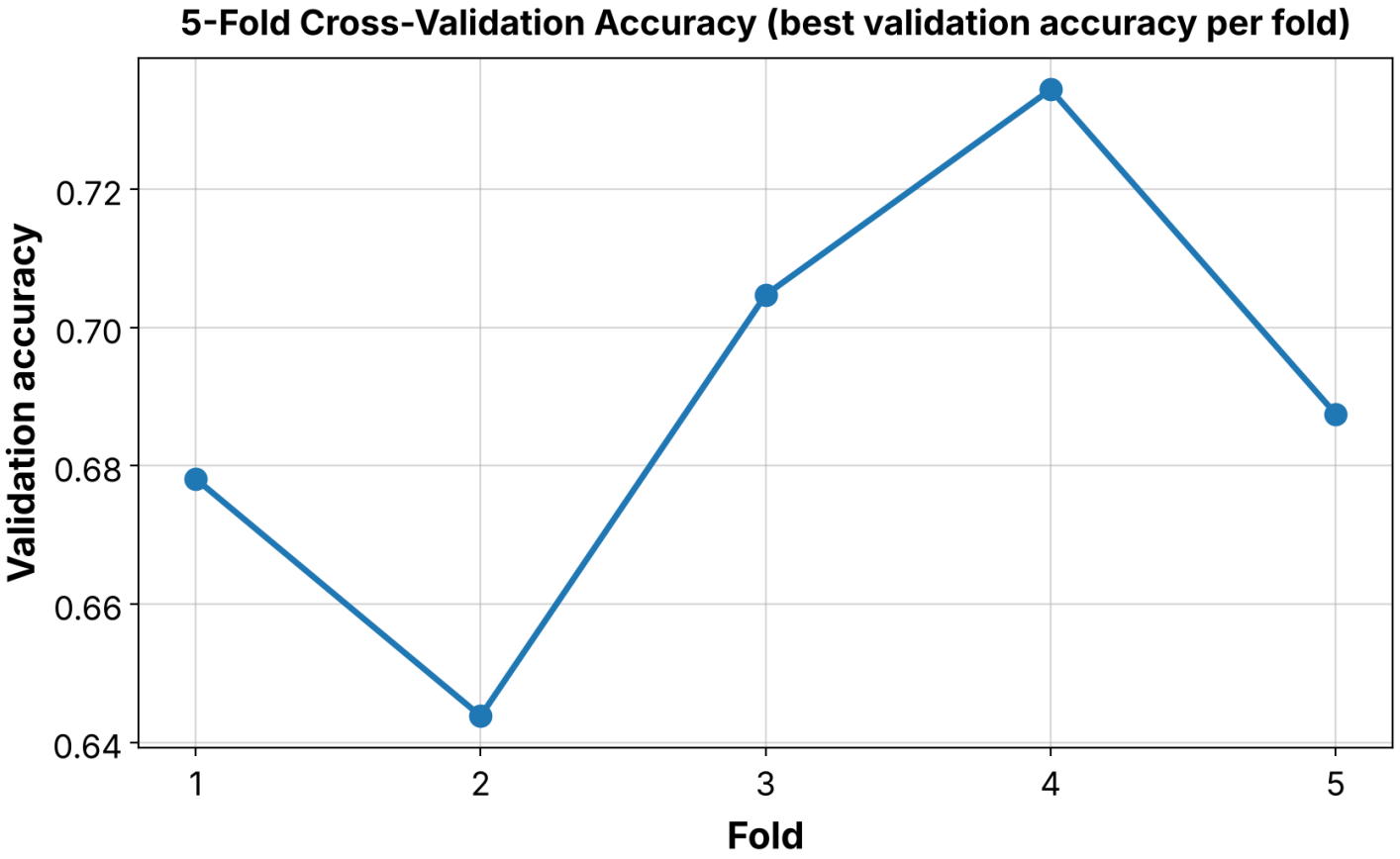
Training / Validation Graphs





38. AlexNet — 5-Fold CV, Direct Classification — graphs (2/2)

Training / Validation Graphs



39. EfficientNet-B3 + Gaussian Naive Bayes

Source notebook: efficinetnet b3 baseline and ml.ipynb (inside Models_DL.zip)

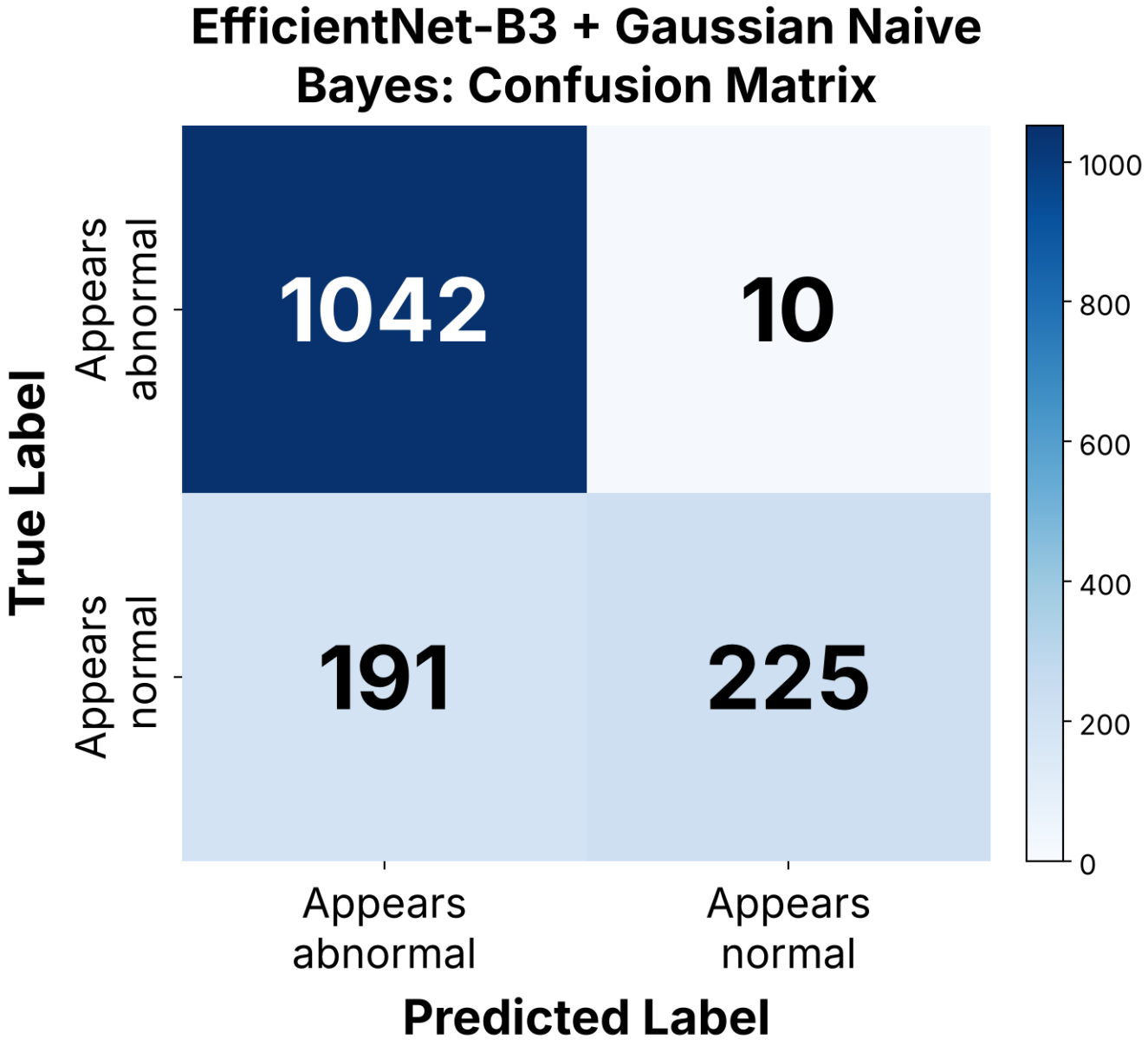
Accuracy: 86.31%

Precision: 87.69%

Recall: 86.31%

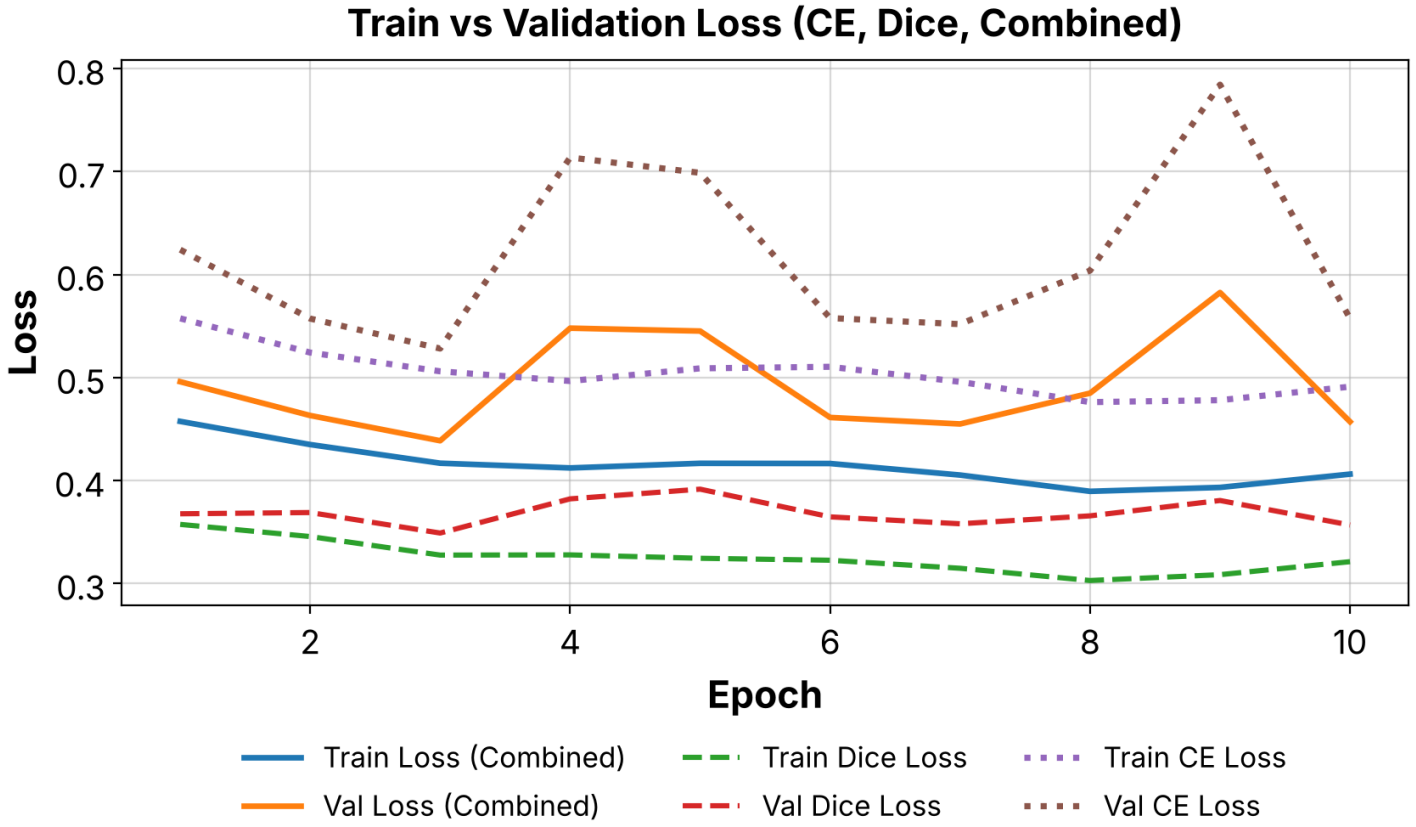
F1-score: 84.95%

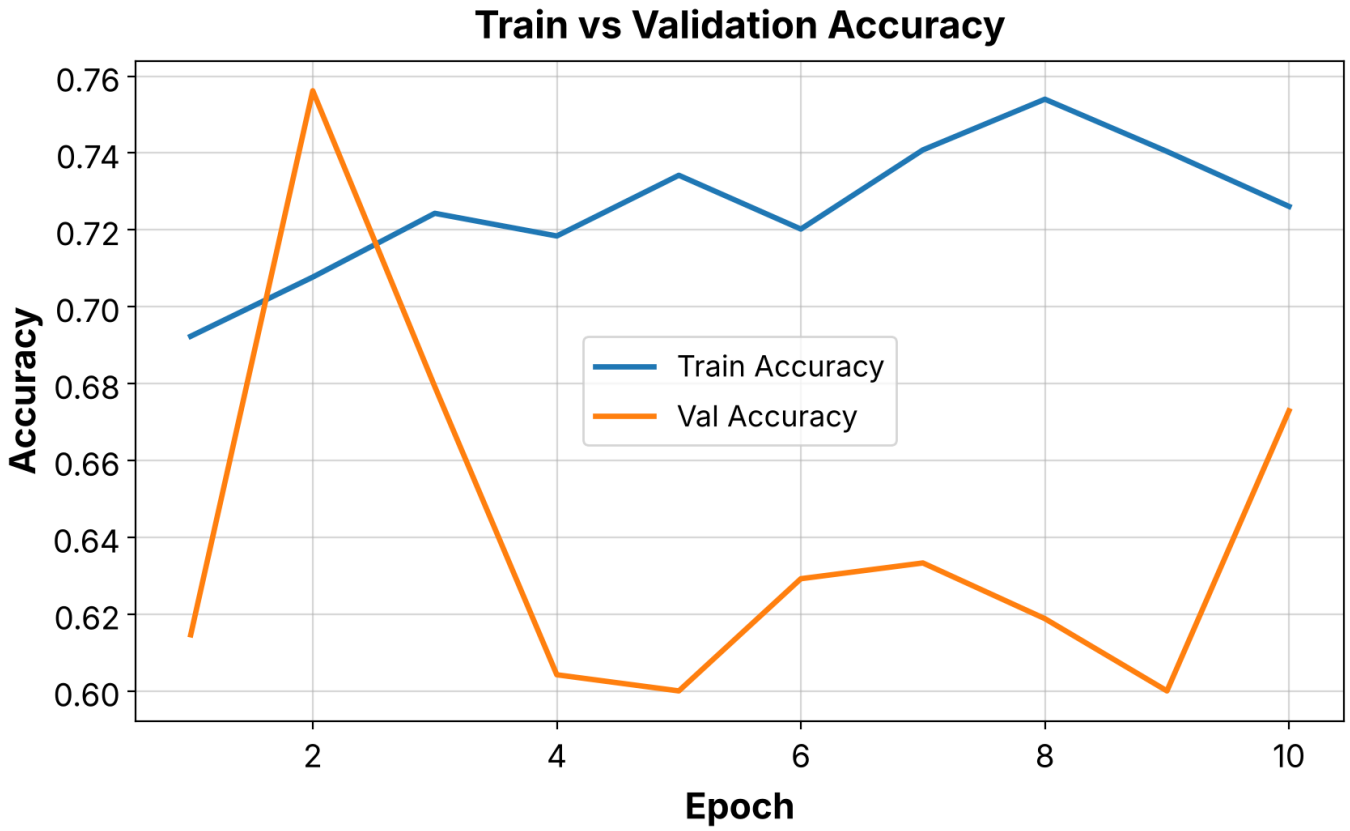
Confusion Matrix



39. EfficientNet-B3 + Gaussian Naive Bayes — graphs

Training / Validation Graphs





40. ResNet18 — Direct Classification (AdaBoost section run)

Source notebook: ResNet with ML.ipynb (inside Models_DL.zip)

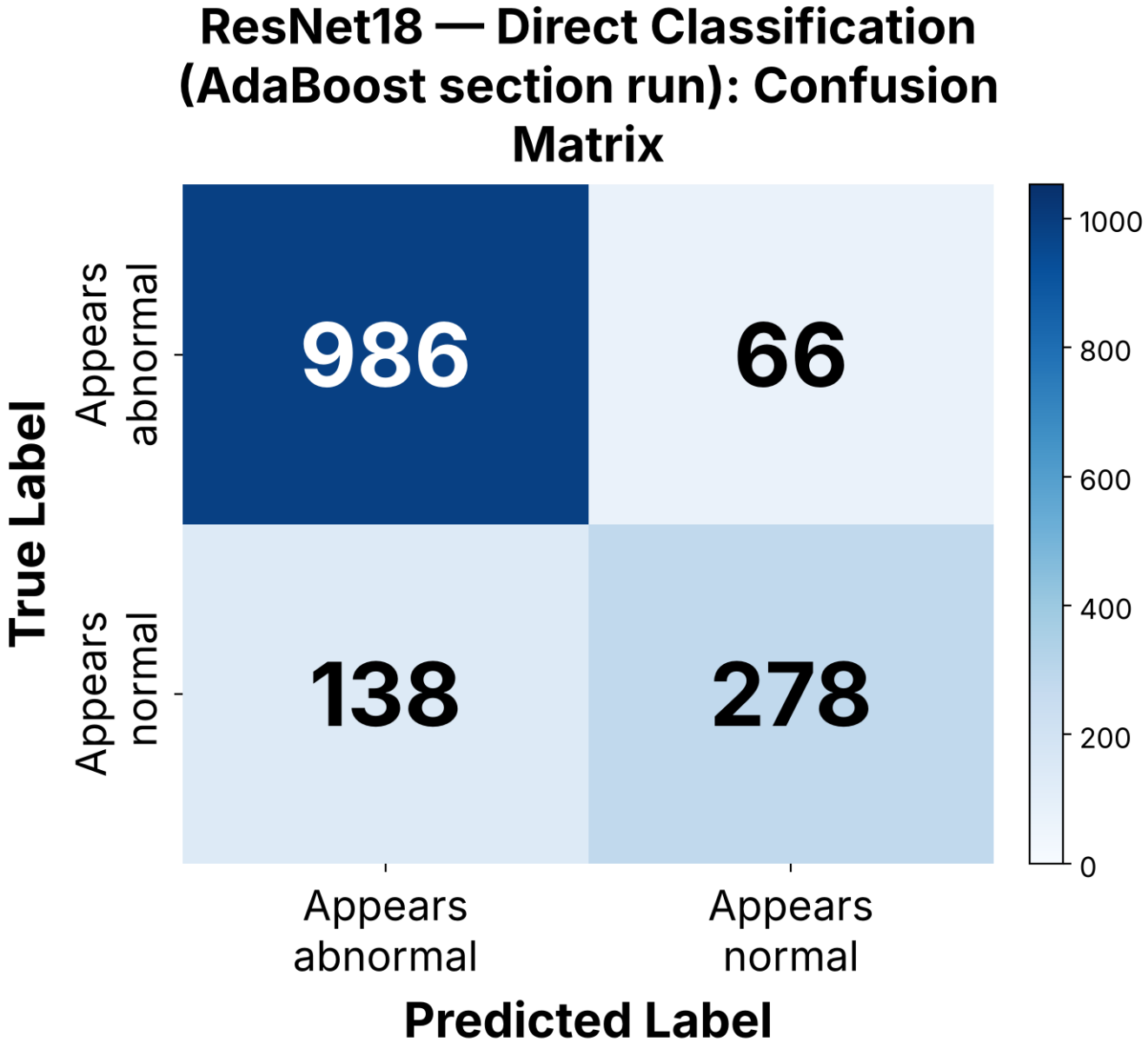
Accuracy: 86.10%

Precision: 85.76%

Recall: 86.10%

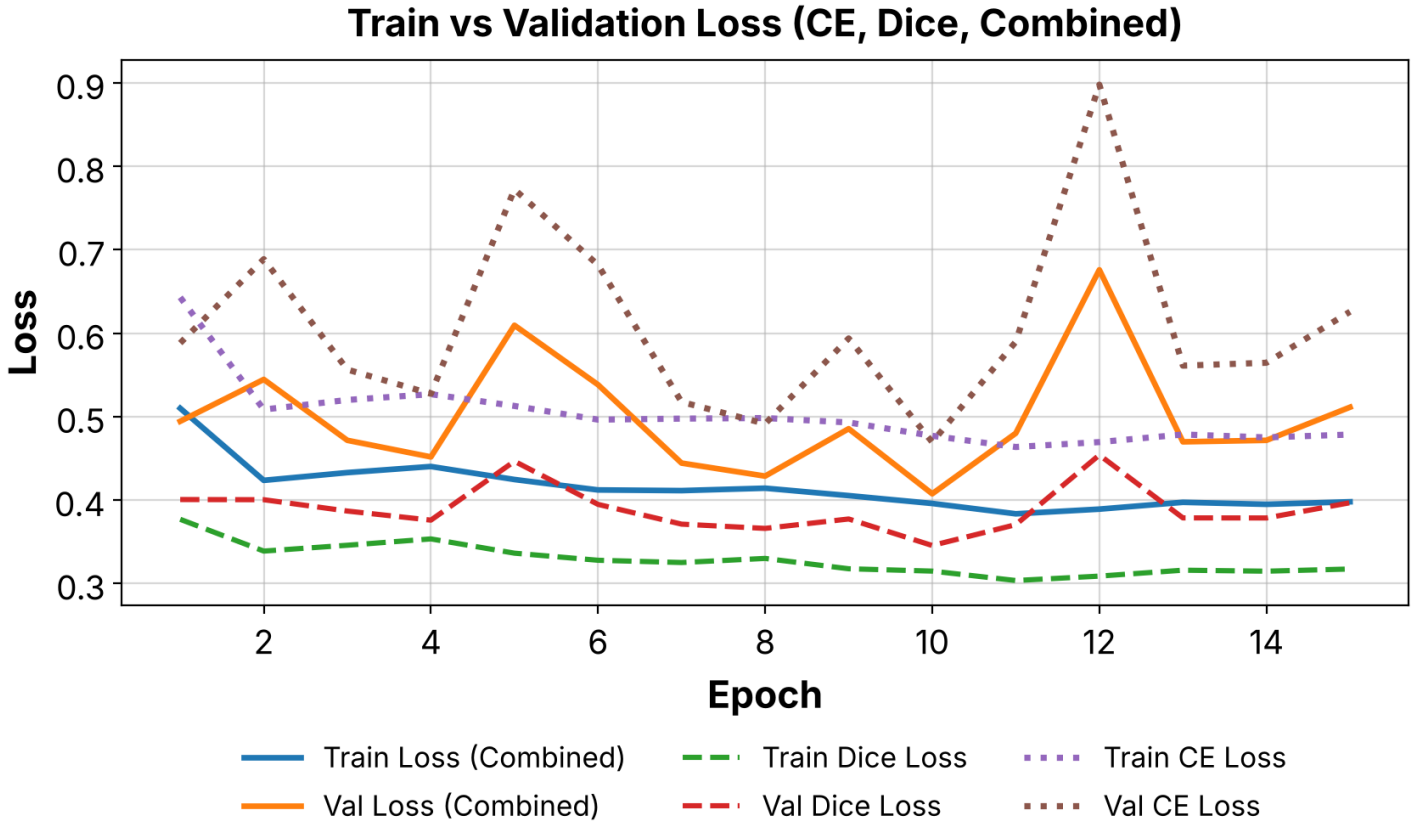
F1-score: 85.68%

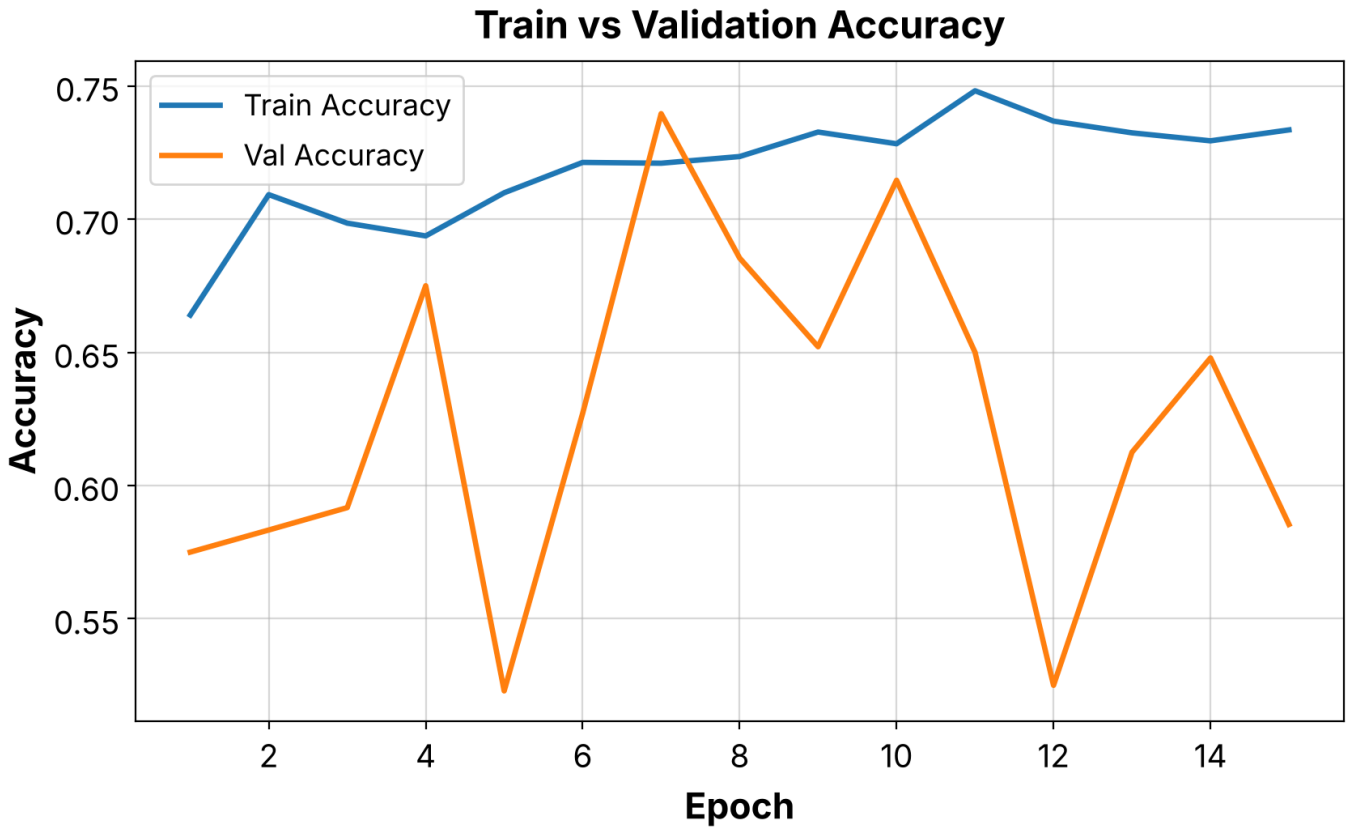
Confusion Matrix



40. ResNet18 — Direct Classification (AdaBoost section run) — graphs

Training / Validation Graphs





41. AlexNet + XGBoost

Source notebook: Xgboost and SVM and all 4 models.ipynb (inside Models_DL.zip)

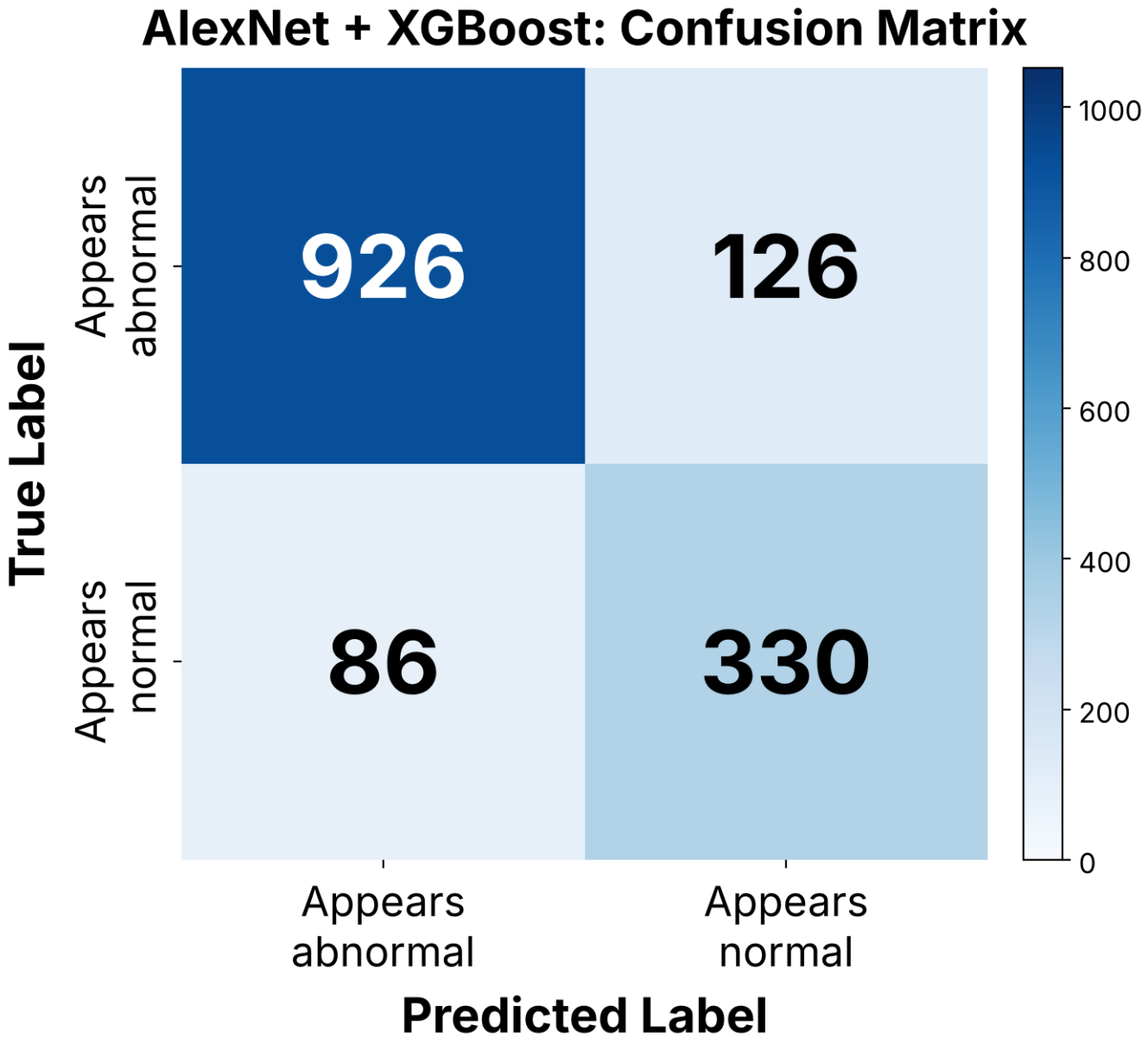
Accuracy: 85.56%

Precision: 86.08%

Recall: 85.56%

F1-score: 85.75%

Confusion Matrix



No training graphs: this notebook reloads an already-trained network and prints no epoch log, so there is nothing to plot for this row.

42. AlexNet — Direct Classification (Dice Loss baseline)

Source notebook: AlexNet with Dice Loss (Baseline).ipynb (inside Models_DL.zip)

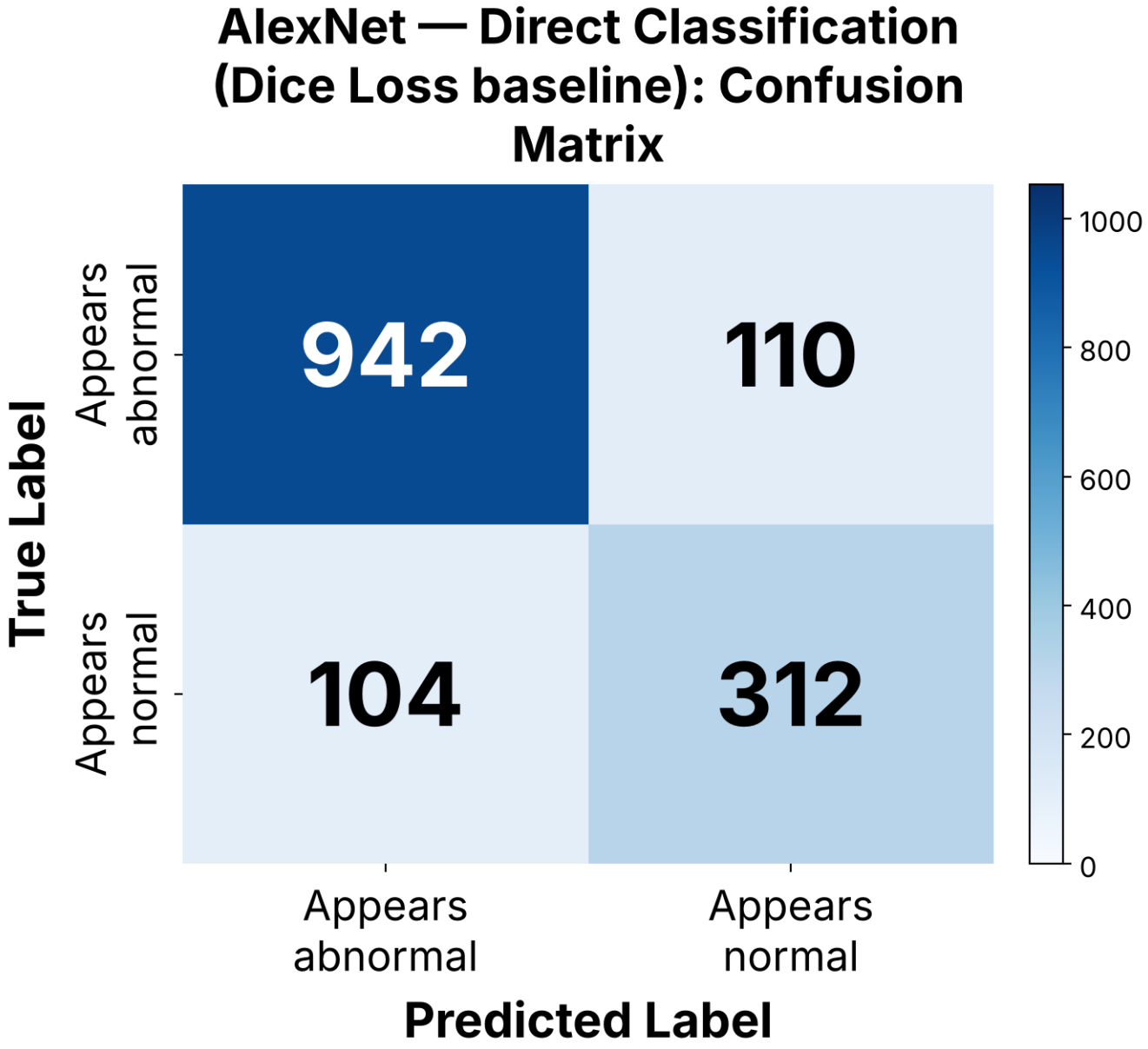
Accuracy: 85.42%

Precision: 85.49%

Recall: 85.42%

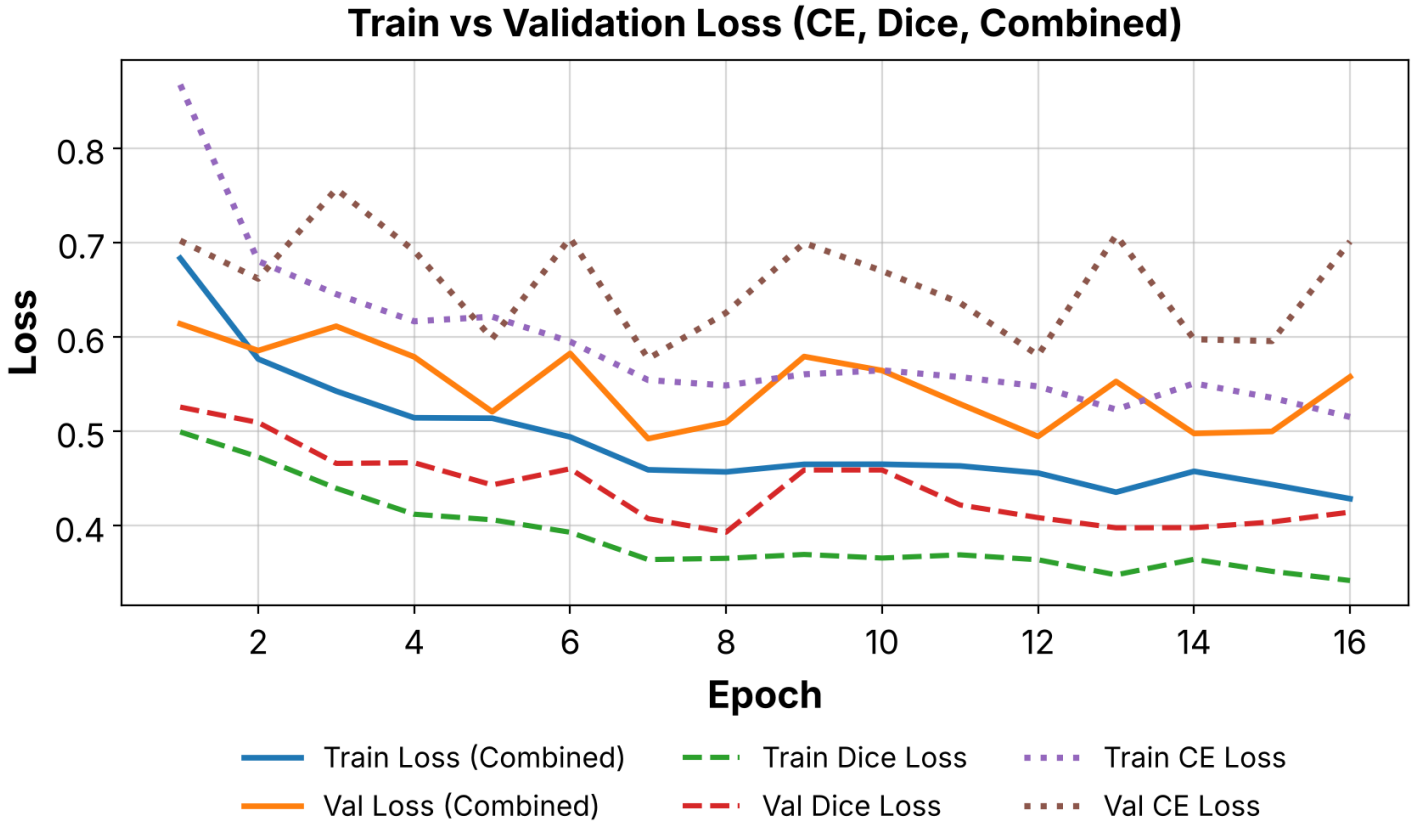
F1-score: 85.45%

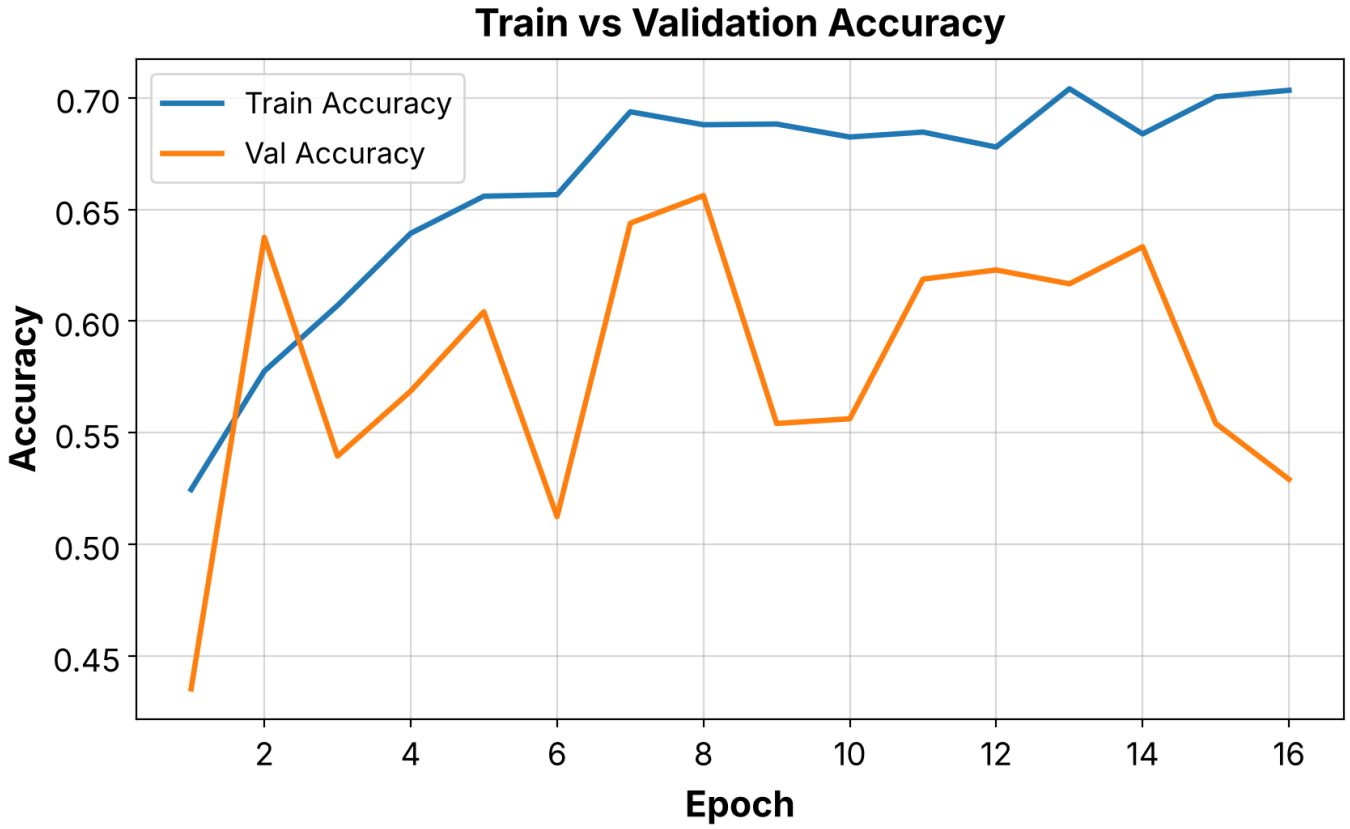
Confusion Matrix



42. AlexNet — Direct Classification (Dice Loss baseline) — graphs

Training / Validation Graphs





43. VGG16 — Direct 5-fold CV with preprocessing

Source notebook: MobileNet with Preprocessing.ipynb (inside Models_DL.zip)

Accuracy: 85.35%

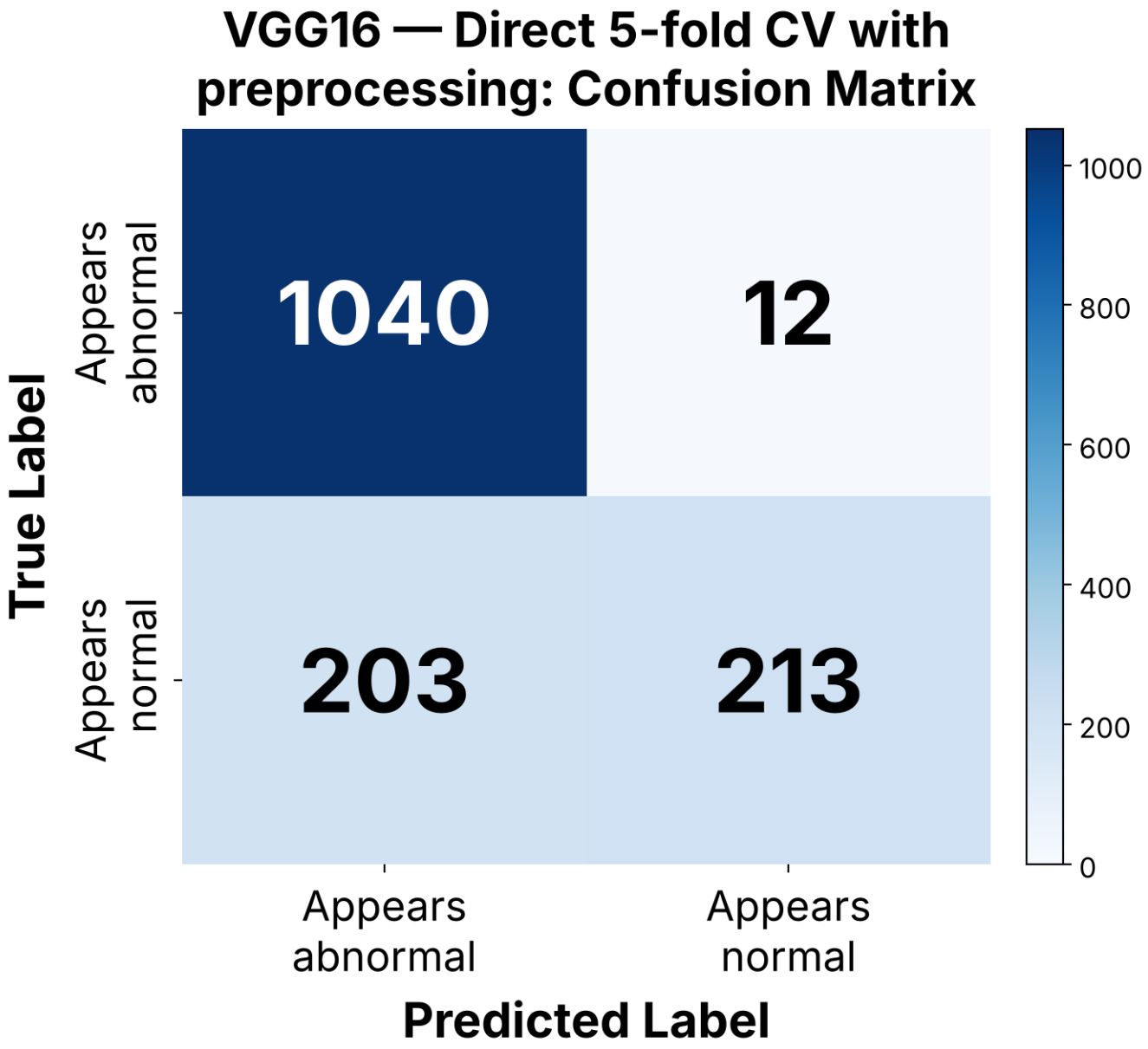
Precision: 86.79%

Recall: 85.35%

F1-score: 83.78%

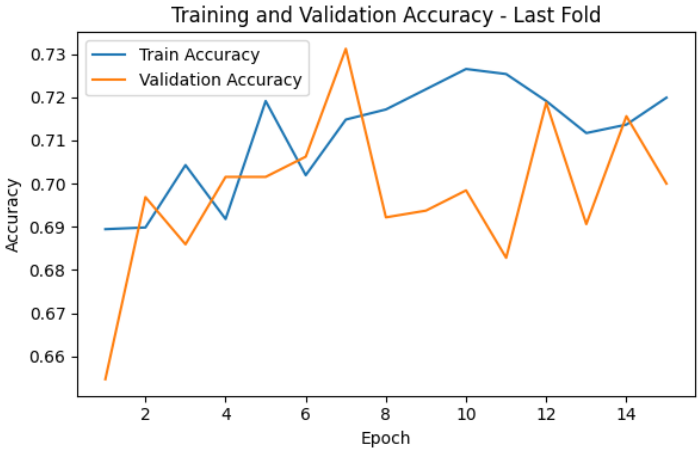
Note: the notebook prints 82.53% as its test / “Final Test Accuracy”, which does not agree with the notebook’s own printed confusion matrix and weighted recall (this printed number is not the test accuracy). The test accuracy shown here is the one implied by the confusion matrix and equals the weighted recall.

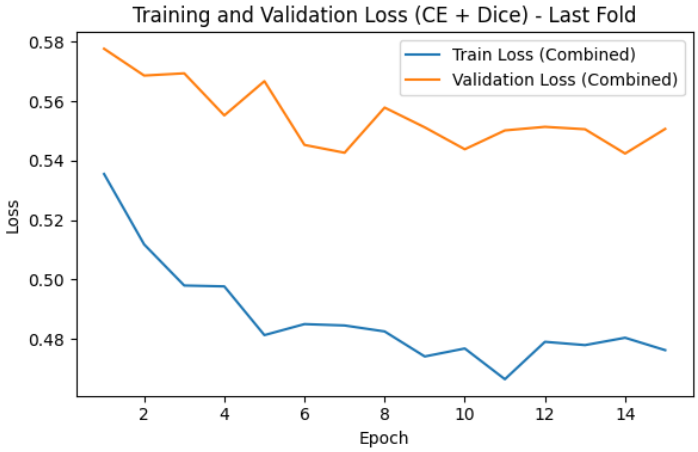
Confusion Matrix

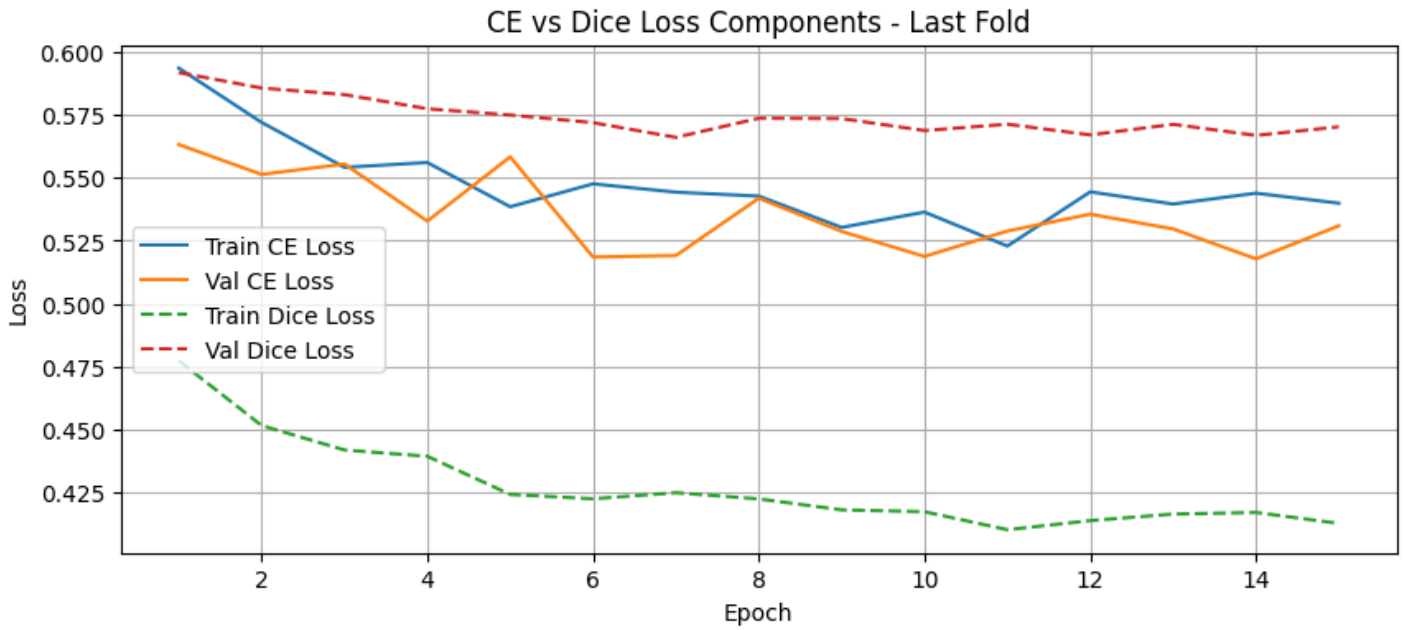


43. VGG16 — Direct 5-fold CV with preprocessing — graphs (1/2)

Training / Validation Graphs

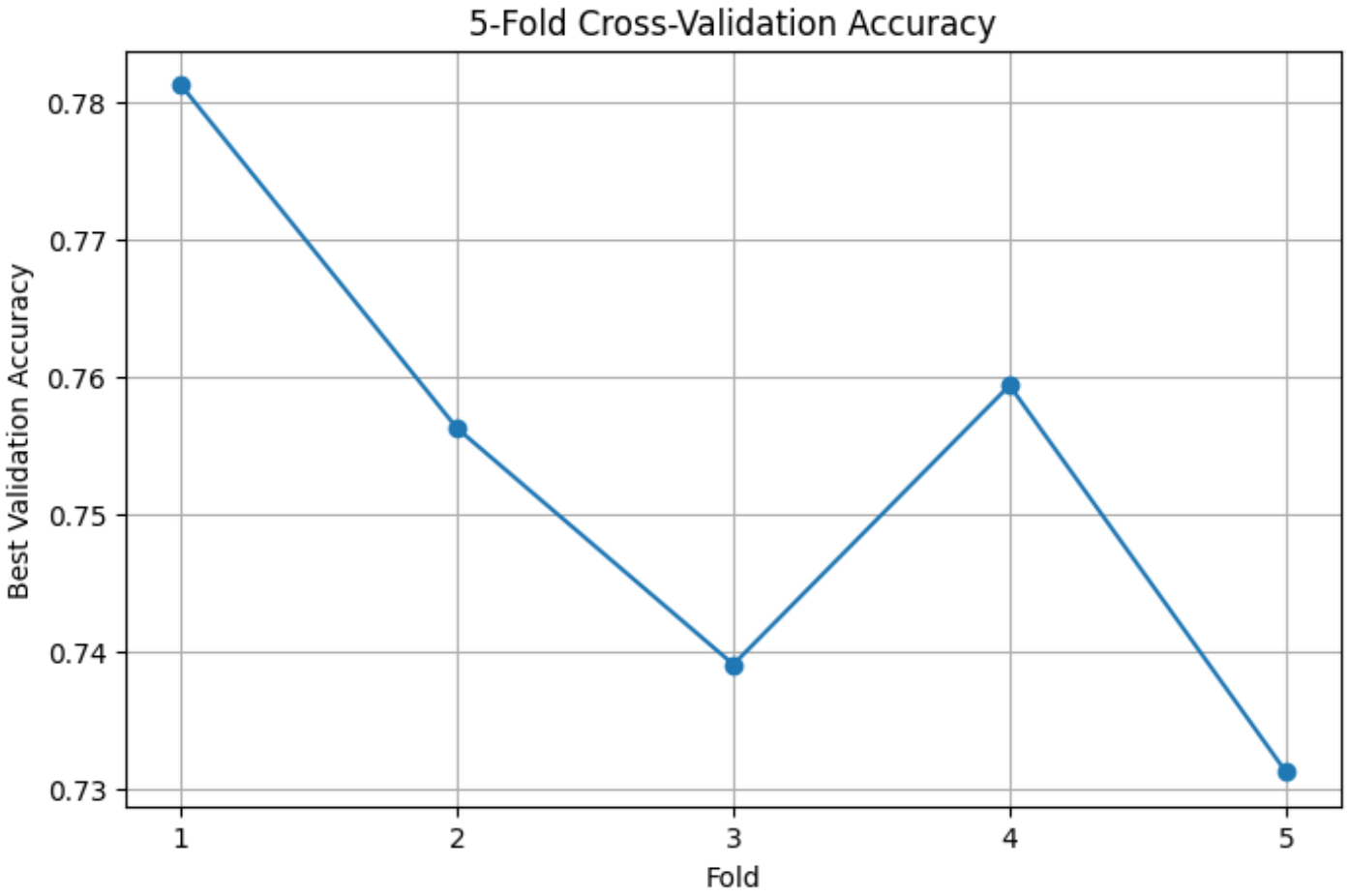






43. VGG16 — Direct 5-fold CV with preprocessing — graphs (2/2)

Training / Validation Graphs



44. InceptionV3 + KNN

Source notebook: Inception v3 baseline and ml.ipynb (inside Models_DL.zip)

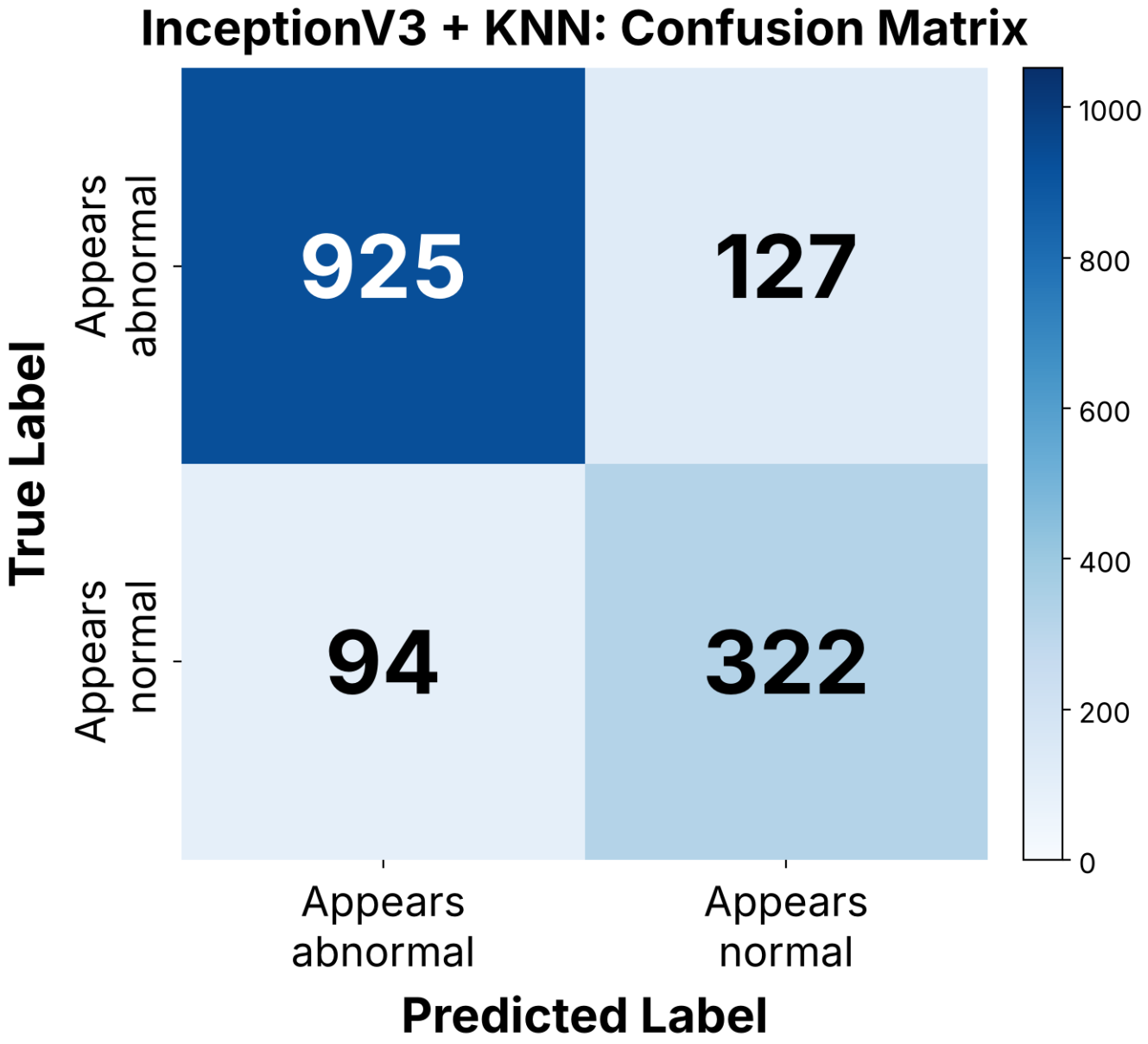
Accuracy: 84.95%

Precision: 85.37%

Recall: 84.95%

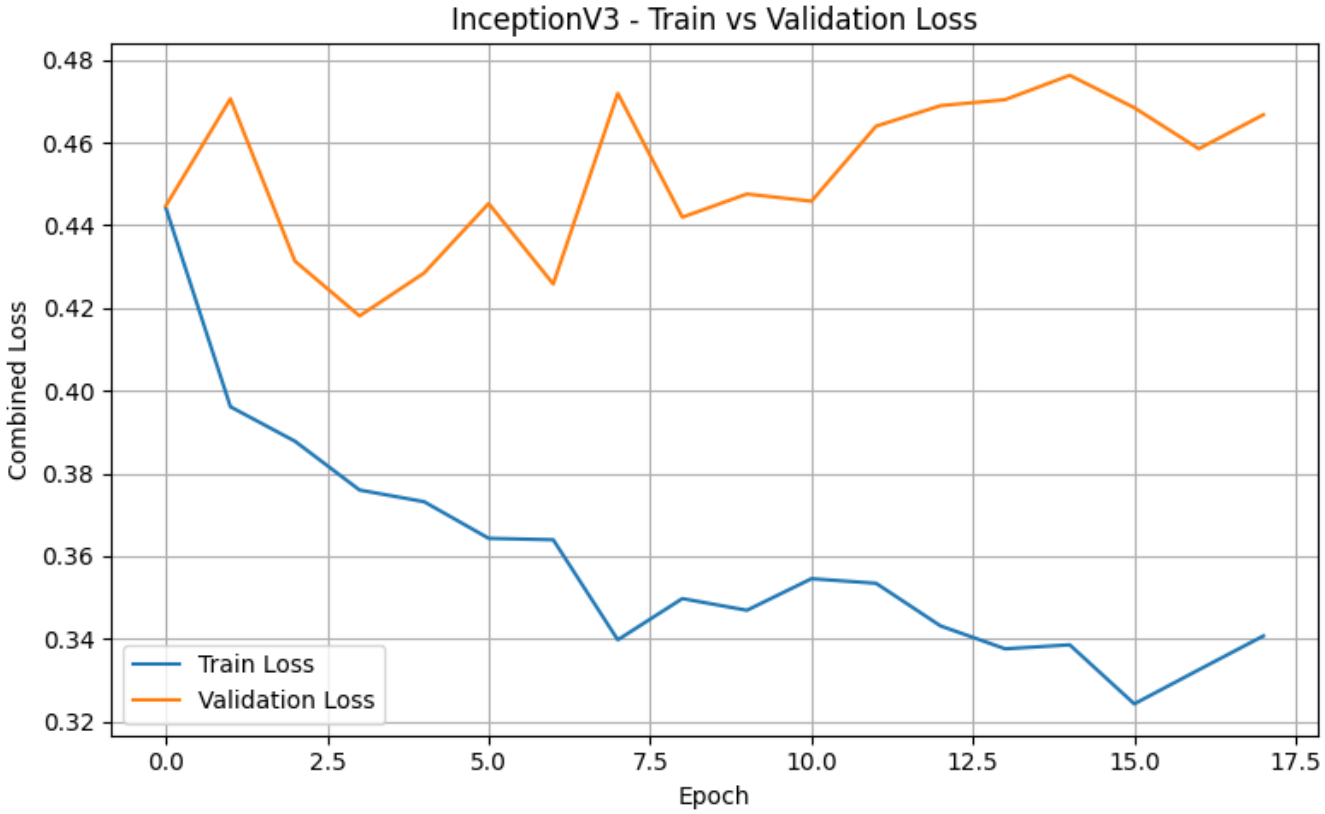
F1-score: 85.11%

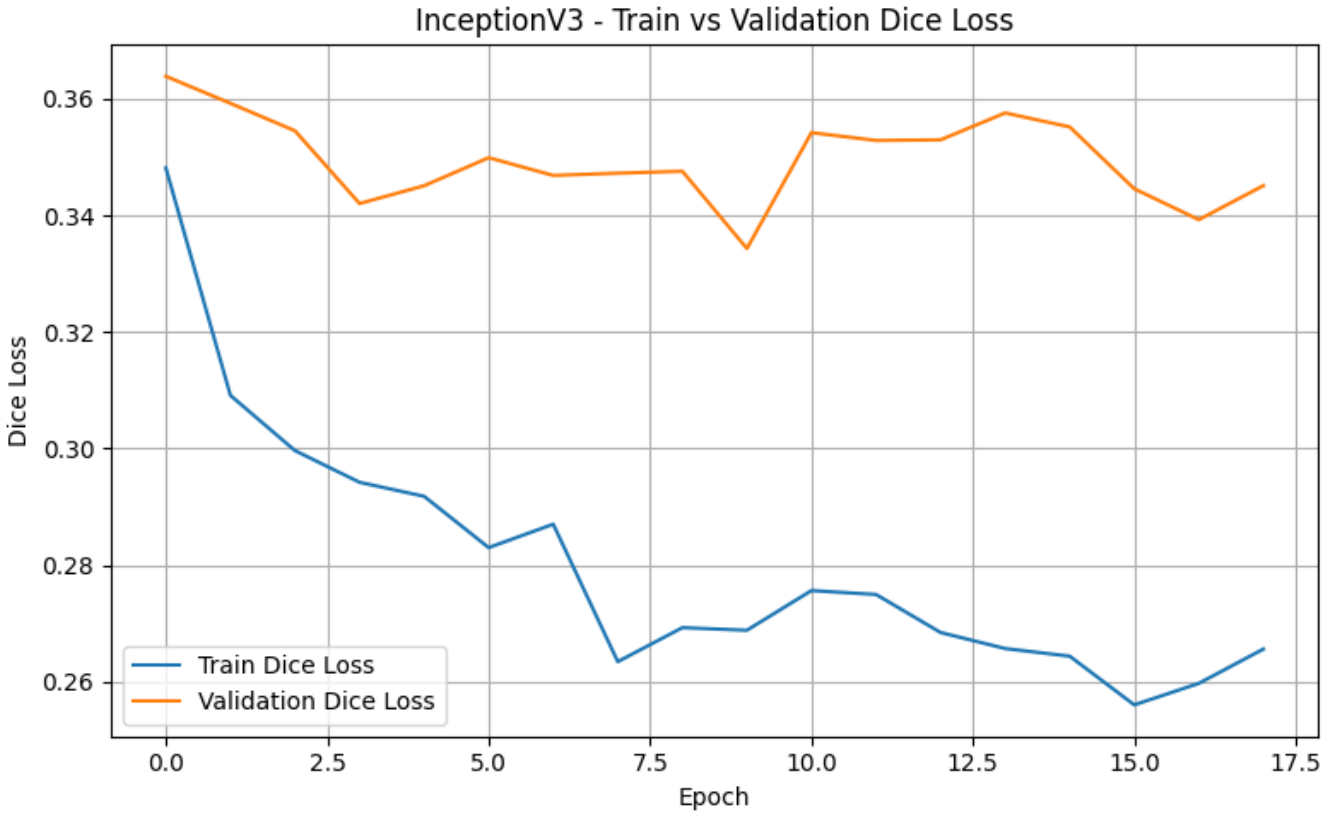
Confusion Matrix



44. InceptionV3 + KNN — graphs (1/2)

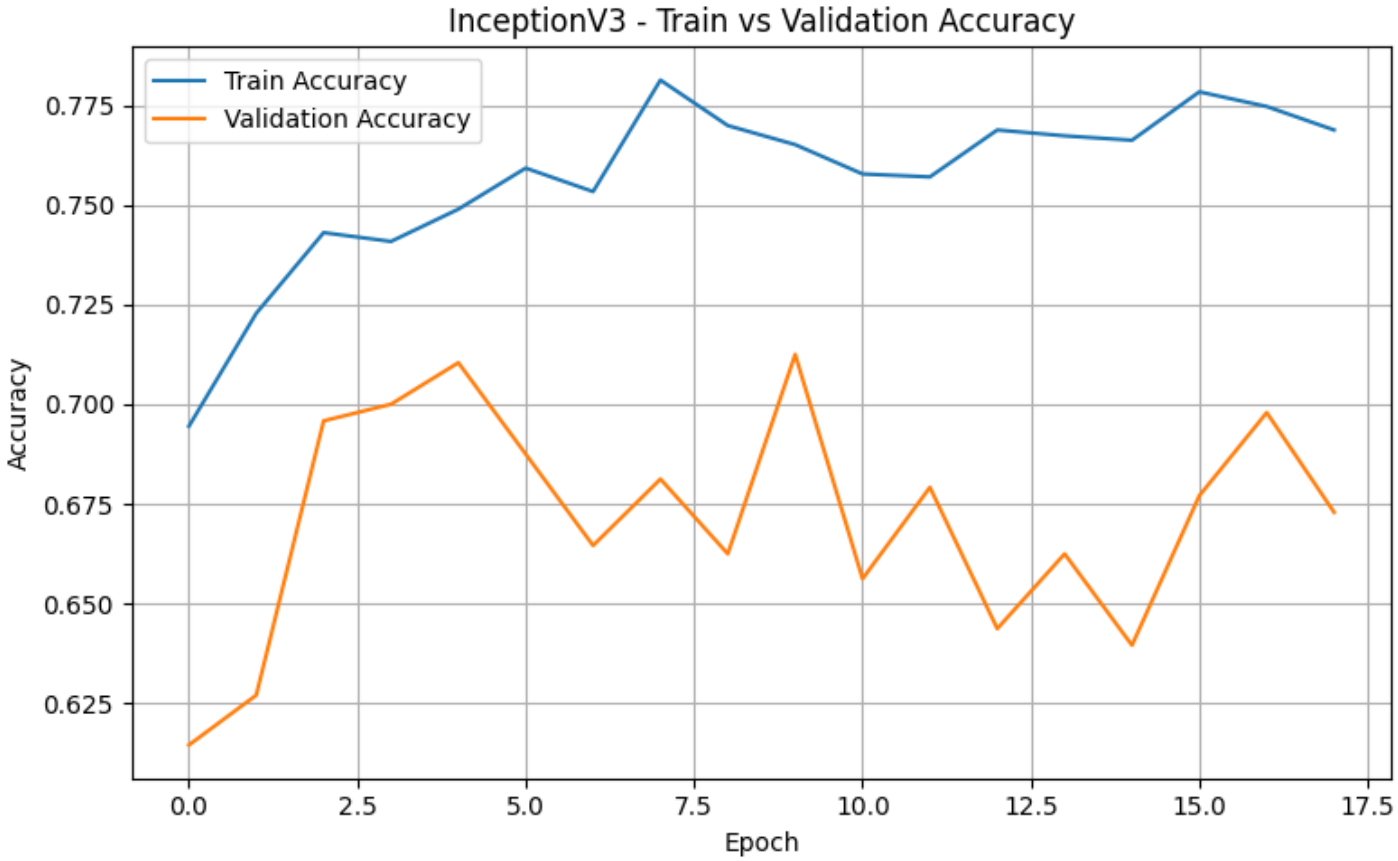
Training / Validation Graphs





44. InceptionV3 + KNN — graphs (2/2)

Training / Validation Graphs



45. AlexNet + SVM

Source notebook: Xgboost and SVM and all 4 models.ipynb (inside Models_DL.zip)

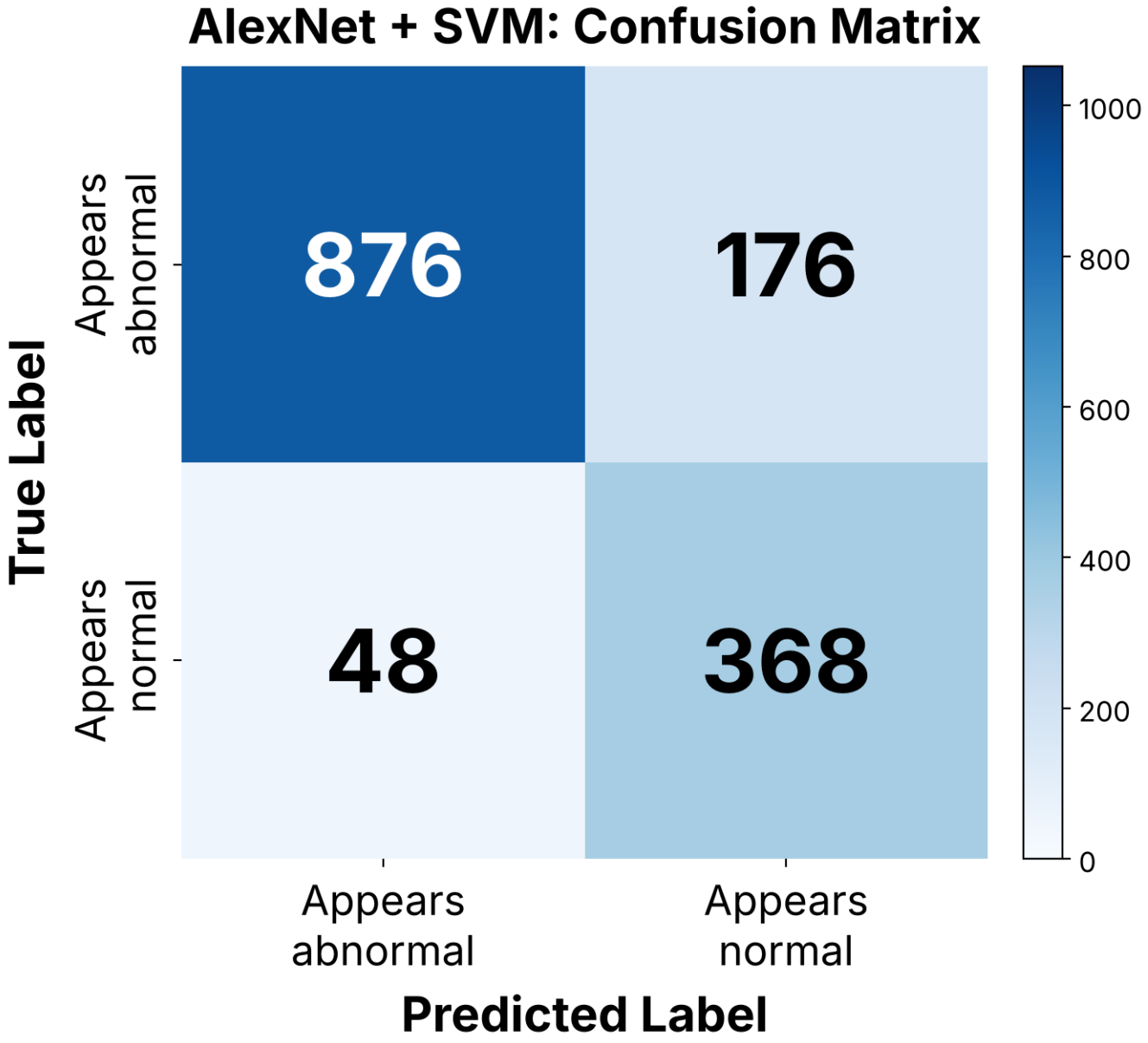
Accuracy: 84.74%

Precision: 87.11%

Recall: 84.74%

F1-score: 85.26%

Confusion Matrix



No training graphs: this notebook reloads an already-trained network and prints no epoch log, so there is nothing to plot for this row.

46. VGG16 — 5-Fold CV, Direct Classification

Source notebook: Vgg kfold.ipynb (inside Models_DL.zip)

Accuracy: 84.20%

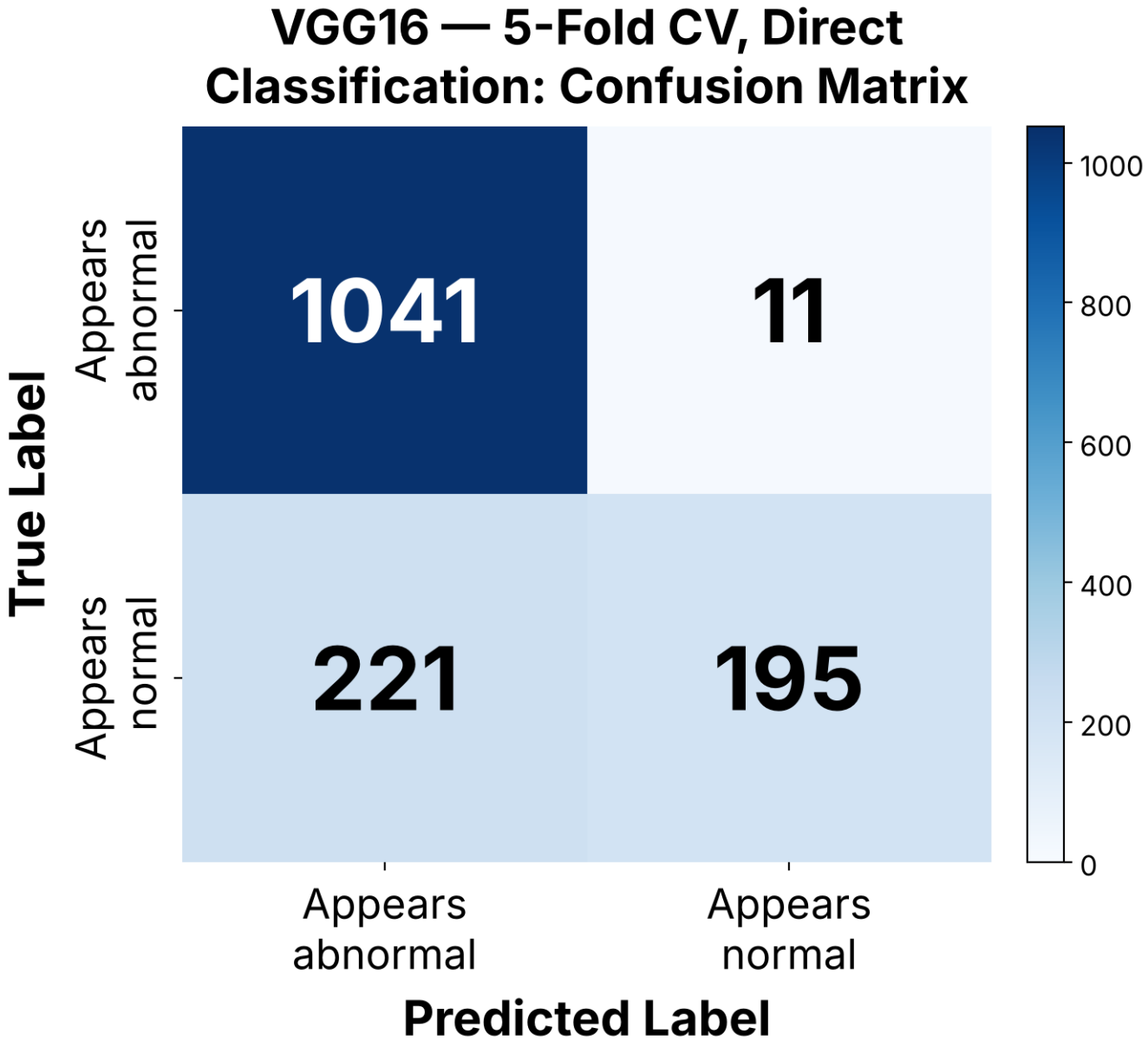
Precision: 85.94%

Recall: 84.20%

F1-score: 82.25%

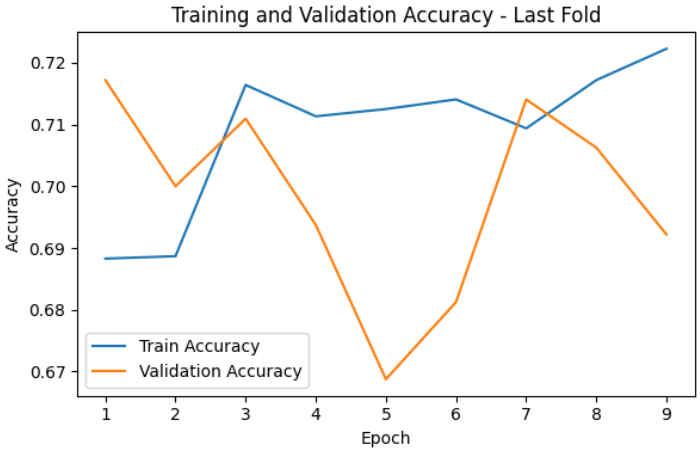
Note: the notebook prints 81.06% as its test / “Final Test Accuracy”, which does not agree with the notebook’s own printed confusion matrix and weighted recall (this printed number is not the test accuracy). The test accuracy shown here is the one implied by the confusion matrix and equals the weighted recall.

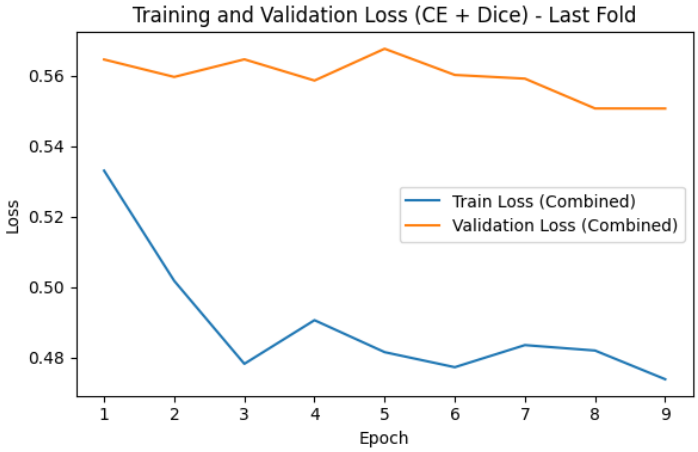
Confusion Matrix

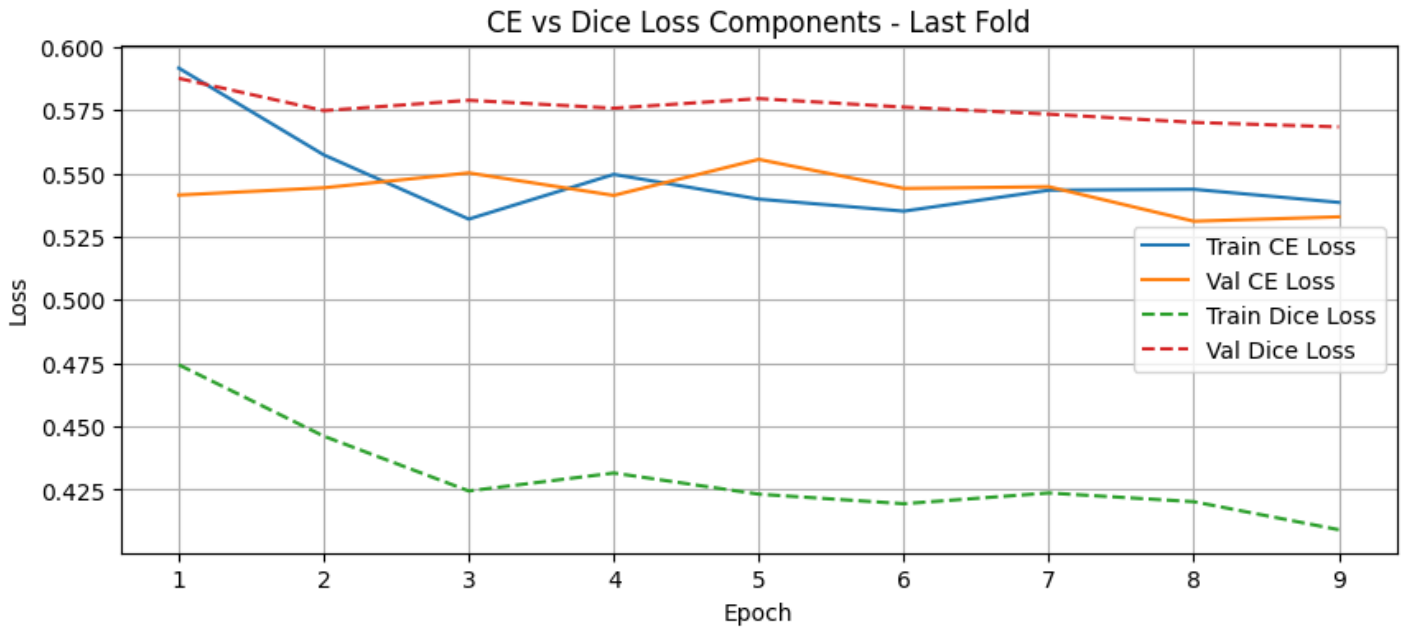


46. VGG16 — 5-Fold CV, Direct Classification — graphs (1/2)

Training / Validation Graphs

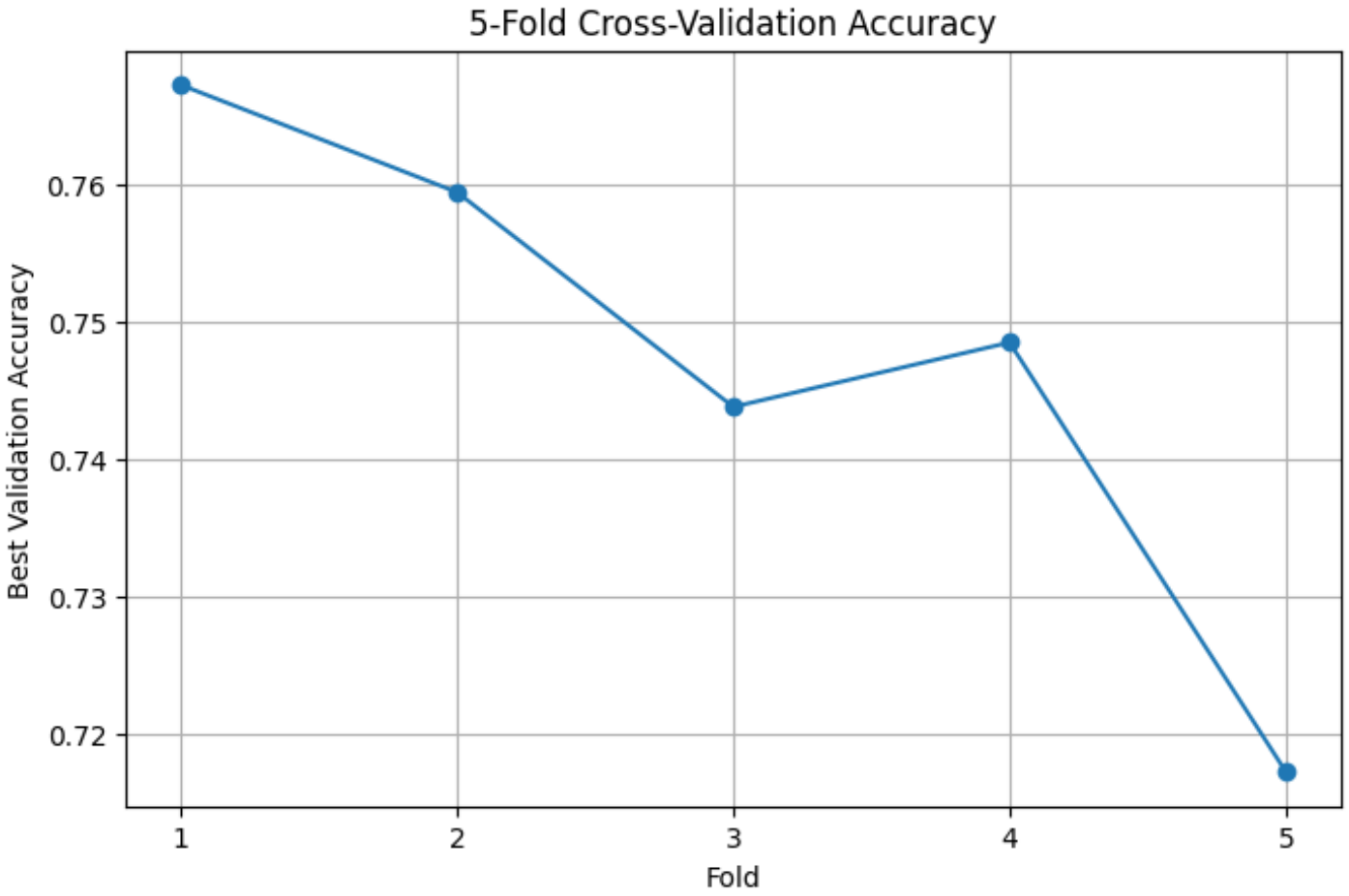






46. VGG16 — 5-Fold CV, Direct Classification — graphs (2/2)

Training / Validation Graphs



47. ResNet18 — Direct Classification (Dice Loss baseline)

Source notebook: ResNet with Dice Loss (BaseLine).ipynb (inside Models_DL.zip)

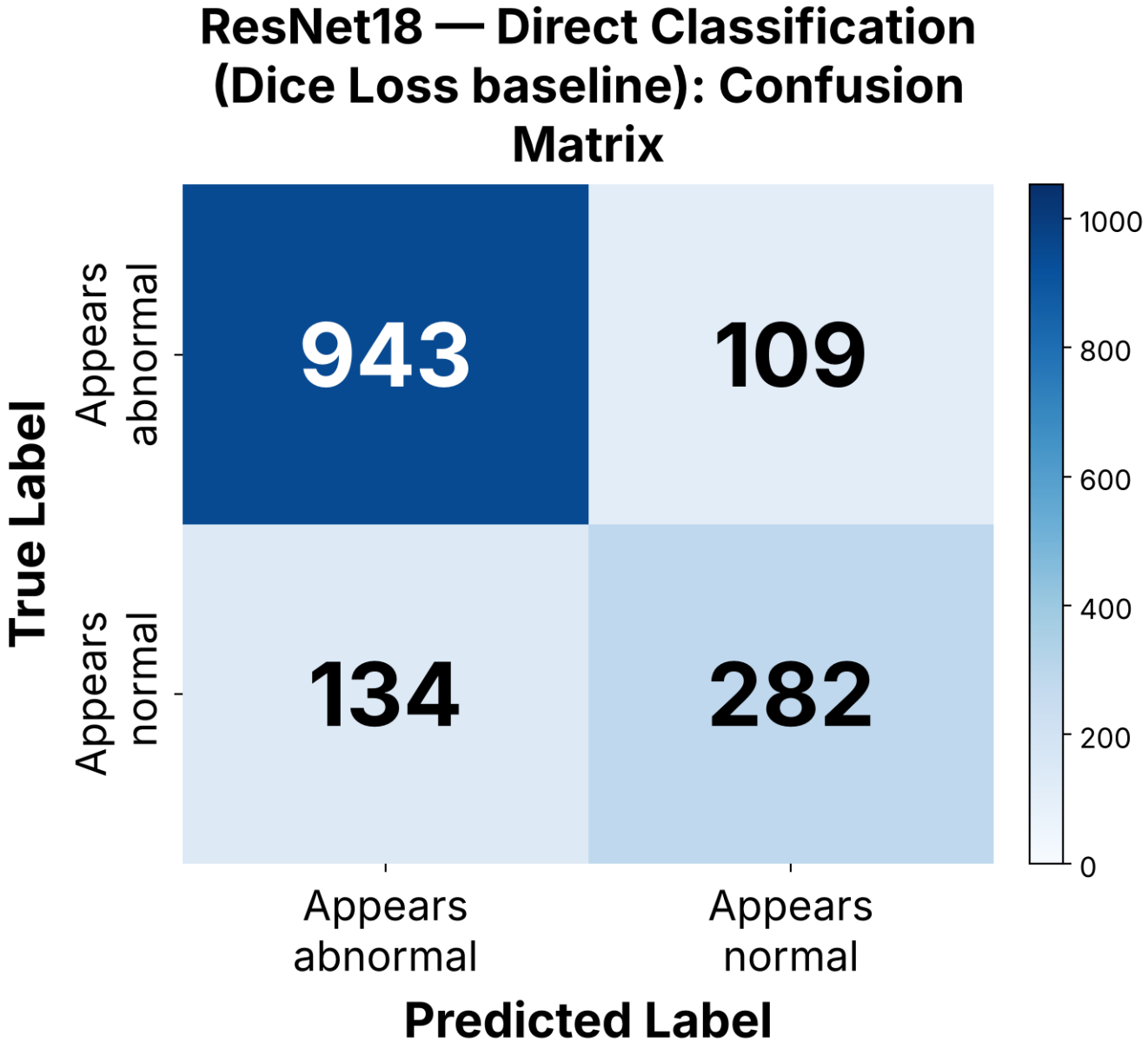
Accuracy: 83.45%

Precision: 83.18%

Recall: 83.45%

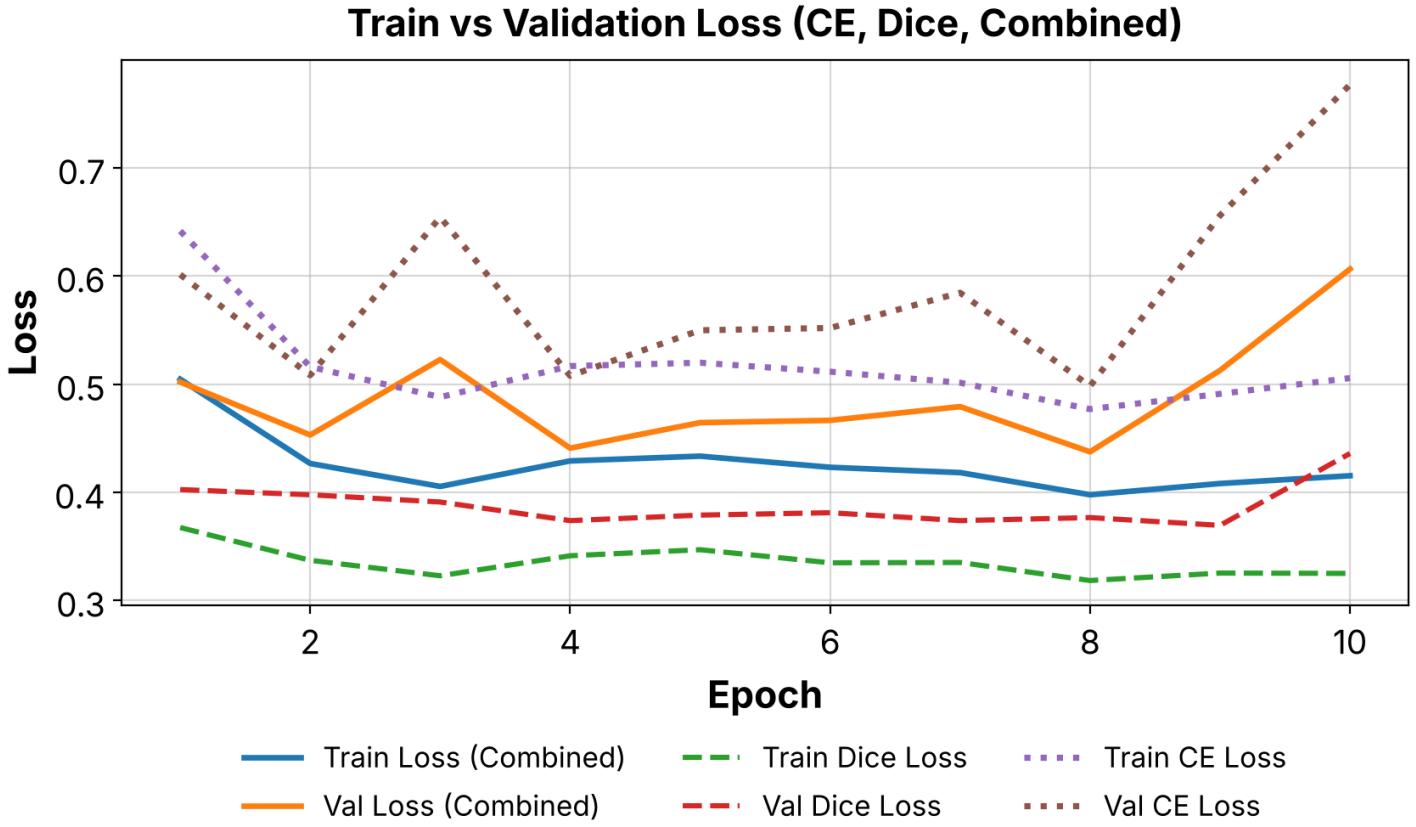
F1-score: 83.29%

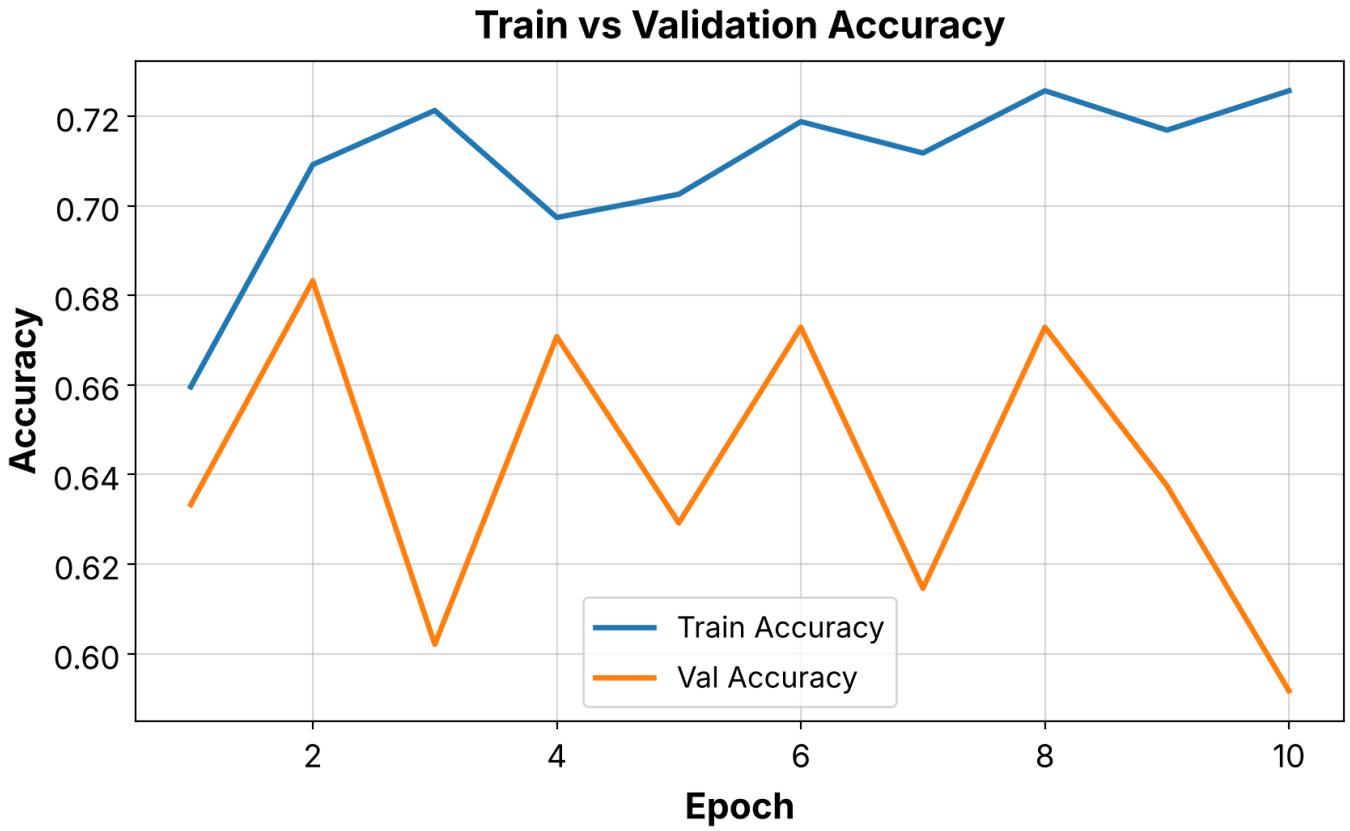
Confusion Matrix



47. ResNet18 — Direct Classification (Dice Loss baseline) — graphs

Training / Validation Graphs





48. InceptionV3 + Gaussian Naive Bayes

Source notebook: Inception v3 baseline and ml.ipynb (inside Models_DL.zip)

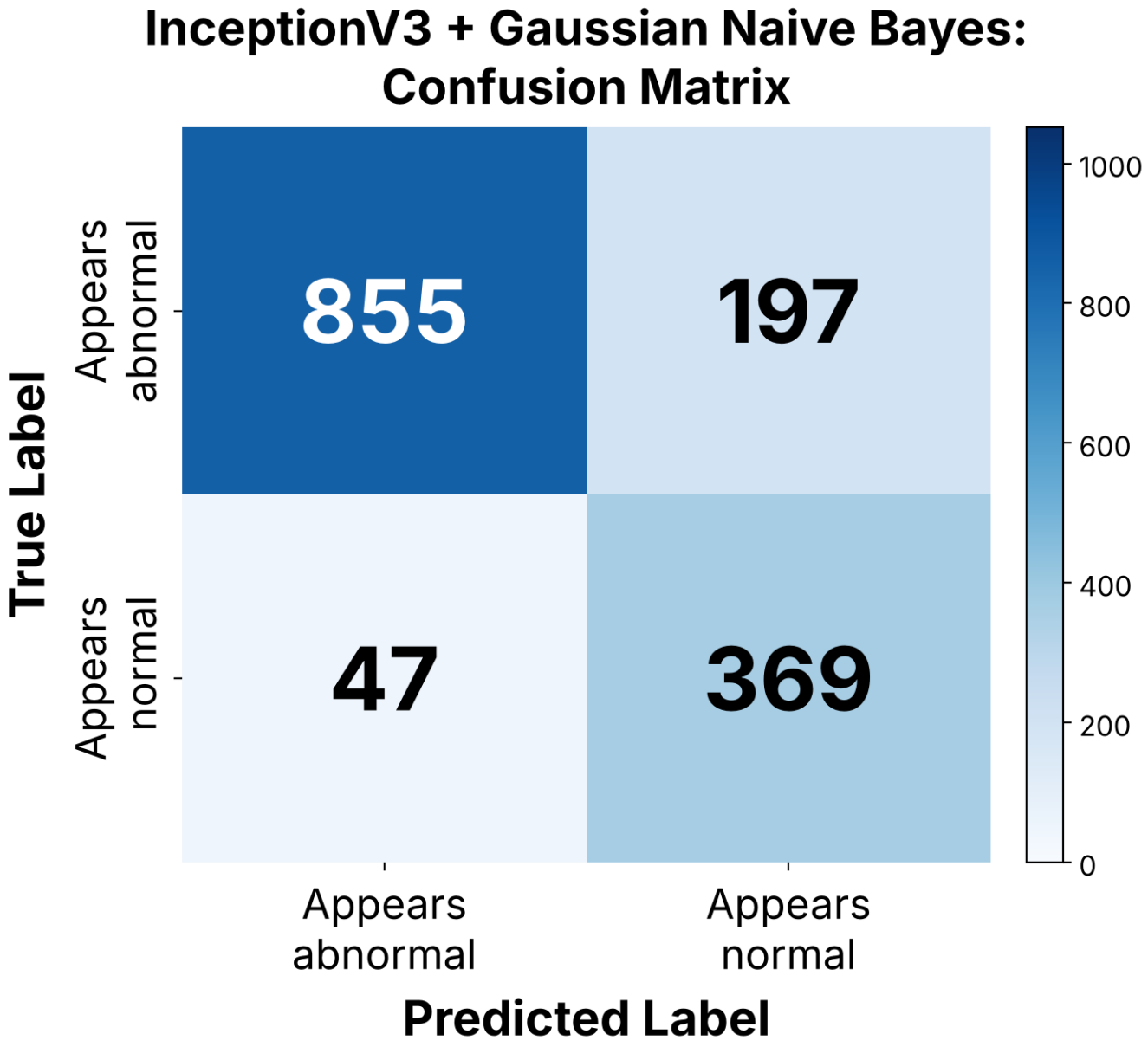
Accuracy: 83.38%

Precision: 86.40%

Recall: 83.38%

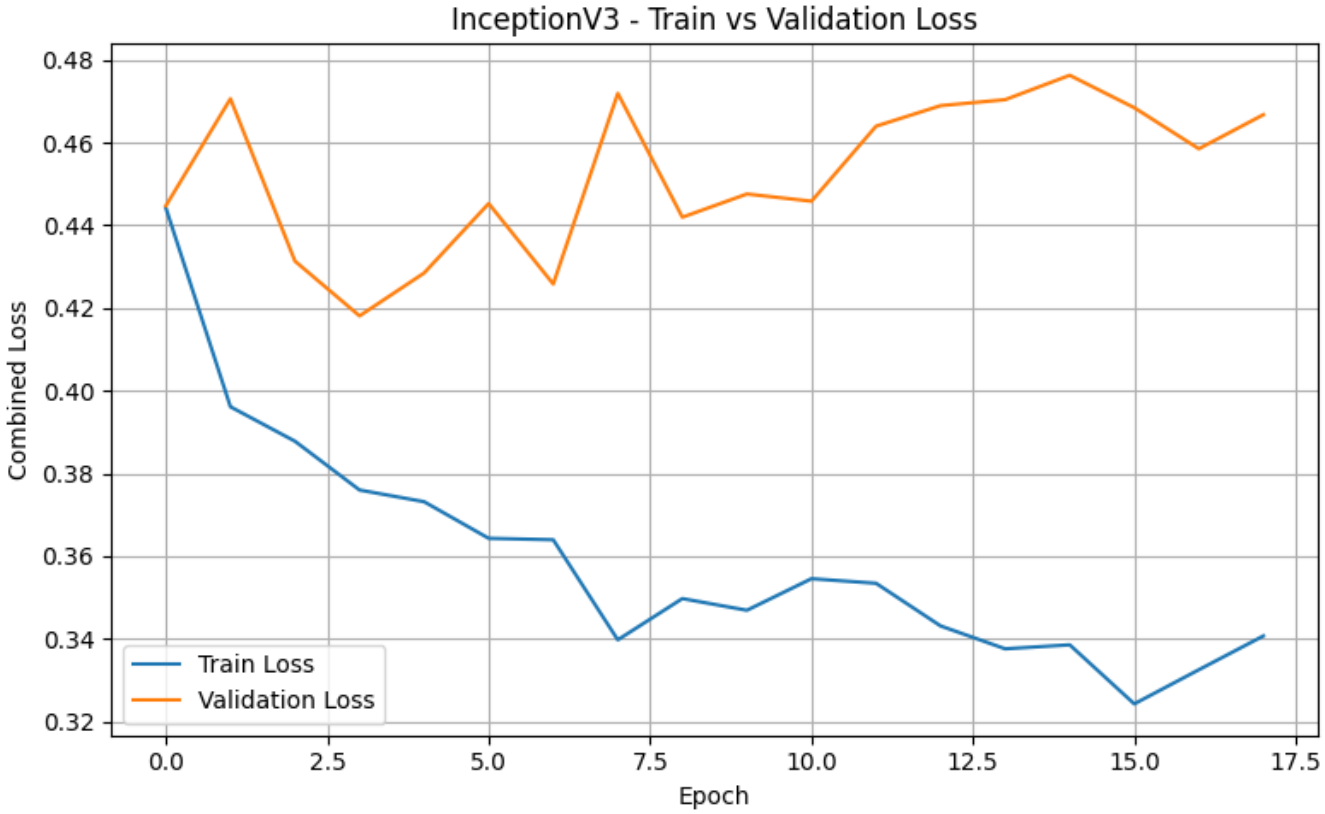
F1-score: 84.01%

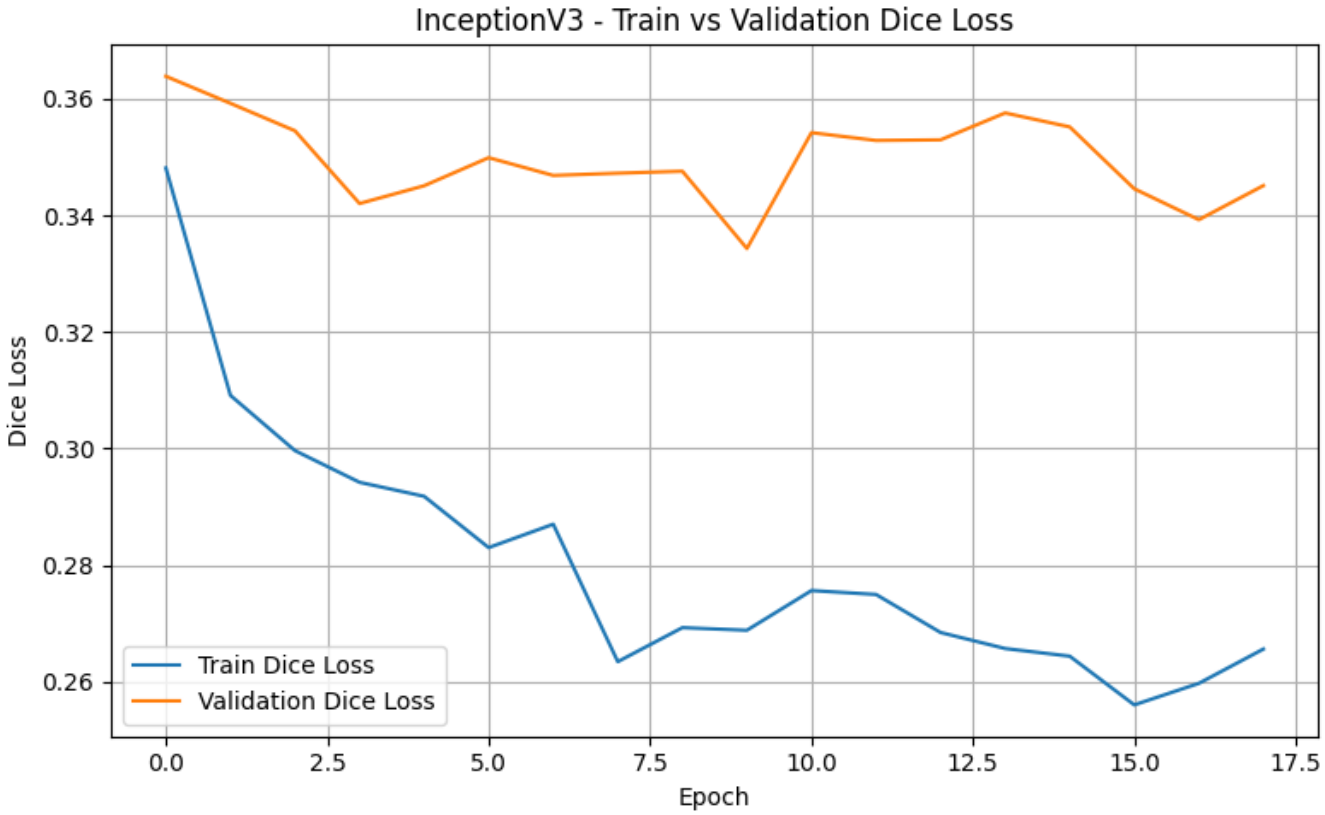
Confusion Matrix



48. InceptionV3 + Gaussian Naive Bayes — graphs (1/2)

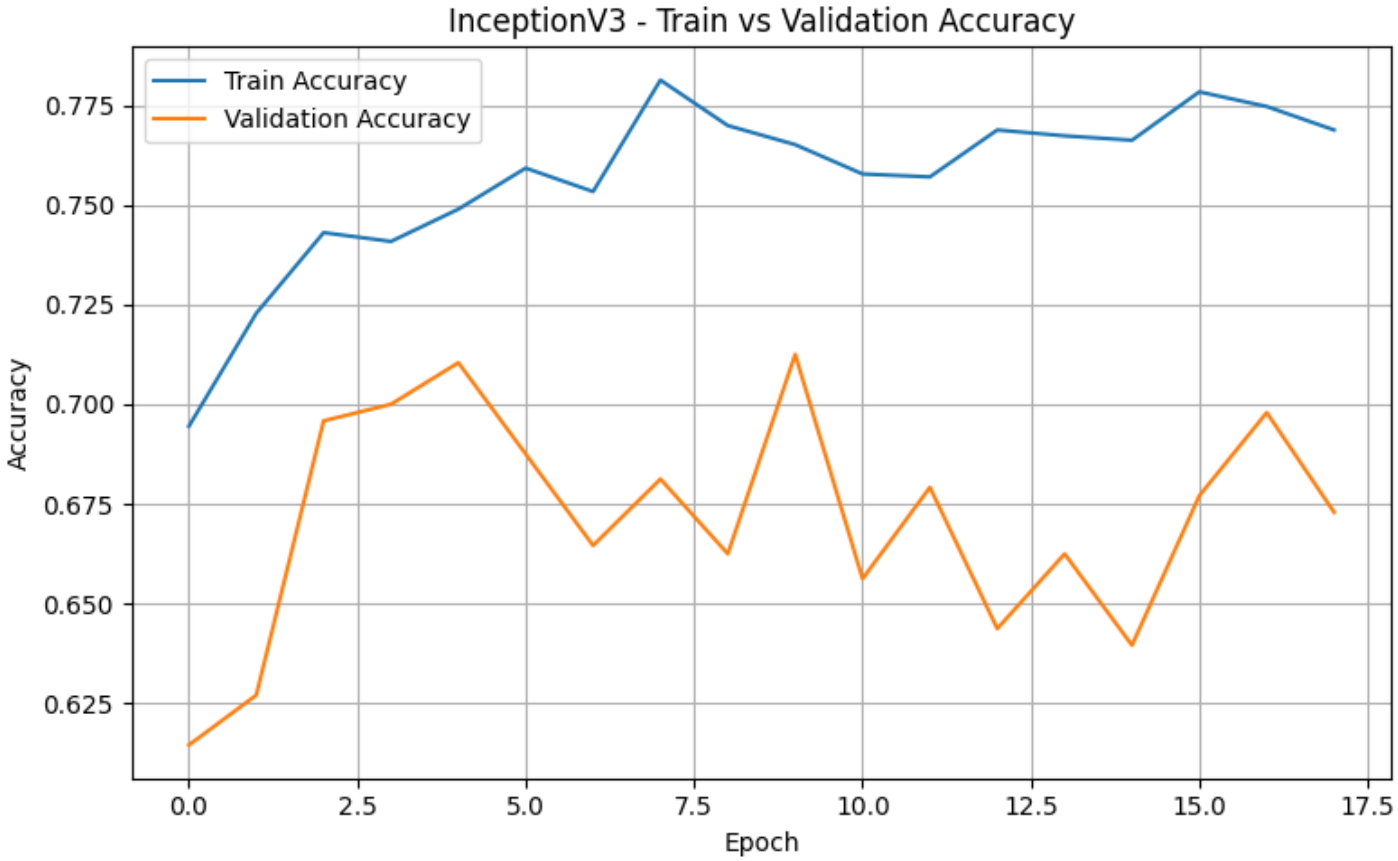
Training / Validation Graphs





48. InceptionV3 + Gaussian Naive Bayes — graphs (2/2)

Training / Validation Graphs



49. VGG16 + Gaussian Naive Bayes

Source notebook: Vgg with ML.ipynb (inside Models_DL.zip)

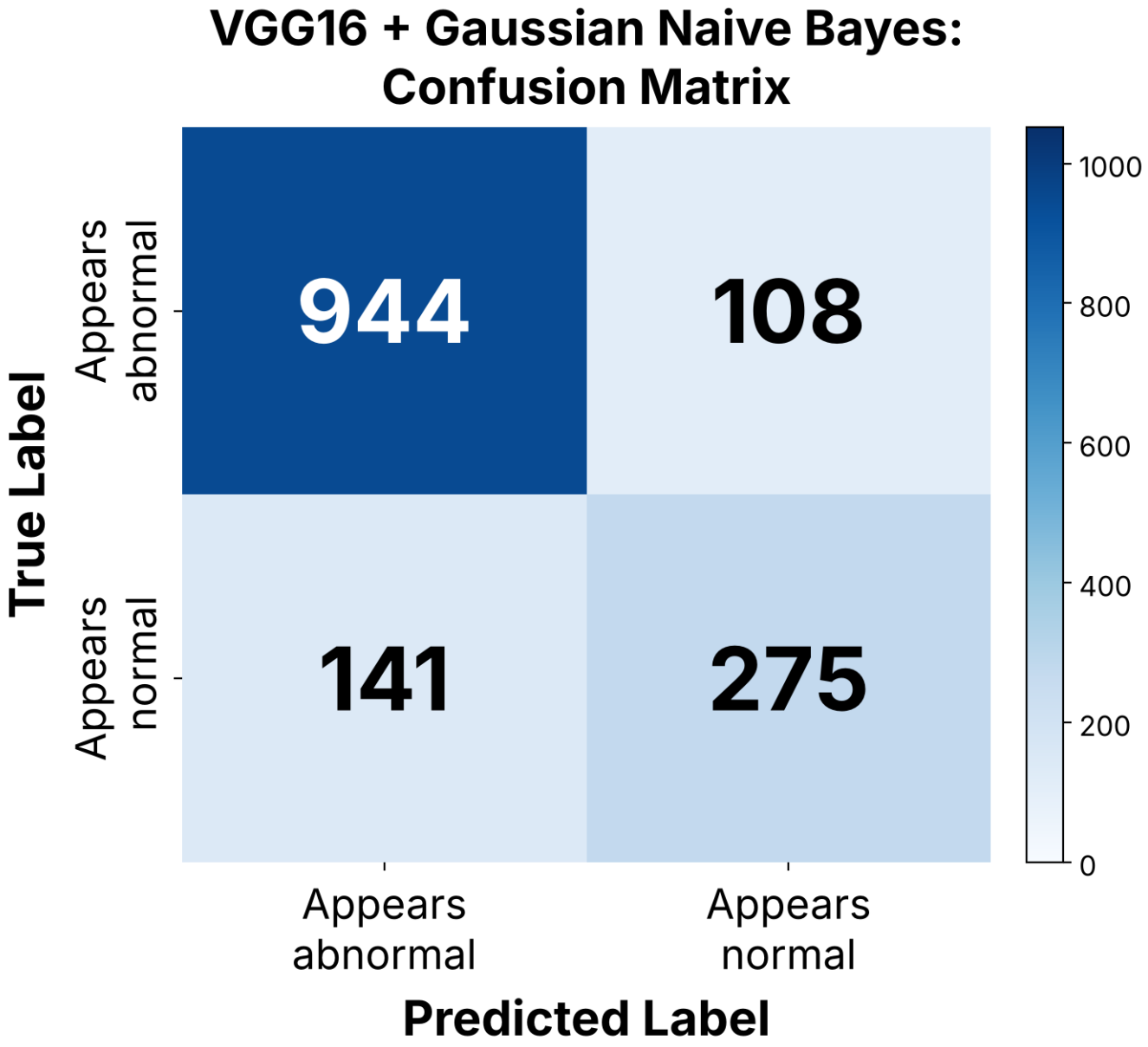
Accuracy: 83.04%

Precision: 82.70%

Recall: 83.04%

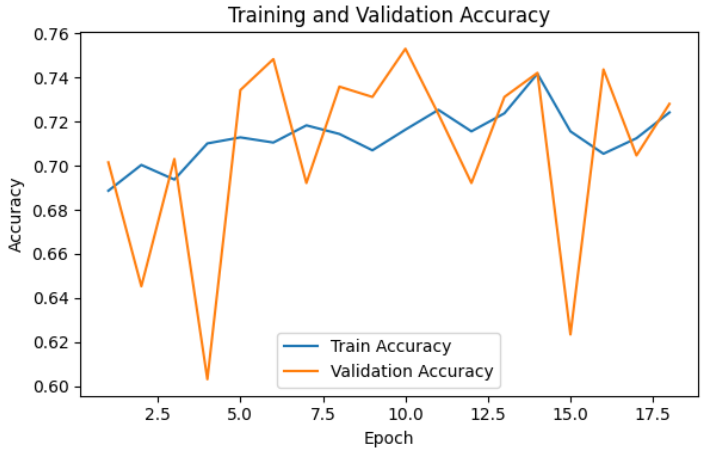
F1-score: 82.82%

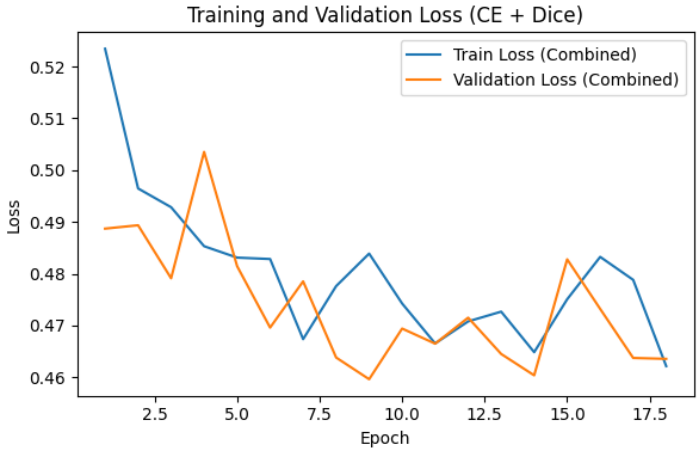
Confusion Matrix

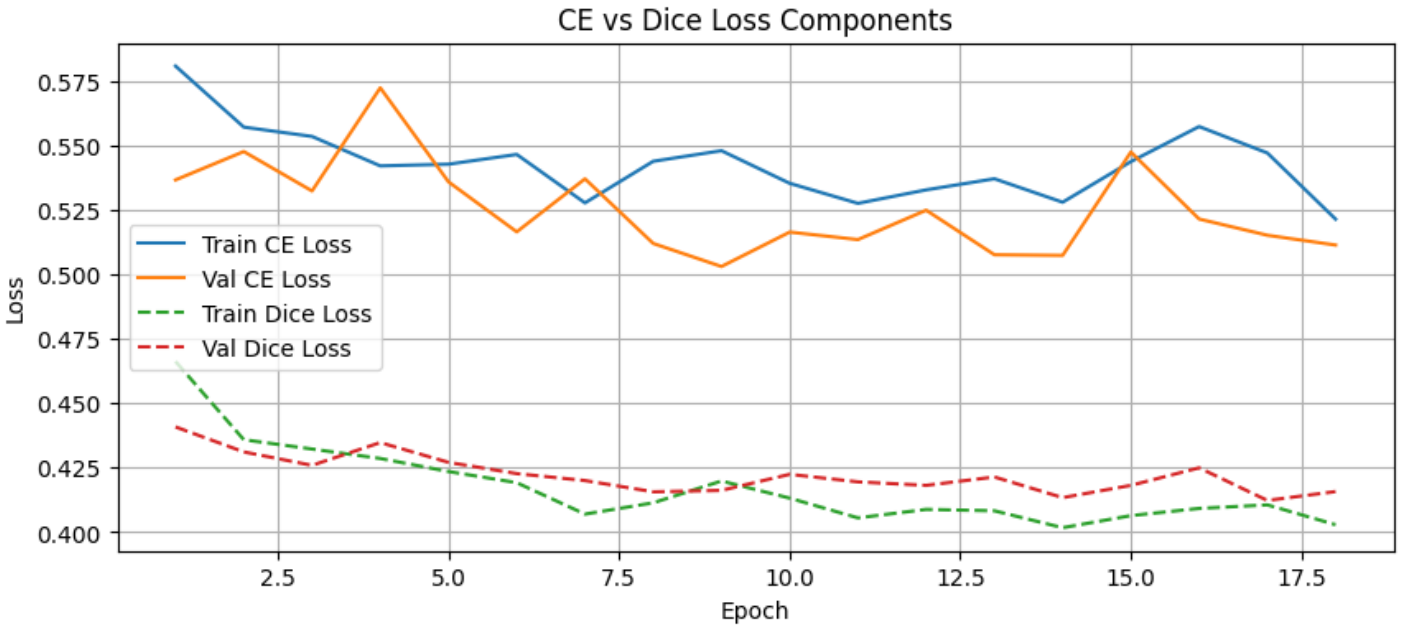


49. VGG16 + Gaussian Naive Bayes — graphs

Training / Validation Graphs







50. ResNet18 — 5-Fold CV, Direct Classification

Source notebook: ResNet wthi Kfold.ipynb (inside Models_DL.zip)

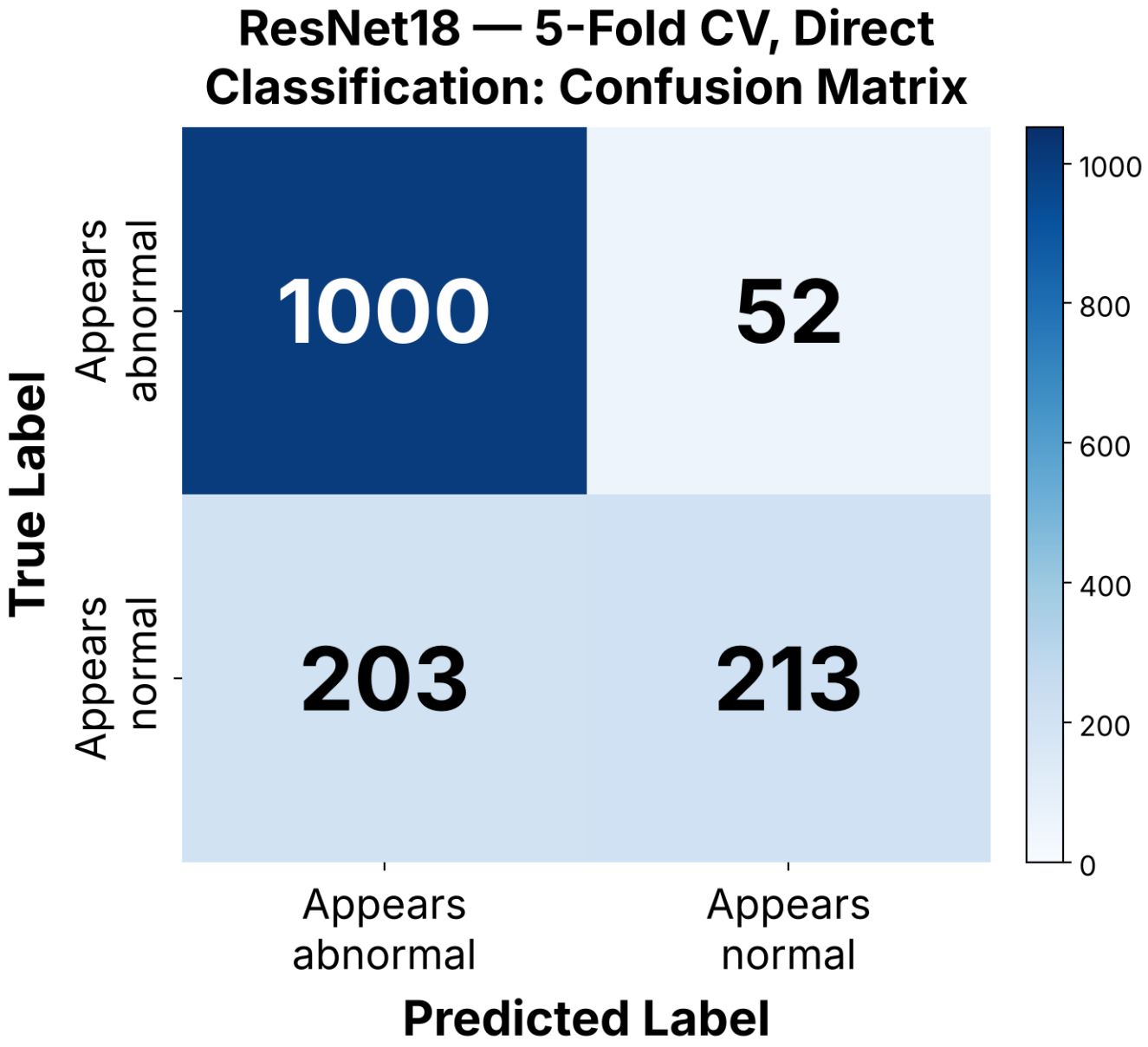
Accuracy: 82.63%

Precision: 82.35%

Recall: 82.63%

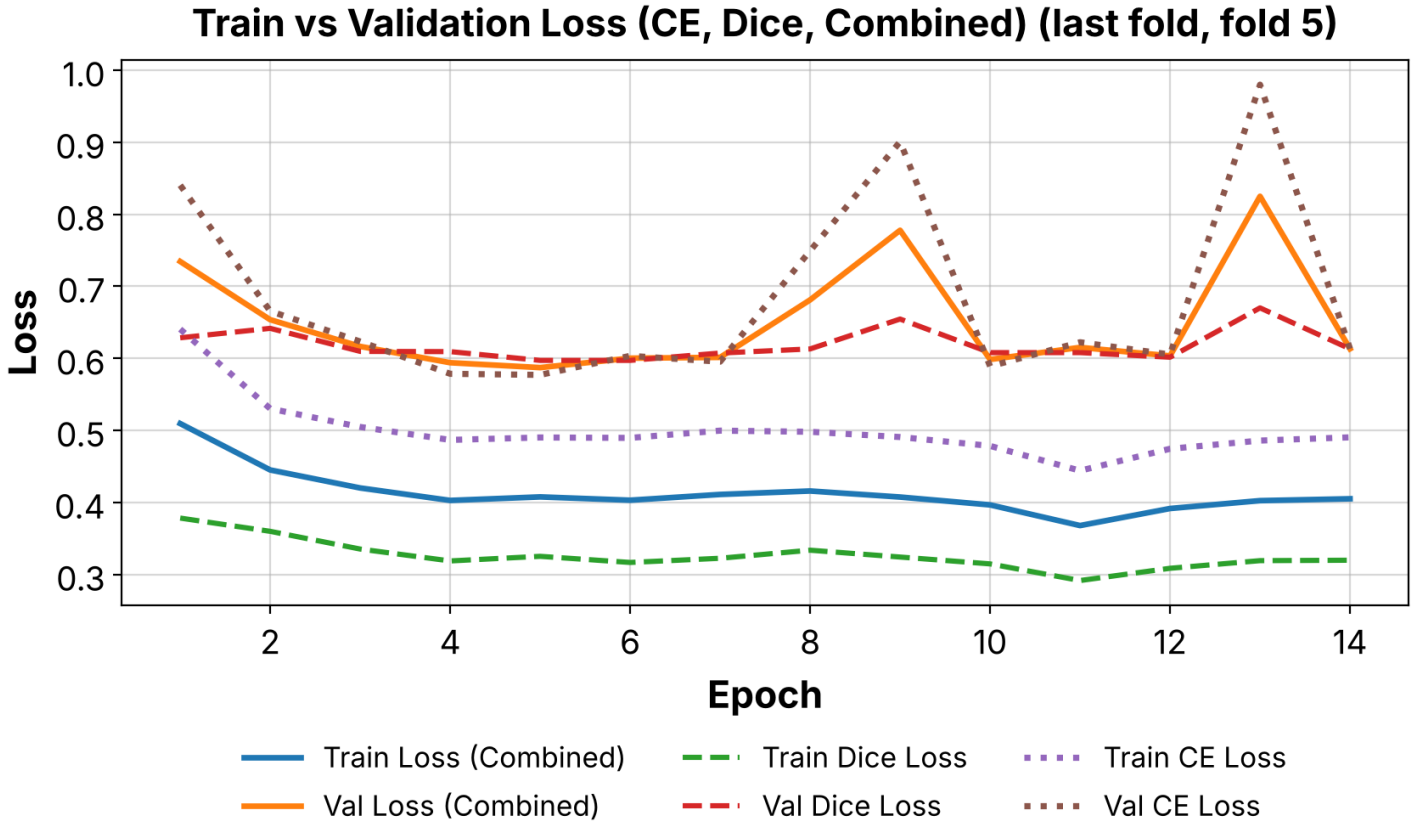
F1-score: 81.29%

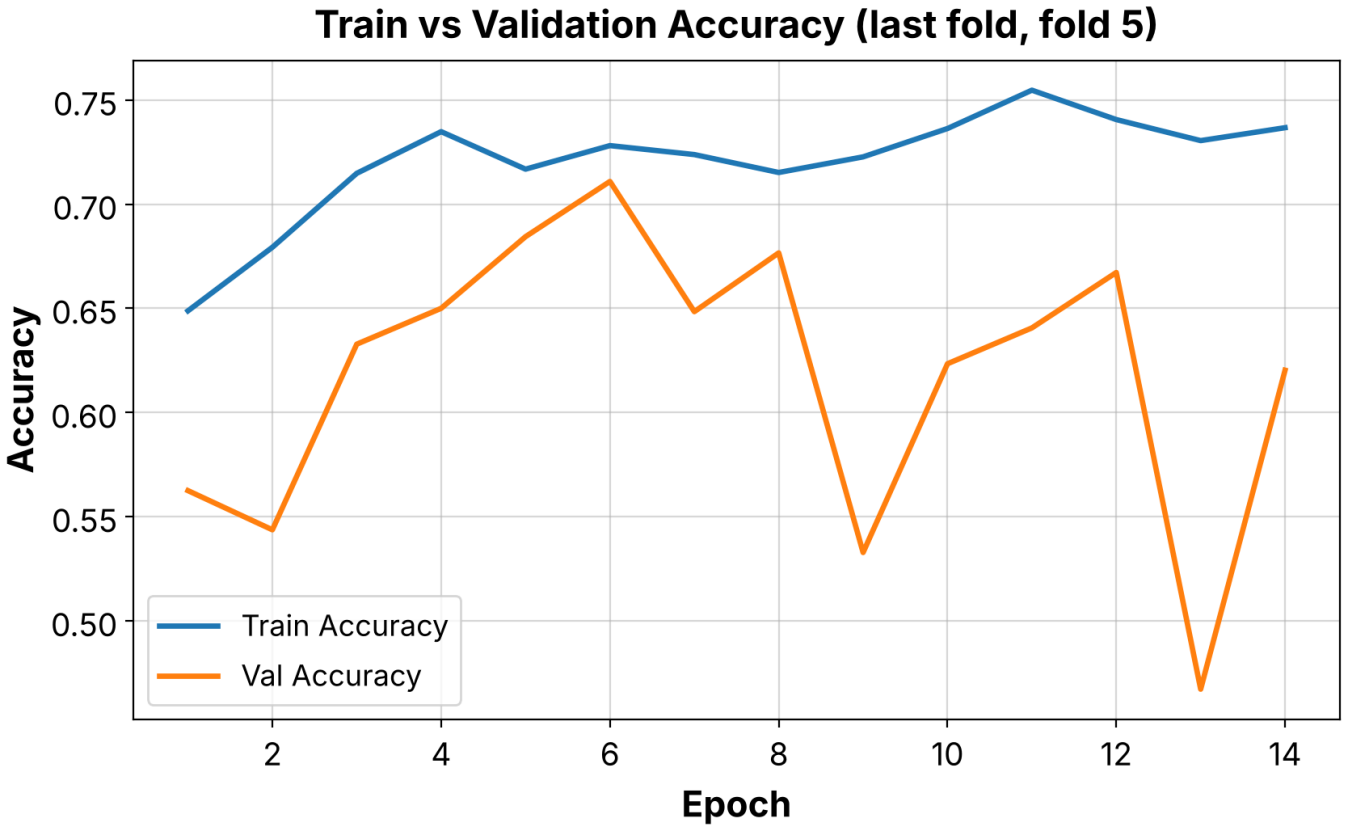
Confusion Matrix



50. ResNet18 — 5-Fold CV, Direct Classification — graphs (1/2)

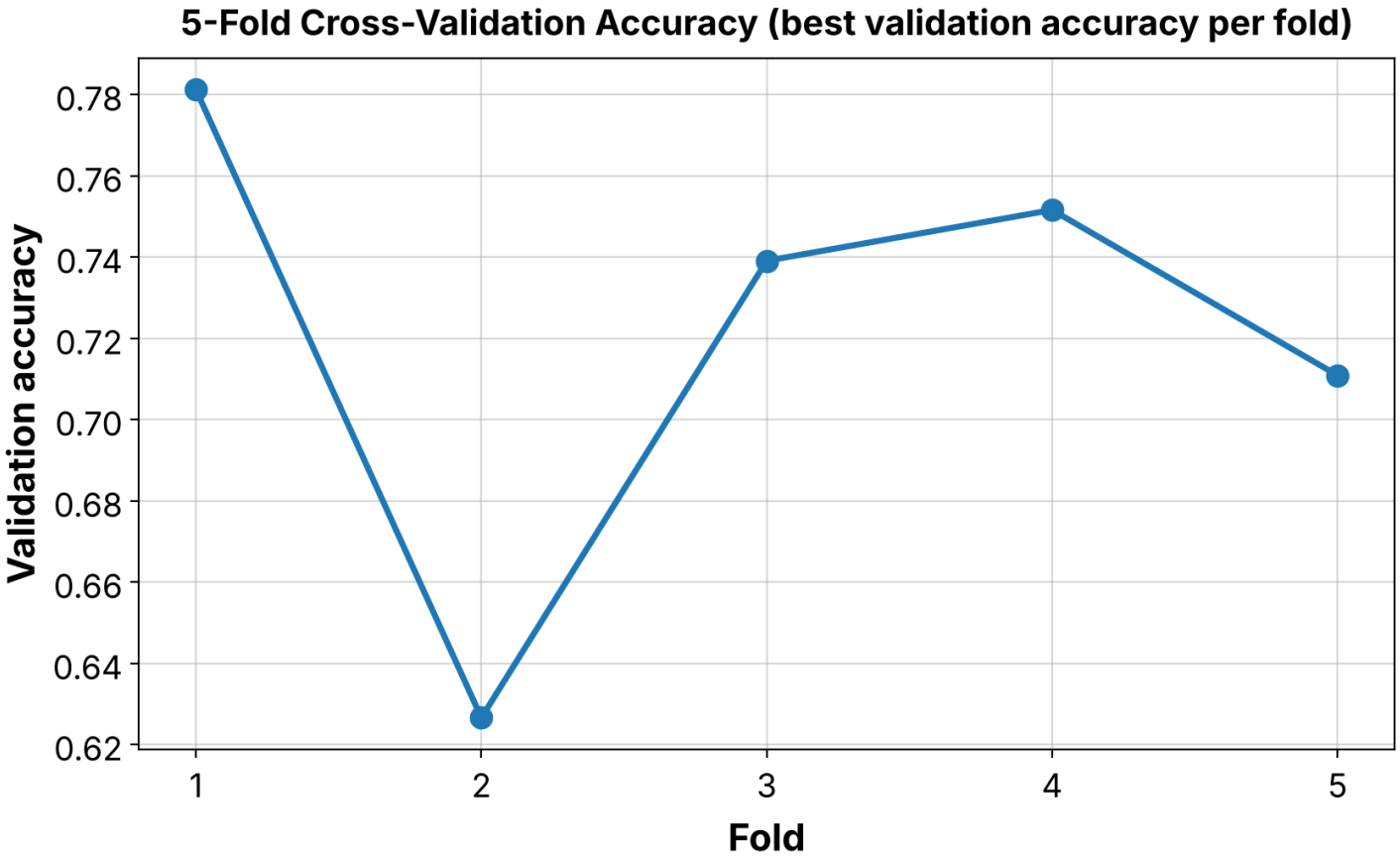
Training / Validation Graphs





50. ResNet18 — 5-Fold CV, Direct Classification — graphs (2/2)

Training / Validation Graphs



51. AlexNet + KNN (second run; notebook section titled "Adaboost")

Source notebook: AlexNet with ML.ipynb (inside Models_DL.zip)

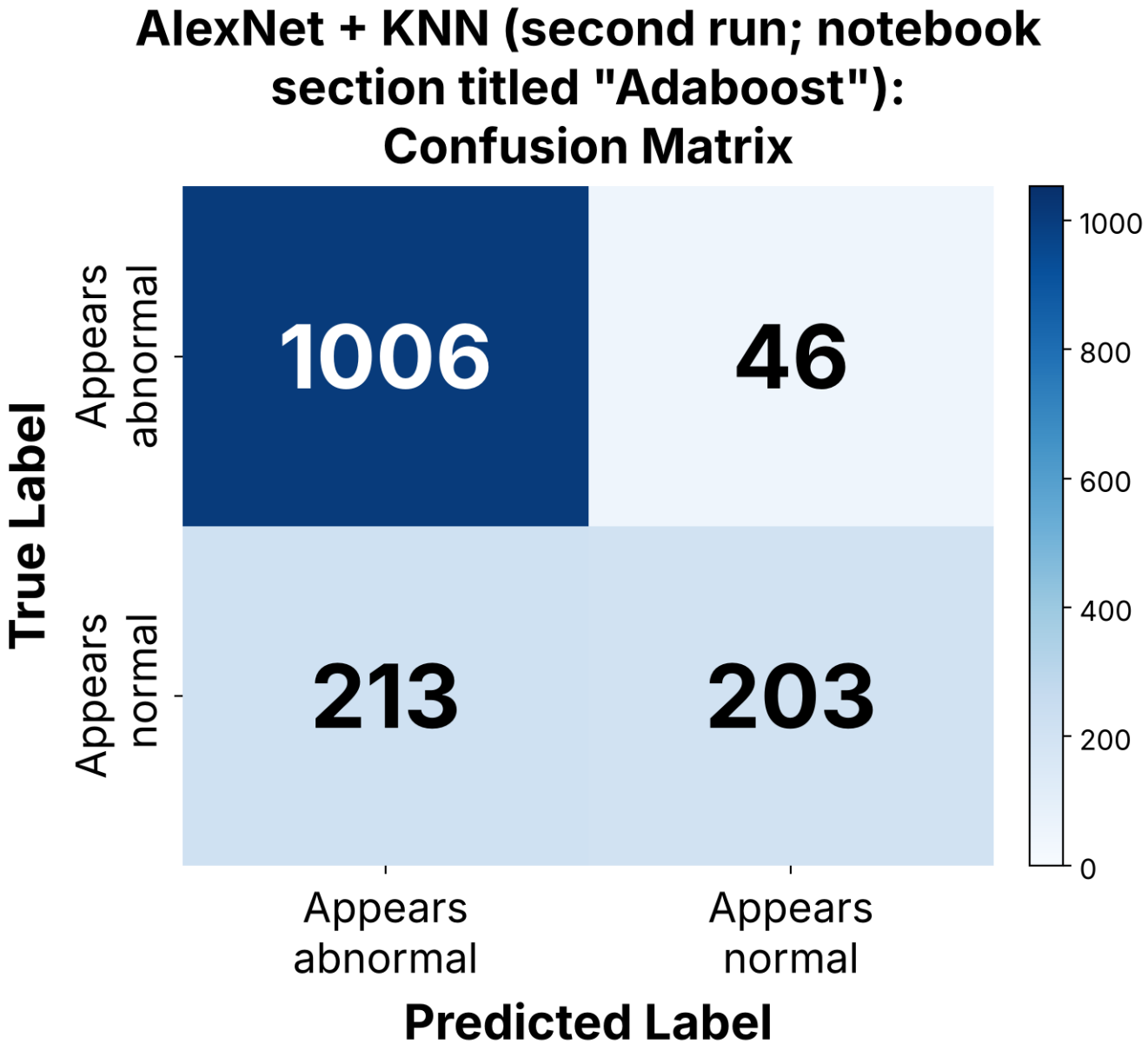
Accuracy: 82.36%

Precision: 82.24%

Recall: 82.36%

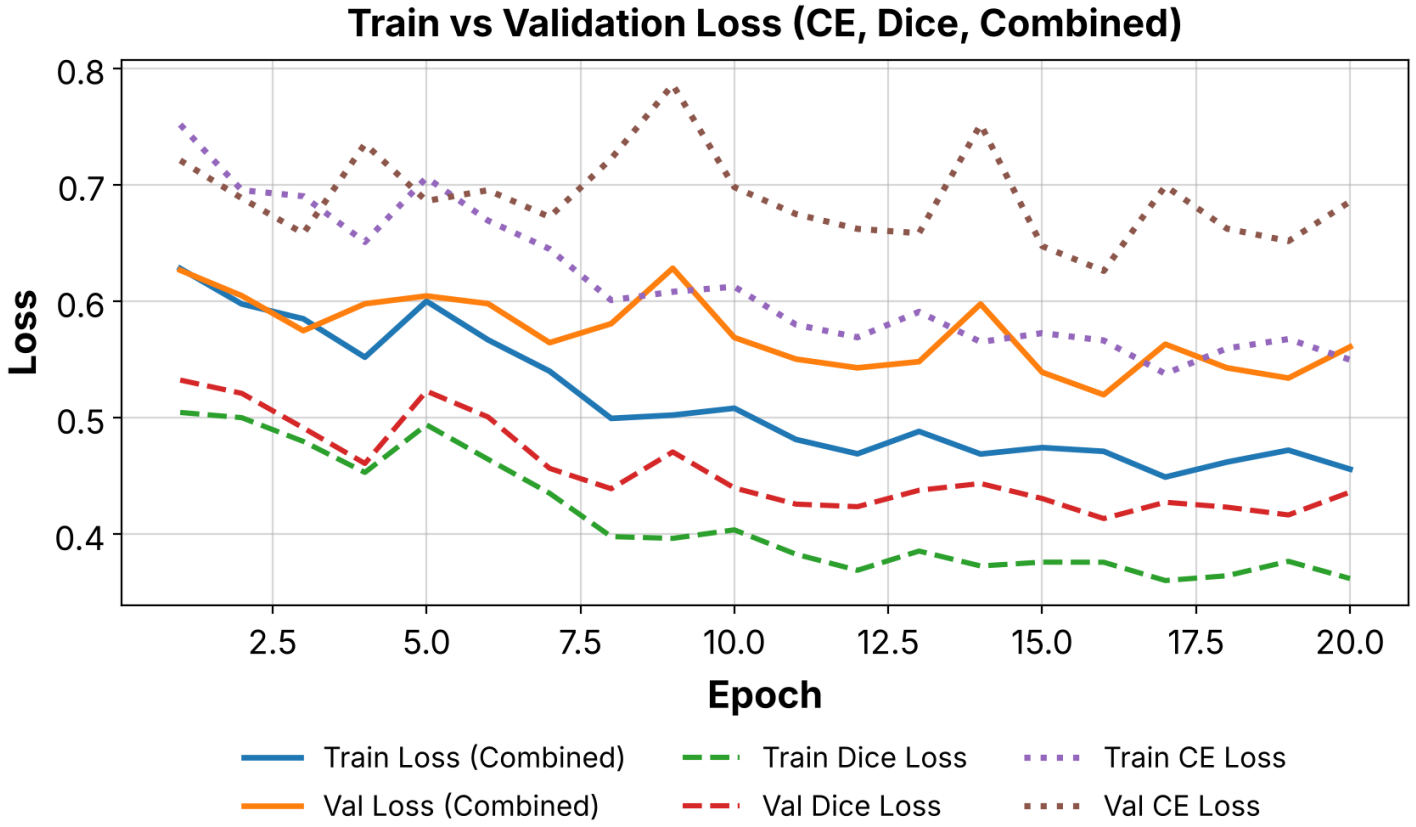
F1-score: 80.79%

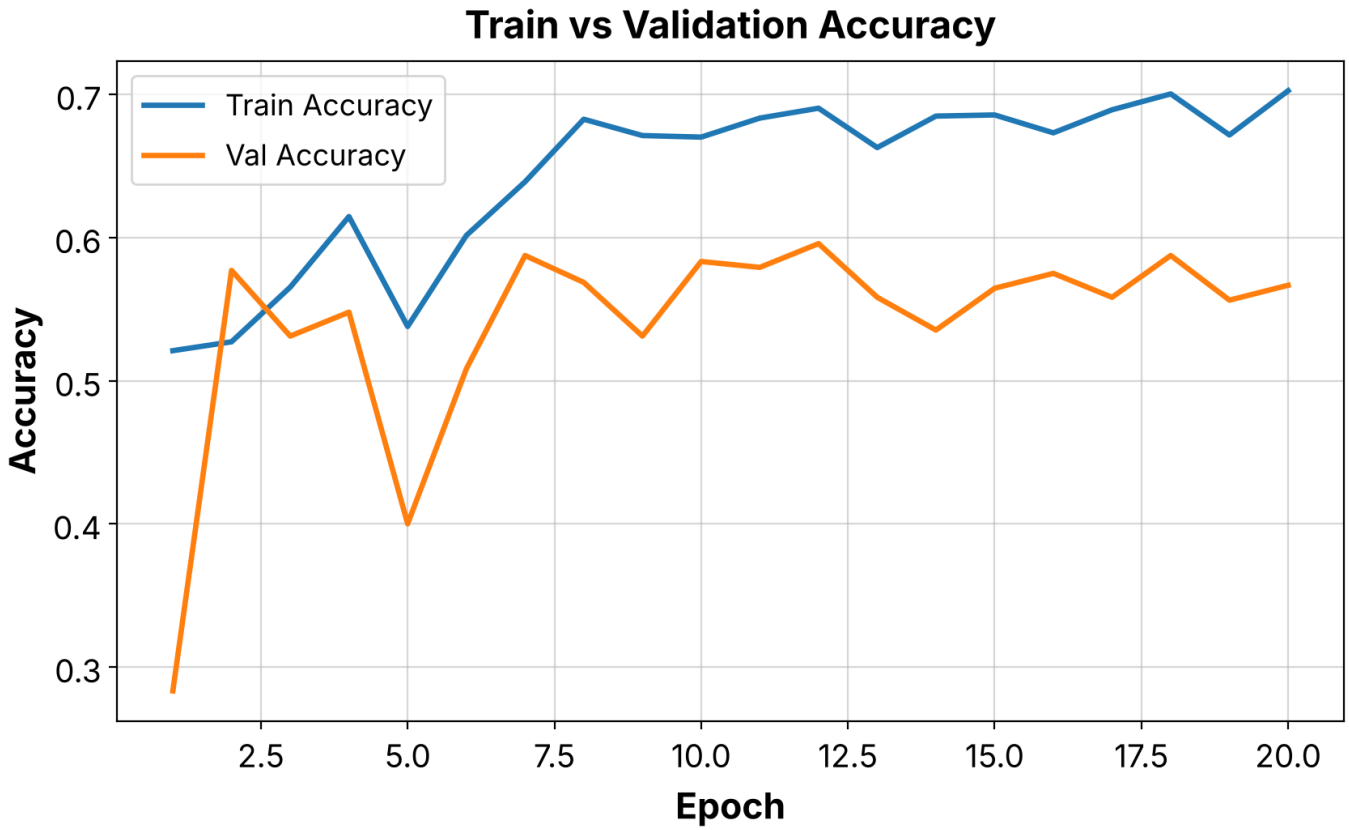
Confusion Matrix



51. AlexNet + KNN (second run; notebook section titled "Adaboost") — graphs

Training / Validation Graphs





52. ResNet18 — Direct Classification (KNN section run)

Source notebook: ResNet with ML.ipynb (inside Models_DL.zip)

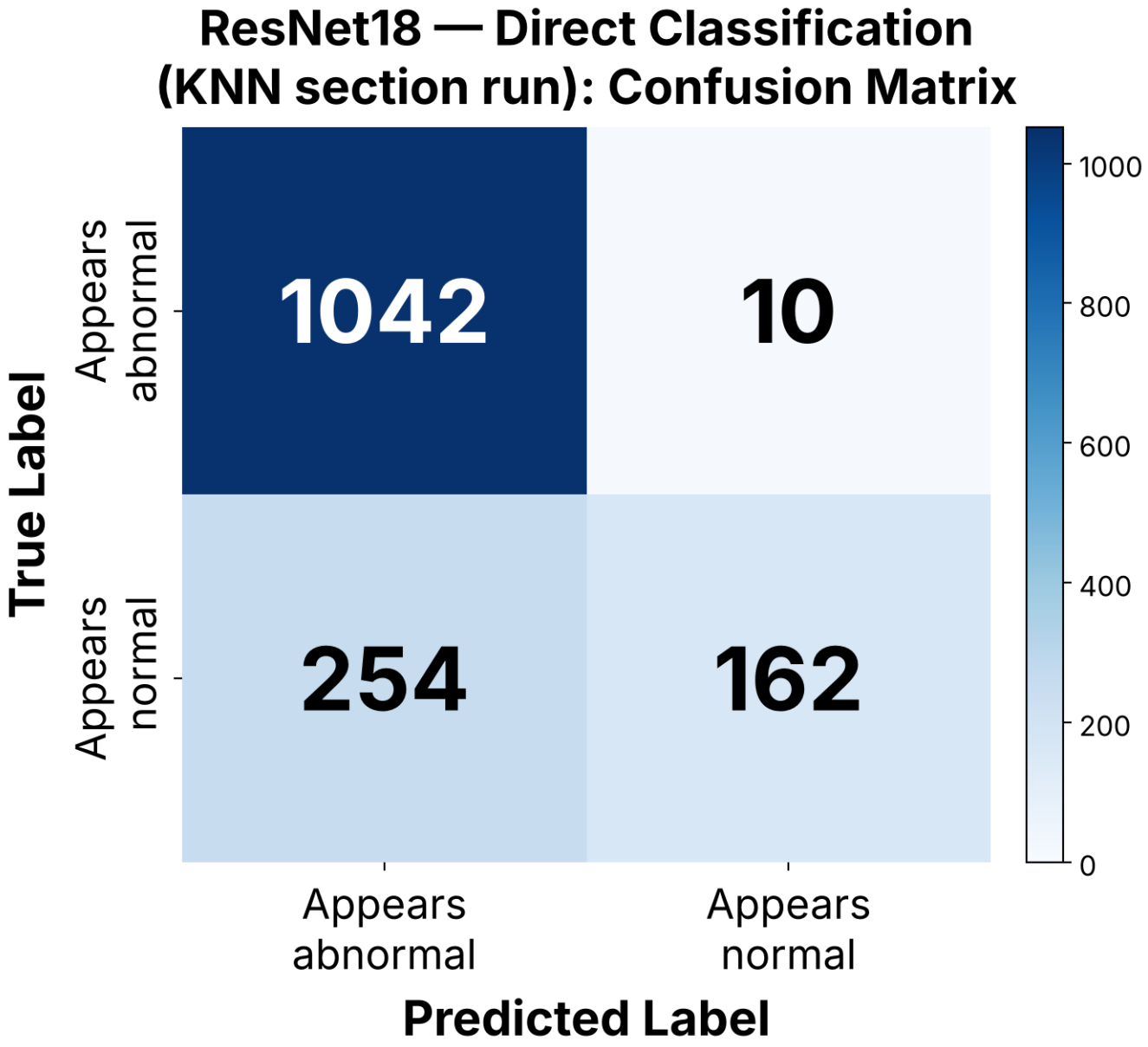
Accuracy: 82.02%

Precision: 84.31%

Recall: 82.02%

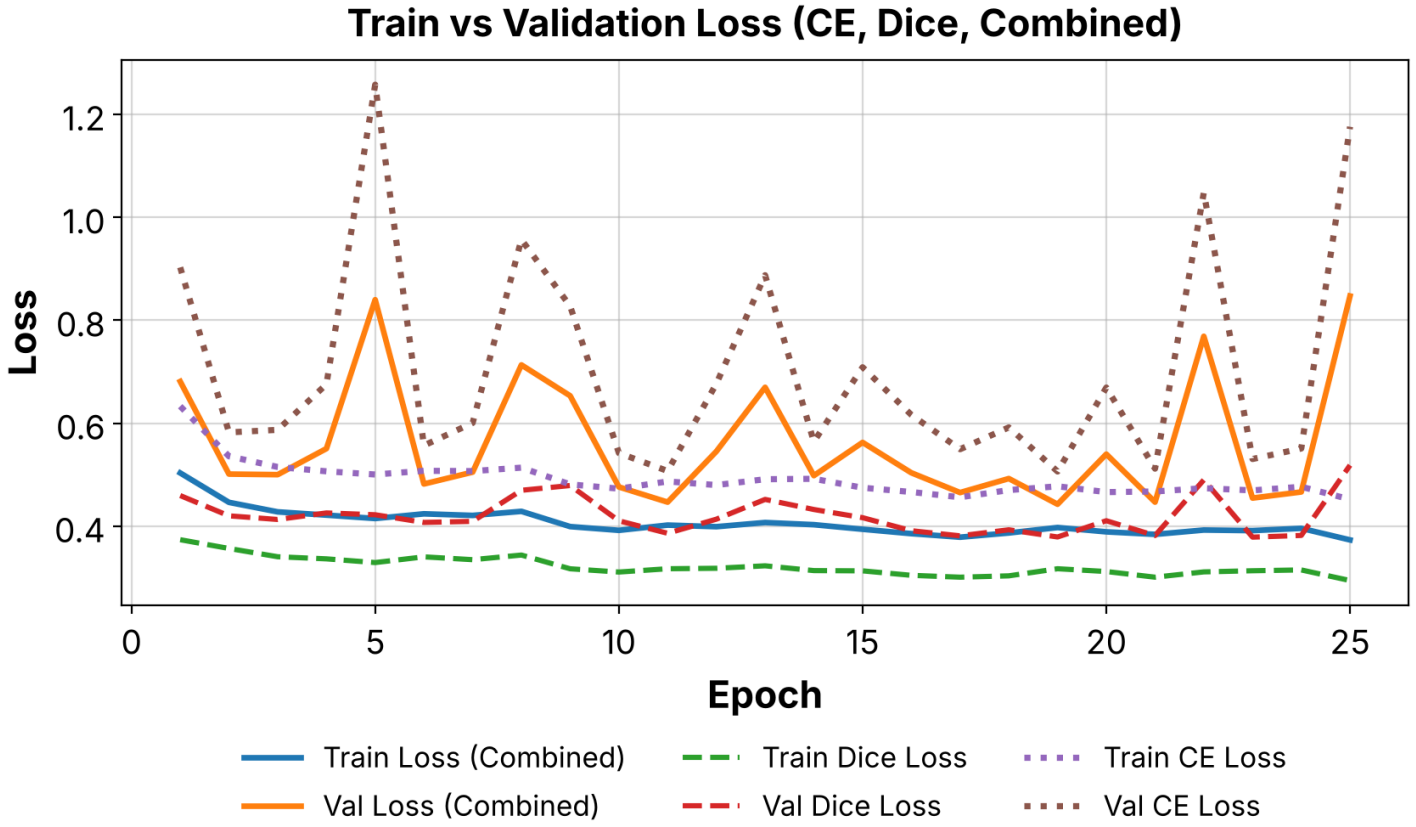
F1-score: 79.22%

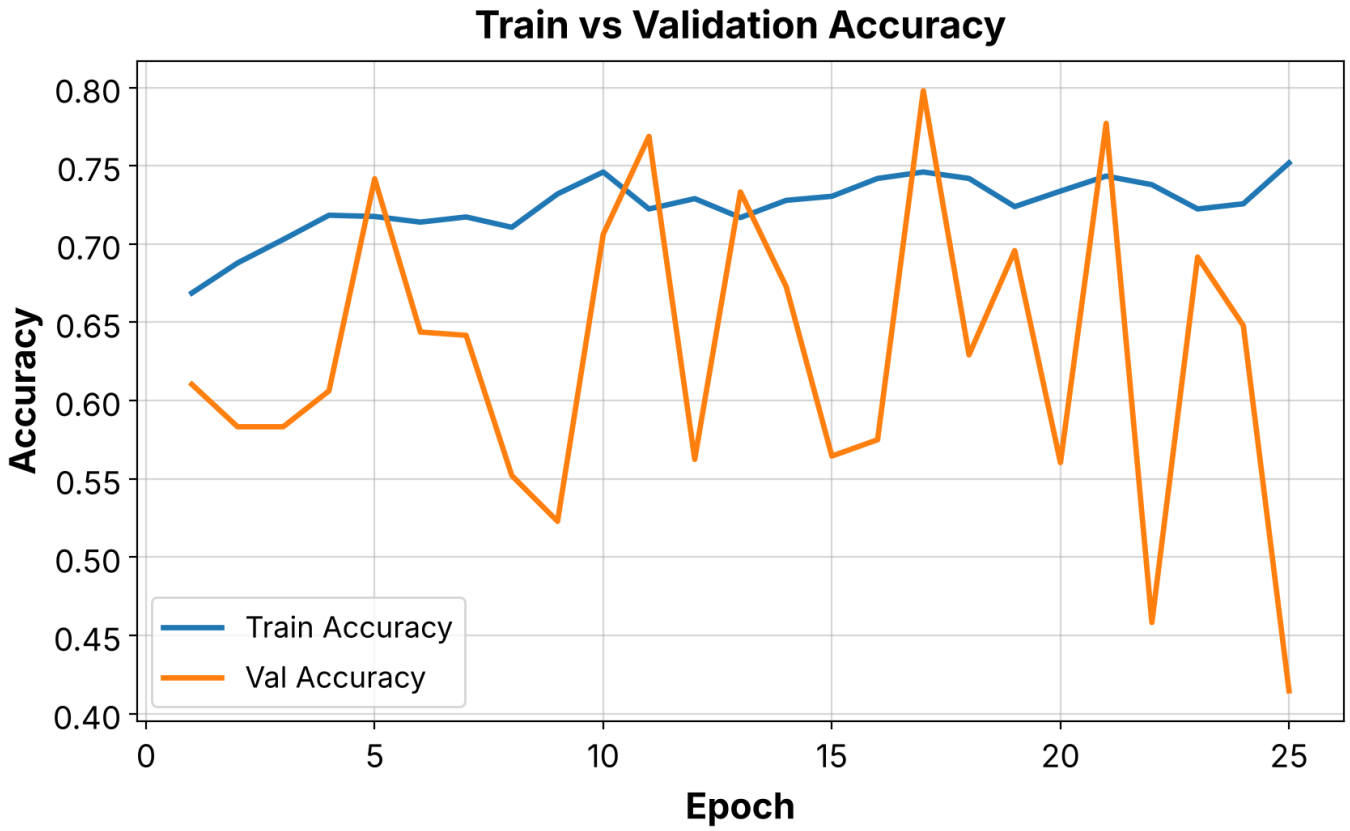
Confusion Matrix



52. ResNet18 — Direct Classification (KNN section run) — graphs

Training / Validation Graphs





53. AlexNet + Random Forest

Source notebook: AlexNet with ML.ipynb (inside Models_DL.zip)

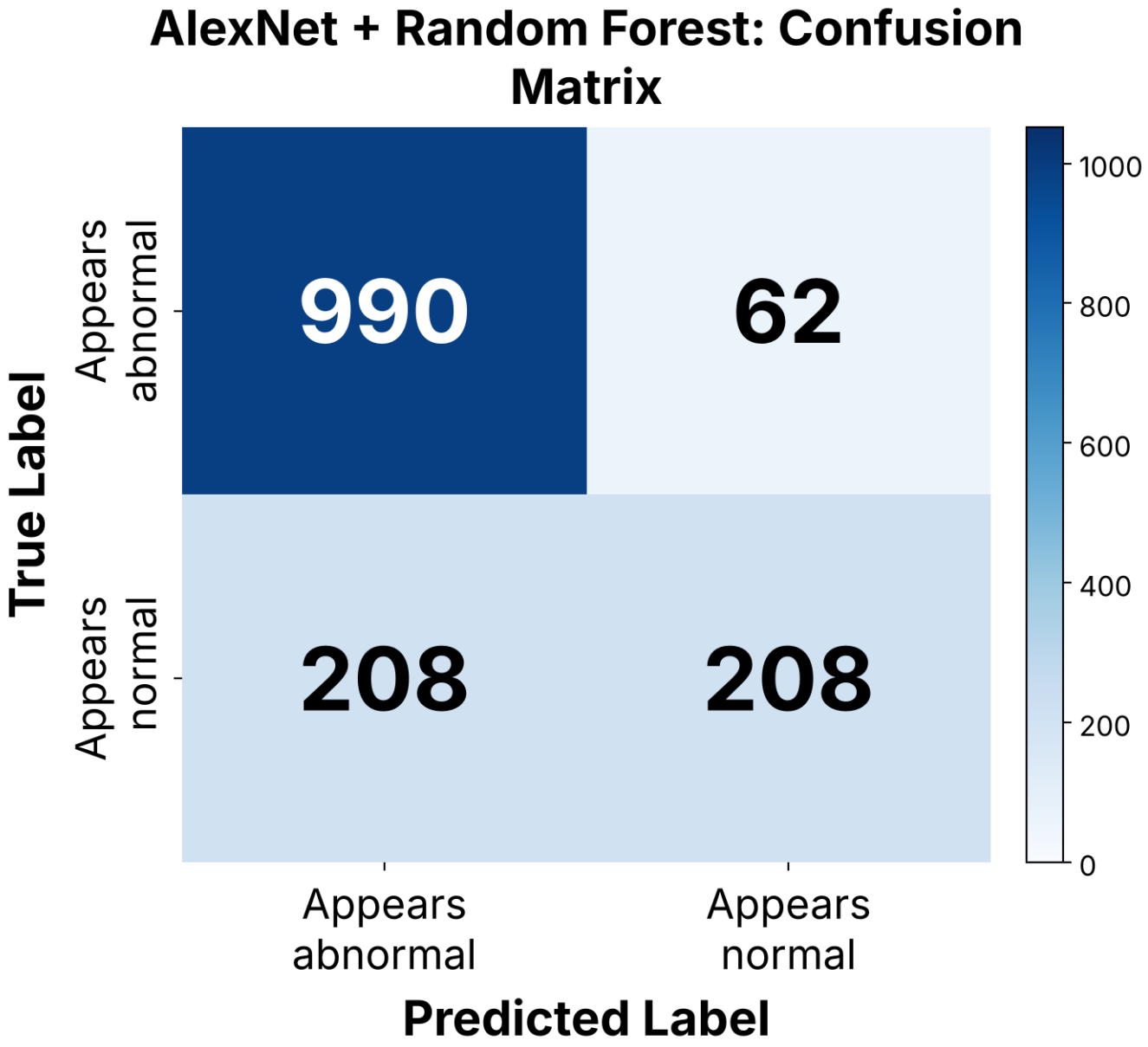
Accuracy: 81.61%

Precision: 81.05%

Recall: 81.61%

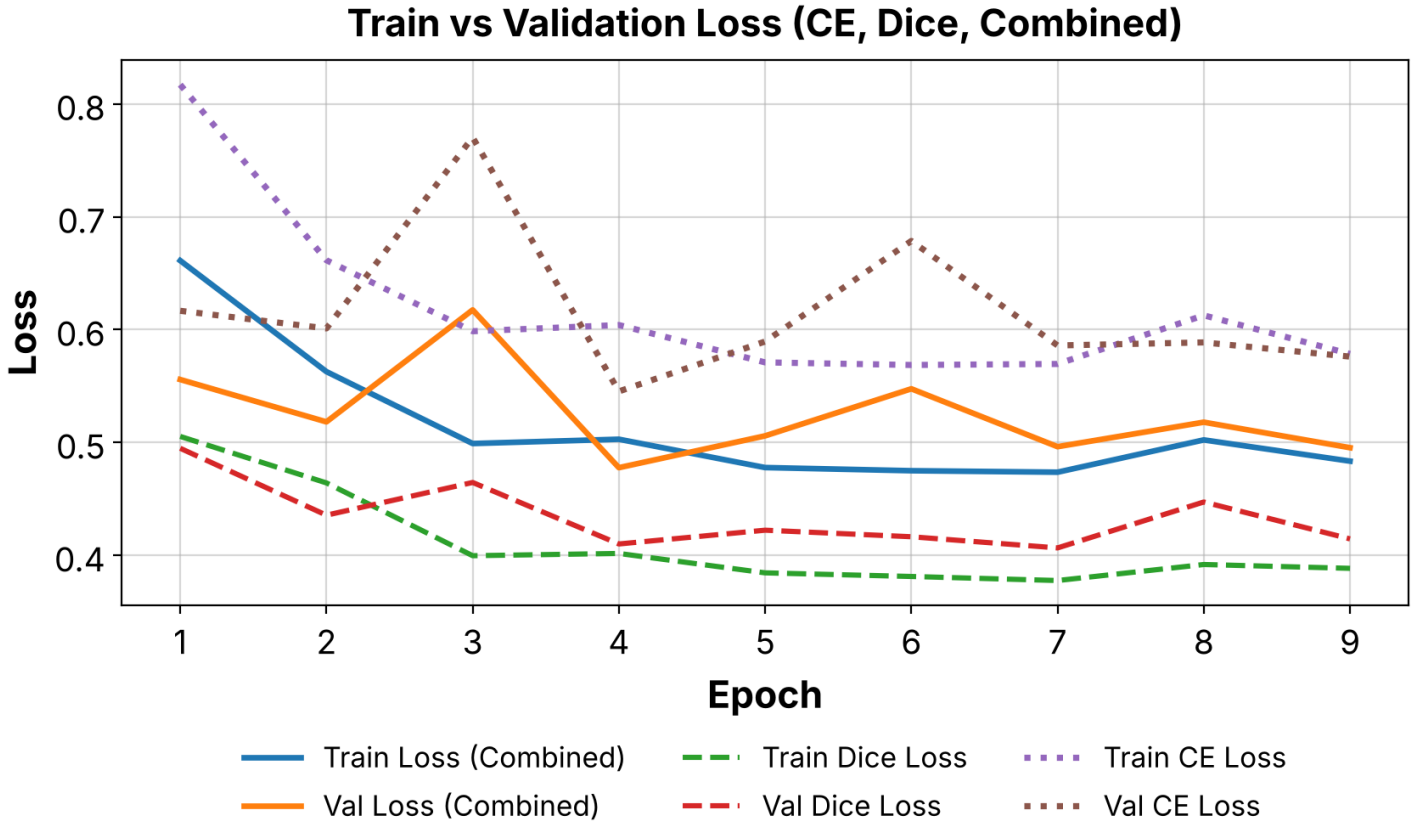
F1-score: 80.25%

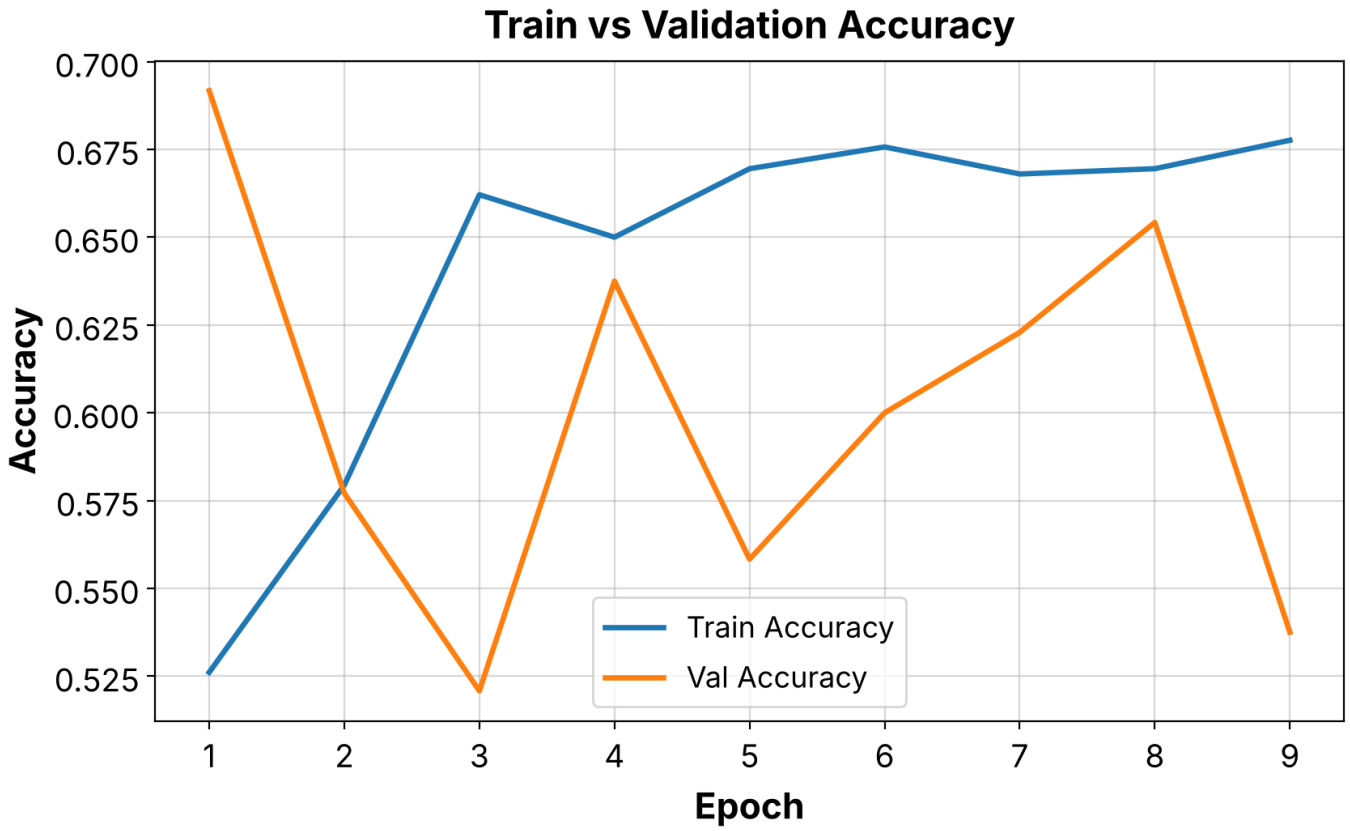
Confusion Matrix



53. AlexNet + Random Forest — graphs

Training / Validation Graphs





54. AlexNet + KNN

Source notebook: AlexNet with ML.ipynb (inside Models_DL.zip)

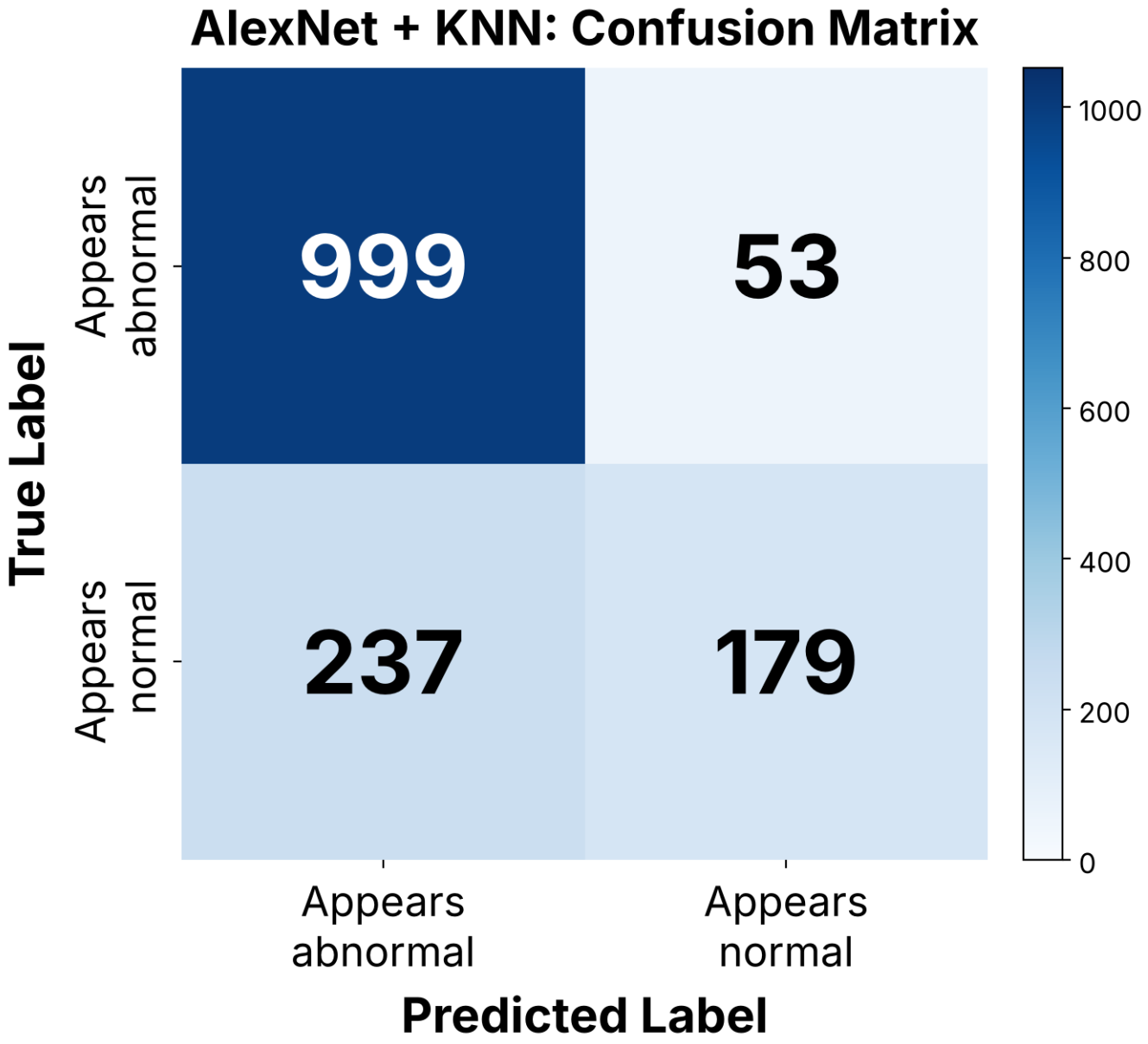
Accuracy: 80.25%

Precision: 79.79%

Recall: 80.25%

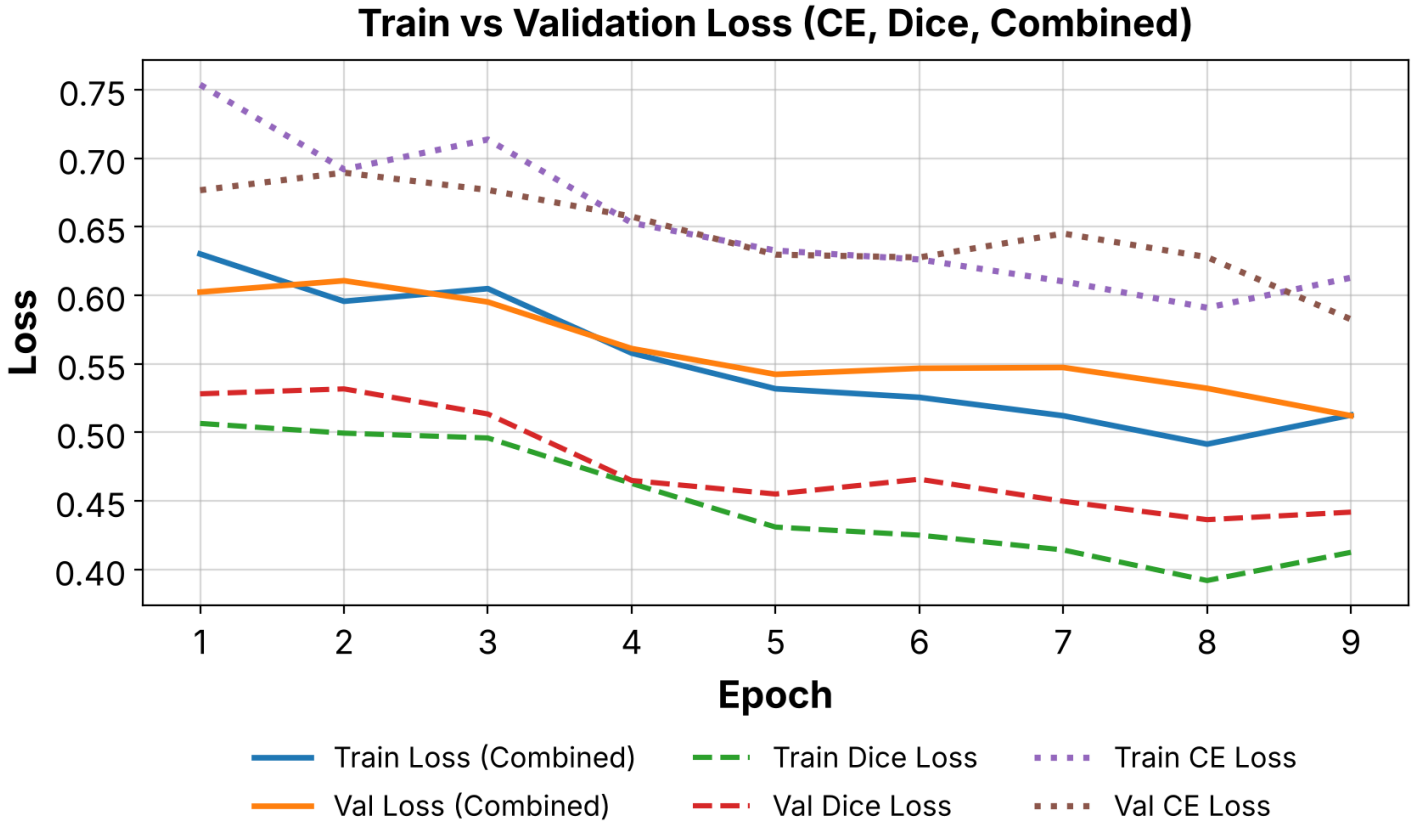
F1-score: 78.23%

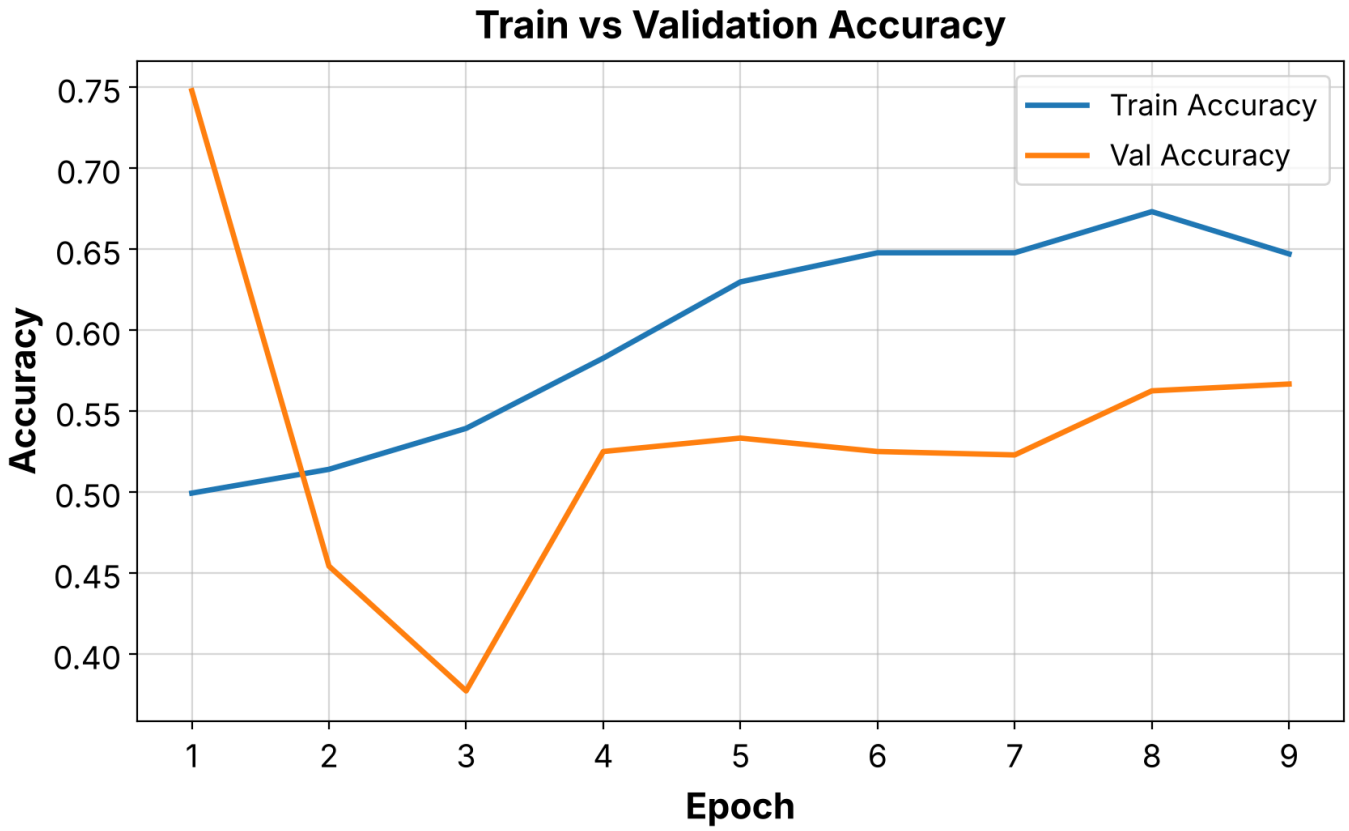
Confusion Matrix



54. AlexNet + KNN — graphs

Training / Validation Graphs





55. VGG16 + AdaBoost

Source notebook: Vgg with ML.ipynb (inside Models_DL.zip)

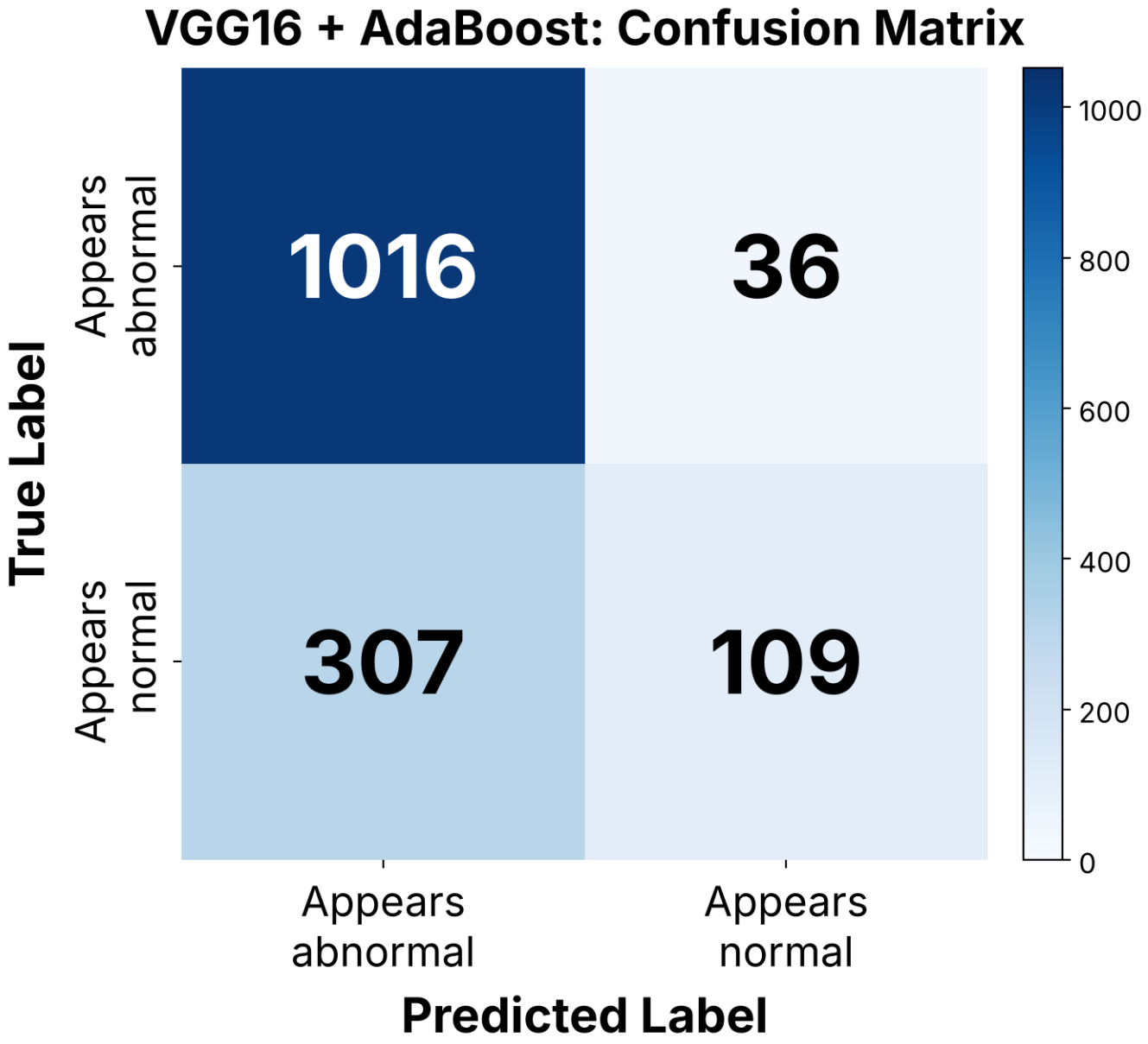
Accuracy: 76.63%

Precision: 76.34%

Recall: 76.63%

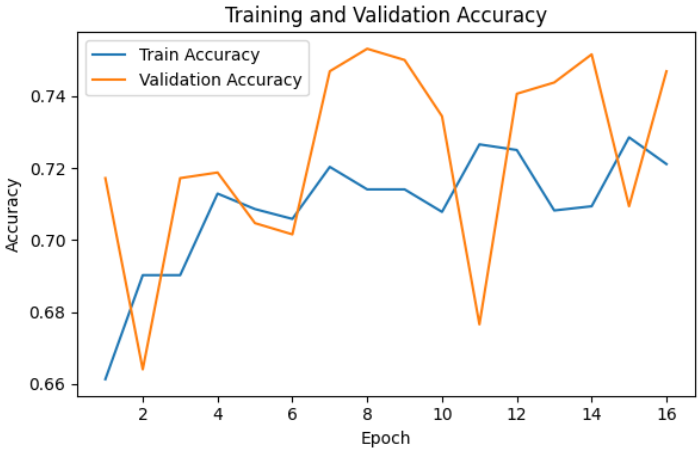
F1-score: 72.32%

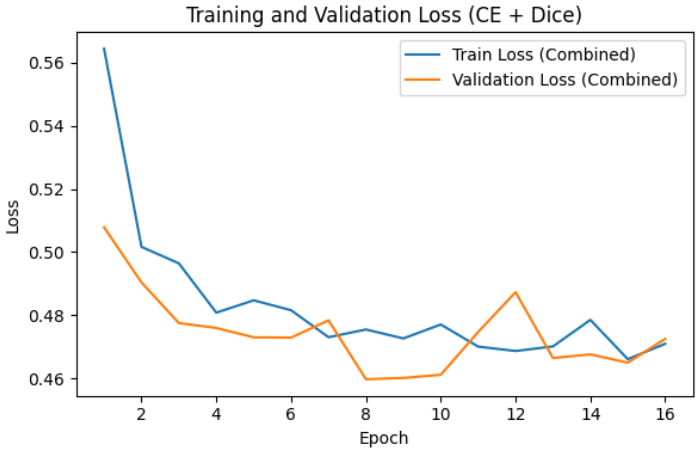
Confusion Matrix

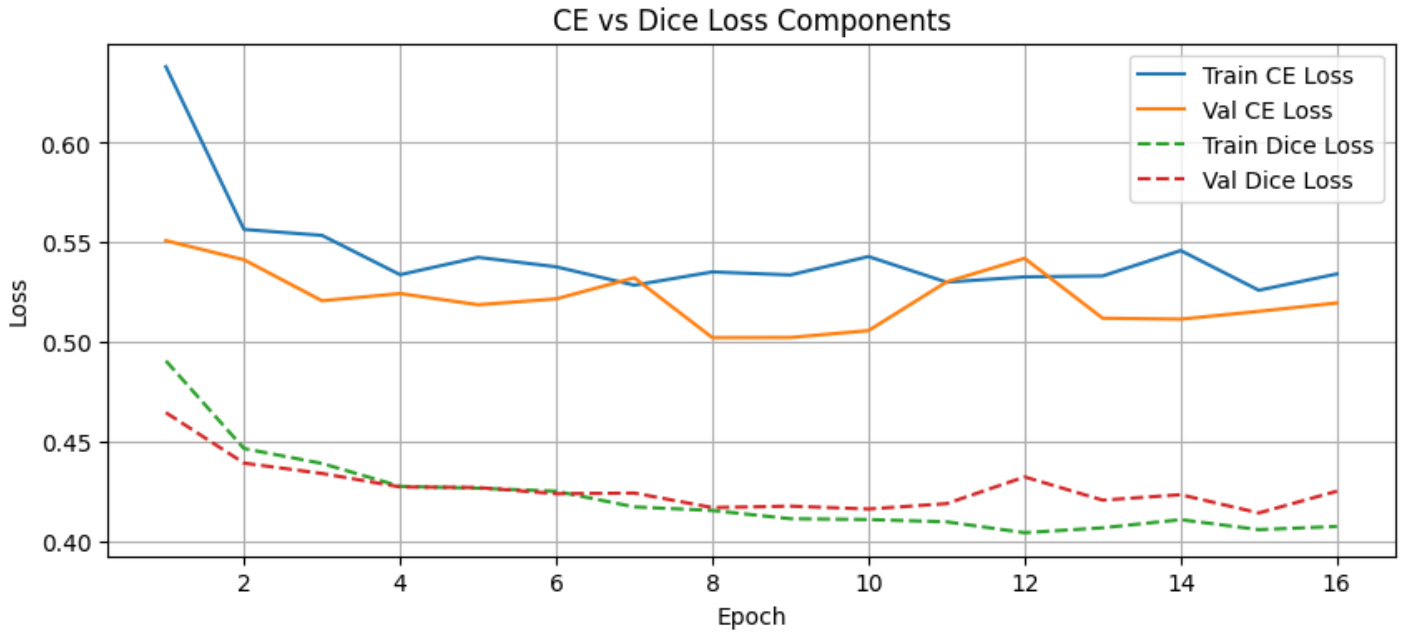


55. VGG16 + AdaBoost — graphs

Training / Validation Graphs







56. AlexNet + Gaussian Naive Bayes

Source notebook: AlexNet with ML.ipynb (inside Models_DL.zip)

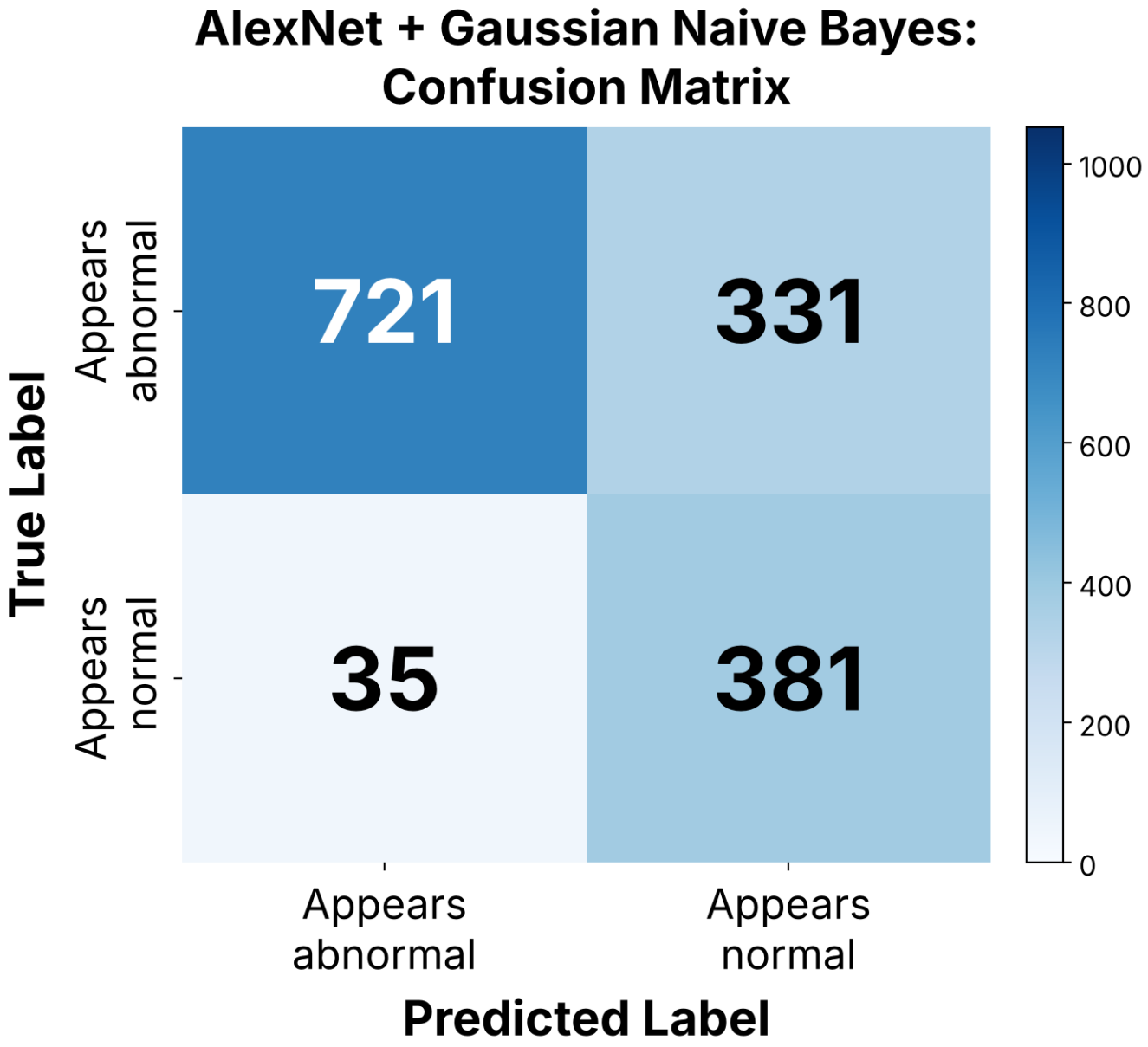
Accuracy: 75.07%

Precision: 83.51%

Recall: 75.07%

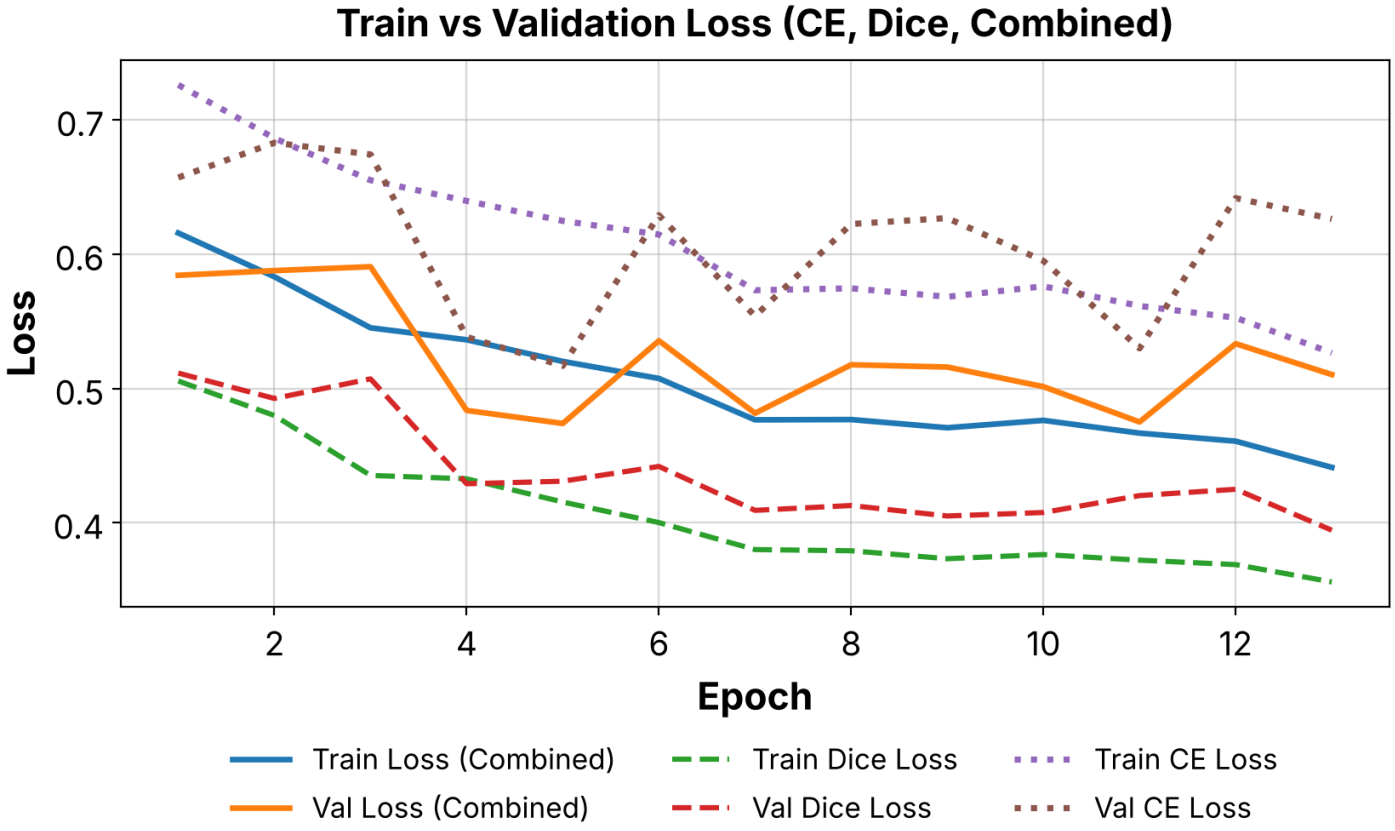
F1-score: 76.30%

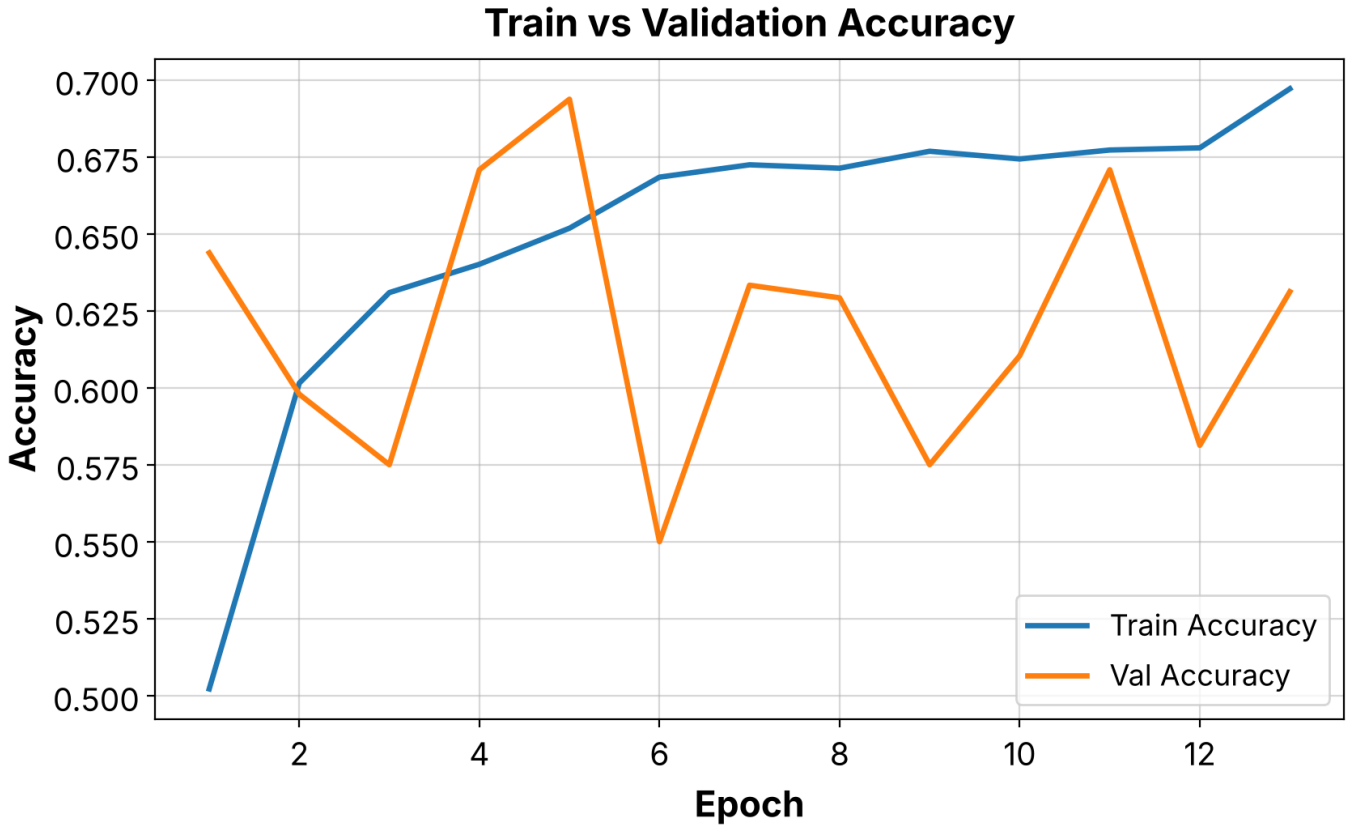
Confusion Matrix



56. AlexNet + Gaussian Naive Bayes — graphs

Training / Validation Graphs





57. VGG16 + Random Forest

Source notebook: Vgg with ML.ipynb (inside Models_DL.zip)

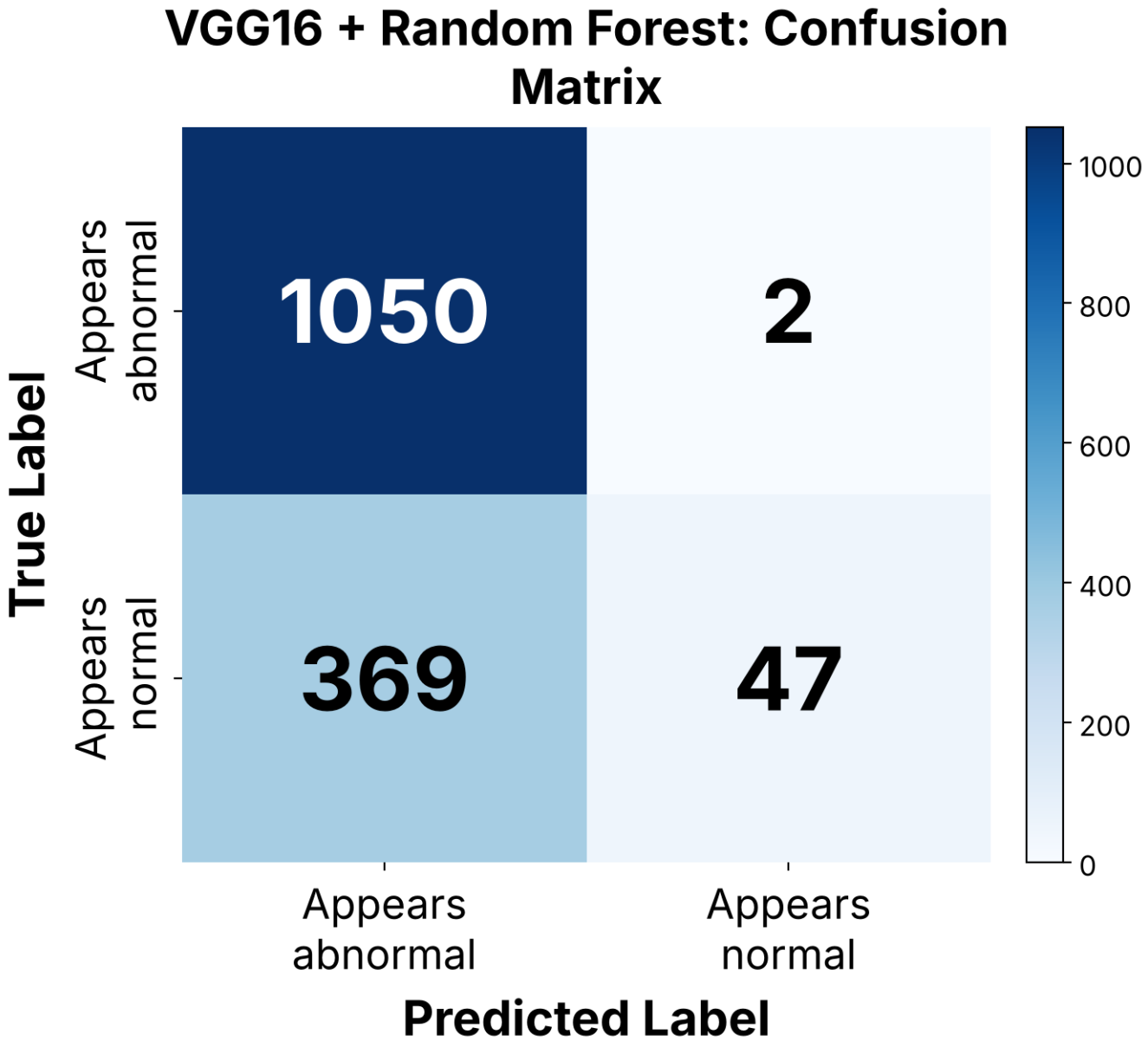
Accuracy: 74.73%

Precision: 80.21%

Recall: 74.73%

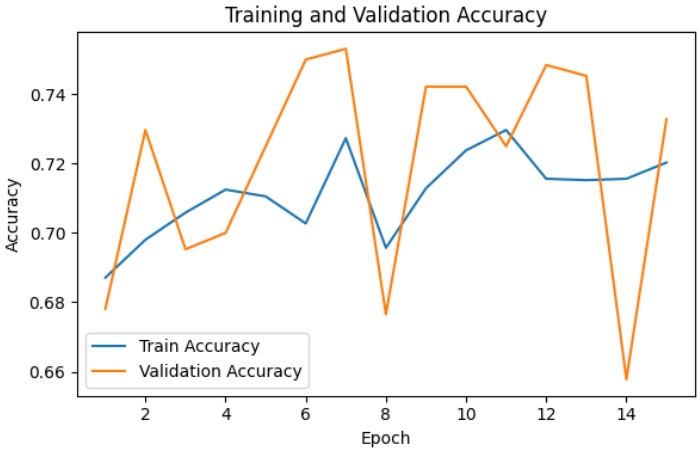
F1-score: 66.63%

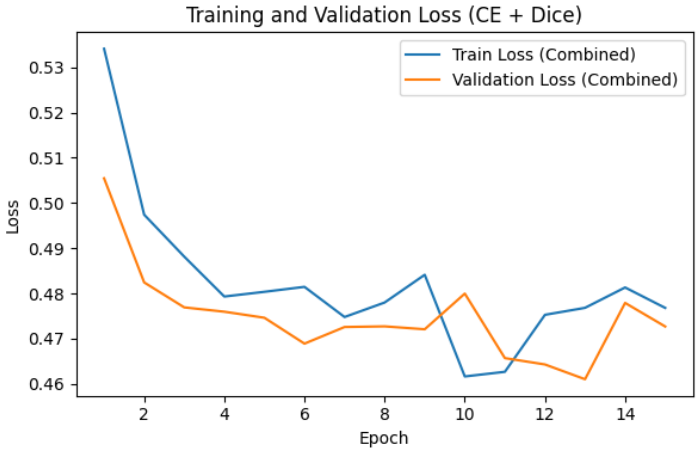
Confusion Matrix

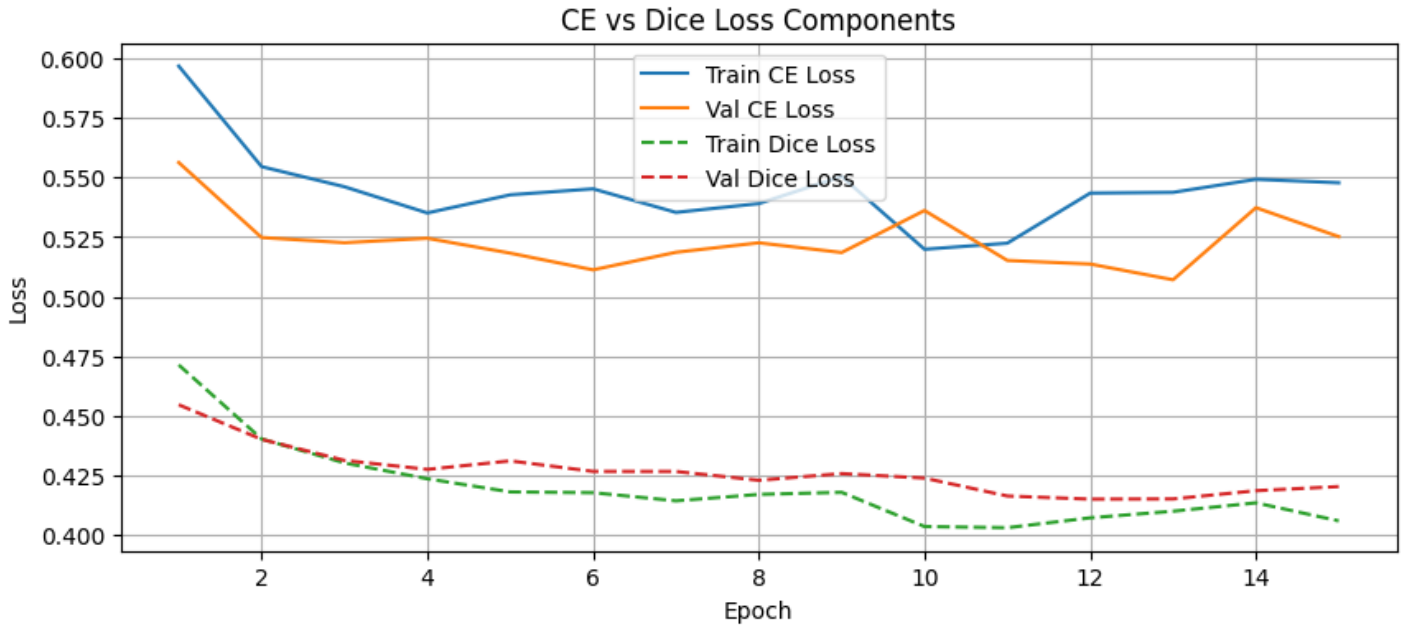


57. VGG16 + Random Forest — graphs

Training / Validation Graphs







58. ResNet18 + Gaussian Naive Bayes

Source notebook: ResNet with ML.ipynb (inside Models_DL.zip)

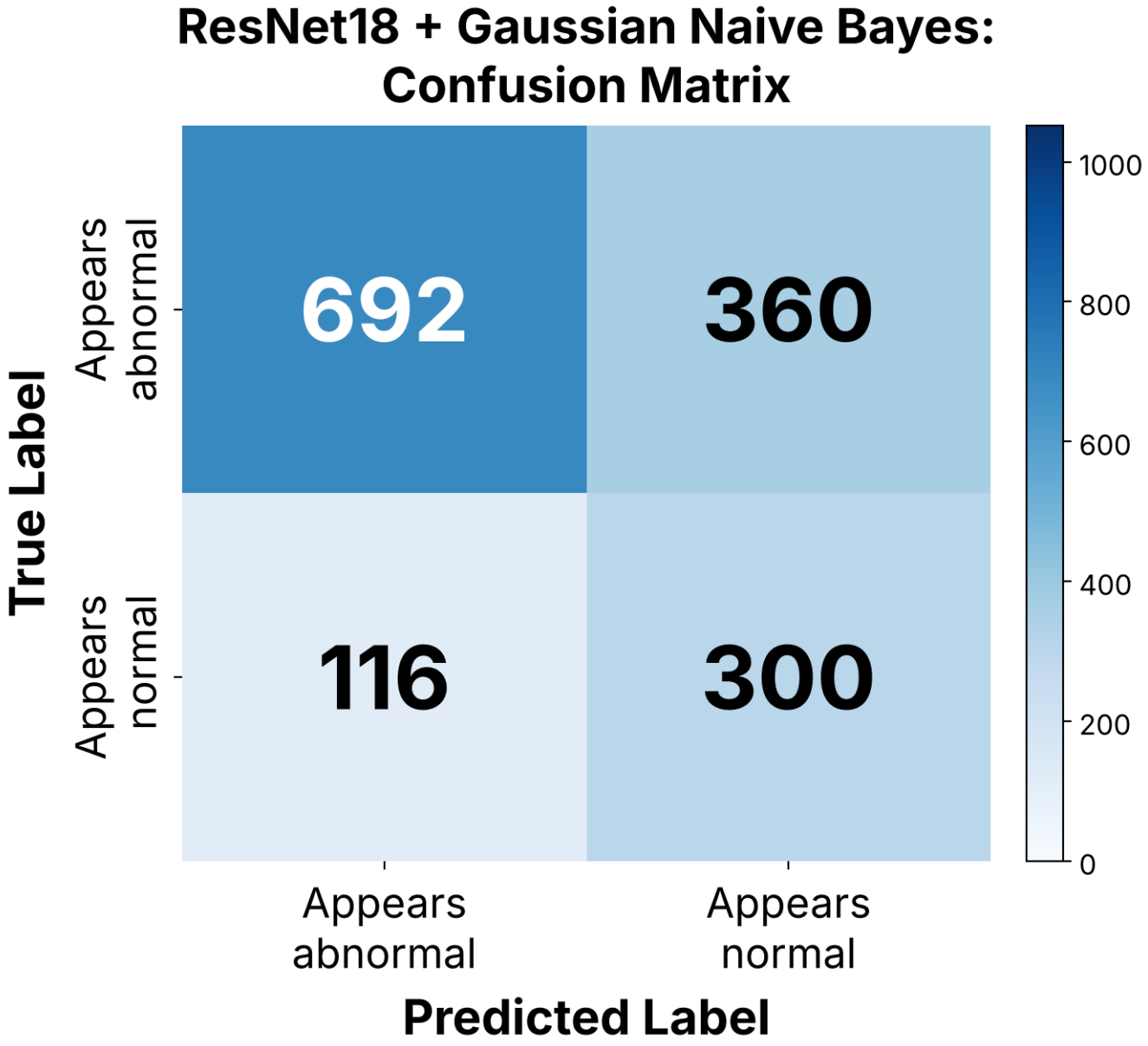
Accuracy: 67.57%

Precision: 74.25%

Recall: 67.57%

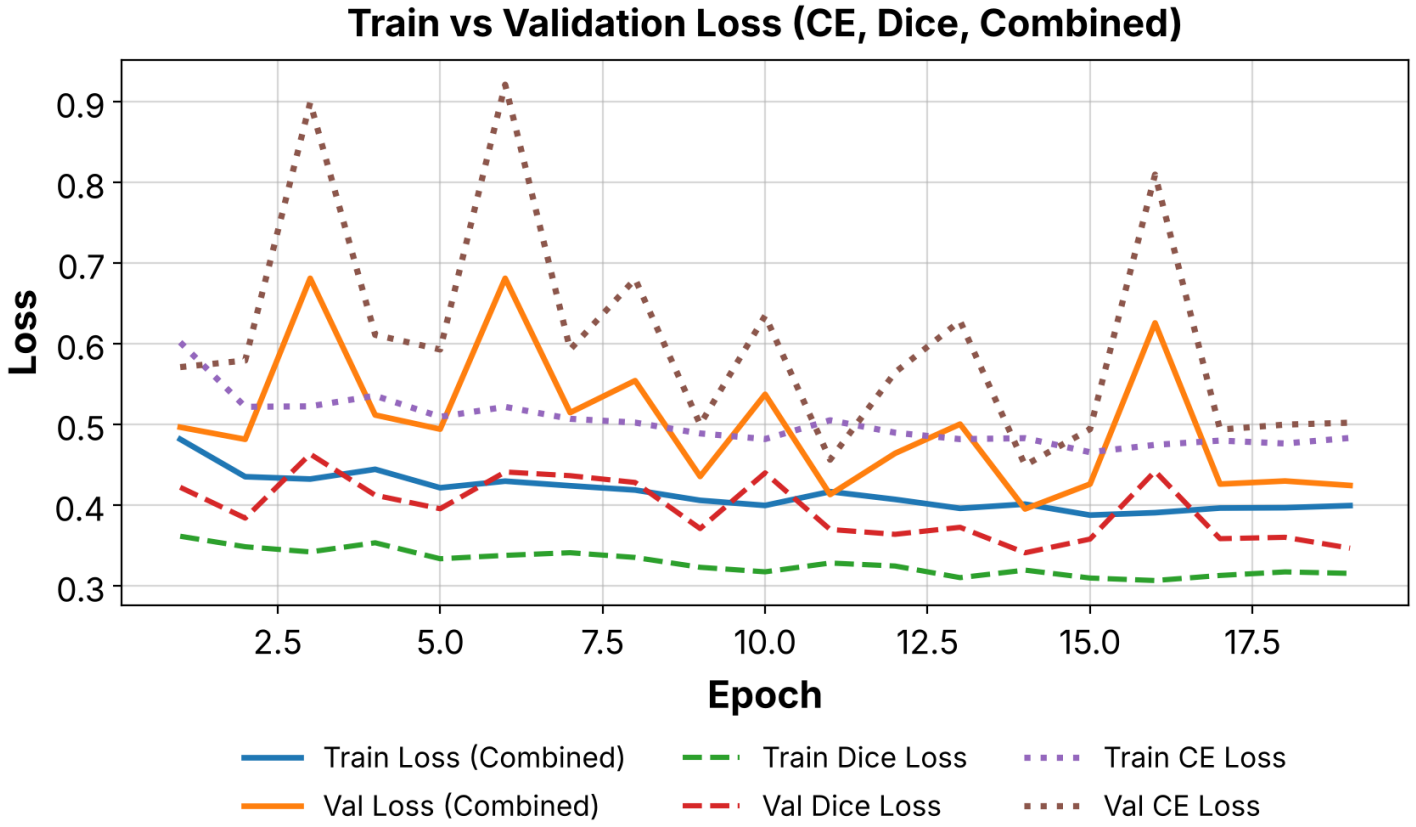
F1-score: 69.12%

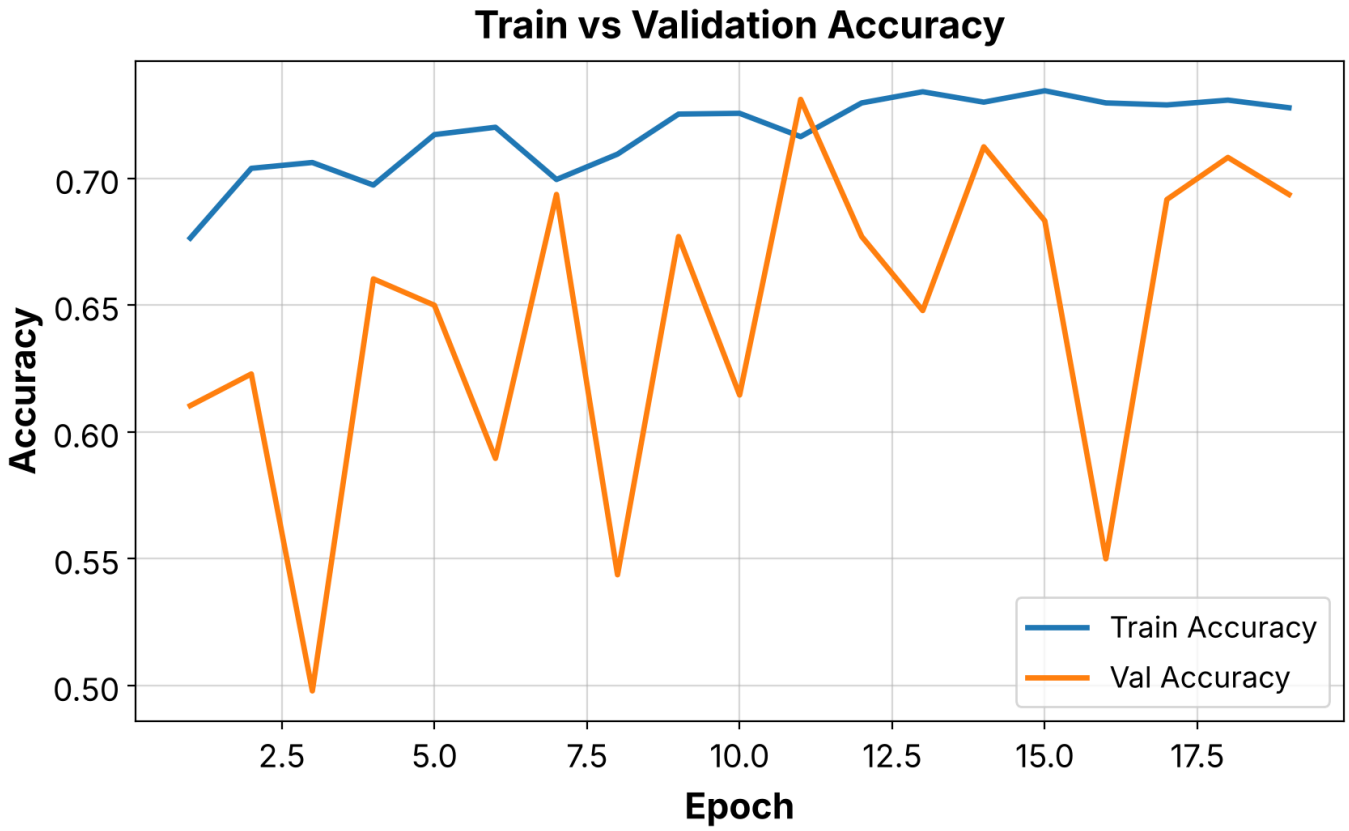
Confusion Matrix



58. ResNet18 + Gaussian Naive Bayes — graphs

Training / Validation Graphs





3. Notes

- Sources: the notebooks of Models_DL.zip (17 notebooks). Every value was read from the notebooks' saved outputs; nothing was re-run.
- Source column: "cell N" is the position of the cell inside the .ipynb file counted from 0 over all cells. Some notebooks also contain their own "# CELL n" comments, which use a different numbering and are not used here.
- Confusion matrices: copied from the notebook output where it prints a matrix; where only an image or per-class report exists, the counts were recovered exactly from per-class recall × support (1,052 / 416) and cross-checked with precision and accuracy. All matrices use the same colour map and the same fixed colour scale (0–1,052 images).
- Graphs: only values printed in the notebooks' training logs are plotted — nothing is estimated, smoothed or interpolated. Where the notebook printed a complete epoch log the graph is redrawn from it (with Train/Val CE curves added); the notebook's own original figures are shown wherever they exist. "CE" is torch.nn.CrossEntropyLoss; the combined loss is 0.5 × CE + 0.5 × Dice loss. In all scripts the validation images carry the training augmentations, so validation accuracy is lower and noisier than test accuracy.
- Where a notebook prints a test accuracy that disagrees with its own confusion matrix and weighted recall (Vgg kfold prints 81.06% against a recall of 84.20%; Vgg with Dice Loss prints 88.84%, which is the best validation accuracy), the table uses the accuracy implied by the confusion matrix, which equals the weighted recall. The page of such a row says so.
- The 5-fold section inside MobileNet with Preprocessing.ipynb trains torchvision VGG16 (not MobileNetV2) in its code, so it is listed as VGG16 — Direct 5-fold CV with preprocessing. The notebook prints 82.53% while its confusion matrix and weighted recall give 85.35%; the table uses 85.35%.
- AlexNet with ML.ipynb: the last section is titled "Adaboost" but the code trains KNN; it is listed as AlexNet + KNN (second run).
- Eight rows (VGG16 / ResNet18 / MobileNetV2 / AlexNet + SVM and + XGBoost) come from "Xgboost and SVM and all 4 models.ipynb", which reloads already-trained networks and prints no epoch log, so these rows have no training graphs.
- The Inception notebook prints only the combined loss and accuracy in its epoch log (CE and Dice are not logged separately), so its graphs are the notebook's original figures.